

Chapter 11

A Zoo of Groups, the Hunt

In line with the promise made in the **Purpose** chapter, we will now enter the group-theoretic zoo and begin looking at various concrete examples of groups the Indiana Jones style - while doing so we will be wearing two hats, one hat will read *a blackboard theoretician* and the other hat will read *a field practitioner*.

On this short tour our main job will be the development of the group-detection skill. See that creature lurking in the distance over there? Is it a wolf? Is it a fox? Is it a dog? What is it? Do we or do we not add this animal to our group-theoretic collection?

The act of juggling both the theoretical and the concrete bits and pieces while answering the above questions should help the readers reconcile and settle in their minds the main ideas of what a mathematical group is, what works for such a group and, just as, if not more, importantly, what does *not* work for a group - it is often quite interesting and educational to understand exactly *why* a candidate set and a candidate operation fail to form a group, what was the main reason for such a failure?

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Roughly speaking, a fair amount of problems in mathematics can be sorted into two neat piles:

- a problem to find (compute) and
- a problem to prove

The problems in each of the above piles, besides sometimes being tightly interconnected, have their own characteristic texture, level of interest and cultural or pedagogical value.

Compare, for example, the following three problems *to find*:

- find the area of a planar triangle, given the lengths of its base and height
- a rigid sphere of radius r was dropped on the floor and shattered into pieces. Given only the Euclidean compass, straight edge and a sheet of paper, find the length of the sphere's radius r (with mathematical precision)
- find the minimal single-variable polynomial with integer coefficients, whose roots are the following four real numbers:

$$\cos^4 \frac{\pi}{9}, \cos^4 \frac{2\pi}{9}, \cos^4 \frac{3\pi}{9}, \cos^4 \frac{4\pi}{9}$$

As another example, consider the following two problems *to prove*:

- using the method of mathematical induction, prove that the sum of the first n consecutive natural numbers is such and such
- prove that n great circles, no three of which intersect at any one point, partition a sphere into $n^2 - n + 2$ spherical regions

In that respect, the theory of groups is no exception from the other, vaguely defined, mathematical disciplines and it has a huge number of very interesting problems *to find* and problems *to prove*.

However.

Logically, it makes sense to *first* spend a reasonable amount of time on putting the newly learned information about the various sets of group axioms to a practical use and begin developing the skill of *recognizing a group* in the zoo of *sets* and the respective *operations* defined on these sets.

Intuitively speaking, the next time the set *foo* and the operation *bar* roll across our faces, will we recognize that it was *a group* such and such that did it? Or will we ignorantly shrug it off and keep walking blissfully unaware of what just happened?

The next time we get hit by a truck that reads the set *booh* and the operation *yah*, will we see it for what it is, a group or not a group? Or will we simply run the other way?

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On a more technical note, the modern theory of groups is shrapneled with literally hundreds and hundreds of highly specific terms, all which, of course, mean something and pertain to certain features and properties of groups. Before we begin comprehending this group-theoretic lexicon, it is only beneficial to, right away, look at many many samples of groups and, thus, develop a corpus of concrete material on which the relevant abstract concepts can be tested.

The manner in which we will be developing the said corpus of concrete groups will be characteristic of *working with groups* in general.

As such, the upcoming exercises in this section of the book will be more of a problem *to prove* variety for the following reason.

Many, but not all, traditional textbooks on the subject formulate this type of problems as follows:

show that the set this and that and the operation such and such, defined on the said set, is a group

But in the **What Makes It Perfect** chapter we promised our readers that one way to develop a proper mathematical culture in students is to frame the problem statements of the relevant exercises in a way that does not reveal the answer in any way, shape or form.

Why does framing the problem statements of the relevant exercises in a way that does not reveal the answer in any way, shape or form facilitates the development of a proper mathematical culture in students?

Framing the problem statements of the relevant exercises in a way that does not reveal the answer in any way, shape or form facilitates the development of a proper mathematical culture in students because this is exactly a scenario that is faced by the research mathematicians on the daily bases – solving open-ended problems.

Think about it.

As people who plan on *moving the proverbial needle*, we will be operating at the very edge of the existing textbook knowledge staring into the abyss of the unknown and we will be constantly facing the situation when not only a ready-made cookie-cutting recipe for a given problem does not exist, but the very *tools* that can make such a recipe a reality may not exist also.

What then?

Aha.

It will then be *our task* to create the next bit of the-not-yet-existing-textbook-knowledge and it may be on us to create the said tools, possibly from scratch.

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In order to succeed in such an endeavor, however, we should be able to reason and operate independently, from square one and our own.

That is exactly the skill that is developed by the exercises that do not reveal the answer in any way, shape or form - for in this case the students are nurtured and shepherded into the needed mode of thinking independently, from first principles and relying only on their own creative juices.

To that end, most of the upcoming exercise will be framed in a way that facilitates the development of the above skill:

does the set foo and the operation bar , defined on that set, form a group? Yes or no? You tell us

Once in a while we will slip in there an even more devious problem where we will remove the name of the candidate operation altogether:

can the n -th roots of unity form a group?

In general, such a question is a challenge because in order to answer it in the positive, we have to find a proper operation under which the set of the n -th roots of unity is a group and in order to answer such a problem in the negative, we will have to generate a convincing argument that at least under *these popular* operations the set of n -th roots of unity definitely cannot form a group.

Either way, in order to address the above questions we will have to step into the grown-up shoes and deal with the said *open-endedness* of the way in which the problems are formulated.

That is the idea and that is the agenda.

It is expected, then, that during the solution of the upcoming batch of exercises our readers will juggle the technical information from *some* already studied set of group axioms and reason from first principles on their own, thus developing the ability to generate the correct proofs.

We will begin by studying the examples of groups that many readers should feel comfortable with - the groups that are based on the *numeric* sets, such as integers, rational numbers, real numbers and complex numbers under the two popular operations of multiplication and addition.

Initially, honoring the spirit of this text, we will be very slow and thorough, but after several detailed solutions of the group-detection exercises, we will be providing less and less hand-holding and guidance and eventually we will converge into a mostly grown-up narrative of cutting to the chase right off the bat.

At one point, we will ask our readers *to invent/discover* a group on their own based on the fodder supplied by us, a set of just two numbers, separately, a planar triangle and four dedicated points associated with such a triangle and, separately, a couple of straight lines and an equilateral hyperbola,

because there is a great educational value not only in the process of *solving* problems but in the converse process of *constructing* interesting mathematical problems as well.

Exercises 1: Numeric Sets Under Addition and Multiplication

In the majority of the upcoming exercises we will be asked to answer the following question **Q**:

Q: is the set S and the operation $\circ$ defined on S a group?

The intended, mathematically sound, answer to such a question has absolutely no room for *guessing* the one, out of the two, *yes* or *no*, possible, answers.

On the contrary.

The intended answer to the question **Q** amounts to actually generating, however small and simple but still, a *correct mathematical proof* that the candidate set and the candidate operation defined on the elements of that set *is or is not* a group.

In other words, the only proper solution of each and every exercise in this section that benefits a reader is a correct yourself-generated proof that clearly demonstrates that the candidate duo *is* a group or is *not* a group. In the latter case it is also a good idea to list or enumerate *all* the reasons *why* the candidate pair is *not* a group.

In the introduction section of this chapter we already explained *why* all of such exercises have the open-ended formulation - because such a formulation of the relevant exercises develops and nurtures in the students the desired mental habits of:

- canceling the preconceived notions
- generating their own lines of correct and mathematically sound reasoning
- deducing the answer(s) from first principles
- analyzing a given problem starting from its foundation
- not relying on and avoiding the wholesale cookie-cutting recipes
- searching for a genuine solution path
- combining the relevant units of technical knowledge in, perhaps, novel or unexpected ways (improvising) and overall

which amount to *thinking like* a mathematician.

Since now we are in the possession of no less than *five* types or five families of group axioms, one way to solve the next batch of problems pivots on the idea of:

filtering the incoming matter through the chosen set of group axioms

If the result of that filtering process is *positive* then the answer to the question **Q** is also *positive* - yes, the candidate set S and the candidate operation $\circ$ defined on the elements of S is a group.

If the result of that filtering process is *negative* then the answer to the question **Q** is also *negative* - no, the candidate set S and the candidate operation $\circ$ defined on the elements of S is *not* a group.

The said result of the said filtering process is obtained by the means of answering the following, sample, set of *yes/no* questions:

- **Q1:** does the candidate operation $\circ$ act on (precisely) *two* elements of the candidate set S ?
- **Q2:** is it the case that for any two elements a, b of the set S there exists a uniquely determined element c of S such that $a \circ b = c$?
- **Q3:** is the candidate operation $\circ$ associative?
- **Q4:** is it the case that under the candidate operation $\circ$ the candidate set S possesses the unique double-sided identity element e such that for all the elements a of S the relationships $a \circ e = e \circ a = a$ hold?
- **Q5:** is it the case that under the candidate operation $\circ$ every element a of the candidate set S possesses a unique double-sided inverse element b such that $a \circ b = b \circ a = e$?

Only if the answers to *all* of the above sample questions are a *yes*, will the result of the said filtering process be *positive* and the answer to the question **Q** will be a *yes* - the candidate set S and the candidate operation $\circ$ defined on S is a group.

However, as soon as the answer to any single question above is a *no*, the result of the said filtering process will be short-circuited into *negative* and the answer to the question **Q** will be a *no* - the candidate set S and the candidate operation $\circ$ defined on S is *not* a group.

In that case, the first question answered in the negative reveals the earliest, in some sense, reason why the proposed set S under the proposed operation $\circ$ is not a group. It is, nonetheless, educational to go further and list *all* the reasons why the proposed set S under the proposed operation $\circ$ is not a group.

The above sample set of group-filtering questions is based on the set of the group axioms discussed in the **Take 1** section of the **Group Axioms** chapter. That set of the group axioms demands or assumes both the existence and the uniqueness of the double-sided identity element and of the double-sided inverses for each element of the candidate set under the candidate operation.

But we are the smart bears now. We are equipped with *four* more sets of group axioms.

Across these four sets of group axioms, we generated the relevant proofs of the double-sidedness and of the uniqueness of the respective elements of a group and we are not afraid to use these four alternative sets of the groups axioms in the above filtering process.

If we decide to do that then we will have to adjust the filtering questions accordingly:

- **Q4:** is there a left identity and
- **Q5:** is there a left inverse for each element of the candidate set under the candidate operation?

and so on.

In any case, regardless of which specific set of group axioms we decide to use in the above filtering process, the characteristic property of such a process is the following:

- if the answer to the *current* question is *a yes* then such a positive answer does not automatically guarantee that the candidate duo *is* a group and we have, potentially, more work to do but
- as soon as the answer to the current question is *a no* then such an answer is enough to stop the said filtering process altogether and announce that nope, the candidate duo is not a group

For example, say, we have answered the first two questions above, the question **Q1** and the question **Q2**, in the positive.

At that point in the process we do know that we have, as per **Definition 10**, an operation that is binary or that the proposed set is *closed* under the proposed operation.

However, the fact that the candidate operation $\circ$ defined on the candidate set S is binary alone is not enough to deterministically answer the question **Q** overall - we still have more work to do and more intermediate questions to answer.

More technically, if we agree to designate the pair of a set S and an operation $\circ$ defined on S as $\langle S; \circ \rangle$ then the answer to the question **Q** can be obtained by stringing the answers to the questions **Q1-Q5** together with the logical AND operators as follows:

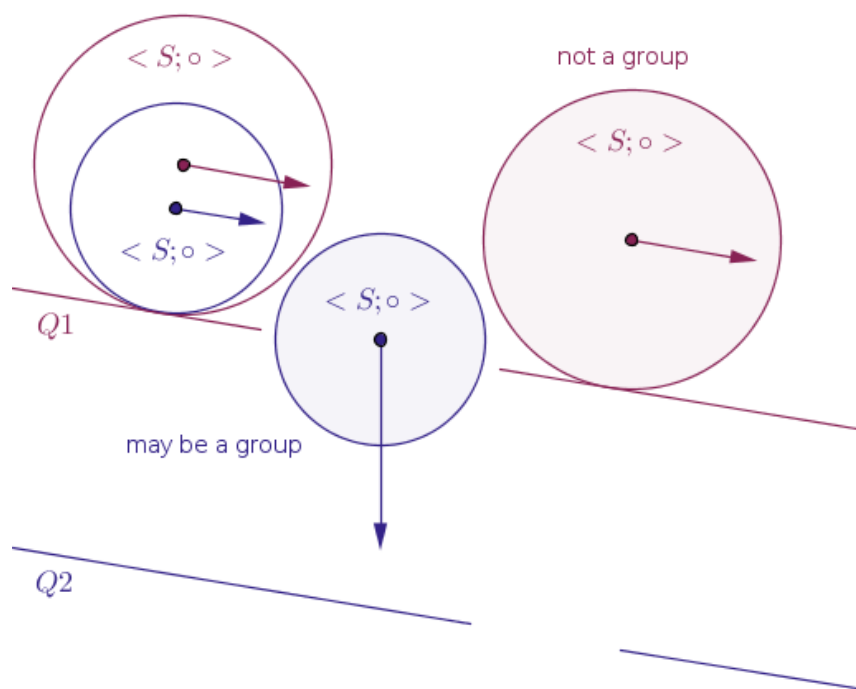
```
if ( Q1 AND Q2 AND Q3 AND Q4 AND Q5 )
{
    yay, the (unordered) pair  $\langle S; \circ \rangle$  IS a group
}
else
{
    nope, the (unordered) pair  $\langle S; \circ \rangle$  is NOT a group
}
```

More loosely, the above group-filtering process can be associated with a toy model of sorting coins by their linear size by rolling the said coins down a series of inclined planes that have the appropriately sized openings in them.

We, first, roll the model candidate $\langle S; \circ \rangle$ coins down the inclined plane **Q1** that has the **Q1**-opening in it that is just big enough to let through only the coins that answer the question **Q1** in the positive.

If the candidate $\langle S; \circ \rangle$ coin, such as the coin shown in red in the diagram below, is *too big* for the **Q1**-opening, that is if that coin does not answer the question **Q1** in the positive, then such a coin will keep on rolling down the inclined plane **Q1** without falling through the **Q1**-opening and will be routed to the exit from the sorting device that reads *This Coin Is Not a Group*.

If the candidate $\langle S; \circ \rangle$ coin, such as the coin shown in blue in the diagram below, is *small enough*, that is if that coin does answer the question **Q1** in the positive, then such a coin will fall through *into* the **Q1**-opening down onto the inclined plane **Q2** below (Figure 11.1):



where the entire process will repeat again.

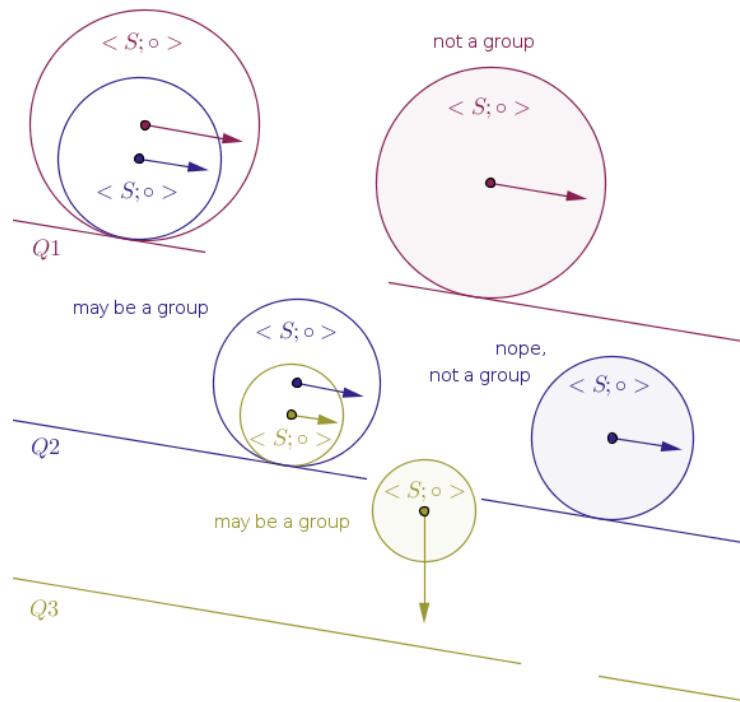
If such a coin is too big for the **Q2**-opening, that is if that coin does not answer the question **Q2** in the positive, then it will roll right over that opening and will be routed to the exit from the sorting device that reads *This Coin Is Not a Group*.

If a candidate a coin, such as the coin shown in green in the diagram below, is *small enough*, that is if that coin does answer the question **Q2** in the positive, then such a coin will fall through into the **Q2**-opening onto the inclined plane **Q3** below where the entire process will repeat again and so on.

We, thus, see that as the coins, read *unordered pairs* $\langle S; \circ \rangle$, roll along more and more inclined planes, some of these coins will be kicked out and routed to the *Not A Group* exit of our group-theoretic sorting contraption, while some of these coins, which, as it turns out, possess more and more restrictive properties, will be allowed to compete and stay in the game a little longer.

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But at each inclined plane Q_k each coin, group candidate, will be tested for its size (Figure 11.2):



It follows, then, that only the coins that will fall through into the **Q5**-opening in the inclined plane **Q5** will be the coins that possess the most restrictive properties that we are looking for and, hence, it is only these coins that will be routed to the exit of this sorting device that reads *This Coin Is a Group*, while all the other coins will be routed to the exit of this sorting device that reads *This Coin Is Not a Group* (Figure 11.3):



With the preliminaries out of the way, let us see how this filtering process may work in practice.

Exercise 11.1: is the set of integers $\mathbb{Z}$ coupled with the operation of *multiplication* $\cdot$ defined on the elements of that set a group?

Solution: of this exercise amounts to trawling the given set $\mathbb{Z}$ and the given operation $\cdot$ defined on that set through the series of questions shown above.

Only if the answer to *all* of the above questions is a *yes* then the answer to this exercise is also a *yes*.

However, as soon as the answer to any of the above questions is a *no*, then the answer to this exercise can be given right away - *no*, the given set and the given operation defined on that set is not a group.

Q1. Taking on the matters from the first bar, does the operation of multiplication $\cdot$ defined on the set of integers $\mathbb{Z}$ act on exactly *two*, no more and no less, elements?

Yes, the operation of multiplication $\cdot$ defined on the set of integers $\mathbb{Z}$ acts on exactly two, no more and no less, elements k, m of that set:

$$k \circ m = k \cdot m = km$$

So far, so good - since the answer to this question is a *yes*, we move onto the next question.

By analogy, this model coin $\langle \mathbb{Z}; \cdot \rangle$ managed to fall through the **Q1**-opening in the **Q1** inclined plane down onto the **Q2** inclined plane.

Q2. Is it the case that for any two elements k, m of the set $\mathbb{Z}$ there exists a uniquely determined element n of $\mathbb{Z}$ such that $k \circ m = n$?

Yes, sure - a product of any two integers is, again, an integer and that integer is uniquely determined for the fixed input integers k and m . For example, $3 \cdot 15 = 45$ and the fact that other integers, such as 5 and 9 can multiply to 45 does not concern us - what matters here is that the fixed input integers 3 and 15 multiply *to one and only one integer*, 45, and not to multiple integers.

We observe in passing that such a favorable outcome, when a proposed operation takes two elements of a proposed set and returns an element of *the same* set, is not guaranteed and cannot be taken for granted - in the upcoming exercises we will come across certain sets and certain operations defined on these sets that will take us out of the parent set.

So far, so good - since the answer to this question is a *yes*, we move onto the next question.

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By analogy, this model coin $\langle \mathbb{Z}; \cdot \rangle$ managed to fall through the **Q2**-opening in the **Q2** inclined plane down onto the **Q3** inclined plane.

Q3. Is the operation of multiplication defined on the set of integers associative?

Yes, it is. For any three integers k, l, m it is the case that:

$$k \cdot (l \cdot m) = (k \cdot l) \cdot m$$

So far, so good - since the answer to this question is a *yes*, we move onto the next question.

By analogy, this model coin $\langle \mathbb{Z}; \cdot \rangle$ managed to fall through the **Q3**-opening in the **Q3** inclined plane down onto the **Q4** inclined plane.

Q4. Does the set of integers possess the unique (double-sided) identity element under the operation of multiplication?

In other words, does there exist a unique integer e such that for any integer k it is the case that:

$$e \cdot k = k \cdot e = k$$

?

Yes, sure. The integer $e = 1$ is the sought-after unique identity element under the operation of multiplication, since for any integer k it is the case that:

$$1 \cdot k = k \cdot 1 = k$$

Separately, the element $e = 1$ is often referred to as *the multiplicative identity*.

So far, so good - since the answer to this question is a *yes*, we move onto the next question.

Would you look at that? This feisty little pair is quite nimble, isn't it? The finish line is already looming on the horizon. We can practically smell the victory in the air.

By analogy, this model coin $\langle \mathbb{Z}; \cdot \rangle$ managed to fall through the **Q4**-opening in the **Q4** inclined plane down onto the, final, **Q5** inclined plane.

Q5. Is it the case that *every* integer possesses a unique (double-sided) multiplicative inverse?

Nope, it is not. Bummer! The multiplicative identity element 1 is the only element of the set of integers that possesses such an inverse:

$$1 \cdot 1 = 1$$

and that is it.

The element 0, when multiplied by any integer k , does produce an integer:

$$0 \cdot k = k \cdot 0 = 0 \neq 1$$

but that integer is never the multiplicative identity 1 and for all the *other* integers, that is the integers $n \neq 0$ and $n \neq 1$, the multiplicative inverse, x , simply does not exist because the following equations:

$$n \cdot x = 1, \quad x \cdot n = 1$$

have no solutions in the integers.

Thus, the answer to the question posed in this case is:

no, the set of integers $\mathbb{Z}$ under the operation of multiplication is not a group

and we see exactly *what* was the impenetrable barrier to entry into the group membership in this case - keeping the axiom of unrestricted group division in mind, we realize that exactly because the operation of *division* over the set of integers is not *always* possible, the set of integers coupled with the operation of multiplication is not a group. $\square$

In this particular case we came awfully close to answering all the group-detection questions in the positive, to only be cut down by *the last* one - the absence of multiplicative inverses for every integer is what snatched this pair from the jaws of success.

Exercise 11.2: is the set of rational numbers $\mathbb{Q}$ coupled with the operation of *multiplication* $\cdot$ defined on the elements of that set a group?

Solution: of this exercise is modeled after the solution of the previous exercise. Here, however, we will provide less commentary, hoping that our readers will step in on their own.

We, thus, see that the answers to all the first four questions, **Q1-Q4**, are a *yes* with $e = 1 = \frac{1}{1}$ being the candidate multiplicative identity element.

Therefore, the overall answer to the current question, again, hinges on the answer to the question **Q5**:

does *every* rational number possess a multiplicative inverse?

Well, actually *almost* all rational numbers do possess a unique multiplicative inverse - the only exception here is the rational number *zero* because the result of the multiplication of any rational number by zero is zero, which is not the identity element 1.

In other words, it is *not* the case that *every* rational number possesses a unique multiplicative inverse.

Game over:

no, the set of rational numbers $\mathbb{Q}$ under the operation of multiplication is not a group

Observe that while in the set of *integers* only one element, 1, *did* possess a unique multiplicative inverse and the rest of the elements did *not*, in the set of *rational* numbers only one element, 0, does *not* possess a multiplicative inverse and the rest of the *non zero* elements p/q do possess such a unique inverse:

$$\left(\frac{p}{q}\right)^{-1} = \frac{q}{p}$$

such that:

$$\frac{p}{q} \cdot \left(\frac{p}{q}\right)^{-1} = \left(\frac{p}{q}\right)^{-1} \cdot \frac{p}{q} = 1$$

which is to say that:

$$\frac{p}{q} \cdot \frac{q}{p} = \frac{q}{p} \cdot \frac{p}{q} = 1$$

It stands to reason now, that if we throw the element called *zero* from the set of rational numbers then such a reduced or diminished set coupled with the operation of multiplication *will* be a group and we get a cheap ride on the next exercise. $\square$

Exercise 11.3: is the set of *non zero* rational numbers $\mathbb{Q} \setminus \{0\}$ coupled with the operation of *multiplication* defined on the elements of that set a group?

Solution: by the results of **Exercise 11.2:**

yes, the set of non zero rational numbers $\mathbb{Q} \setminus \{0\}$ under the operation of multiplication is a group

We remember that, according to **Definition 15**, this particular group is *infinite*.

Moreover, because for any two non zero rational numbers a, b it is the case that $a \cdot b = b \cdot a$, as per **Definition 17**, this group is *commutative*. $\square$

Exercise 11.4: is the set of *negative* rational numbers $\mathbb{Q}^-$ coupled with the operation of *multiplication* $\cdot$ defined on the elements of that set a group?

Solution: noting that the set of negative rational numbers $\mathbb{Q}^-$ does not contain the number zero, we answer the first question, **Q1**, in the positive.

Q1. Does the operation of multiplication $\cdot$ defined on the set of negative rational numbers $\mathbb{Q}^-$ act on exactly *two*, no more and no less, elements?

Yes, the operation of multiplication $\cdot$ defined on the set of negative rational numbers $\mathbb{Q}^-$ acts on exactly two, no more and no less, elements $-(a/b), -(c/d)$ of that set:

$$-\frac{a}{b} \circ \left(-\frac{c}{d}\right) = -\frac{a}{b} \cdot \left(-\frac{c}{d}\right)$$

So far, so good - since the answer to this question is a *yes*, we move onto the next question.

Q2. Is it the case that the product of *any* two negative rational numbers is, again, a negative rational number?

Nope, it is not. In fact, the product of any two *negative* rational numbers is a *positive* rational number:

$$-\frac{a}{b} \circ \left(-\frac{c}{d}\right) = -\frac{a}{b} \cdot \left(-\frac{c}{d}\right) = \frac{ac}{bd} > 0 \notin \mathbb{Q}^-$$

and since no positive rational number belongs to the set of negative rational numbers, the candidate operation of multiplication defined on the set of negative rational numbers, as per our **Definition 10**, is *not* binary (we remind our readers that we will introduce the concept of *closure* of a subset under a binary operation when we will be studying the so-called *subgroups of a group*).

Game over:

no, the set of negative rational numbers $\mathbb{Q}^-$ under the operation of multiplication is not a group

$\square$

Answer the following questions on your own:

- does the set in **Exercise 11.4** under its operation possess a unique double-sided multiplicative identity element?

- Is the operation defined over the given set associative?
- Does every negative rational number have a unique double-sided multiplicative inverse?

Write down your replies.

Exercise 11.5: is the set of *positive* rational numbers $\mathbb{Q}^+$ coupled with the operation of *multiplication* defined on the elements of that set a group?

Solution: noting that the set of positive rational numbers $\mathbb{Q}^+$ does not contain the number zero, we see that the answers to all five questions listed above are a *yes*.

Q1: the operation of multiplication defined on the set of positive rational numbers acts on exactly two, no more and no less, elements.

Q2: the product of *any* two positive rational numbers is, again, a positive rational number, which is an element of the parent set. Moreover, for any two fixed positive rational numbers their product is uniquely determined.

Q3: the operation of multiplication of positive rational numbers *is* associative.

Q4: under the operation of multiplication the set of positive rational numbers does possess the unique double-sided identity element, $e = 1$, since 1 is a positive rational number and since any ordered product of 1 and any positive rational number is, again, that positive rational number itself.

Q5: *every* positive rational number p/q does possess a unique double-sided inverse element under the proposed operation of multiplication, q/p , such that:

$$\frac{p}{q} \cdot \frac{q}{p} = \frac{q}{p} \cdot \frac{p}{q} = 1 = e$$

Hence:

yes, the set of positive rational numbers $\mathbb{Q}^+$ under the operation of multiplication is a group

According to **Definition 15**, this particular group is *infinite* and, according to **Definition 17**, this group is *commutative*. $\square$

Exercise 11.6: is the set of just two integers $\{-1, +1\}$ coupled with the operation of *multiplication* defined on the elements of that set a group?

Solution: we already understand that the operation of multiplication in this case acts on exactly two, no more and no less, elements.

Thus, a *yes* for the question **Q1**.

A brute-force combinatorial enumeration of all the ordered pairs of elements of this particular set is merciful enough:

$$-1 \cdot (-1) = 1$$

$$1 \cdot 1 = 1$$

$$-1 \cdot 1 = 1 \cdot (-1) = -1$$

Hence, since the, uniquely defined, results of all such products belong to the parent set also, a *yes* for the question **Q2**.

The operation of multiplication of integers is associative - a *yes* for the question **Q3**.

The integer $e = 1$ in this case is the unique (double-sided) multiplicative identity element, as was already shown by the above *products-of-all-ordered-pairs-of-elements* exercise, a *yes* for the question **Q4**.

By the same exercise we also see right away that not only *every* element of the given set $\{-1, +1\}$ does possess a unique (double-sided) multiplicative inverse, but every element of that set is its own inverse!

Where else have we come across the same phenomenon already?

Correct - in the early examples of:

- the duplex light switch group
- the tire rotation group and
- the group of rigid symmetries of a non-square rectangle

every element within each such group is its own inverse. We will see more examples of this phenomenon in the near future.

In any case, a *yes* for the question **Q5** and, therefore:

yes, the set of two integers $\{-1, +1\}$ under the operation of multiplication is a group

According to **Definition 15**, this particular group is *finite* and according to **Definition 16**, its order is equal to 2.

As per **Definition 17**, this group is *commutative*. $\square$

Exercise 11.7: is the set of just one integer $\{0\}$ coupled with the operation of *multiplication* $\cdot$ defined on that set a group?

Solution: the operation of multiplication in this case acts on exactly two, no more and no less, elements.

Thus, a *yes* for the question **Q1**.

A brute-force combinatorial enumeration of all the ordered pairs of elements of this particular set is simplicity itself:

$$0 \cdot 0 = 0$$

Hence, since the results of all such products belong to the parent set also, a *yes* for the question **Q2**.

The operation of multiplication of integers is associative - a *yes* for the question **Q3**.

The integer $e = 0$ in this case is the unique multiplicative identity element, as was already shown by the above *products-of-all-ordered-pairs-of-elements* exercise, a *yes* for the question **Q4**.

Note here that by the **Axiom 2** in most of the families of the group axioms that we have looked at so far, the candidate set must possess the identity element. In other words, no identity element - no group. It follows, then, that if a candidate set wants to be a group in a happy marriage with a certain operation then that set must contain *at least one element*.

By the same exercise, we also see right away that not only *every* element of the set $\{0\}$ does possess a unique multiplicative inverse, but every element of that set is its own inverse. Again.

A *yes* for the question **Q5** and, therefore:

yes, the set of one integer $\{0\}$ under the operation of multiplication is a group

According to **Definition 15**, this particular group is *finite* and, according to **Definition 16**, its order is equal to 1. As per **Definition 17**, this group is *commutative*. $\square$

Exercise 11.8*: is the set of these four complex numbers $\{-1, -i, 1, i\}$ coupled with the operation of *multiplication* $\cdot$ defined on that set a group?

Solution: in **Exercise 8.16** we already showed that the operation of multiplication defined on the elements of the given set *is* binary - the product of *any* two elements of that set is, again, a unique element of the same set.

Thus, a *yes* for both questions **Q1** and **Q2**.

The operation of multiplication defined on the set of complex numbers is associative - a *yes* for the question **Q3**.

The multiplicative identity element of that set under the given operation is, clearly, the complex number $e = 1$, as is evident from the result of **Exercise 8.16** - a *yes* for the question **Q4**.

The elements $+1$ and -1 of the given set are their own inverses:

$$-1 \cdot (-1) = e = 1$$

$$1 \cdot 1 = e = 1$$

and the elements $+i$ and $-i$ of the given set are the inverses of *each other*:

$$i \cdot (-i) = (-i) \cdot i = e = 1$$

A *yes* for the question **Q5** and, hence:

yes, the set of four complex numbers $\{-1, -i, 1, i\}$ under the operation of multiplication is a group

According to **Definition 15**, this particular group is *finite* and, according to **Definition 16**, its order is equal to 4. As per **Definition 17**, this group is *commutative*. $\square$

The readers who are familiar with the so-called *n-th roots of unity* will readily recognize the fourth roots of unity in the above set.

Consider this exercise as a hint for one of the upcoming exercises.

We undertook such a detailed analysis of the possible answers to the question **Q** in order for the readers to begin developing a *realistic* feel for what does it actually mean when we say that sometimes answering the question **Q** amounts to a fair amount of tedium and that it would be nice if we could reduce the number of tests that we have to make in order to deterministically resolve the question **Q** one way or the other.

In the next few exercises we will consider some popular numeric sets under the operation of *addition*.

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Since we know that the operation of addition defined on the elements of popular numeric sets acts on exactly two, no more and no less, elements, we, right away, know that the answer to the question **Q1** is a *yes*.

We also know that the operation of addition over *numbers* is associative - the answer to the question **Q3** is, thus, a *yes* also.

The identity element of a candidate group under addition must be such a number e that the following relationships:

$$a + e = e + a = a$$

hold for *all* the numbers a of the candidate set.

Clearly, the role of such an identity element will be played by the number 0. Thus, $e = 0$ in this case is the so-called *additive identity* and we will only have to check whether the number 0 is the element of the suggested set.

An inverse x of a given number a under the operation of addition is a number that satisfies the following relationships:

$$x + a = 0 = a + x$$

implying that the only choice for such an additive inverse is the number $x = (-a)$:

$$a + (-a) = 0 = (-a) + a$$

and we will only have to check whether every number of the suggested set does possess a number taken with the minus sign in front of it.

Let us see how this will work in practice.

Exercise 11.9: is the set of integers $\mathbb{Z}$ coupled with the operation of *addition* $+$ defined on that set a group?

Solution: since the sum of any two integers is, again, a uniquely determined integer, it follows that the addition over integers is a *binary* operation, as per **Definition 10**.

The number zero is an integer and any integer taken with the minus sign in front of it is, again, an integer.

Thus, since all the group axioms covered in, say, the **Take 1** discussion of the **Group Axioms** chapter, are satisfied, we conclude that:

yes, the set of integers $\mathbb{Z}$ under the operation of addition is a group

By **Definition 15**, this particular group is *infinite* and by **Definition 17**, this group is *commutative*. $\square$

Exercise 11.10: is the set of rational numbers $\mathbb{Q}$ coupled with the operation of *addition* $+$ defined on the elements of that set a group?

Solution: since the sum of any two rational numbers is, again, a uniquely determined rational number, it follows that addition over rational numbers is a *binary* operation, as per **Definition 10**.

The number zero is a rational number and any rational number taken with the minus sign in front of it is, again, a rational number.

Thus, since all the group axioms covered in, say, the **Take 1** discussion of the **Group Axioms** chapter, are satisfied, we conclude that:

yes, the set of rational numbers $\mathbb{Q}$ under the operation of addition is a group

By **Definition 15**, this particular group is *infinite* and by **Definition 17**, this group is *commutative*. $\square$

Exercise 11.11: is the set of *non-zero* rational numbers $\mathbb{Q} \setminus \{0\}$ coupled with the operation of *addition* $+$ defined on the elements of that set a group?

Solution: since the proposed set does not contain the number zero, it follows that under the proposed operation the set of non-zero rational numbers does not have the identity element:

nope, the set of non-zero rational numbers $\mathbb{Q} \setminus \{0\}$ under the operation of addition is not a group $\square$

Exercise 11.12: is the set of *positive* rational numbers $\mathbb{Q}^+$ coupled with the operation of *addition* $+$ defined on the elements of that set a group?

Solution: noting that the set of positive rational numbers $\mathbb{Q}^+$ does not contain the number zero, from the solution of the previous exercise it follows that:

nope, the set of positive rational numbers $\mathbb{Q}^+$ under the operation of addition is not a group $\square$

Exercise 11.13: is the set of *non-negative* rational numbers $\mathbb{Q}_0^+$ coupled with the operation of *addition* $+$ defined on the elements of that set a group?

Solution: noting that even though the sum of any two non-negative rational numbers is a uniquely determined non-negative rational number and even though the set of non-negative rational numbers $\mathbb{Q}_0^+$ does contain the number zero, the proposed set under the proposed operation does not contain the additive inverses, since no single *negative* rational number belongs to the set $\mathbb{Q}_0^+$:

nope, the set of non-negative rational numbers $\mathbb{Q}_0^+$ under the operation of addition is not a group $\square$

Exercise 11.14: is the set of just two integers $\{-1, +1\}$ coupled with the operation of *addition* $+$ defined on the elements of that set a group?

Solution: since the number zero is not the element of the proposed set, the game is over right there:

nope, the set of two integers $\{-1, +1\}$ under the operation of addition is not a group $\square$

Answer the following questions on your own:

- Is the proposed operation in **Exercise 11.14** over its set binary?
- Is that operation associative?
- Does every element of the proposed set have a unique additive inverse?

Write your answers down.

Exercise 11.15: is the set of just one integer $\{0\}$ coupled with the operation of *addition* $+$ defined on that set a group?

Solution: since the sum of two zeros is, again, zero, which is the element of the given set:

$$0 + 0 = 0$$

the proposed operation over the proposed set, as per **Definition 10**, is binary.

The additive identity element in this case is zero, $e = 0$, and the number zero is its own inverse. Hence:

yes, the set of one integer $\{0\}$ under the operation of addition is a group

By **Definition 15**, this particular group is *finite* and, by **Definition 16**, the order of that group is equal to 1. As per **Definition 17**, this group is *commutative*. $\square$

In the next discussion we will keep investigating the numeric sets under the two popular operations of addition and multiplication.

Exercises 2: More Numeric Sets Under Addition and Multiplication

The purpose of this discussion is to let the readers hone the group-detection skill on their own using the solutions of the previous exercises as a guide - in this section we will omit the detailed proofs and will only show the correct final answers with a few relevant details sprinkled here and there.

In the first part of the upcoming exercises we will study the formation of the relevant numeric groups under the operation of *addition* and in the second part of the upcoming exercises we will study the formation of the relevant numeric groups under the operation of *multiplication*.

Since the operations of *addition* and *multiplication* of numbers *are* associative, there is no need to explicitly verify the associativity property of these operations.

While for the negative answers below we give only *the earliest* reason why the candidate set/operation unordered pair is not a group, the readers should really enumerate *all* such reasons.

As a short technical reminder:

- *positive* numbers a are all the numbers that are *strictly greater than zero*, $a > 0$
- *non-negative* numbers a are all the numbers that are greater than *or equal to zero*, $a \geq 0$
- *non-positive* numbers a are all the numbers that are smaller than *or equal to zero*, $a \leq 0$
- *negative* numbers a are all the numbers that are *strictly smaller than zero*, $a < 0$

Addition

Deduce on your own whether the following sets of numbers form a group under the operation of *addition*, taking it that for the numbers a, b under consideration:

$$a \circ b = a + b$$

for the duration of the next twelve exercises.

Exercise 11.16: $\mathbb{R}$, the set of all real numbers.

Answer: *yes*, $\mathbb{R}$ under addition *is* a group: the sum of any two real numbers is, again, a uniquely determined real number, the unique additive identity element $e = 0$ exists and the unique additive inverses $(-a)$ exist for all real numbers a . This group is infinite and commutative. $\square$

Exercise 11.17: $\mathbb{R} \setminus \{0\}$, the set of all non-zero real numbers.

Answer: *no*, $\mathbb{R} \setminus \{0\}$ under addition is *not* a group: the candidate operation is not binary, explain *why* (hint: use **Definition 10**). $\square$

Exercise 11.18: $\mathbb{R}^+$, the set of all positive real numbers.

Answer: *no*, $\mathbb{R}^+$ under addition is *not* a group: in $\mathbb{R}^+$ no additive identity element exists. $\square$

Exercise 11.19: $\mathbb{R}_0^+$, the set of all non-negative real numbers.

Answer: *no*, $\mathbb{R}_0^+$ under addition is *not* a group: the sum of any two non-negative real numbers is, again, a uniquely determined non-negative real number, the unique additive identity element $e = 0$ exists, but the unique additive inverse exists only for the element 0 and no other, positive, real number has an additive inverse, since the sum of any two positive real numbers is never zero. $\square$

Exercise 11.20: $\{2^n\}$, $n = 1, 2, 3, \dots$, the set of all positive integral powers of two.

Answer: *no*, $\{2^n\}$, $n = 1, 2, 3, \dots$ under addition is *not* a group: the candidate operation is not binary, explain *why*. $\square$

Exercise 11.21: $\{2^k\}$, $k \in \mathbb{Z}$, the set of all integral (negative, zero and positive) powers of two.

Answer: *no*, $\{2^k\}$, $k \in \mathbb{Z}$ under addition is *not* a group: the candidate operation is not binary, explain *why*. $\square$

Exercise 11.22*: $\mathbb{C}$, the set of all complex numbers.

Answer: *yes*, $\mathbb{C}$ under addition *is* a group: the sum of any two complex numbers is, again, a uniquely determined complex number, the unique additive identity element $e = 0 + i0 = 0$ such that:

$$z + 0 = 0 + z = z$$

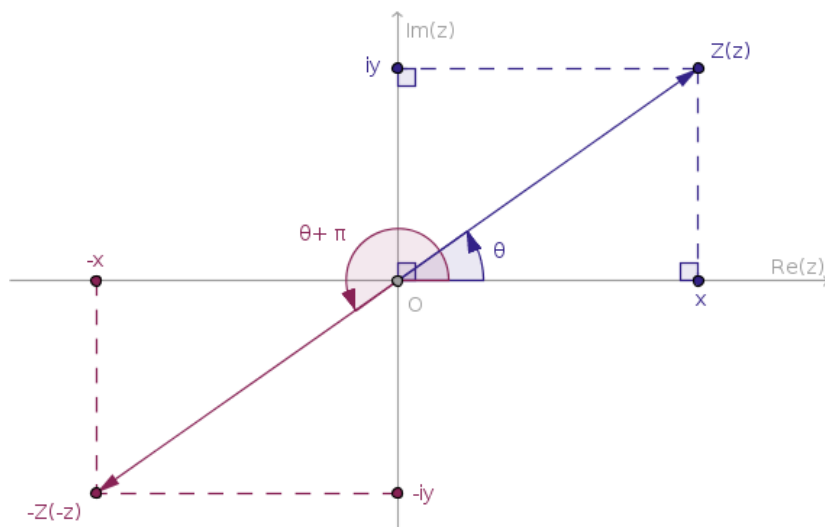
for all complex numbers z exists (in $\mathbb{C}$) and the unique additive inverse $(-z)$ such that:

$$z + (-z) = (-z) + z = 0 + i0 = 0 = e$$

for any complex number z exists as well.

Geometrically, the additive complex inverses are *centrally symmetric* images of each other with respect to the origin O .

That is: the points in the complex plane $Z(z)$, $O(0)$ and $-Z(-z)$ are collinear and the origin O cuts the line segment defined by the points Z and $-Z$ exactly in half (Figure 11.4):



This group is infinite and commutative. $\square$

Exercise 11.23*: $\mathbb{C} \setminus \{0\}$, the set of all non-zero complex numbers.

Answer: *no*, $\mathbb{C} \setminus \{0\}$ under addition is *not* a group: the candidate operation is not binary, explain why on your own. $\square$

Exercise 11.24*: $\mathbb{C}, |z| = 1$, the set of all complex numbers with the modulus equal to 1.

Answer: *no*, $\mathbb{C}, |z| = 1$ under addition is *not* a group: the candidate operation is not binary, explain why. $\square$

Exercise 11.25*: $\{a + ib\}, a, b \in \mathbb{Z}$, the set of all complex numbers whose real and imaginary parts are *integers*.

Answer: *yes*, $\{a + ib\}, a, b \in \mathbb{Z}$ under addition *is* a group: the sum of any two complex numbers whose real and imaginary parts are integers is, again, a uniquely determined complex number whose real and imaginary parts are integers, the unique additive identity element $e = 0 = 0 + i \cdot 0$ exists and the

unique additive inverses $-(a + ib)$ exist for all the complex numbers of the form $a + ib$, $a, b \in \mathbb{Z}$, see Figure 11.4 above.

This group is infinite and commutative. $\square$

Exercise 11.26*: $\{a + ib\}$, $a, b \in \mathbb{Q}$, the set of all complex numbers whose real and imaginary parts are *rational numbers*.

Answer: *yes*, $\{a + ib\}$, $a, b \in \mathbb{Q}$ under addition *is* a group: the sum of any two complex numbers whose real and imaginary parts are rational numbers is, again, a uniquely determined complex number whose real and imaginary parts are rational numbers, the unique additive identity element $e = 0 = 0 + i \cdot 0$ exists and the unique additive inverses $-(a + ib)$ exist for all the complex numbers of the form $a + ib$, $a, b \in \mathbb{Q}$, see Figure 11.4 above.

This group is infinite and commutative. $\square$

Exercise 11.27*: $\{a + ib\}$, $a, b \in \mathbb{Q} \setminus \{0\}$, the set of all complex numbers whose real and imaginary parts are *non-zero rational numbers*.

Answer: *no*, $\{a + ib\}$, $a, b \in \mathbb{Q} \setminus \{0\}$ under addition is *not* a group: the candidate operation is not binary, explain *why*. $\square$

Multiplication

Deduce on your own whether the following sets of numbers form a group under the operation of *multiplication*, taking it that for the numbers a, b under consideration:

$$a \circ b = a \cdot b$$

for the duration of the next twelve exercises.

Exercise 11.28: $\mathbb{R}$, the set of all real numbers.

Answer: *no*, $\mathbb{R}$ under multiplication is *not* a group: no multiplicative inverse for the element 0 of $\mathbb{R}$ exists. $\square$

Exercise 11.29: $\mathbb{R} \setminus \{0\}$, the set of all non-zero real numbers.

Answer: *yes*, $\mathbb{R} \setminus \{0\}$ under multiplication *is* a group: the product of any two non-zero real numbers is, again, a uniquely determined non-zero real number, the unique multiplicative identity element $e = 1$ exists (in $\mathbb{R} \setminus \{0\}$) and the multiplicative inverses $1/a$ exist for all non-zero real numbers a .

This group is infinite and commutative. $\square$

Exercise 11.30: $\mathbb{R}^+$, the set of all positive real numbers.

Answer: *yes*, $\mathbb{R}^+$ under multiplication *is* a group: the product of any two positive real numbers is, again, a uniquely determined positive real number, the unique multiplicative identity element $e = 1$ exists (in $\mathbb{R}^+$) and the unique multiplicative inverses $1/a$ exist for all positive real numbers a .

This group is infinite and commutative. $\square$

Exercise 11.31: $\mathbb{R}_0^+$, the set of all non-negative real numbers.

Answer: *no*, $\mathbb{R}_0^+$ under multiplication is *not* a group: no multiplicative inverse for the element 0 of $\mathbb{R}_0^+$ exists. $\square$

Exercise 11.32: $\{2^n\}$, $n = 1, 2, 3, \dots$, the set of all positive integral powers of two.

Answer: *no*, $\{2^n\}$, $n = 1, 2, 3, \dots$ under multiplication is *not* a group: no multiplicative identity element in that set exists. $\square$

Exercise 11.33: $\{2^k\}$, $k \in \mathbb{Z}$, the set of all integral (negative, zero and positive) powers of two.

Answer: *yes*, $\{2^k\}$, $k \in \mathbb{Z}$ under multiplication *is* a group: the product of any two integral powers of two is, again, a uniquely determined integral power of two, the unique multiplicative identity element $e = 2^0 = 1$ in that set exists and the unique multiplicative inverses 2^{-k} exist for all the elements 2^k , $k \in \mathbb{Z}$.

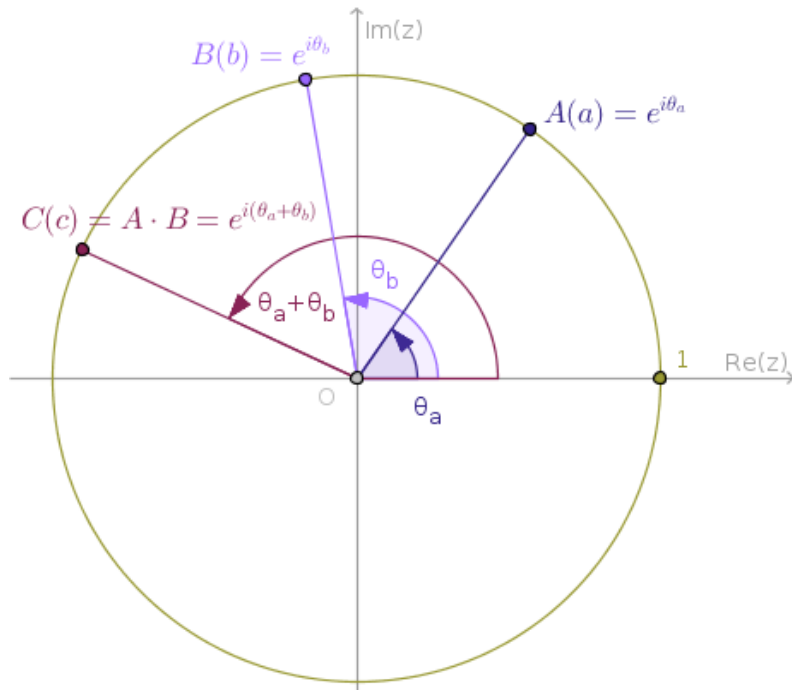
This group is infinite and commutative. $\square$

Exercise 11.34*: $\mathbb{C}$, the set of all complex numbers.

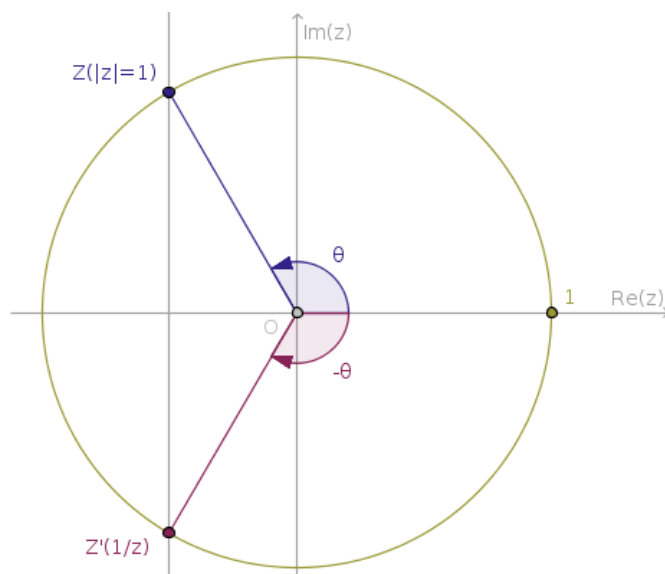
Answer: *no*, $\mathbb{C}$ under multiplication is *not* a group: no multiplicative inverse for the complex number 0 exists. $\square$

Exercise 11.35*: $\mathbb{C}, |z| = 1$, the set of all complex numbers with the modulus equal to 1.

Answer: yes, $\mathbb{C}, |z| = 1$ under multiplication *is* a group: the product of any two complex numbers a, b with the modulus 1 is, again, a uniquely determined complex number $c = a \cdot b = b \cdot a$ with the modulus 1 (Figure 11.5):



The unique multiplicative identity element $e = 1, |1| = 1$ in this case exists and the unique multiplicative inverses $1/z = e^{-i\theta}$ (Figure 11.6):



exist for all complex numbers $z = e^{i\theta}$ such that $|z| = 1$.

This group is infinite and commutative. $\square$

We remind our readers that geometrically the inverse $1/z$ of a non-zero complex number z can be obtained as the result of a commutative composition of the following two geometric transformations of the complex plane:

- reflection of a point in the real axis and
- the inversion of the plane with respect to the origin-centered unit circle with positive power

In the case of the above exercise the inversion of the plane with respect to the origin-centered unit circle with positive power will leave a point located on the periphery of the said circle *fixed*.

Exercise 11.36*: $\{a + ib\}, a, b \in \mathbb{Z}$, the set of all complex numbers whose real and imaginary parts are *integers*.

Answer: *no*, $\{a + ib\}, a, b \in \mathbb{Z}$ under multiplication is *not* a group: no multiplicative inverse of the element 0 of the candidate set exists. $\square$

Exercise 11.37*: $\{a + ib\}, a, b \in \mathbb{Q}$, the set of all complex numbers whose real and imaginary parts are *rational numbers*.

Answer: *no*, $\{a + ib\}, a, b \in \mathbb{Q}$ under multiplication is *not* a group: no multiplicative inverse of the element 0 of the candidate set exists. $\square$

Exercise 11.38*: $\{a + ib\}, a, b \in \mathbb{Q} \setminus \{0\}$, the set of all complex numbers whose real and imaginary parts are *non-zero rational numbers*.

Answer: *yes*, $\{(a + ib) \neq 0\}, a, b \in \mathbb{Q}$ under multiplication *is* a group: the product of any two non-zero complex numbers with rational real and imaginary parts is, again, a uniquely determined non-zero complex number with rational real and imaginary parts, the unique multiplicative identity element $e = 1 = 1 + i \cdot 0$ in the candidate set exists and the unique multiplicative inverses:

$$\frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2}, \quad a, b \in \mathbb{Q}, \quad (a + ib) \neq 0$$

exist for all complex numbers of the form $(a + ib) \neq 0, a, b \in \mathbb{Q}$ because:

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- the square of any non zero rational number p/q is still a uniquely determined non-zero rational number, since the square of a non-zero integer is still a uniquely determined non-zero integer:

$$\frac{p^2}{q^2} \in \mathbb{Q}$$

- the sum of any two non-zero rational numbers a/b and c/d is a uniquely determined non-zero rational number (justify on your own):

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + cb}{bd} \in \mathbb{Q}$$

- and the ratio of any two non zero rational numbers a/b and c/d is a uniquely determined non-zero rational number (justify on your own):

$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{cb} \in \mathbb{Q}$$

Verify on your own that the product of any complex number of the requested shape $(a + ib) \neq 0$ and its proposed inverse is indeed the unique multiplicative identity element $e = 1$:

$$(a + ib) \left[\frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2} \right] = \left[\frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2} \right] (a + ib) = 1$$

This group is infinite and commutative. $\square$

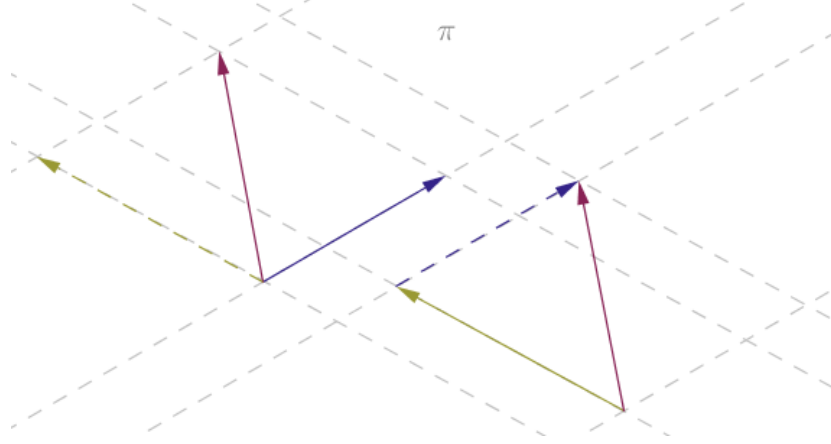
Exercises 3: Vectors, XOR, Linear Real And Linear-Fractional Complex Functions

In this batch of exercises we will switch gears and look at the examples of non-numeric groups – if a certain exercise in this discussion is too difficult and is beyond your current level of mathematical knowledge then you can skip that exercise and come back to it later, when you familiarize yourself with the more advanced material.

Exercise 11.39: is the set of all co-planar vectors coupled with the operation of addition defined on these vectors a group or not?

Solution: in this problem we are given the set of vectors that belong to one and the same plane.

In **Exercise 8.11** we already showed that the operation of addition defined on the set of co-planar vectors *is binary*: as per **Definition 10**, that means that the sum of any two co-planar vectors is a vector that belongs to the same, parent, plane (Figure 11.7):



Therefore, the operation of addition of co-planar vectors does not take us out of the parent set of all co-planar vectors.

In terms of the notion of a *closure of a set* under an operation, the above observation can be stated as follows: *the set of co-planar vectors is closed under addition*.

Moreover, the operation of addition of co-planar vectors is associative.

The unique additive identity element e in this case is the zero vector: $e = \vec{0}$, since it is the case that for any vector $\vec{a}$ the following relations hold:

$$\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$$

The unique additive inverse of any given vector $\vec{a}$ is the vector that:

- is located in the same plane
- has the same length $|\vec{a}|$ as the original vector and
- has the direction that is *opposite* to the direction of the original vector, $-\vec{a}$, so that $\vec{a} + (-\vec{a}) = -\vec{a} + \vec{a} = \vec{0}$

Thus, the answer to this problem is: *yes, the set of all co-planar vectors under the operation of addition is a group*. $\square$

Exercise 11.40: is the set of finite binary strings of fixed length $n > 0, n \in \mathbb{N}$ coupled with the XOR operation defined on such strings a group or not?

Solution: a finite binary string of fixed length $n > 0, n \in \mathbb{N}$ is simply a contiguous or a non-interruptible linear collection of 0s and 1s jammed together tight, with no gaps and no overlaps:

100111

in such a way that the total number of the said 0s and 1s is equal to n .

The above description of what a binary string is tells us that geometrically *a string* is a formation that is strictly linear and, explicitly, non-circular.

The perceived individual *positions* into which the zeros and the ones in a binary string are distributed are often referred to as *bits*, for **b**inary **d**igits. A synonym for a binary *string* is *an array* filled with zeros and ones in some order.

The minimalistic functional portrait of the XOR operation as it acts on a single pair of bits, highlighted as a three-story stack of such bits of the same color, is shown below (Figure 11.8):

XOR	0	1	1	0
	0	1	0	1
	0	0	1	1

From the above table of all the possible linear permutations of zeros and ones sprinkled across just two bits it follows that the said XOR operation evaluates to:

- the non-defined value of *True*, designated as 1, only when the contents of the bits under the consideration *differ* or *do not agree* and
- when these contents are the same or do agree, then the said XOR operation evaluates to the non-defined value of *False*, designated as 0

For example:

```

XOR:
10110
11001
-----
01111

```

and so on.

In **Exercise 8.14** we already showed that over the set of binary strings of finite fixed length $n > 0$ the XOR operation is binary - the result of the application of the XOR operator to any two binary strings of finite fixed length $n > 0$ is, again, a finite binary string of the same length n .

Such a result is obtained by the consecutive application of the XOR operation to all n corresponding pairs of bits of the input strings one by one.

So far so good.

In order to investigate the associativity of the XOR operation as it applies to finite binary strings of an arbitrary length $n > 1$, we first want to understand whether the said XOR operation is associative when it is applied to just a single pair of bits, driven by the above table.

The definition of the associativity of a binary operation involves exactly three elements or exactly three bits in this case.

Each such bit can take on only one of only two possible values, zero or one.

Hence, the revealing question here is:

in how many ways can three distinct elements be arranged in a linear fashion with repetitions allowed?

In a slightly more official way:

how many linear r -permutations of n distinct types of items exist under the assumption that each type of item is available in an infinite supply?

In our case $r = 3$, for three *positions*, and $n = 2$ (zero and one).

Well, the first position in a linear r -permutation can be filled in n ways.

Regardless of the choices made for the candidates that now populate the first position in our permutation, the second position in such a permutations can also be filled in exactly n different ways as well and so on.

Therefore, by the Multiplication Counting Principle, the number of linear r -permutations of n distinct types of items each of which is available in an infinite supply is given by:

$$\underbrace{n \cdot n \cdot \dots \cdot n}_r = n^r$$

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In our case, then, we will have to examine:

$$n^r = 2^3 = 8$$

different linear 3-permutations of 2 distinct *types* of items, taking it that $\circ = \text{XOR}$ as per the diagram shown in Figure 11.8 above:

$$0 \circ (0 \circ 0) = 0 = (0 \circ 0) \circ 0$$

$$0 \circ (0 \circ 1) = 1 = (0 \circ 0) \circ 1$$

$$0 \circ (1 \circ 0) = 1 = (0 \circ 1) \circ 0$$

$$0 \circ (1 \circ 1) = 0 = (0 \circ 1) \circ 1$$

$$1 \circ (0 \circ 0) = 1 = (1 \circ 0) \circ 0$$

$$1 \circ (0 \circ 1) = 0 = (1 \circ 0) \circ 1$$

$$1 \circ (1 \circ 0) = 0 = (1 \circ 1) \circ 0$$

$$1 \circ (1 \circ 1) = 1 = (1 \circ 1) \circ 1$$

Aha.

We now discovered and proved, via a non-elegant brute-force enumeration, that the operation of XOR applied to the *individual* pairs of bits *is* associative.

But how is the result of the XOR operation applied to finite binary strings of length $n > 1$ formed?

The result of the XOR operation applied to finite binary strings of length $n > 1$ is formed by applying the results of the XOR operation carried out against the individual pair bits one by one for all n such pairs.

Say, we now have three 2-strings a, b and c such that:

$$a = a_1a_1, \quad b = b_1b_2, \quad c = c_1c_2$$

meaning that each such string consists of only two bits $a_{1,2}$ and $b_{1,2}$ and $c_{1,2}$, where the symbols a_k, b_k and c_k simply mean *either* 0 *or* 1.

Then, what is the meaning of the following expression: $a \circ (b \circ c)$?

And does such an expression even have a meaning in the first place?

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Well, we already know that:

$$b \circ c = (b_1 b_2) \circ (c_1 c_2) = (b_1 \circ c_1)(b_2 \circ c_2)$$

which is simply a re-write of the above *definition* of the rule according to which the operation XOR acts on *compound* binary strings of length $n > 1$, that is all.

We, thus, move further by unwinding the result of XORing the above expression with the string a :

$$a \circ (b \circ c) = [a_1 \circ (b_1 \circ c_1)][a_2 \circ (b_2 \circ c_2)]$$

But we already showed that when the operation of XOR is applied to the *individual* pairs of bits, such as the pairs shown above, it is certainly associative.

Therefore, we can write:

$$[a_1 \circ (b_1 \circ c_1)][a_2 \circ (b_2 \circ c_2)] = [(a_1 \circ b_1) \circ c_1][(a_2 \circ b_2) \circ c_2]$$

which is to say that:

$$[(a_1 \circ b_1) \circ c_1][(a_2 \circ b_2) \circ c_2] = [(a_1 \circ b_1)(a_2 \circ b_2)] \circ (c_1 c_2)$$

which further means that:

$$a \circ (b \circ c) = (a \circ b) \circ c$$

for the binary strings of length 2.

Repeating the above steps verbatim for the binary strings of length $n > 2$ is just a tedious exercise that fundamentally relies on the fact that the XOR operation is associative for the individual pairs of bits.

Thus, we conclude that the operation of XOR defined on binary strings of length $n > 1$ is associative.

Does the proposed set possess the unique identity element e under the proposed operation such that:

$$a \text{ XOR } e = e \text{ XOR } a = a$$

no matter the contents of the binary string a of an arbitrary, but fixed, length $n > 0$?

In order to answer that question, let us go back to our earlier, definition-driven, observation: a zero anywhere in the proposed identity element e will pass the peer 0 into the result unmolested and it will flip the peer 1 into 1.

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Thus, we reason, the unique exclusive OR identity element over the set of binary strings of finite, fixed, length n must be the zero-string of the same length n :

$$e_n = \underbrace{00 \dots 0}_{n \text{ zeros}}$$

For example, for $n = 5$ we have:

$$\begin{array}{rcl} & \text{XOR:} & \\ 10110 & = & a \\ 00000 & = & e \\ \hline 10110 & = & a \end{array}$$

Great. The unique identity element in this case does exist.

What about the unique (double-sided) inverses?

Again, we go back to the very definition of the operation at hand - we wish to find such a mystery binary string x_n that will flip any *given* binary string a_n into the identity element e_n via the XOR operation:

$$x_n \circ a_n = a_n \circ x_n = e_n$$

What could that mystery element x_n be?

Well, if we XOR the given binary string a_n *with itself* then there will be absolutely no difference in the contents of the corresponding pairs of bits, right?

$$\begin{array}{rcl} & \text{XOR:} & \\ 10110 & = & a \\ 10110 & = & x \\ \hline 00000 & = & e \end{array}$$

In that case the XOR operation will flip the results of all the individual pairs of bits into a zero, which will result in the identity element e_n :

$$a_n \circ a_n = e_n = \underbrace{00 \dots 0}_{n \text{ zeros}}$$

In other words, it is the case that for any binary string of fixed length $n > 0$ a unique inverse element under the XOR operation exists and that inverse is that element itself.

We, thus, conclude that:

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yes, the set of finite binary strings of fixed length $n > 0, n \in \mathbb{N}$ under the XOR operation is a group

Such a group is finite, commutative and its order is 2^n .

The commutativity of this group follows from the diagram shown in Figure 11.8, since $1 \text{ XOR } 0 = 1$
 $= 0 \text{ XOR } 1$.

The order of this group can be deduced via the Multiplication Counting Principle or **MCP** for short - there are only 2 choices, 0 or 1, for the contents of the first bit in a string. Regardless of that actual choice of a particular binary value, there are exactly 2 choices, still 0 or 1, for the contents of the second bit in a string.

By the said **MCP**, there are $2 \cdot 2$ distinct ways in which the first 2 bits of a string can be populated and so on - all n bits of a string can be populated with 0s and 1s in:

$$\underbrace{2 \cdot 2 \cdot \dots \cdot 2}_{n \text{ twos}} = 2^n$$

distinct ways.

In this particular group every element is its own inverse. $\square$

Consider the specific XOR group on the binary strings of length $n = 2$.

The order of that group is $2^2 = 4$ with $00 = e$ being the group's unique identity elem

The remaining three, non-identity, elements of this particular group are:

$$01, 10, 11$$

We already proved that every element of that group is its own inverse:

$$\begin{aligned} 01 \text{ XOR } 01 &= 00 = e \\ 10 \text{ XOR } 10 &= 00 = e \\ 11 \text{ XOR } 11 &= 00 = e \end{aligned}$$

What can be said about the product of any two distinct non-identity elements of that group?

Well, the answer to that question is computed in a straightforward manner:

$$\begin{aligned} 01 \text{ XOR } 10 &= 11 = 10 \text{ XOR } 01 \\ 01 \text{ XOR } 11 &= 10 = 11 \text{ XOR } 01 \\ 10 \text{ XOR } 11 &= 01 = 11 \text{ XOR } 10 \end{aligned}$$

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In other words, the product of any two distinct non-identity elements of that group is equal to the remaining, third, non-identity element of the same group.

Have we seen such groups already?

Yes.

Where?

Hint1: we have studied exactly *three* such groups already. In **Exercise 5.1.2** we gave another *six* examples of such groups and in the next discussion we will meet *two* more such groups.

Hint 2: construct the multiplication table of that group and study its colorful trace when the symbols are lifted from that table.

Answer: for a reference with a correct response, if we designate the elements of the XOR group on the binary strings of fixed length $n = 2$ as follows:

$$e = 00, x = 01, y = 10, z = 11$$

then the multiplication table of that group will be (Figure 11.9):

XOR	$e = 00$	$x = 01$	$y = 10$	$z = 11$
$e = 00$	e	x	y	z
$x = 01$	x	e	z	y
$y = 10$	y	z	e	x
$z = 11$	z	y	x	e

telling us that what we have here is yet another incarnation of the Klein four-group.

Which, hopefully, improves our initial and non-formal understanding of the concept of *group isomorphism*: on the surface certain groups may come across as being descriptively quite different but upon a deeper inspection we can discover that such seemingly different groups are not that different after all because *the structure* of such groups is one and the same.

Exercise 11.41*: is the set of all linear single-variable real-to-real functions coupled with the operation of composition defined on these functions a group or not?

Solution: in **Exercise 8.17** we saw that a linear single-variable real-to-real function $y(x)$ is symbolized as follows:

$$y(x) = ax + b, \quad a \neq 0 \quad (1)$$

where otherwise the symbols a and b represent arbitrary *constant* real numbers and the symbol x represents an independent real variable.

In the said exercise we also proved that the operation of composition defined on the set of such linear functions *is binary*: a substitution of an arbitrary linear function into an arbitrary linear function results in a linear function:

$$(ax + b) \circ (cx + d) = (ac)x + (ad + b) \quad (2)$$

Does such a substitution represent an operation that is necessarily *associative*?

Even though in the respective branch of mathematics it is proven that the composition of any three single-variable real-to-real appropriately defined functions $f(x)$, $g(x)$ and $h(x)$ *is* associative:

$$(f \circ g) \circ h = f \circ (g \circ h)$$

and in the long run and in the steady state of doing the grown-up mathematics we would normally not explicitly verify the associativity of the composition of the *concrete* functions, at this stage of our discourse it is educational to look at the by-hand mechanics of such a verification process.

To that end, in order to investigate the associativity of the composition of, specifically, *linear* functions, we introduce yet another linear function:

$$ex + f \quad (3)$$

construct *two* compositions of the composition shown in (2) with the function shown in (3):

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$$[(ax + b) \circ (cx + d)] \circ (ex + f) ? (ax + b) \circ [(cx + d) \circ (ex + f)] \quad (4)$$

and compare the results - if the said results are equal then the candidate operation *is* associative and if the said results do not match then the candidate operation is not associative and the game will be over right then and there.

In the composition of functions $g \circ f$ we first *execute* the function f and only then we *execute* the function g , taking the result or the output of the execution of the function f as the input.

To that end, we, first, replace the composition in the square brackets on the LHS of (4) with its RHS image shown in (2):

$$[(ax + b) \circ (cx + d)] \circ (ex + f) = ((ac)x + (ad + b)) \circ (ex + f)$$

and then we substitute the expression $ex + f$ into the variable x bound to the factor ac above:

$$((ac)x + (ad + b)) \circ (ex + f) = ac(ex + f) + ad + b$$

obtaining for the result:

$$[(ax + b) \circ (cx + d)] \circ (ex + f) = (ace)x + (acf + ad + b) \quad (5)$$

which is certainly a linear single-variable real-to-real function since $a, c, e \neq 0$.

Next, for the rightmost composition on the RHS of (4) we have:

$$(cx + d) \circ (ex + f) = c(ex + f) + d = (ce)x + (cf + d)$$

and, consequently, we find for the entire RHS of (4):

$$(ax + b) \circ ((ce)x + (cf + d)) = a(ce)x + a(cf + d) + b$$

which is to say that:

$$(ax + b) \circ [(cx + d) \circ (ex + f)] = (ace)x + (acf + ad + b) \quad (6)$$

Since the RHSs of (5) and (6) match, it follows that the operation of composition defined on the set of linear single-variable real-to-real functions *is* associative.

What could be the unique compositional identity element in this set?

Let us think together - by definition, such an identity element must be a mystery linear function $ux + v$, $u \neq 0$ such that $(ax + b) \circ (ux + v) = ax + b$.

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Is it possible to unmask the values of the mystery numbers u and v ?

If we spell out the definition of a composition of two linear functions shown on the LHS of the above expression then we shall have:

$$(ax + b) \circ (ux + v) = (au)x + av + b$$

But the above expansion must be equal to the original linear function:

$$(au)x + av + b = ax + b$$

Thus, if there is any hope for the existence of the compositional identity element in this set under this operation then it must be the case that:

$$av = 0, \quad u = 1 \neq 0$$

and, since by the definition of a linear function we have it that $a \neq 0$, it follows that $v = 0$, necessarily.

Thus, in this case the unique compositional identity element e exists and is equal to the following linear function:

$$e = 1 \cdot x + 0 = x$$

The verification of such a claim is more than straightforward:

$$(ax + b) \circ x = x \circ (ax + b) = ax + b$$

for any linear function $ax + b, a \neq 0$.

What about the unique compositional inverses?

Again, let us think - if there is any hope for such an inverse to exist then it must be the case that:

$$(ax + b) \circ (px + q) = x$$

for any linear function $ax + b, a \neq 0$.

Can we lift the veil of secrecy that covers the real numbers p and q ?

Let us see. On the one hand, we have:

$$(ax + b) \circ (px + q) = a(px + q) + b = (ap)x + (aq + b)$$

On the other hand, it must be the case that:

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$$(ap)x + (aq + b) = x$$

from where it follows that:

$$ap = 1 \text{ and } aq + b = 0$$

But since the function $ax + b$ is a good linear citizen, we know that $a \neq 0$.

Hence, the following numbers, each of which has the inverse $1/a$ baked into it, are always well-defined and exist:

$$p = \frac{1}{a}, \quad q = -\frac{b}{a}$$

and we, indeed, have a unique compositional inverse:

$$\frac{1}{a}x + \left(-\frac{b}{a}\right)$$

that is a good linear function citizen because $1/a \neq 0$ for all $a \neq 0$.

The verification of such a claim is more than straightforward. For example:

$$(ax + b) \circ \left(\frac{1}{a}x + \left[-\frac{b}{a}\right]\right) = x - b + b = x = e$$

As such, the answer to this problem is:

yes, the set of all linear single-variable real-to-real functions under the operation of composition is a group

□

Note that instead of the purely *group-theoretic* way of unmasking the potential inverses of the linear functions, we could have *cheated* a bit by stepping away from the purist line and solving the following equation:

$$y = ax + b$$

for x :

$$x = \frac{1}{a} \cdot y + \left(-\frac{b}{a}\right)$$

with the consequent mechanical and formal swap of x and y in the above result:

$$y = \frac{1}{a} \cdot x + \left(-\frac{b}{a} \right)$$

We could then claim, and verify, right away that the above function represents the unique, well-defined, compositional inverse of an arbitrary linear function $ax + b, a \neq 0$.

Exercise 11.42:** is the set of all linear fractional single-variable complex-to-complex functions of the first kind coupled with the operation of composition defined on these functions a group or not?

Solution: the said set of *linear fractional* functions of the first kind is defined as the set of functions $w(z)$ of a complex variable z such that:

$$w(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0 \quad (7)$$

where a, b, c and d are constant complex numbers locked into a relationship that honors the $ad - bc \neq 0$ constraint, which means that our *linear fractional* function of the first kind is not a constant, $w \neq \text{const}$, since if it is, then it degenerates into the previously studied *linear* function of the first kind that has the following shape: $f(z) = az + b, a \neq 0$.

In the above definition of the linear fractional function of the first kind shown in (7) it is assumed that if $c \neq 0$ then:

$$\lim_{z \rightarrow \infty} w(z) = \frac{a}{c}$$

$$\lim_{z \rightarrow -\frac{d}{c}} w(z) = \infty$$

where the point $z_0 = -d/c$ represents the pole of the linear fractional function of the first kind of order one, and if $c = 0$ then:

$$\lim_{z \rightarrow \infty} w(z) = \infty$$

Thus, our linear fractional function of the first kind is defined everywhere in the *extended* complex plane - the original complex plane equipped with the point at infinity ∞ .

We already know that the operation of composition, or the *followed by* process, acts on exactly two, no more and no less, elements: the target linear fractional function $v(z)$:

$$v(z) = \frac{ez + f}{gz + h}$$

and the source linear fractional function $w(z)$:

$$w(z) = \frac{az + b}{cz + d}$$

that is substituted into the independent variable portion of the target function $v(z)$ (Figure 11.10):

$$v(z) \circ w(z) = \frac{ez + f}{gz + h} \quad w(z) = \frac{az + b}{cz + d}$$

resulting in the following tower of fractions (Figure 11.11):

$$v(z) \circ w(z) = \frac{e \frac{az+b}{cz+d} + f}{g \frac{az+b}{cz+d} + h}$$

in a sample $v(z) \circ w(z)$ composition.

In **Exercise 8.18** we already showed that such a composition represents an operation that is necessarily *binary* - as per our **Definition 10**, such a composition is, again, a linear fractional function of the first kind.

A composition of the linear fractional functions of the first kind *is* associative - even though this fact follows from a general theorem, it is still possible, but, admittedly, tedious, to verify this claim by hand in exactly the same way in which we have done that in the previous exercise.

The unique compositional identity element for any linear fractional function of the first kind clearly exists and is equal to z :

$$z = \frac{1 \cdot z + 0}{0 \cdot z + 1}, \quad ad - bc = 1 \cdot 1 - 0 \cdot 0 = 1 \neq 0$$

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since the substitution of the variable z into an arbitrary linear fractional function of the first kind $w(z)$ and vice versa results in the said function $w(z)$:

$$w(z) \circ z = z \circ w(z) = w(z)$$

However, we encourage our readers to go through the motions of a purely group-theoretic reasoning for the recovery of such an identity element on their own - assume that such an identity $e(z)$:

$$e(z) = \frac{a_e z + b_e}{c_e z + d_e}$$

exists and then attempt to recover the values of the constants a_e, b_e, c_e and d_e by demanding that it must be the case that:

$$w(z) \circ e(z) = w(z)$$

meaning that it must be the case that:

$$w \circ e = \frac{a \cdot \frac{a_e z + b_e}{c_e z + d_e} + b}{c \cdot \frac{a_e z + b_e}{c_e z + d_e} + d} = \frac{az + b}{cz + d}$$

Multiply the above LHS numerator and the denominator through by $(c_e z + d_e)$:

$$\frac{aa_e z + ab_e + bc_e z + bd_e}{ca_e z + cb_e + dc_e z + dd_e} = \frac{az + b}{cz + d}$$

Collect the terms in the result that are z -bound and z -free:

$$\frac{(aa_e + bc_e)z + (ab_e + bd_e)}{(ca_e + dc_e)z + (cb_e + dd_e)} = \frac{az + b}{cz + d}$$

Demand that the factors of z and the z -free terms in the respective numerators and denominators above must match. For example, for the proposed compositional identity to exist, it must be the case that $aa_e + bc_e = a$.

Conclude that:

$$a_e = 1, b_e = 0, c_e = 0, d_e = 1, a_e d_e - b_e c_e = 1 \neq 0$$

and so on.

The unique compositional inverse of a linear fractional function of the first kind can be quickly recovered by solving the following equation for z :

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$$w = \frac{az + b}{cz + d}$$

by clearing its denominator:

$$w \cdot (cz + d) = az + b$$

collecting all the z -terms of the result on the same side of the equal sign:

$$wcz - az = -dw + b$$

and isolating the said variable z :

$$z = \frac{dw - b}{-cw + a}, \quad ad - bc \neq 0 \quad (8)$$

It is, however, quite instructional to obtain the same result in a purely group-theoretic fashion by assuming that such an inverse:

$$w^{-1}(z) = \frac{pz + q}{rz + s}, \quad ps - qr \neq 0$$

must be such a linear fractional function of the first kind that, say:

$$w(z) \circ w^{-1}(z) = e(z)$$

implying that it should be the case that:

$$\frac{a \cdot \frac{pz+q}{rz+s} + b}{c \cdot \frac{pz+q}{rz+s} + d} = z$$

Multiply the above LHS numerator and the denominator through by $(rz + s)$:

$$\frac{apz + aq + brz + bs}{cpz + cq + drz + ds} = z$$

Collect the z -bound and the z -free terms of the result together:

$$\frac{(ap + br)z + (aq + bs)}{(cp + dr)z + (cq + ds)} = z$$

Clear the LHS denominator above one more time:

$$(ap + br)z + (aq + bs) = (cq + ds)z + (cp + dr)z^2$$

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It now follows that for the sought-after compositional inverse to exist:

- the free-floating summand on the LHS above must vanish:

$$aq + bs = 0 \quad (9)$$

- the factors of the above summands linear in z must match:

$$ap + br = cq + ds \quad (10)$$

- and the factors of the above summand quadratic in z must vanish:

$$cp + dr = 0 \quad (11)$$

But wait a second.

Across the *three* equations above, namely, across the equations (9, 10, 11) we have *four* variables to recover!

Clearly, this must be a dead end!

Well. Not really.

Think: how truly independent our complex constant numbers a, b, c, d or p, q, r, s are?

We know that such numbers must honor the $ps - qr \neq 0$ constraint and, thus, it can never be the case that all such constants are zero at once: $p = q = r = s = 0$.

Therefore, at least one such constant is never zero.

Hence, we can always divide the numerator and the denominator of the given linear fractional function of the first kind by such a non-zero constant and obtain the *three* ratios instead.

For example, assume that it is the case that, say, $p \neq 0$:

$$\frac{pz + q}{rz + s} = z = \frac{z + \frac{q}{p}}{\frac{r}{p}z + \frac{s}{p}}$$

Thus, we can divide the equation (11) through by p and find:

$$\frac{r}{p} = -\frac{c}{d} \quad (12)$$

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We can divide the equation (9) through by p :

$$a \frac{q}{p} + b \frac{s}{p} = 0 \quad (13)$$

and do the same with the equation (10) via (12):

$$a - \frac{bc}{d} = c \frac{q}{p} + d \frac{s}{p} \quad (14)$$

Solving the pair of equations (13, 14) for the ratios q/p and s/p , we, trivially, find:

$$\frac{q}{p} = -\frac{b}{d}, \quad \frac{s}{p} = \frac{a}{d} \quad (15)$$

We now reverse engineer the results in (13) and (15):

$$p = d, q = -b, r = -c, s = a$$

where we, admittedly, do have some freedom in attaching the positive and negative signs to such constants:

$$w^{-1}(z) = \frac{dz - b}{-cz + a}, \quad ad - bc \neq 0 \quad (16)$$

in agreement with the result established by different means.

In any case, the unique compositional inverses, shown in (16), for all the elements from the set of linear fractional functions of the first kind exist.

In order to verify our find, we compute, say, the following composition:

$$w \circ w^{-1} = \frac{a \frac{dz-b}{-cz+a} + b}{c \frac{dz-b}{-cz+a} + d}$$

which, at this point, after doing all of the above work, is more than straightforward. That is, in the above tower of fractions we multiply both the numerator and denominator by $(-cz + a)$:

$$w \circ w^{-1} = \frac{adz - ab - bcz + ab}{cdz - cb - cdz + ad}$$

observing that all the free-floating summands in the numerator and the z -bound summands in the denominator of the result cancel out:

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$$w \circ w^{-1} = \frac{adz - bcz}{ad - cb} = z = e(z)$$

and so on.

We, hence, conclude that:

yes, the set of linear fractional functions of the first kind under the operation of composition is a group

This particular group is infinite but it is *not* commutative - see if you can prove that claim on your own.

We remember that in order to invalidate a universal claim, it is sufficient to produce just one measly counterexample.

Well, here is one. Let:

$$w(z) = \frac{1}{z}, \quad v(z) = z + 1$$

Prove on your own that the functions $w(z)$ and $v(z)$ above are indeed linear fractional functions of the first kind.

Then, on the one hand, for the composition $w \circ v$ we have:

$$w \circ v = \frac{1}{z} \circ (z + 1) = \frac{1}{z + 1}$$

and, on the other hand, for the composition $v \circ w$ we have:

$$v \circ w = (z + 1) \circ \frac{1}{z} = \frac{1}{z} + 1$$

But, clearly, it is not the case that for all z the results of the above two compositions are equal:

$$\frac{1}{z + 1} \neq \frac{1}{z} + 1$$

The fact that the results of the above two compositions do match for just two *specific* values of z :

$$z_{1,2} = -\frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

is irrelevant:

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$$w \circ v \neq v \circ w$$

Therefore, it is not the case that for any two linear fractional functions of the first kind w and v the following relation holds:

$$w \circ v = v \circ w$$

As such, this particular group is *not* commutative. $\square$

Extra For Experts

Observe that the set of all linear fractional single-variable complex-to-complex functions of the *second* kind:

$$w(z) = \frac{a\bar{z} + b}{c\bar{z} + d}, \quad ad - bc \neq 0$$

coupled with the operation of composition defined on these functions does *not* form a group because any odd number of conjugations of a conjugate $\bar{z}$ removes the said conjugation from the variable z :

$$\overline{\bar{z}} = z, \quad \overline{\overline{\bar{z}}} = \bar{z}, \quad \dots$$

and, thus, alters the shape of the result that does not match the original. In other words, the set of all linear fractional single-variable complex-to-complex functions of the *second* kind is not closed under composition

Exercises 4: Rigid Planar Motions

In this discussion, whose results will come in handy later on, we will investigate whether the certain sets of rigid *planar motions* form a group under composition.

In the **Rectangle, Anyone?** discussion we gave a somewhat loose **Definition 8** of a *rigid motion* because the objective of that discussion was a rigorous enough introduction of an early example of a group and not the generation of a water-tight proof.

Now, however, the time to get formal and rigorous has come and below we give an updated definition of a rigid planar motion that is suited exactly for the proof-generation purposes.

The two technical key features of a planar motion that will allow us to resolve the above question are the demands:

- for a *bijection* of a plane onto itself and
- for *the conservation of the distance* between any two preimage points under the given transformation

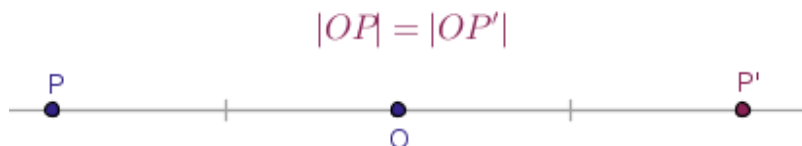
Officially,

Definition 18: a rigid motion of the plane $\mathbb{R}^2$ is such a bijection of that plane onto itself $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ under which the distance between any two preimage points $A, B \in \mathbb{R}^2$ and the distance between their respective image points $f(A)$ and $f(B)$ is one and the same:

$$|AB| = |f(A)f(B)| \quad (17)$$

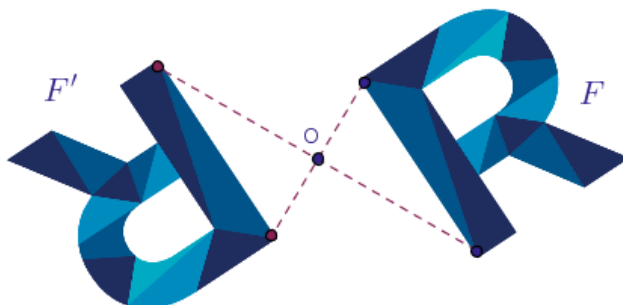
In order to give our readers some concrete substance to which the upcoming reasoning can be anchored, below we define such fundamental transformations of the plane as *rotation*, *parallel translation* and *reflection in a straight line and a point*.

Definition 19.1: a reflection of the plane, p , with respect to a *fixed point*, O , also known as *central symmetry*, is a bijection of the plane p onto itself that takes every point P of the plane p into its image point P' such that 1) the points P, O and P' are collinear and 2) the distance from the center of symmetry O to the preimage point P , $|OP|$, is equal to the distance from the center of symmetry O to the image point P' , $|OP'|$ (Figure 11.12):



The centrally symmetric image P' of the point P can be constructed with Euclidean tools, a compass and a straightedge, alone (see if you can invent such a construction).

In order to obtain the centrally symmetric image F' of a figure F , every point of the figure F must be reflected in a given center of symmetry as explained above (Figure 11.13):

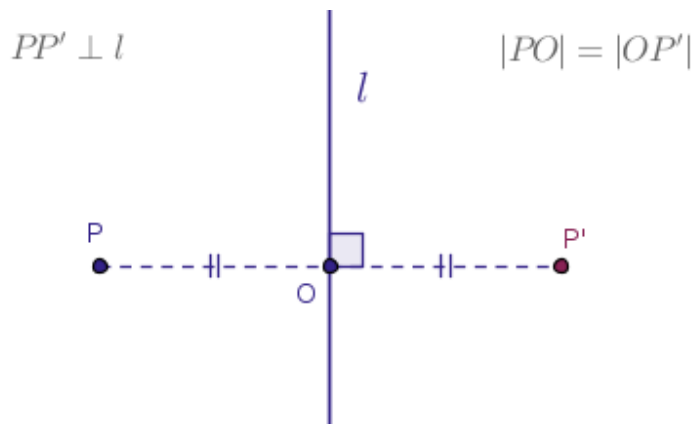


From the above definition of central symmetry of the plane it follows that if the point P' is the image of the point P under that transformation then, conversely, the point P is the image of the point P' under the same transformation and, in general, it is said that the points P and P' in that case are the centrally symmetric images of each other.

Under the transformation of central symmetry the only point that remains fixed is the center of symmetry or the point with respect to which the plane is being reflected, every other point of the plane is displaced in a non-trivial fashion.

Central symmetry takes circles into circles and straight lines into straight lines. Every straight line that passes through the center of symmetry is taken by that symmetry into itself.

Definition 19.2: a *reflection* of the plane, p , in a fixed straight line, l , is a bijection of the plane, p , onto itself under which every point P of the plane p is taken into its image point P' such that the line segment PP' 1) is perpendicular to the straight line l and 2) is cut by l exactly in half (Figure 5.3.16):



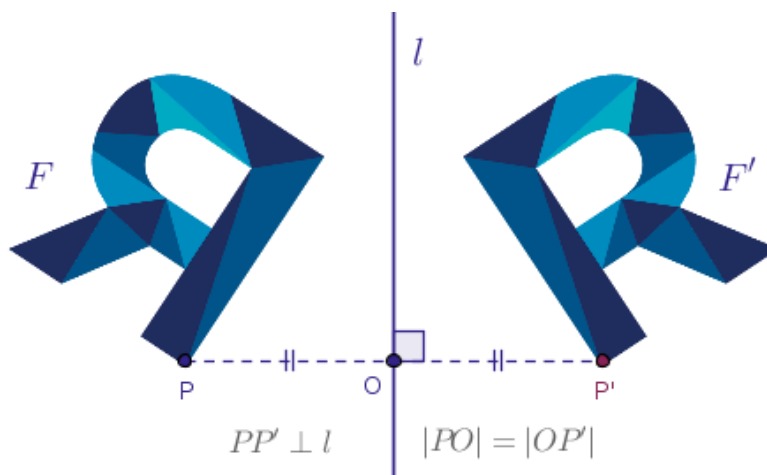
From the above definition of reflection of the plane in a (fixed) straight line it follows that the straight line that passes through the points P and P' intersects the straight line l under a right angle at the point O in such a way that the distance from the point P to the point O is equal to the distance from the point O to the point P' .

Consequently, the reflective image P' of the point P can be constructed with Euclidean tools, a compass and a straightedge, alone (see if you can invent such a construction).

Moreover, from the above definition of reflection of the plane in a straight line it follows that if the point P' is the image of the point P under that transformation then, conversely, the point P is the image of the point P' under the same transformation.

In general, then, it is said that the points P and P' in that case are the reflective images of each other.

In order to obtain the reflective image F' of a figure F , every point of the figure F must be reflected in a given straight line as explained above (Figure 5.3.17):

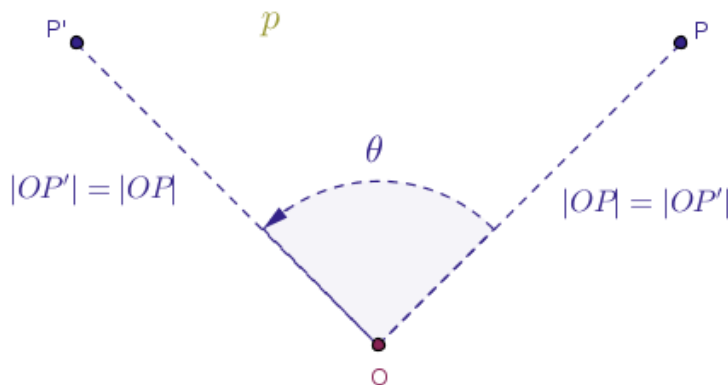


Under a reflection of the plane in a (fixed) straight line the only points that remain fixed are the points that belong to the straight line of symmetry with respect to which the plane is being reflected, every other point of the plane is displaced in a non-trivial fashion.

Reflection of the plane in a straight line takes straight lines into straight lines and circle into circles.

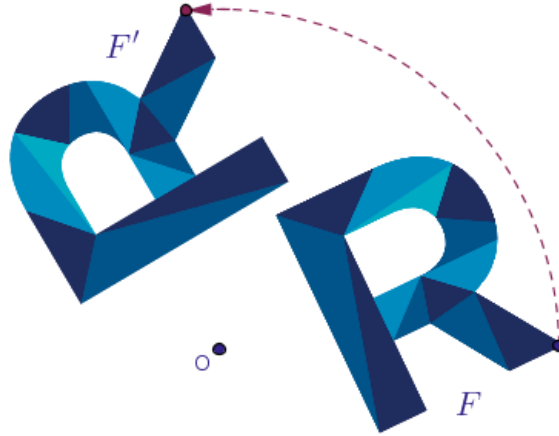
Definition 19.3: a *rotation* of the plane p about a fixed point, *the center* of rotation, O , by a directed angle θ is a bijection of the plane, p , onto itself under which every point P of p is taken into a point P' in p in such a way that the distance between the center of rotation, O , and the point P , $|OP|$, is equal to the distance between the image point P' and O , $|OP'|$ ($|OP| = |OP'|$) and the magnitude and the direction of the angle POP' is equal to the magnitude and the direction of the angle θ ($\angle POP' = \theta$).

The above notation $\angle POP' = \theta$ implies not only the equality of the *magnitudes* of the angles at hand but also the equality of the *directions* of these angles (Figure 5.3.5):



The rotational image P' of the point P can be constructed with the Euclidean compass alone (see if you can invent such a construction).

In order to obtain the rotational image F' of a figure F , every point of the figure F must be rotated as explained above (Figure 11.14):



From the above definition of rotation of the plane it follows that the only fixed point under any non-trivial rotation of the plane is the center of rotation – every other point of the said plane under a non-trivial rotation undergoes a non-trivial transformation (displacement).

Moreover, if the point P' is the image of the point P under the rotation of the plane about the center O by the angle θ then the point P is the image of the point P' under the rotation of the same plane about the same the center O but by the angle $-\theta$.

Definition 19.3: a *parallel translation* ($\mathcal{T}_{\vec{a}}$) of the plane p along a given fixed vector $\vec{a}$ is a bijection of the plane p onto itself that takes every point P of the plane p into the image point P' in p such that:

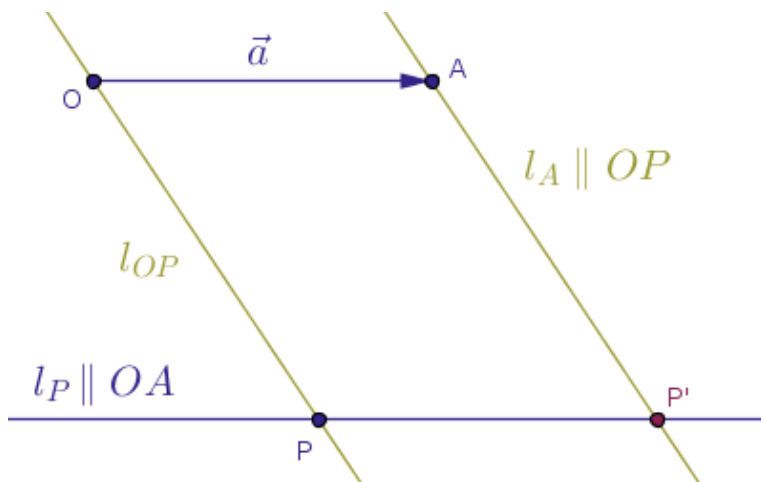
$$\overrightarrow{PP'} = \vec{a}$$

In the above definition a *fixed* vector simply means that neither the direction nor the size of the vector at hand changes.

From the above definition of a parallel translation it follows that the image point P' of the respective preimage point P under such a translation can be obtained with Euclidean tools alone as follows:

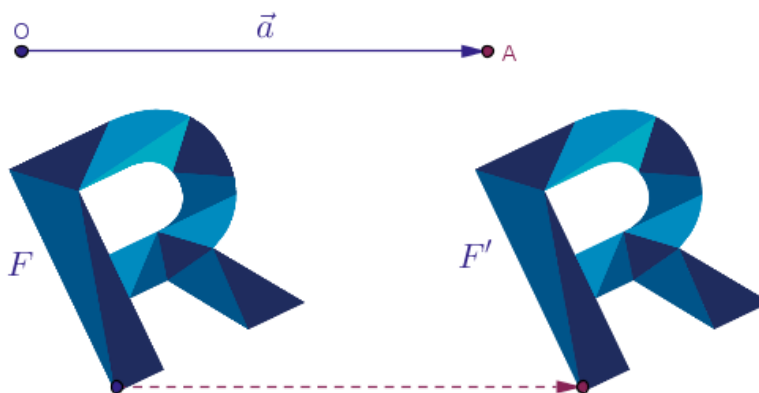
- draw a straight line l_{OP} through the points O and P
- draw a straight line l_A through the point A parallel to the straight line l_{OP}
- draw a straight line l_P through the point P parallel to $\vec{a}$
- the intersection of the straight lines l_A and l_P locates the unique image P' of the original point P under the parallel translation specified

Observe that with respect to the point P , its image point P' is located *in the same direction* as the end point A of the given vector a is located with respect to its origin point O . Moreover, the distance $|PP'|$ is equal to the length $|\vec{a}|$ of the given vector a (Figure 11.15):



The translational image P' of the point P can be constructed with the Euclidean tools, a compass and a straightedge, alone (see if you can invent such a construction).

In order to obtain the translational image F' of a figure F , every point of the figure F must be translated as explained above (Figure 11.16):



Note also that under every non-trivial parallel translation of the plane *every* (!) point of the parent plane is moved in a non-trivial fashion. Which is to say that a parallel translation of the plane does not have any fixed points.

Parallel translation takes circles into circles and straight line into straight lines as well.

For completeness, the above transformations of the plane have a number of interesting properties. For example, a composition of two consecutive reflections of the plane in two (fixed) *parallel* straight lines

is equivalent to a particular parallel translation of the said plane and a composition of two consecutive reflections of the plane in two (fixed) *intersecting* straight lines is equivalent to a particular *rotation* of the said plane, etc.

In addition, both the Euclidean and the complex planes can be reflected not only in a (fixed) point and a (fixed) straight line but in a (fixed) circle, parabola and hyperbola and such reflections of the plane can be exported into the Euclidean 3-space naturally, where the said space can be reflected also in a (fixed) plane, sphere, ellipsoid, circular and elliptic paraboloids and one- and two-sheeted hyperboloids and so on.

It is beneficial to read the above definitions at least once and play with them a bit before tackling the next exercise because:

- *rotations*
- *translations*
- *straight-line symmetries* and
- *central symmetries*

are the specific examples of planar motions - if in the upcoming proofs something becomes not clear then the readers should fall back on these examples of planar motions and trace the troublesome line of reasoning via them.

Exercise 11.43: is the set of rigid planar motions under composition a group or not?

Solution: As usual, one way to answer the above question is to trawl the definition of a rigid planar motion shown above, see **Definition 18**, through a particular set of group axioms.

Say, we will use the popular set of group axioms shown in the **Take 1** discussion of the **Group Axioms** chapter.

We already know that a composition of planar motions acts on exactly two, no more and no less, such motions.

Is It Binary?

Is a composition of two rigid planar motions is, again, a rigid planar motion?

In other words, is the operation of composition defined on the set of rigid planar motions binary?

In order to answer that question, we have to investigate whether a composition of rigid planar motions is bijective and whether such a composition conserves the distance between any two of its preimages.

To that end, we address the first question: is a composition of two rigid planar motions *bijective*?

Asked differently, is a composition of two rigid planar motions *surjective* and *injective*?

Injectivity

Well, from **Definition 6** we remember that a mapping is injective if under that mapping to distinct preimages there correspond distinct images.

Thus, let us consider two rigid planar motions f and g , such as *a parallel translation* along a fixed vector and *a rotation* about a fixed point, for example.

Further, agreeing that X is an arbitrary point in the parent plane, let us carry out the planar motion f first, followed by the planar motion g next:

$$(g \circ f)(X) = g(f(X))$$

In that case a point A of the parent plane will be taken by the motion f into a uniquely defined point $f(A)$ in the same plane and then that point $f(A)$ will be taken by the motion g into a uniquely defined point $g(f(A))$ also in the same plane.

Moreover, any two *distinct* points $A \neq B$, by **Definition 18**, will be taken by the motion f into the points $f(A)$ and $f(B)$ that are distinct because under any bijection, which f certainly is, to distinct preimages there correspond distinct images.

Thus, the distinct points $f(A)$ and $f(B)$, by the same **Definition 18**, will be taken by the rigid motion g into the points $g(f(A))$ and $g(f(B))$ that are also distinct.

Therefore, any two distinct points $A \neq B$ will be taken by the composition $g \circ f$ into distinct points $g(f(A))$ and $g(f(B))$, which, in turn, means that the composition $g \circ f$ is *injective*.

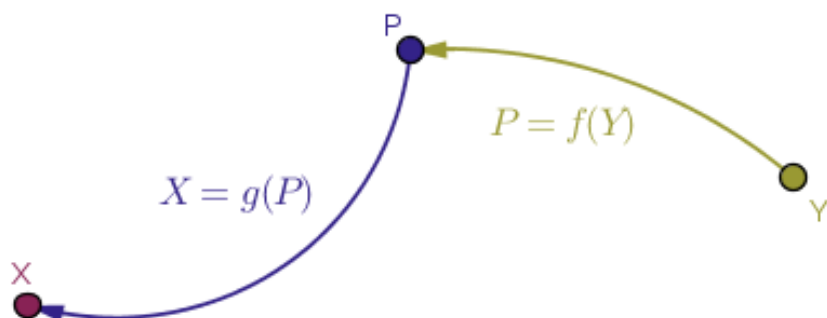
Surjectivity

From **Definition 5** we remember that a mapping is surjective if under that mapping every image has at least one preimage.

Which, as our wishful thinking runs in this particular case, means that into *every* point of a transformed plane the composition $g \circ f$ should take *a certain* point of the parent plane.

As such, if X is an arbitrary point in the parent plane then, by **Definition 18**, there exists a point P in the same plane that is taken into X by the motion g because that motion is surjective.

There also exists a point Y in the same plane that is taken into P by the motion f because that motion is also surjective (Figure 11.17):



Therefore, for every image point X in the parent plane there exists a preimage point Y in the same plane such that $X = g(f(Y))$, implying that the composition $g \circ f$ is also *surjective*.

But now we showed that that composition $g \circ f$ is both surjective and injective. Hence, the composition $g \circ f$ is bijective.

Next, we address the second question: does that composition conserve the distance between the corresponding pairs of its image points?

It sure does - by the above definition, the distance between the points A and B is equal to the distance between the points $f(A)$ and $f(B)$:

$$|AB| = |f(A)f(B)|$$

as per (17), and by the same definition the distance between the points $f(A)$ and $f(B)$ is preserved by the motion g :

$$|AB| = |f(A)f(B)| = |g(f(A))g(f(B))|$$

Therefore, the composition $g \circ f$ also preserves the distance between any of its two preimage points and, hence, such a composition is also a *rigid motion of the plane*.

Is It Associative?

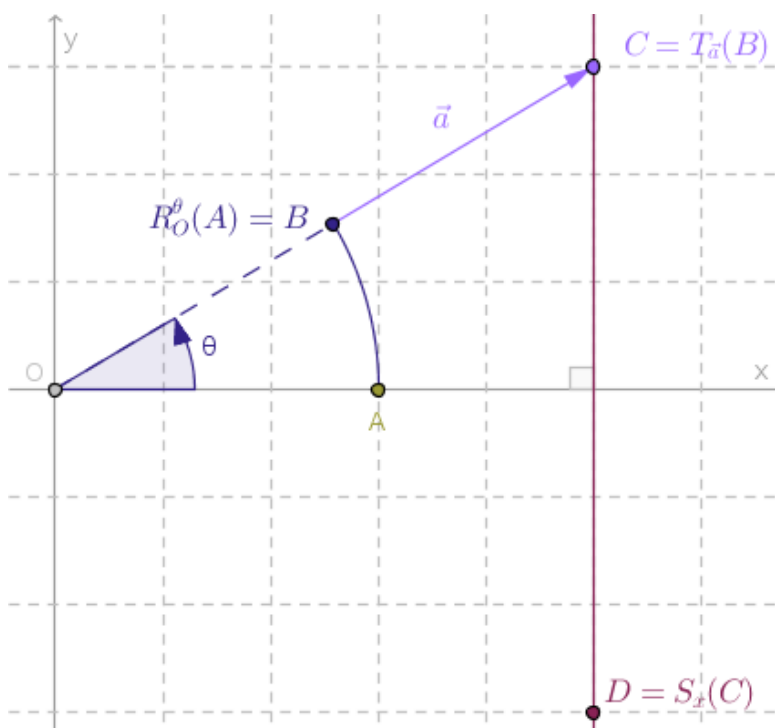
Again, assume that f , g and h are three rigid planar motions. Will it be the case that:

$$(h \circ g) \circ f = h \circ (g \circ f)$$

?

In order to answer that question, let us experiment with some sample points: let the motion f take a point A into a point B , let the motion g take the point B into a point C and let the motion h take the point C into a point D .

For example, say, f is the rotation about origin, g is a translation along a vector $\vec{a}$ and h is the reflection in the x -axis (Figure 11.18):



and the result of the following composition:

$$[h \circ (g \circ f)](A) = h(g(f(A))) = D$$

is shown in the diagram above.

If we trace the mechanics of the following composition:

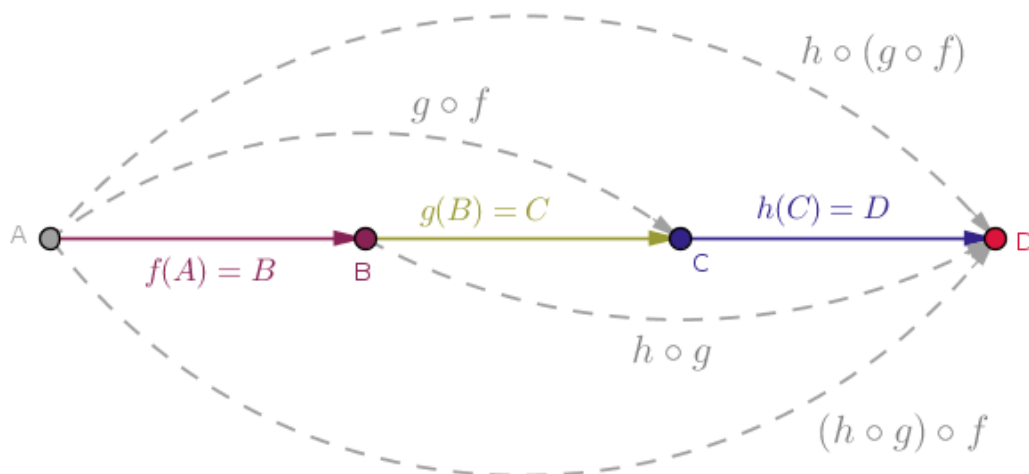
$$[(h \circ g) \circ f](A) = (h \circ g) \circ f(A) = h(g(f(A))) = D$$

then we shall find exactly the same result, the point D , as before.

Commentary: keep in mind the fact that in the above composition of three motions the overall order of the said motions in both *expressions* remains the same - the motion g is positioned *to the right* of the motion h and the motion f is positioned *to the right* of the motion g .

Whether a composition of two motions a and b is *commutative* or not is a totally different story - most compositions (of rigid motions) are actually not commutative: $a \circ b \neq b \circ a$.

Rewriting the above in a more general way, on the one hand, the composition $g \circ f(A) = g(f(A)) = g(B) = C$ will take the point A into the point C and the composition $h \circ (g \circ f(A)) = h(g(B)) = h(C) = D$ will take the intermediate point C into the point D , thus, taking the original point A into the point D (Figure 11.19):



On the other hand, the composition $h \circ g(B) = h(g(B)) = h(C) = D$ will take the point B into the point D and, thus, the composition $(h \circ g) \circ f(A) = D$ will also take the original point A into the point D because, first, the result of $f(A)$ will be the point B and, second, the point B will be acted on by the composition $(h \circ g)$, which will take it into D .

Thus, the composition of rigid planar motions *is* associative.

Commentary: note that in the above argument we, technically, did not *really* use the fact that the symbols f , g and h stood specifically for rigid planar motions. We defined a rigid planar motion as a bijection that has a certain, distance-preserving, property. A bijection is a specific type of a mapping, see our **Definition 4**. It turns out that the definition of a mathematical mapping has enough power baked into it in order to make a *composition* of mappings associative right out of the box.

In other words, if a certain operation defined on the elements of two sets is a *mapping* then the associativity of a composition of such mappings necessarily follows. In a traditional text that fact is proven shortly after the definition of a mapping is given. In our text, however, we delayed the discussion of the concept of associativity to later chapters in order to conform to the overall beat and flow of the discourse.

Thus, in general, a rigid planar motion is a mapping and, hence, a composition of rigid planar motions is associative.

Identity Element

Does the unique identity rigid motion of the plane exist?

Yes sure - the identity motion of a plane e is a motion that takes every point X of the parent plane back into the same point: $X = e(X)$.

Such a transformation of a plane satisfies **Definition 18** because that transformation is bijective, since to every image point X its preimage X clearly exists (surjectivity) and to distinct points $X \neq Y$ there correspond the same distinct points $X \neq Y$ (injectivity), and the distance $|XY|$ between any two points in the parent plane will, also clearly, be preserved by the identity transformation:

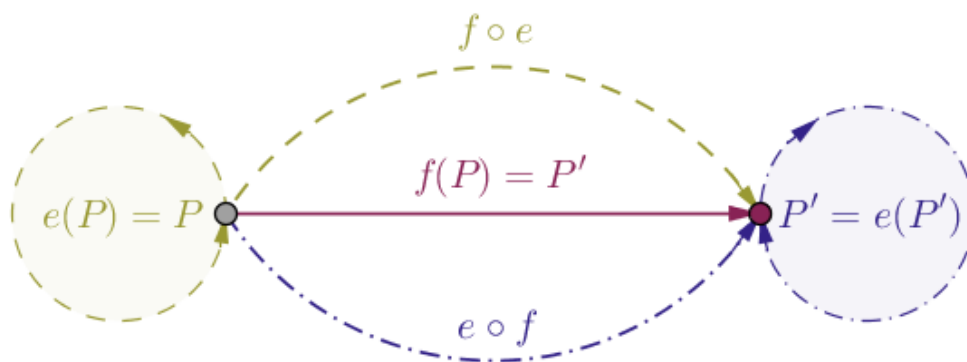
$$|XY| = |e(X)e(Y)| = |XY|$$

Therefore, such a transformation *is a rigid planar motion*.

Moreover, for every transformation f it will also be the case that:

$$e \circ f = f \circ e = f$$

since taking a preimage point P *somewhere* by f , into the point P' , and then leaving its image there via e is the same as taking a preimage point P into itself via e and then moving it into the same *somewhere*, the point P' , by f (Figure 11.20):



Lastly, we want to see if planar motions have well-defined inverses.

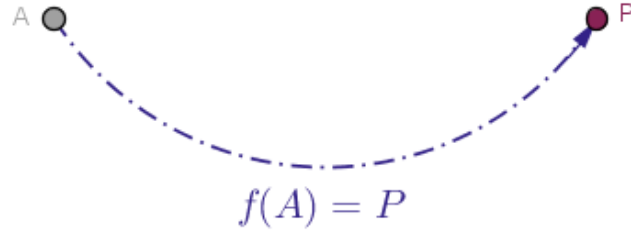
Inverse Elements

By definition, we seek such a rigid motion of the plane g that, given a direct rigid motion f , will take *all* the image points of f back into their preimages:

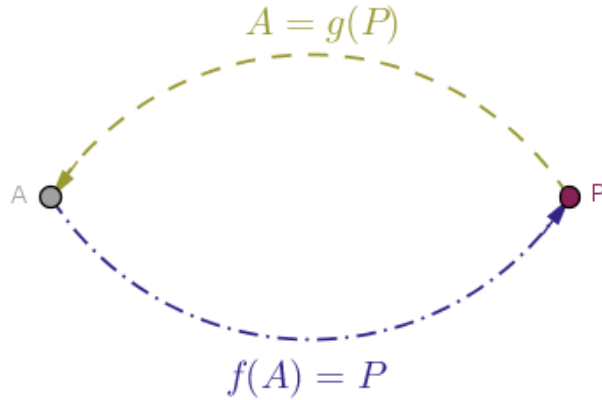
$$g \circ f = f \circ g = e$$

regardless of the order in which the candidate inverse motion g is carried out.

If we *define* such a candidate, for now, inverse planar motion g in such a way that, given an arbitrary point A in the parent plane and a motion f that takes A into a unique point $f(A) = P$ (Figure 11.21):



the operation g takes the point $f(A) = P$ back into the point A (Figure 11.22):



then can it be shown that the said operation g is indeed a *rigid planar motion* and that it is the inverse of f ?

In other words, can it be shown that g is bijective, that it preserves the distance between any of its two preimage points and that the required relationships:

$$g \circ f = f \circ g = e$$

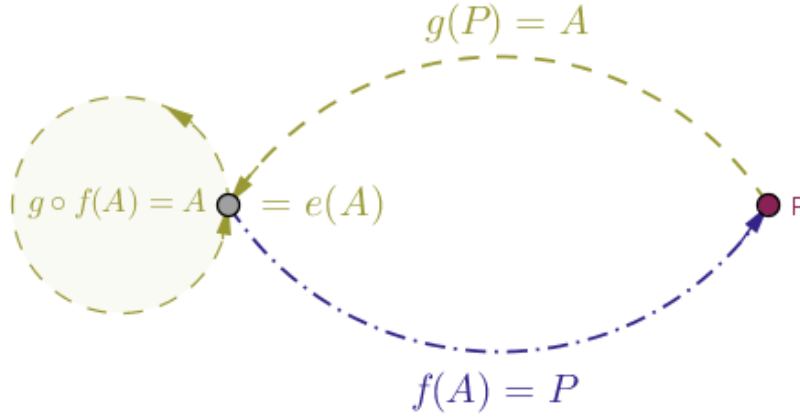
for it hold?

Well, we know that the planar motion f takes an arbitrary point of the parent plane A into a unique point $f(A) = P$. With respect to the that motion f the point A plays the role of a *preimage* and the point P plays the role of *the image* of A under f .

Thus, if we first act on such a point A with the motion f and then on the point $f(A)$ we act with the candidate inverse motion g then we shall obtain the original point A , implying that the said composition, $g \circ f$, does amount to the identity transformation:

$$[g \circ f](A) = g(f(A)) = g(P) = A = e(A) = e$$

depicted on the left side in the diagram below (Figure 11.23):

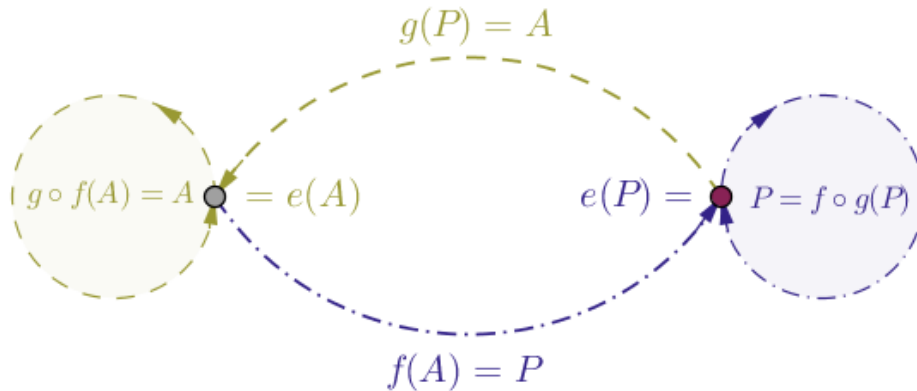


But since the image of the point A under the motion f can be *any* point of the parent plane, meaning that any point in such a plane can be labeled as $f(A)$, it follows that for *any* point in the plane it can be said that it is into that point the candidate motion g takes a *certain* point of the plane.

In other words, since an arbitrary point in the plane can be named as either A or P , if we first act on a point $P = f(A)$ with the candidate motion g and then on the point $g(P)$ we act with the motion f then we shall obtain the original point P , implying that the said composition, $f \circ g$, does amount to the identity transformation:

$$[f \circ g](P) = f(g(P)) = f(A) = P = e(P) = e$$

depicted on the right side in the diagram below (Figure 11.24):



This means that both the motion $g \circ f$ and the motion $f \circ g$ take an arbitrary point of the given plane back into itself: $g \circ f = f \circ g = e$.

Is the proposed motion bijective?

Well, can it so happen that the same point $P = f(A)$ is labeled in many ways, as $f(A)$ and as $f(B)$ perhaps?

No, that can never happen because by definition, the motion f takes different preimage points into different image points, since f is injective. Therefore, any point of the parent plane is the image of one and only one point, A , and, thus, can be labeled in one way only, $f(A)$.

Also, is it the case that into *every* point of the parent plane the motion g takes *some* point of that plane or is it the case that every image, A , under g has a preimage, $f(A)$? Yes, sure since into the point A there taken the point $f(A)$.

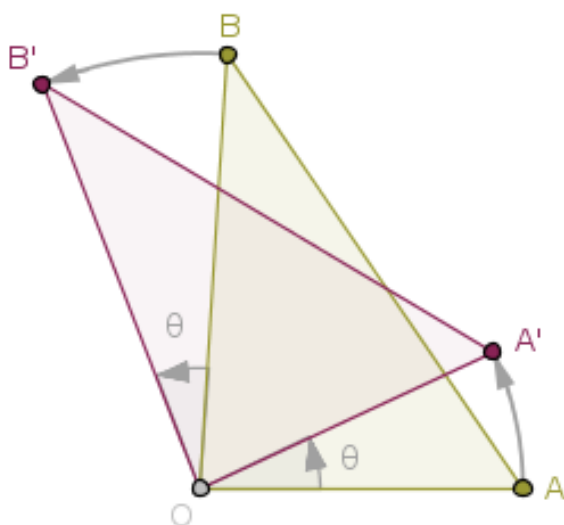
Lastly, since the distance between the points $f(A)$ and $f(B)$ is equal to the distance between the points A and B , the converse is also true - the distance between the points A and B , the images under the candidate motion g , is equal to the distance between the points $f(A)$ and $f(B)$, the preimages under g .

All in all, we have shown that the candidate operation g defined above is a planar motion that is the inverse of the motion f .

Thus, the set of rigid planar motions under the operation of composition *is* a group. $\square$

With a little bit more, purely geometric, work it can be *proved* that the above popular *individual* transformations of a plane *are* planar motions.

For example, under a rotation of the plane about a fixed point O by a directed angle θ the *distinct* points A and B that are *not* collinear with O will be taken into their images A' and B' (Figure 11.25):



in such a way that the center of rotation, the point O , will be its own image, the distances $|OA|$ and $|OA'|$ will be equal one another, $|OA| = |OA'|$, the distances $|OB|$ and $|OB'|$ will be equal one another as well, $|OB| = |OB'|$, and the corresponding angles will be equal to the angle of rotation θ :

$$\angle AOA' = \angle BOB' = \theta$$

But that means that the angles AOB and $A'OB'$, not shown, are also equal one another:

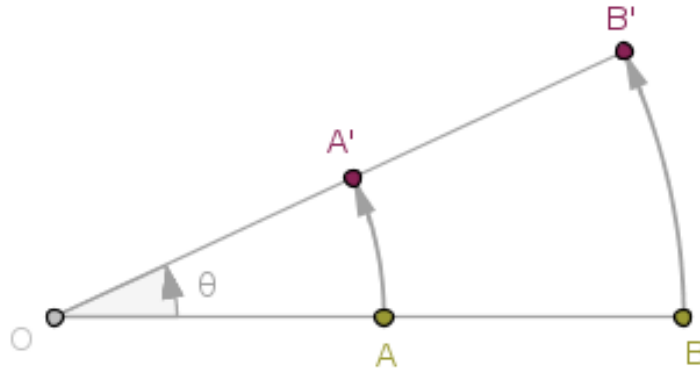
$$\angle AOB = \theta + \angle A'OB = \angle A'OB + \theta = \angle A'OB'$$

since the angle $A'OB$ is one and the same. Therefore, by **SAS**, the triangles AOB and $A'OB'$ are *congruent*. But in congruent triangles against equal interior angles there lie the sides of equal length. Hence:

$$|AB| = |A'B'|$$

If the distinct points A and B are collinear with the center of rotation then there are only two cases to consider.

In the first case, the points A and B are located *on the same side* of the center of rotation O (Figure 11.26):



meaning that these points will be taken by the said rotation into the points A' and B' such that that:

$$|OA| = |OA'| \text{ and } |OB| = |OB'|$$

but this time the image points A' and B' will also be collinear and these images will be located on the same side of the fixed point O because the triangles AOA' and BOB' will *share* the same interior angle at O because:

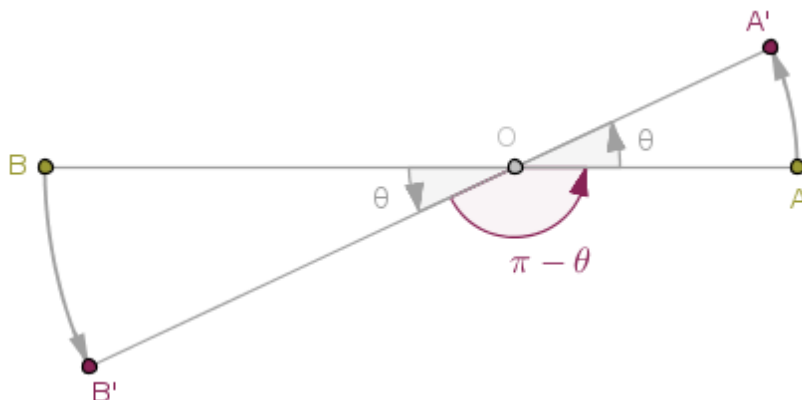
$$\theta = \angle AOA' = \angle BOB'$$

by construction.

But then, clearly:

$$|AB| = |OB| - |OA| = |OB'| - |OA'| = |A'B'|$$

In the second case, the center of rotation is positioned *between* the points A and B as shown below (Figure 11.27):



in which case the angle $B'OA$ will be equal to $\pi - \theta$, since the points A, O, B are collinear by assumption, which, in turn, implies that the angle $B'OA'$ is equal to π , which, in turn, implies that the points A', O, B' are also collinear and that the fixed point O is still *between* these image points.

But, then, clearly:

$$|AB| = |OA| + |OB| = |OA'| + |OB'| = |A'B'|$$

Either way, the distance between any two points in a given plane under rotation will be equal to the distance between the images of these points - remember that the center of rotation O under a rotation is taken into itself:

$$R_O^\theta(O) = O' = O$$

and, thus, $|OO| = 0 = |OO'|$.

Hence, a rotation of the plane about a fixed point by a directed angle is a rigid planar motion.

A Technical Wrinkle

We have to understand that the above theorem that we just proved in Exercise 11.43 casts its net very wide and it pertains to *all* planar motions, such as rotations, translations and reflections in a straight line, *taken in concert* as one whole.

Whether such motions taken one-at-a-time *individually* form a group under composition is a totally different story and the respective analysis of these cases must be undertaken anew each time for each such planar motion looked at in an isolation.

In that respect the group-forming situation here is rather delicate.

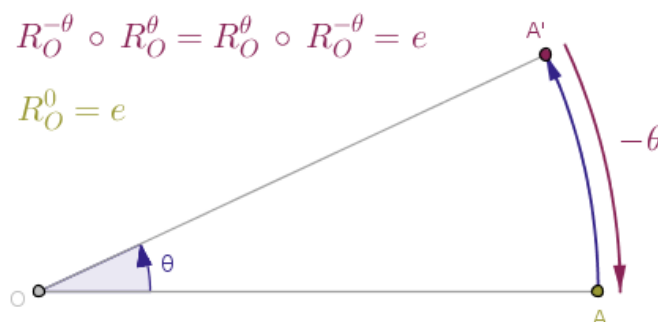
For example, it turns out that two consecutive *reflections* in two intersecting straight lines amount to a *rotation* - thus, a composition of two reflections in two intersecting straight lines is still a planar motion, as is claimed by our theorem, but that conclusion only holds when we consider *all* planar motions.

In other words, a composition of any two planar motions is still a planar motion.

However, if we limit ourselves to *just and only* reflections in a straight line and nothing else then the set of such individual motions under composition already does *not* form a group. See below*.

Bearing in mind the above observations, we conclude that when it comes to rotations of the plane and nothing else then only the set of all rotations of the plane about *one and the same* fixed point under composition is a group.

The identity transformation of that group is the rotation about its center by 0 radians and the inverse of any rotation of the plane by an angle θ is the rotation of the same plane about the same center by the angle $-\theta$ (Figure 11.28):



We also understand that that group is infinite and commutative, since rotating the plane about a fixed point by the angle θ_1 first and by the angle θ_2 second is exactly the same as rotating that plane about the same fixed point by the angle θ_2 first and by the angle θ_1 second.

Why is that?

Because a composition of any two rotations of the plane about one and the same fixed point can be reduced to a *sum of two corresponding real numbers*.

For example, if we name these real numbers as θ_1 and θ_2 then:

$$R_O^{\theta_2} \circ R_O^{\theta_1} = R_O^{\theta_1 + \theta_2}$$

but we know that the operation of addition of real numbers is commutative:

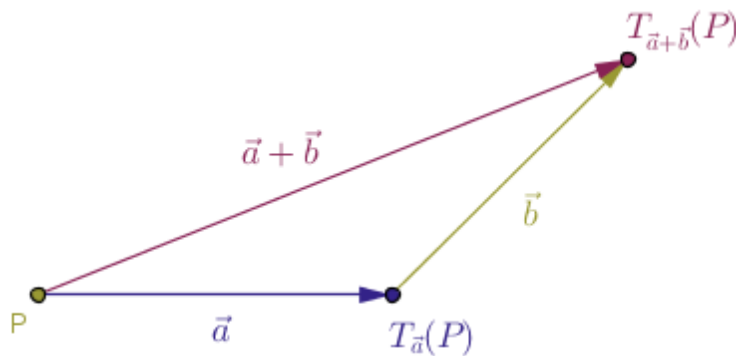
$$\theta_1 + \theta_2 = \theta_2 + \theta_1$$

Note carefully that the set *all* rotations of the plane, that is a set of rotations of the plane about *distinct* points, is *not* a group under composition of rotations because it may very well happen that a rotation of the plane about a point A followed by a rotation of the same plane about a point $B \neq A$ will be a *parallel translation* and not a rotation.

Consequently, rotations of the plane are not closed under composition.

Likewise, it can be shown that a parallel translation of the plane is also a motion of the plane and that such translations under composition do form a group.

A parallel translation of the plane along a vector $\vec{a}$ followed by a parallel translation of the same plane along a vector $\vec{b}$ is equivalent to a parallel translation of the said plane along the vector $\vec{a} + \vec{b}$ (Figure 11.29):



from where we see that the group of parallel translations of the plane is infinite and commutative because the operation of addition of coplanar vectors is commutative since the relation:

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

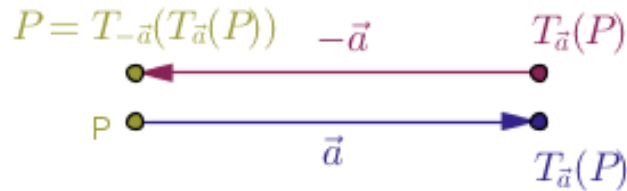
holds for all vectors $\vec{a}$ and $\vec{b}$.

The identity motion of that group is the translation along the zero vector $\vec{0}$.

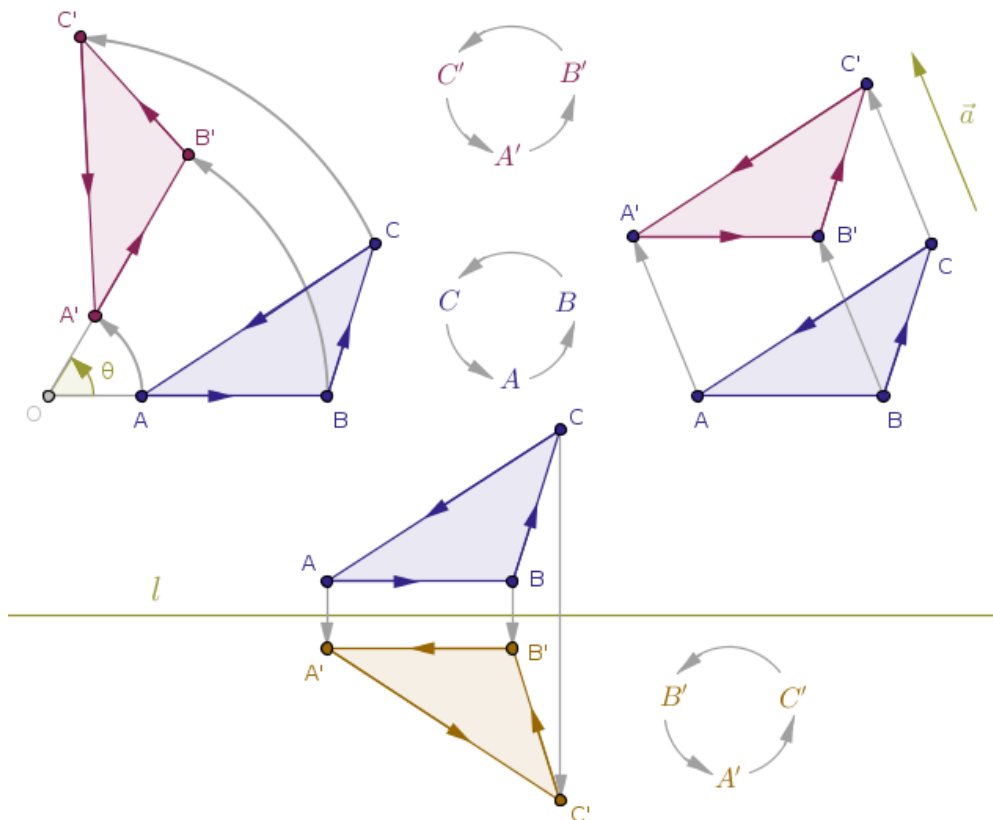
The inverse of any parallel translation along the vector $\vec{a}$ is the translation along the vector $-\vec{a}$ (Figure 11.30):

$$T_{\vec{0}} = e$$

$$T_{-\vec{a}} \circ T_{\vec{a}} = T_{\vec{a}} \circ T_{-\vec{a}} = e$$



*Observe that while *parallel translations* of the plane and *rotations* of the plane about *one and the same* fixed point preserve the so-called *orientation* of their preimages, *reflections in a straight line* do not - that is why *the individual* set of reflections of the plane in a straight line under composition is *not* a group (Figure 11.31):



Namely, say, the vertices of a sample preimage triangle in the parent plane are named in a way that they form the following cycle if we were to walk around the periphery of that triangle in such a way that our *left* hand, when stretched out, is always **inside** of the said triangle, while our *right* hand is always **outside** of the said triangle:

after visiting the vertex named A, we will visit the vertex named B, after which we will visit the vertex named C, after which we will visit the vertex named A

It turns out that the images of the said sample triangle, and of other planar figures, under rotations about the same point and parallel translations keep their orientation, meaning that the order in which we visit the vertices of a preimage figure and the order in which we visit the respective vertices of the image figure under these motions will be one and the same, so long that in both cases we traverse the periphery of the said figures in the same direction as shown in Figure 11.31 above.

In other words, if, always walking along the perimeter of our sample triangle counterclockwise, we discover the vertices of that preimage triangle in the $A \rightarrow B \rightarrow C \rightarrow A$ cycle then we will discover the vertices of its image triangle in the, same, $A' \rightarrow B' \rightarrow C' \rightarrow A'$ cycle for rotations and translations.

However, under the reflections in a straight line, l , as shown in the diagram above, we will discover the vertices of the image triangle in a *different* $A' \rightarrow C' \rightarrow B' \rightarrow A'$ cycle, which, effectively, takes us out of the parent set of orientation-preserving motions and becomes the barrier-to-entry for reflections in a straight line to form a group under composition.

Take a good mental note of this orientation preservation and non-preservation phenomenon, as we will use it very soon.

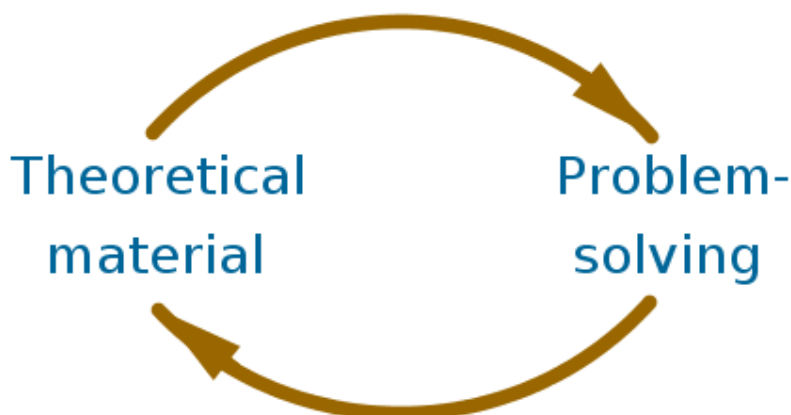
Exercises 5: Invent A Group On 2 And On 4 Symbols

It is safe to state that in pretty much all the participating countries, over pretty much the entire practicing time period, pretty much all the mainstream textbooks on mathematics, theoretical physics and computer science are built around the core process of *solving* problems.

The discipline and the regimen of solving problems has at least two purposes.

The first purpose of solving technical problems in any mathematical domain is to integrate the possibly dry, detached and abstract theoretical material into a practical context where the students, while expanding their web of technical knowledge, get to feel and experience with their own skin what a given theory is all about and what the tactical might of the developed theory has to offer, can do and can achieve.

During the solution of the domain-specific technical problems the back propagation process actually delivers the miracle of the *improved understanding* of the relevant theoretical material, which, in turn, enables the students to take on deeper and more difficult problems, which, through the back propagation channel, improve the understanding of the theoretical material, which, in turn ... (Figure 11.32):



The second purpose of solving technical problems in any mathematical domain is to develop and improve the students' problem-solving skills and tactical prowess.

A sobering question in that respect, however, is: solving the problems constructed by *whom*? *Where* did these problems come from? Where do the interesting *mathematical* problems *live*?

Ah.

Such problems are born in the minds of certain people who possess a great skill, an immaculate taste for and the needed experience in successful creation of the said problems.

It seems that it is actually easier, in some sense, to become a reasonable problem-solver than it is to become a reasonable problem-maker or a reasonable problem-constructor.

In other words, while the processes of problem-making and problem-solving seem to be symmetrical, the effort that goes into each is, likely, not.

Be that as it may, there is a great learning benefit to the process of problem-**construction** for, at least, the following reasons.

We rarely can create a reasonably interesting mathematical problem on the first try.

Most of time the situation with problem-construction is quite the opposite - before we manage to come up with a half-way decent mathematical problem, we will try and fail many, many times.

The iterative *try-and-fail* problem-making process is where the educational gold is because in order to engage in such a process we have to be in a firm command of the local material at the minimum.

In other words, if we are *not* in a firm command of the local material then there is absolutely no hope in creating a half-way decent mathematical problem.

But even if we *are* in a firm command of the local material then that fact alone does not guarantee that we will be able to eek out a decent problem in mathematics (or theoretical physics, or computer science) for an extra skill is needed to interpret a given set of mathematical facts and wrap them into a nice and enticing mysterious setting.

It is exactly during the problem-construction time when we kick the various technical ideas around in our mind:

- Will this work?
- Will that work?
- How about this? Nah.
- May be this? Not really.
- Wait, what if we do that? No, this sucks. Hm.
- What about this? Well, maybe ...

that the magic of learning and material reconciliation takes place.

To that end, in this discussion we shall attempt to do just that - we will give our readers some gentle hints and then we will ask them to invent a group based on the fodder supplied.

We will then suggest our version of the problems and we, as always, will be more than open and more than happy to hear from our readers about their own creative inventions.

Admittedly, constructing a decent mathematical problem is a tall order and it will be a difficult task for many readers.

However, there is little to nothing to be gained by doing only easy things and a lot can be learned from attempting to do difficult things even if we, initially, do not succeed at them.

Exercise 11.44*: on your own, invent a group on *two* symbols that is based on the result of **Exercise 11.6**, where we showed that the numeric set $\{-1, +1\}$ under multiplication is a finite, commutative, group of order 2.

Solution: if we slow down a bit, we may recall that in **Exercise 11.7** we showed that zero under *multiplication* is a finite, commutative, group of order 1 and in Exercise 11.15 we showed that zero under *addition* is also a finite, commutative, group of order 1.

A Zoo of Groups

We already know that we can construct the multiplication tables of these groups in which their elements are spelled out explicitly and in which their elements are abstracted in the group-theoretic fashion as follows (Figure 11.33):

$\times$	0
0	0

$+$	0
0	0

$\circ$	e
e	e

If we repeat the above experiment for the multiplicative group from **Exercise 11.6**, then we shall have (Figure 11.34):

$\times$	1	-1
1	1	-1
-1	-1	1

$\circ$	e	a
e	e	a
a	a	e

Now we exercise the idea put forward in the **Abstraction** chapter:

- we started with some specific elements, the numbers -1 and $+1$, and a specific operation, the multiplication of numbers
- we abstracted both and, next,
- we want to find some *other* material that *also* fits the newly found abstraction

In some, perhaps philosophical sense, we live in a quite remarkable universe in which the above *concrete-abstract-go-wide* maneuver is possible!

For example, how about the **O**dd and **E**ven numbers under the operation of addition?

Will they form a group that is *essentially the same* as the group found in **Exercise 11.6**?

Well.

Let us see if the newly found entities **O** and **E** are closed under addition.

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In other words, let us investigate whether adding these entities together in various order-sensitive ways will always produce one of these entities:

- $E + E = E$
- $E + O = O + E = O$
- $O + O = E$

It turns out that yes, the new entities are closed under addition.

Moreover, the unique additive identity in this candidate group will be the element **Even**, and the additive inverse of the element **Odd** will be that same element **Odd**.

What else?

Well, in **Exercise 11.40** we already showed that the set of binary strings of length $n > 0, n \in \mathbb{N}$ under the XOR operation is a group. Hence, a binary string of length $n = 1$ under the XOR operation is *essentially* the same group as the ones shown above (Figure 11.35):

+	<i>E</i>	<i>O</i>		o	<i>e</i>	<i>a</i>		XOR	0	1
<i>E</i>	<i>E</i>	<i>O</i>		<i>e</i>	<i>e</i>	<i>a</i>		0	0	1
<i>O</i>	<i>O</i>	<i>E</i>		<i>a</i>	<i>a</i>	<i>e</i>		1	1	0

As a matter of fact, we can now put together a much stronger line of reasoning here.

Namely.

Say, *two* symbols got together over a nice cup of tea and decided that they want to be a group under a certain operation. What are the possible choices for the interaction of these two symbols under the said operation?

Well, we know that by the popular **Axiom 2**, such a group *must* possess a unique identity element e with no wiggle room there. But that automatically means that there must be exactly one non-identity element, x , in such a group. Moreover, by the demand of the **Axiom 2**, in many popular sets of group axioms that we studied so far, it must be the case that:

$$e \circ x = x \circ e = x$$

besides the fact that $e \circ e = e$, of course, by the very definition of the identity element of a group.

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Thus far, we can already fill out the multiplication table of such a group and in that table we will know all the entries but one (Figure 11.36):

$\circ$	e	x	$\circ$	e	x
e	e	x	e	e	x
x	x	?	x	x	e
$\circ$			$\circ$		
					
					

Think about it - all 3 entries in the multiplication table of this group *must be* exactly e , x and x and the only entry in such a table that is not known to us yet is the result of the group product $x \circ x$.

But is the value of this product open for debate?

No, not really.

Why not?

Because of the **Axiom 3**, according to which *every* element of a group must possess a unique inverse.

We already know that the identity element of any group is its own inverse.

What about the mystery element x ? Is there any wiggle room for the inverse of that element?

Nope. In this particular group there is no wiggle room for the inverse of the element x because the inverse of any element of a group must still be an element of that group. Correct?

And how many elements are in this group? Only two, e and x . Thus, there two and only two candidates for the unique inverse of x . Namely, the inverse of x must be either e or x itself. But if e is the inverse of x then it must be the case that $x \circ e = e$ by definition and that relation is already taken by the (identity)

Axiom 2. Thus, the only choice for the inverse of x is the element x itself:

$$x \circ x = e$$

More formally, if we assume that $x \circ x = x$ then by multiplying that equation from, say, right by x^{-1} we shall arrive at:

$$x \circ x \circ x^{-1} = x \circ e = x = e = x \circ x^{-1}$$

a contradiction, since all the elements in a group must be distinct - meaning that it must be the case that $e \neq x$, and we can fill the multiplication table of this group completely as shown in Figure 11.36 above.

We now understand that the names of the symbols that make up this particular group and the name of the group operation are irrelevant - the only fact that matters here and in general is how the elements of a group interact with each other under the group operation.

For demonstration purposes, if a cherry and a banana form a group under a certain operation in which a cherry is the group's identity then the multiplication table of that group will look like the table shown at the bottom *left* in the diagram above and if a cherry and a banana form a group under a certain operation in which a banana is the group's identity then the multiplication table of that group will look like the table shown at the bottom *right* in the diagram above.

By playing around with just two symbols we arrive at the following, breathtaking, conclusion:

all the groups on two symbols have the same structure, shown in the upper right multiplication table in the Figure 11.36 above

The number 2 happens to be *prime* and there is a much deeper phenomenon at play here. As it turns out, the above conclusion also holds for any (finite) group whose order is a *prime number* but at this stage of our knowledge of the theory of groups we are not ready to prove that fact yet.

In any case, later on in your mathematical career you may learn that the following two planar geometric transformations, a rotation about a fixed point, designated with R , and a symmetry in a fixed straight line, also known as a *reflection* in a straight line, designated with S , interact with each other under the operation of composition as follows:

$$R \circ R = R$$

$$S \circ S = R$$

$$R \circ S = S \circ R = S$$

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Do these two symbols form a familiar group under composition?

Construct the multiplication table of these two symbols, see our sample below as a guide, and name the identity element of this maybe-group.

In order to drive the point home, recall that in our previous discussion we learned that while certain geometric transformations of a plane, known as *rigid planar motions*, do preserve the orientation of figures that these motions act on, there exist such planar motions that do not preserve the orientation of figures that these motions act on (reflections in a straight line, for example).

Let us use just that seemingly trivial fact to construct yet another group on two symbols: let us designate the planar motions that do preserve the orientation of figures with a plus sign, $+$, and let us designate the planar motions that do *not* preserve the orientation of figures with a minus sign, $-$.

Then, a composition of any two orientation-preserving motions and of any two orientation non-preserving motions will result in an orientation-preserving motion, while a composition of any orientation-preserving and an orientation non-preserving motion will result in an orientation non-preserving motion.

The set of such two types of planar motions under composition will form a group (Figure 11.37):

$\circ$	R	S	$\circ$	$+$	$-$
R	R	S	$+$	$+$	$-$
S	S	R	$-$	$-$	$+$

that has a structure that we already discussed.

Congratulations, you now invented your own problem that can be given to other students: do entities this and that form a group under the operation such and such? $\square$

Exercise 11.45*: on your own, invent a group on *four* symbols that is based on the three vertices of an arbitrary planar equilateral triangle and a certain special point associated with that triangle.

Hint 1: physics may help.

Hint 2: recall the concept of *the moment of a force*, which is centered around a force *multiplied* by a distance.

Solution: we already remember that an intrinsic property of matter known as *mass* can be modeled with 1) complex numbers, 2) negative real numbers and 3) positive real numbers.

If mass is modeled with *complex* numbers then it can be shown that the center of mass of two complex point masses can be located *anywhere* in the ambient plane - not necessarily on the straight line that connects the two said complex masses.

If mass is modeled with *negative* real numbers then it can be shown that the center of mass of two real point masses *will be* located in the ambient plane on the straight line connecting the two points but *outside* of the line segment that these two points define.

If mass is modeled with *positive* real numbers then it can be shown that the center of mass of two real point masses *will be* located in the ambient plane on the straight line connecting the two points *inside* of the line segment that these two points define.

Moreover, the notion of *center of mass* can be used to an elegant competitive advantage when evaluating certain finite sums, such as the sum of the dynamically scaled binomial coefficients:

$$\sum_{k=0}^n k \binom{n}{k}$$

for example.

While none of the above ideas seem to matter in the end, we, as mentioned earlier, visit plenty of dead ends in our personal web of knowledge trying to construct a mathematical problem.

Somehow such visits may put us on the right track.

Your trip will be different from mine.

Thus, taking the cue from the concept of the moment of inertia and the center of mass, we drop all the units of measurement and introduce the group operation simply as *the product of one given point by another*, the barycentric style.

Let the vertices of a planar equilateral triangle be A , B and C and let M be the center of mass of that triangle - the hinted at certain special point associated with a planar triangle.

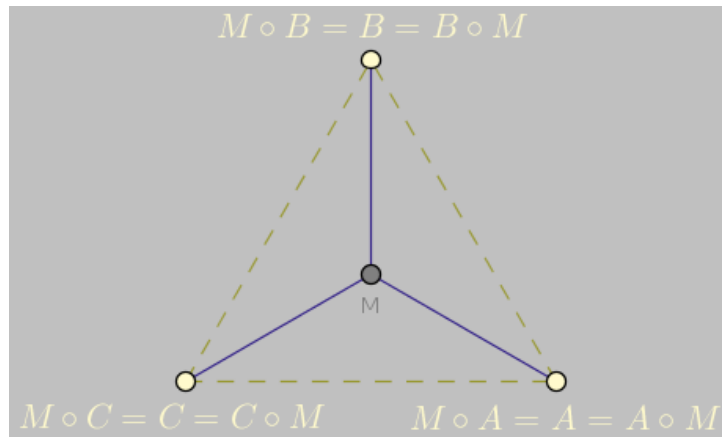
We, next, introduce the following toy model that will help us visualize the group product in this case.

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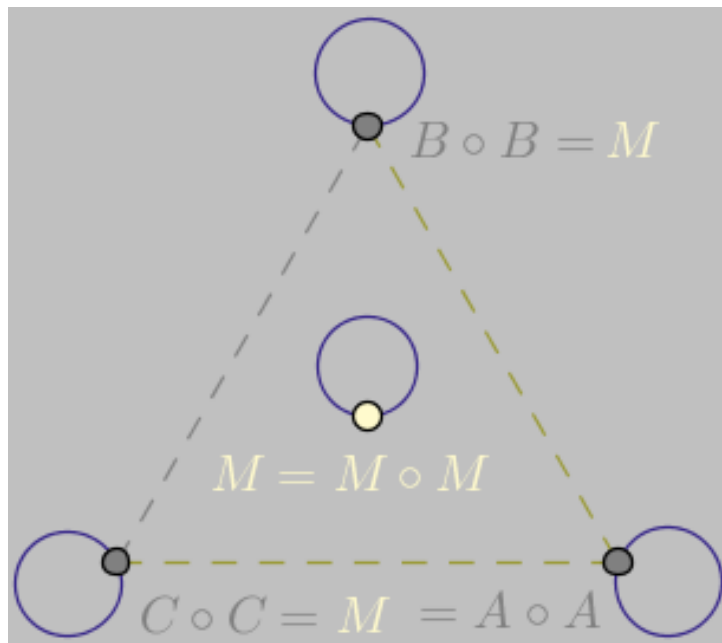
Let us agree to model the given points with the light bulbs and let us agree to model the group product itself with an electrical wire that connects the two light bulbs, read *points*, that comprise the said group product.

The result of a group product will be modeled with a light bulb that is turned on or lit up.

Thus, we suggest that when the center of mass M of a planar equilateral triangle is multiplied by any vertex of that triangle then *that* vertex results. In other words, when we connect the center of mass M of a planar equilateral triangle with *any* of its vertices with an electrical wire, shown in blue in the diagram below, then that vertex lights up, while the center of mass M remains dimmed (Figure 11.38):



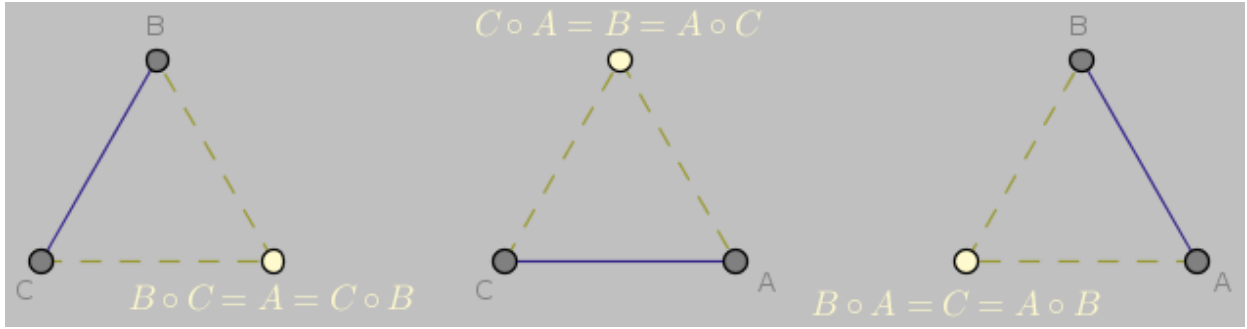
When a vertex of a planar equilateral triangle is multiplied by itself then the center of mass results (Figure 11.39):



In other words, when we short-circuit any vertex of a planar equilateral triangle onto itself then the center of mass M of that triangle lights up.

The product of the center of mass M of the said triangle with itself will be equal to the center of mass M also. In other words, when we short-circuit the center of mass M of a planar equilateral triangle onto itself then that center of mass M lights up.

Lastly, when any two *vertices* of a planar equilateral triangle are multiplied then *the remaining, third, vertex* of that triangle results. In other words, when we hook up any two vertices of a planar equilateral triangle with a blue wire then the remaining vertex of that triangle turns on (Figure 11.40):



We claim that the following set G of the four points associated with the described above planar equilateral triangle:

$$G = \{A, B, C, M\} \quad (18)$$

coupled with the described above operation of multiplication *is* a, finite, group, of order 4.

Proof of claim: such an operation acts on exactly two, no more and no less, elements of the set G .

The result of the product of any two points from the set G is, again, a point from the set G defined in (18).

We quickly see that by our definition of the group multiplication, we have:

$$AM = MA = A, \quad BM = MB = B, \quad CM = MC = C, \quad MM = M$$

which is to say that the center of mass M of the given set is the unique multiplicative identity element $e = M$.

We just as quickly see that every element of the suggested set G is its own inverse:

$$AA = M = e, \quad BB = M = e, \quad CC = M = e, \quad MM = M = e$$

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Thus, we just have to show that the operation of group multiplication defined above is associative and this is where the bulk of the group-theoretic work is hidden because on the surface, since we do not really know the exact, single, algorithm according to which the above results are produced, we have to enumerate and test all:

$$4 \cdot 4 \cdot 4 = 4^3 = 64$$

group products in a brute-force fashion by-hand. In other words, if, for example, in the set of real numbers we know that the operation of addition is associative because of the way the real numbers are constructed, we do not really have to probe all the 3-element additions of real numbers.

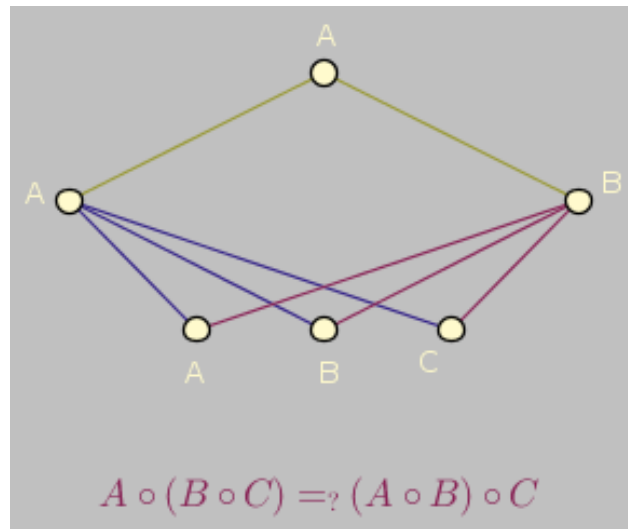
But in our, somewhat artificial, case - we do have to test all the 3-element products, given the absence of a generic algorithm according to which the group products are manufactured.

To that end, in order to minimize the amount of the grunt work that have to do, we observe that, since we already unmasked the unique identity element of the potential group $e = M$, if that identity element shows up anywhere in a 3-element product, such as, say, $A(BM)$, then we do not really have to verify the associativity of such a product because that product will contain only two factors, such as AB , say, and, thus, such a product will always be equal to the product of the remaining factors:

$$A(BM) = AB = (AB)M$$

because $AB = C$ and $CM = C$. Thus, we only want to study the 3-element products comprised of the vertices of a planar equilateral triangle A, B and C .

Next, since we defined a product of two vertices of a planar equilateral triangle in a way that does not favor any specific vertex, without loss of generality we can take the first element of a 3-element product to be A (Figure 11.41):



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The second element of such a product can be either A or, say, B - also without loss of generality.

The third element of such a product in either case can be A or B or C .

Thus, we mechanically read the only *six* 3-element products that we do have to analyze for the associativity off of the above color-coded graph without any difficulties.

Namely, for the green to blue paths we have:

$$AAA, AAB, AAC$$

and for the green to red paths we have:

$$ABA, ABB, ABC$$

Now we go to work, computing the relevant 3-element products *left-to-right* and *right-to-left*:

$$A(AA) = AA = A = AA = (AA)A$$

The above 3-element product is associative - one down, five to go:

$$A(AB) = AC = B = MB = (AA)B$$

The above 3-element product is associative also - two down, four to go:

$$A(AC) = AB = C = MC = (AA)C$$

and so on:

$$A(BA) = AC = B = CA = (AB)A$$

$$A(BB) = AM = A = CB = (AB)B$$

$$A(BC) = AA = M = CC = (AB)C$$

We, thus, have shown that the binary operation of multiplication defined on the above set G is associative.

Hence, sine all the criterions of a group have been satisfied, our claim is true - congratulations, you have constructed yet another problem. $\square$

At the end of the **Difficult, Giant, Leap Forward** chapter we already met the so-called *multiplication table* of a group.

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While we will undertake a deeper study of that group-theoretic tool later, let us construct such a table for this particular group (Figure 11.42):

$\times$	M	A	B	C
M	M	A	B	C
A	A	M	C	B
B	B	C	M	A
C	C	B	A	M

We will come back to this table in an hour or two.

Exercise 11.46*: on your own, invent a group on four symbols that is based on straight lines and hyperbolas.

Hint: two straight lines and one equilateral hyperbola are enough.

Solution: consider the following set G of four (real-to-real, single-variable) functions:

$$G = \left\{ x, -x, \frac{1}{x}, -\frac{1}{x} \right\} \quad (19)$$

defined on the set of real numbers, with the number zero thrown out, under the operation of *composition* defined on these functions.

Graphically, the elements of the set G defined in (19) correspond to two non-shifted straight lines $\pm x$ and two branches of a non-shifted equilateral hyperbola $1/x$.

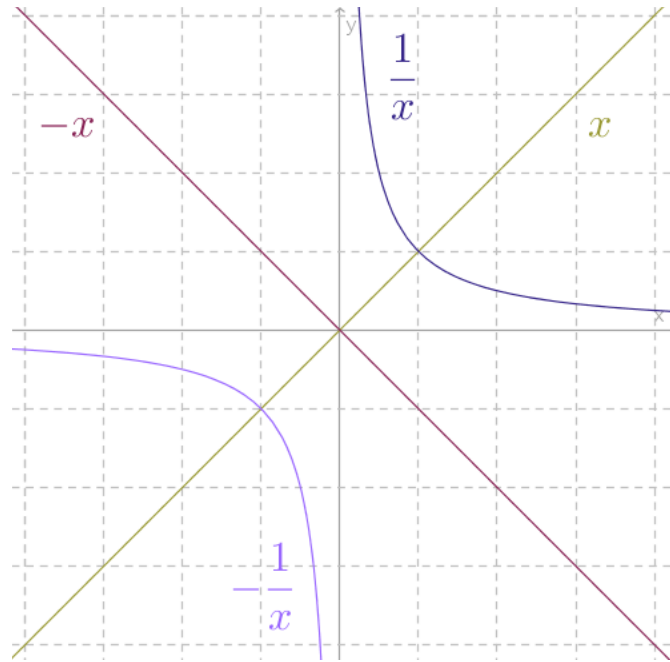
The partial graphs of the above functions are shown below.

We claim that the set G in (19) under the operation of composition *is* a, finite, group, of order 4.

Proof of claim: we know that the operation of composition acts on exactly two, no more and no less, elements or functions.

We also remember that the operation of composition of functions *is* associative.

In order to test whether the composition of any two elements of the set G (Figure 11.43):



belongs back to that set G , designating the given functions for convenience as shown:

$$G = \left\{ m = x, a = -x, b = \frac{1}{x}, c = -\frac{1}{x} \right\} \quad (20)$$

we construct a blank multiplication table of that potential group (Figure 11.44):

$\circ$	m	a	b	c
m				
a				
b				
c				

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and go to work.

We right away see that the composition of the function x with *any* function in G in any order:

$$m \circ m = x \circ x = x = m$$

$$m \circ a = a \circ m = x \circ (-x) = -x \circ x = -x = a$$

$$m \circ b = b \circ m = x \circ \frac{1}{x} = \frac{1}{x} \circ x = \frac{1}{x} = b$$

$$m \circ c = c \circ m = x \circ \left(-\frac{1}{x}\right) = -\frac{1}{x} \circ x = -\frac{1}{x} = c$$

is again the, same peer, element of G (Figure 11.45):

$\circ$	m	a	b	c
m	m	a	b	c
a	a			
b	b			
c	c			

implying that the element $e = m = x \in G$ is our number one candidate for the unique identity of this group.

Is it really?

Well, let us see.

Commentary: note how with four lines of effort above we captured seven entries in the multiplication table of this group.

Next, we also quickly notice that when the remaining functions in G are composed with themselves:

$$a \circ a = -x \circ (-x) = x = m = e$$

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$$b \circ b = \frac{1}{x} \circ \frac{1}{x} = x = m = e$$

$$c \circ c = -\frac{1}{x} \circ \left(-\frac{1}{x}\right) = x = m = e$$

then the candidate identity element $x = m = e$ results.

In other words, in this candidate group every element is its own inverse (Figure 11.46):

$\circ$	m	a	b	c
m	m	a	b	c
a	a	m		
b	b		m	
c	c			m

Observe how gradually but inevitably our multiplication table is populated, leaving less and less wiggle room for the remaining results!

We add the results of two more compositions, $a \circ b$ and $b \circ a$, to our table (Figure 11.47):

$\circ$	m	a	b	c
m	m	a	b	c
a	a	m	c	
b	b	c	m	
c	c			m

by carefully evaluating these compositions symbolically:

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$$a \circ b = -x \circ \frac{1}{x} = -\frac{1}{x} = c$$

and:

$$b \circ a = \frac{1}{x} \circ (-x) = -\frac{1}{x} = c$$

which leaves us with only four more compositions to go.

Next, we evaluate the results of the $a \circ c$ and $c \circ a$ compositions:

$$a \circ c = -x \circ \left(-\frac{1}{x}\right) = \frac{1}{x} = b$$

and:

$$c \circ a = -\frac{1}{x} \circ (-x) = \frac{1}{x} = b$$

and add these results to our work in progress (Figure 11.48):

$\circ$	m	a	b	c
m	m	a	b	c
a	a	m	c	b
b	b	c	m	
c	c	b		m

Lastly:

$$b \circ c = \frac{1}{x} \circ \left(-\frac{1}{x}\right) = -x = a$$

and:

$$c \circ b = -\frac{1}{x} \circ \frac{1}{x} = -x = a$$

takes care of the two remaining, $b \circ c$ and $c \circ b$, compositions (Figure 11.49):

$\circ$	m	a	b	c
m	m	a	b	c
a	a	m	c	b
b	b	c	m	a
c	c	b	a	m

which completes the multiplication table of this group, which, by the way, contains all the information needed for us to conclude, in vacuum, that the four symbols in it do form a group. $\square$

Reflection

Staying true to the ideas laid out in the **What Makes It Perfect** chapter, we now slow down a lot to reflect and retrospect on our problem-construction and problem-solving experience.

I. On the one hand, our toy model of electric wires and light bulbs used for the group of the vertices of a planar equilateral triangle and its center of mass in **Exercise 11.45** was technically relevant enough and precise enough to be meaningful.

On the other hand, in the big scope of things, that toy model was, to an extent, silly and vague.

However:

- some toy models are more successful than others
- we suggest that our readers keep inventing such toy models as they go along
- the ability to invent any relevant enough toy model still requires the basic understanding of the phenomenon being modeled

and that is why the invention of these toys models can have only a positive educational effect and it may be used as the first stepping stone toward *constructing* problems.

II. Based on the sample group of the vertices of a planar equilateral triangle, we observe that the group *operation* and the group *elements* acted on by that operation are, to some extent, very *cheap* - we, almost literally, invented a group out of, almost, nothing.

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Here the adjective *cheap* does not mean *functionally* cheap because the group operation still has to be binary and associative and the group elements acted on by the proposed operation must exhibit the unique identity element and the unique inverses.

Here the adjective *cheap* means *input-material-cheap*, since many wildly different things can be branded into a group operation and a group element. It is this *input-material-cheapness* that should provide another explanation of why the theory of groups has such *an exceptionally wide reach* not only in the modern grown-up mathematics but outside of it as well.

Speaking of which.

III. By now our readers should catch on to the grown-up idea that *any symbols* can represent the elements of a group - in **Exercise 11.45** we used the symbols M, A, B and C to represent the elements of its group and in **Exercise 11.46** we used the symbols m, a, b and c to represent the elements of a seemingly different group.

We specifically avoided representing the elements M and m from each of these groups with the more traditional symbol e , for the identity element of a group.

Why did we do that?

We did that in order to demonstrate one of the fundamental idea of the theory of groups:

*it is not the names of the symbols that matters but **the interaction** of the said symbols under the proposed operation*

Moreover, the *interaction* of the group elements or, to be more technically accurate, *the way in which group elements interact with each other under the group operation* defines *the structure* of this group and comparing and studying *the structures* of groups in general are the activities of interest.

Indeed.

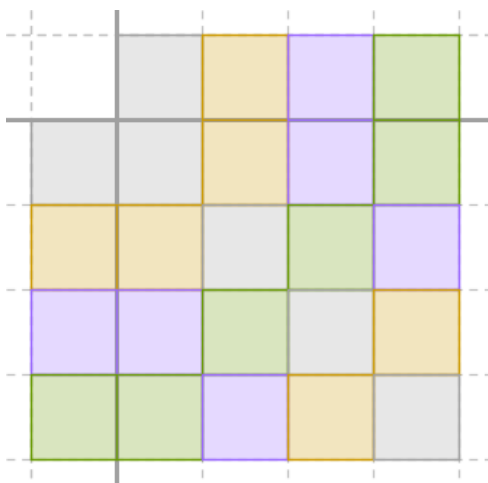
Let us, following the idea already mentioned in the **Difficult, Giant, Leap Forward** chapter, lift the symbols from the two multiplication tables that we constructed in **Exercises 11.45** and **11.46** and discover the fact that the colorful trace left behind by each of **Exercise 11.45** and **Exercise 11.46** groups is actually identical or one and the same.

That fact, in turn, tells us that *structurally* the above two groups are the same despite the fact that the names of the group elements, *points in a plane* and *single-variable real-to-real functions*, and the group operations, a mysterious multiplication of points and a composition of functions, are starkly different.

But wait a second.

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Haven't we already come across some number of groups the colorful trace of whose multiplication tables (Figure 6.13):



is also exactly the same?

We sure have - we have seen the same Klein four-group in:

- the duplex light switch group
- the tire rotations group
- the group of rigid symmetries of a non-square rectangle
- **Exercise 5.1.2** via the *followed by* operation carried out against the *opening* and *closing*:
 - the doors of a coupe
 - the two kitchen drawers
 - the two eyes of a human
 - the two curtains in your living room
 - the two handles of the bathroom faucets
- the binary strings of length $n = 2$ and the XOR operation (recall **Exercise 11.40**)

Hm.

If we explicitly spell out the vocabulary of all these groups in vacuum then we shall have:

- light switch levers, consecutive, flips
- car tires, consecutive, swaps
- non-square rectangle, consecutive, rigid, self-coincidences
- doors of a car, consecutive open/close operations
- kitchen drawers, same as above
- eyes, same as above
- curtains, same as above

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- faucet handles, same as above
- binary strings, length 2, XOR
- vertices, planar, triangle, center of mass, products
- straight-lines, equilateral hyperbola, compositions

In other words, what we have here is a wild vinaigrette of nouns and verbs that make a very good *what-do-they-have-in-common* puzzle that will surely elicit the bewildered looks on the faces of many of your non-indoctrinated friends.

Because now we are the smart bears, we know the answer to that puzzle and we possess enough of the group-theoretic foo in order to state with confidence that:

- these words describe the *isomorphic* instances of the Klein four-group in which:
 - each element is its own inverse and
 - the product of any two distinct non-identity elements produces the remaining, third, non-identity element

Later on, as an exercise, we will prove a theorem that states that if each element of a group is its own inverse then such a group must necessarily be commutative.

In any case, as the food for thought, recall that in the **Abstraction** chapter we observed that any abstraction cleaves a given phenomenon into two parts:

- the irrelevant minutiae and
- the only details that matter

The irrelevant minutiae are thrown out without any regrets, while the only details that matter become the essence of a given phenomenon.

In the context of the theory of groups, as we already see from the solved exercises, *the actual names* of the operations and *the actual names* of the elements that make up a group are absolutely irrelevant and it is *the way in which the said elements interact with each other via the suggested operation* is the only matter of significance and, thus, the only matter that should be studied by:

- divorcing the concrete names of the candidate operations and of the candidate elements, glued together by the said operations, from any context and, consequently,
- by abstracting the group operation into $\circ$ and by abstracting the group elements into a_0, a_1, a_2, a_3 and so on

The fact that in **Exercise 11.45** we did not really reveal the concrete nature of the group operation and kept it quite vague and mysterious was the goal of showing this particular example whose purpose is to impress on the minds of our readers the idea that, in the big scope of things, the exact nature of the

group operation is not that important and that groups can be constructed without the deep and intimate knowledge of how exactly does the proposed group operation *work*.

That idea, further, allows us to make the following observation: if, in a given context, the exact nature of group elements is not important and these group elements interact with each other under the group operation in the same way then multiple groups that are essentially the same (isomorphic) can be treated as indistinguishable or as just one group.

Consequently, by studying the properties of just one such group, we, automatically, study the properties of many such (isomorphic) groups.

Exercises 6: Invent A Group On 6 Symbols

On the wings of the, previous, **Exercise 11.46** we bring a similar exercise whose purpose is to introduce our readers to yet another wrinkle in the grown-up mathematics - doing, sometimes, a fair amount of calculations, the Leonhard Euler (1707–1783) style.

It would be a more than useful way to spend some time reading through some of the Euler's works to see *how* he did what he did, *working in parallel* and trying to keep up with the play of thought of the master.

The amount of the actual calculations that we will do in this exercise is a child's play but to the non-indoctrinated readers it may seem that computing some 36 compositions of functions is a cruel and unusual punishment.

The motivated readers can treat the upcoming exercise as another opportunity to construct a mathematical problem on their own.

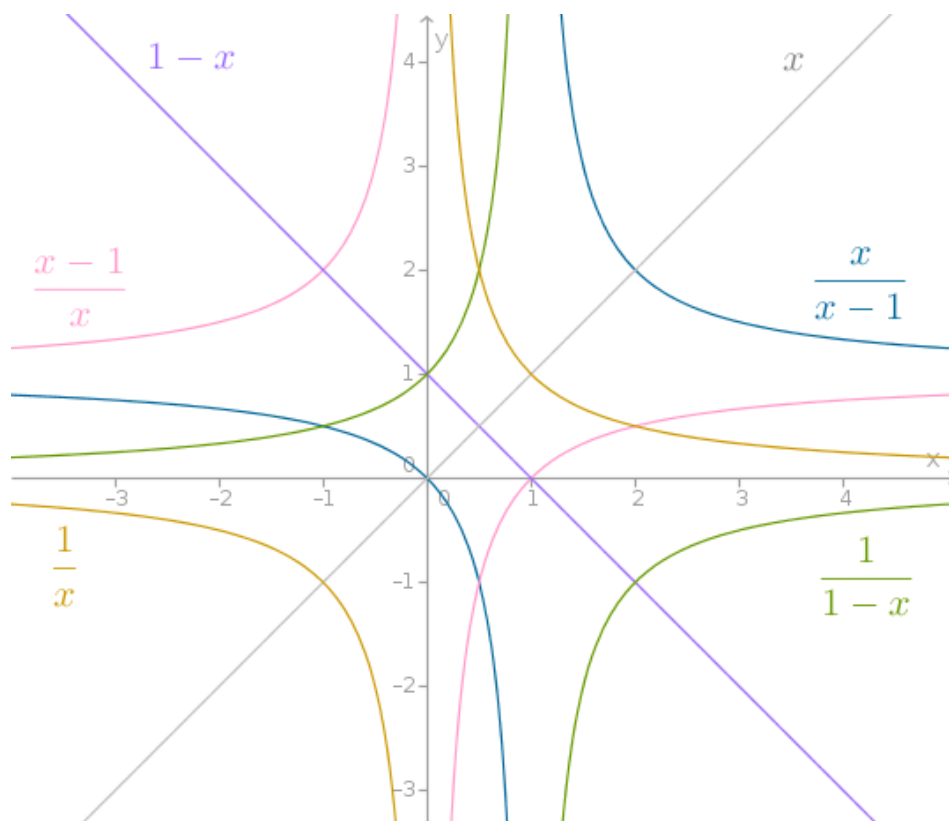
A hint to those brave souls: invent your own group on six symbols based on *equilateral and shifted hyperbolas* and *shifted straight lines*.

Exercise 11.47*: is the set G of the following six, defined on the set of real numbers with 0 and 1 removed, functions:

$$G = \left\{ x, \frac{1}{x}, 1 - x, \frac{1}{1 - x}, \frac{x - 1}{x}, \frac{x}{x - 1} \right\} \quad (21)$$

coupled with the operation of composition a group or not?

Solution: we observe in passing that the elements of the given set G in (21) have the following, partial, visual representations (Figure 11.50):



Now we trawl the proposed set and the proposed operation through a set of popular group axioms.

The operation of composition of functions acts on exactly two, no more and no less, elements.

Check.

We know from the general theory that the composition of functions is an operation that *is* associative.

Check.

We already see that the element x of the given set G is the number one candidate for the unique identity element of this group, if the set G in (21) under the operation of composition is a group, of course, because the composition of that element with any function from G in any order is, again, the element of the given set G .

Check, hopefully.

Thus, naming the functions at hand for convenience as follows:

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$$G = \left\{ e = x, a = \frac{1}{x}, b = 1 - x, c = \frac{1}{1 - x}, d = \frac{x - 1}{x}, f = \frac{x}{x - 1} \right\} \quad (22)$$

we, right away, construct the candidate multiplication table of the candidate pair $\langle G; \circ \rangle$, filling in the 11 **Axiom 2**-driven compositions (Figure 11.51):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a					
b	b					
c	c					
d	d					
f	f					

in a straightforward manner:

$$e \circ e = x \circ x = x = e$$

and:

$$e \circ a = x \circ \frac{1}{x} = \frac{1}{x} = a$$

$$a \circ e = \frac{1}{x} \circ x = \frac{1}{x} = a$$

and:

$$e \circ b = x \circ (1 - x) = 1 - x = b$$

$$b \circ e = (1 - x) \circ x = 1 - x = b$$

and so on. We write down the remaining 6 compositions, thus, arriving at the state of the potential multiplication table of a potential group shown in Figure 11.51 above.

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Next, by analogy with **Exercise 11.46**, we compute the compositions of the functions a, b, c, d, f with themselves, for a total of 5 more compositions:

$$a \circ a = \frac{1}{x} \circ \frac{1}{x} = x = e$$

$$b \circ b = (1 - x) \circ (1 - x) = 1 - (1 - x) = 1 - 1 + x = x = e$$

$$c \circ c = \frac{1}{1-x} \circ \frac{1}{1-x} = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{1-x-1} = \frac{x-1}{x} = d$$

$$d \circ d = \frac{x-1}{x} \circ \frac{x-1}{x} = \frac{\frac{x-1}{x} - 1}{\frac{x-1}{x}} = \frac{x-1-x}{x-1} = \frac{1}{1-x} = c$$

$$f \circ f = \frac{x}{x-1} \circ \frac{x}{x-1} = \frac{\frac{x}{x-1} - 1}{\frac{x}{x-1} - 1} = \frac{x}{x-x+1} = x = e$$

bringing the total number of compositions computed so far to $11 + 5 = 16$ (Figure 11.52):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e				
b	b		e			
c	c				d	
d	d					c
f	f					e

Note that at this stage of our investigation we still cannot say much about the inverses of the elements a, b and f . While it is tempting to claim that the functions a, b and f are their own inverses, we do not have a proof of that claim - maybe the compositions of these functions with the other elements of G will produce the candidate identity element e also?

We do not know. Yet.

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Therefore, we keep computing the remaining compositions of the functions at hand - we decide to do so in a way in which we will gradually fill out the candidate multiplication table of this maybe-a-group in the general top left to bottom right fashion.

This is a completely arbitrary decision that simply helps us to work in a consistent manner.

To that end, we work with the a, b pair of functions:

$$a \circ b = \frac{1}{x} \circ (1 - x) = \frac{1}{1 - x} = c$$

$$b \circ a = (1 - x) \circ \frac{1}{x} = 1 - \frac{1}{x} = \frac{x - 1}{x} = d$$

Hm.

Two more compositions, for a total of $16 + 2 = 18$, and we still haven't stepped out of the given set G . So far so good (Figure 11.53):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c			
b	b	d	e			
c	c			d		
d	d				c	
f	f					e

We have highlighted the new entries in our table above with the respective darker (green and blue) colors.

Next, we work with the a, c pair of functions:

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$$a \circ c = \frac{1}{x} \circ \frac{1}{1-x} = \frac{1-x}{1} = 1-x = b$$

$$c \circ a = \frac{1}{1-x} \circ \frac{1}{x} = \frac{1}{1-\frac{1}{x}} = \frac{x}{x-1} = f$$

Two more compositions, for a total of $18 + 2 = 20$, and we are still in the given set G (Figure 11.54):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b		
b	b	d	e			
c	c	f		d		
d	d				c	
f	f					e

Next, we work with the a, d pair of functions:

$$a \circ d = \frac{1}{x} \circ \frac{x-1}{x} = \frac{x}{x-1} = f$$

$$d \circ a = \frac{x-1}{x} \circ \frac{1}{x} = \frac{\frac{1}{x}-1}{\frac{1}{x}} = 1-x = b$$

Two more compositions, for a total of $20 + 2 = 22$, and we are still in the given set G (Figure 11.55):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	
b	b	d	e			
c	c	f		d		
d	d	b			c	
f	f					e

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Lastly, we try the a, f pair of functions:

$$a \circ f = \frac{1}{x} \circ \frac{x}{x-1} = \frac{x-1}{x} = d$$

$$f \circ a = \frac{x}{x-1} \circ \frac{1}{x} = \frac{\frac{1}{x}}{\frac{1}{x}-1} = \frac{1}{1-x} = c$$

Two more compositions, for a total of $22 + 2 = 24$, and we are still in the given set G (Figure 11.56):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e			
c	c	f		d		
d	d	b			c	
f	f	c				e

and now what do we see?

We see that:

- the candidate identity element e is present in the a -row in the table above exactly and only once and
- the same candidate identity element e is present in the a -column in the table above exactly and only once

But the a -**row** in the table above represents *all* the possible **left**-sided compositions of the element a with the functions of the set G :

$$a \circ e, \quad a \circ a, \quad a \circ b, \quad a \circ c, \quad a \circ d, \quad a \circ f$$

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and the a -**column** in the table above represents *all* the possible **right**-sided compositions of the element a with the functions of the set G :

$$e \circ a, \quad a \circ a, \quad b \circ a, \quad c \circ a, \quad d \circ a, \quad f \circ a$$

In other words, the composition of the given element a with no element other than a produces the candidate identity element e .

Hence, we tentatively claim that the things are shaping up to look good for the unique inverse of the element a - that element a seems to possess a unique inverse and that inverse is the element a itself:

$$a \circ a = e$$

Next, we repeat the above experiment for the function $b = 1 - x$.

For the pair of functions b, c we have:

$$b \circ c = (1 - x) \circ \frac{1}{1 - x} = 1 - \frac{1}{1 - x} = \frac{1 - x - 1}{1 - x} = \frac{x}{x - 1} = f$$

$$c \circ b = \frac{1}{1 - x} \circ (1 - x) = \frac{1}{1 - 1 + x} = \frac{1}{x} = a$$

implying that after the $24 + 2 = 26$ compositions of the functions from G , we are still in G (Figure 11.57):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f		
c	c	f	a	d		
d	d	b			c	
f	f	c				e

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For the pair of functions b, d we have:

$$b \circ d = (1 - x) \circ \frac{x - 1}{x} = 1 - \frac{x - 1}{x} = \frac{x - x + 1}{x} = \frac{1}{x} = a$$

$$d \circ b = \frac{x - 1}{x} \circ (1 - x) = \frac{1 - x - 1}{1 - x} = \frac{x}{x - 1} = f$$

implying that after the $26 + 2 = 28$ compositions of the functions from G , we are still in G (Figure 11.58):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	
c	c	f	a	d		
d	d	b	f		c	
f	f	c				e

Lastly, for the pair of functions b, f we have:

$$b \circ f = (1 - x) \circ \frac{x}{x - 1} = 1 - \frac{x}{x - 1} = \frac{x - 1 - x}{x - 1} = \frac{1}{1 - x} = c$$

$$f \circ b = \frac{x}{x - 1} \circ (1 - x) = \frac{1 - x}{1 - x - 1} = \frac{x - 1}{x} = d$$

implying that after the $28 + 2 = 30$ compositions of the functions from G , we are still in G (Figure 11.59):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d		
d	d	b	f		c	
f	f	c	d			e

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We see that so far:

- the candidate identity element e is present in the b -row in the table above exactly and only once and
- the same candidate identity element e is present in the b -column in the table above exactly and only once

In other words, the composition of the given element b with no element other than b produces the candidate identity element e .

Therefore, we tentatively reason that the element b does possess a unique inverse and that inverse is the element b itself:

$$b \circ b = e$$

Next, we repeat the above experiment for the function $c = 1/(1 - x)$.

For the pair of functions c, d we have:

$$c \circ d = \frac{1}{1-x} \circ \frac{x-1}{x} = \frac{1}{1-\frac{x-1}{x}} = \frac{x}{x-x+1} = x = e$$

$$d \circ c = \frac{x-1}{x} \circ \frac{1}{1-x} = \frac{\frac{1}{1-x} - 1}{\frac{1}{1-x}} = 1 - 1 + x = x = e$$

implying that after the $30 + 2 = 32$ compositions of the functions from G , we are still in G (Figure 11.60):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	
d	d	b	f	e	c	
f	f	c	d			e

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For the pair of functions c, f we have:

$$c \circ f = \frac{1}{1-x} \circ \frac{x}{x-1} = \frac{1}{1-\frac{x}{x-1}} = \frac{x-1}{x-1-x} = 1-x = b$$

$$f \circ c = \frac{x}{x-1} \circ \frac{1}{1-x} = \frac{\frac{1}{1-x}}{\frac{1}{1-x}-1} = \frac{1}{1-1+x} = \frac{1}{x} = a$$

implying that after the $32 + 2 = 34$ compositions of the functions from G , we are still in G (Figure 11.61):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	
f	f	c	d	a		e

We, thus, see that:

- the candidate identity element e is present in the c -row in the table above exactly and only once and
- the same candidate identity element e is present in the c -column in the table above exactly and only once

In other words, the composition of the given element c with no element other than d produces the candidate identity element e .

Therefore, we tentatively reason that the element c does possess a unique inverse and that inverse is the element d :

$$c \circ d = d \circ c = e$$

Lastly, for the pair of functions d, f we have:

$$d \circ f = \frac{x-1}{x} \circ \frac{x}{x-1} = \frac{\frac{x}{x-1} - 1}{\frac{x}{x-1}} = \frac{x - x + 1}{x} = \frac{1}{x} = a$$

$$f \circ d = \frac{x}{x-1} \circ \frac{x-1}{x} = \frac{\frac{x-1}{x} - 1}{\frac{x-1}{x}} = \frac{x-1}{x-1-x} = 1-x = b$$

implying that after the $34 + 2 = 36$ compositions of the functions from G , we are still in G (Figure 11.62):

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

which completes our jigsaw puzzle.

We see that:

- the candidate identity element e is present in the d -row and in the f -row in the table above exactly and only once and
- the same candidate identity element e is present in the d -column and in the f -column in the table above exactly and only once

We, hence, conclude that:

- the unique inverse of the element d is indeed the element c : $d \circ c = c \circ d = e$ and
- that the unique inverse of the element f is that element f itself: $f \circ f = e$

Thus, the given pair $\langle G; \circ \rangle$ does satisfy all the group axioms:

- the operation of composition on the set G in (21) is binary: the composition of any two functions from G is again a function from G
- the operation of composition of functions is associative
- the unique compositional identity element of this group is $e = x$ and
- each function from G has a unique inverse unmasked above

and the overall answer to this exercise is:

yes, the set G in (21) under the operation of composition is a, finite, group, of order 6

Note that since, for example:

$$b \circ c = f \neq a = c \circ b$$

this group is *not* commutative. $\square$

As mathematically inclined thinkers, we normally try to avoid tedious and, perhaps, lengthy computations. Sometimes, however, such a tedium seems to be unavoidable and in this exercise we caught a glimpse of a very, very mild version of what such a tedium may look like.

Separately, the group that we discovered in this exercise has a dedicated name and is known as the **anharmonic group**. The anharmonic group is closely related to the notion of a cross-ratio ω of four complex numbers a, b, c and d :

$$\omega = \frac{a - c}{b - c} \div \frac{a - d}{b - d} = \frac{(a - c)(b - d)}{(b - c)(a - d)}$$

each of which can be represented by a corresponding point in the complex plane: $A(a), B(b), C(c)$ and $D(d)$.

Those readers how solved **Exercise 11.42** correctly on their own may realize that all the functions from the set G shown in (21) are none other than the real versions of the linear fractional function of the first kind also known as a *Möbius transformation*.

Exercises 7: Rotational Symmetry Of Regular 3- And 4-gons

For the consequent development of the material in this space let us consider a few more examples of non-numeric groups that will pave the way for the study of the so-called *cyclic groups* and *subgroups of a group*.

A Zoo of Groups

Because the exercises in this discussion will be hands-on, for best results it pays to have access to a pencil, a compass, a straightedge, a pair of scissors, a push pin or two, some heavy or project paper to cut the regular n -gons from and some cardboard paper to use as a workbench, into which it is perfectly safe to stick the push pins.

For a top notch experience a piece of clear, see-through thin but firm sheet of plastic will do very well.

Omnia Mea Mecum Porto

Staying true to our original promise to introduce the readers to the way in which the grown-up mathematics is done, in this discussion we learn another important, *omnia mea mecum porto*, lesson.

The phrase *omnia mea mecum porto* translates from Latin into English as *all that is mine I carry with me*.

In mathematics (and theoretical physics and computer science) that phrase means that mathematical results do not just grow on trees and that such results are not shipped to us wholesome from a parallel universe by superior beings. All such results must be and are generated by us - there are no gifts in exact sciences and if we need something then we have to produce it. In other words, we are the sole carriers of all things mathematics.

In this particular exercise, before we put scissors to a paper, we have to generate the relevant algorithm on our own.

Exercise 11.48: using the Euclidean tools (a compass and a straightedge) alone:

- construct a regular 3-gon (an equilateral triangle)
- construct the center, O , of a regular 3-gon
- construct the bisectors of the interior angles of a regular 3-gon

Using the construction algorithm thus generated, cut a physical copy (a model) of a regular 3-gon out of either a heavy or project paper or a sheet of thin but firm, preferably transparent, plastic.

Solution:

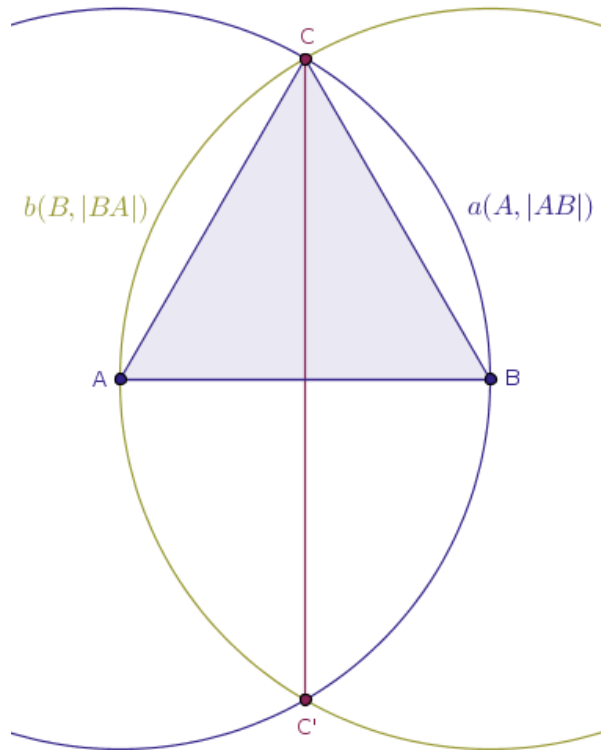
I. Mark and connect two distinct points $A \neq B$ with an arbitrary non-zero distance between them. That distance, $|AB|$, will be the length of the sides of the sought-after regular 3-gon and the points A and B will right away represent the two vertices of that 3-gon (Figure 11.63):



II. Draw a (blue) circle $a(A, |AB|)$ centered at the point A with the radius $|AB|$ and a (green) circle $b(B, |BA|)$ centered at the point B with the radius $|BA|$.

Mark the two points where the circles a and b intersect as C and C' and connect them with the (red) line segment CC' .

Connect the point C with the points A and B (Figure 11.64):



The point C thus constructed is the remaining vertex of an equilateral triangle and the highlighted in solid blue triangle ABC thus constructed is the sought-after regular 3-gon.

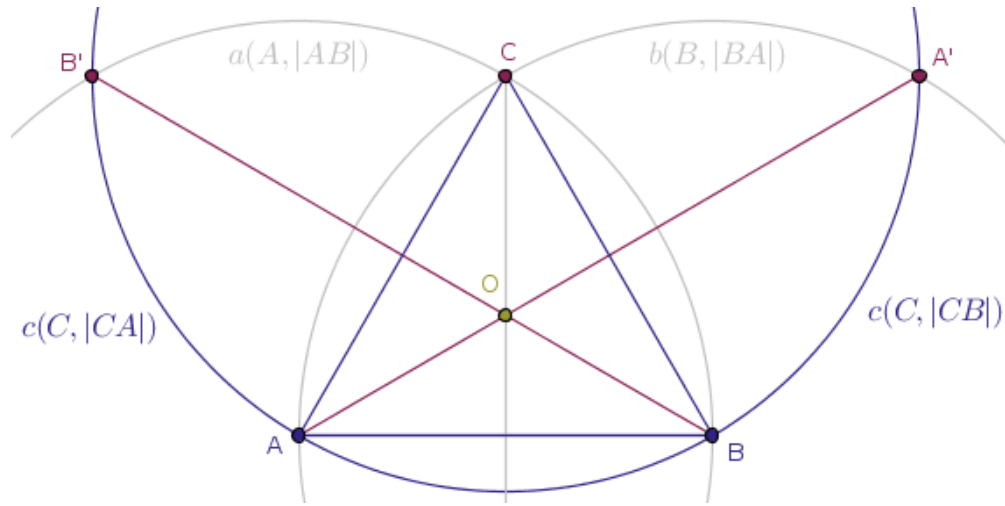
The line segment CC' thus constructed will house both the center of the regular 3-gon, O , and the bisector of its interior angle at the vertex C .

III. Repeat the above steps by drawing the circle $c(C, |CB| = |CA|)$ centered at the vertex C with the radius $|CB| = |CA| = |AB|$ until that circle c intersects:

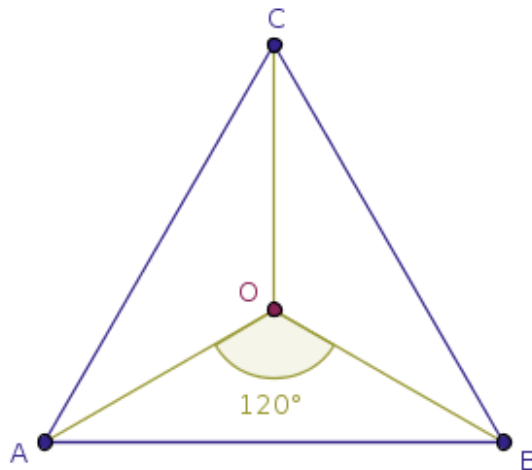
- the circle a - at the additional point B' and
- the circle b - at the additional point A'

Connect the points B and B' and connect the points A and A' .

All three line segments AA' , BB' and CC' will be concurrent at the point O (Figure 11.65):



The point O thus constructed is the sought-after center of the regular 3-gon and the line segments AA' , BB' and CC' contain the sought-after bisectors of the interior angles at the vertices A , B and C respectively (Figure 11.66):



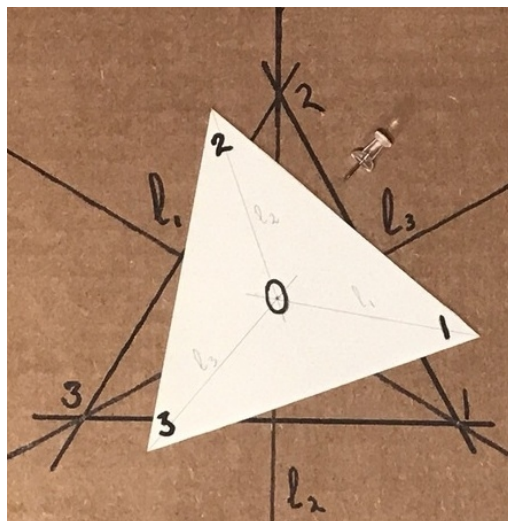
Cut a physical model of a regular 3-gon thus constructed from paper or, clear, plastic and mark its vertices with 1, 2 and 3:

- on one side of the transparent piece of plastic or
- on both sides of paper, making sure that the names of these vertices on either surface agree - a vertex of your regular 3-gon that carries the name of 1 on one side of the paper should also be named as 1 *on the opposite* side of that paper and so on

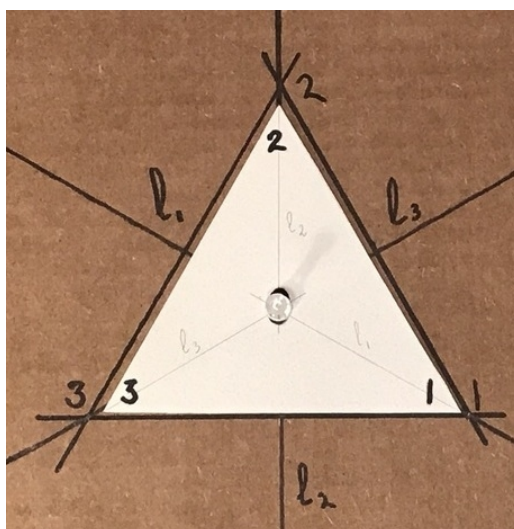
A Zoo of Groups

Put the newly obtained regular 3-gon on the cardboard workbench and trace the triangle's silhouette on it.

Jot down the names of the vertices of the *fixed* or *reference* triangle on the cardboard in accordance with the names of the moving triangle and, anticipating the future work, draw all three *fixed* or *stationary* axes of symmetry $l_{1,2,3}$ of the equilateral triangle on the cardboard - later on we will be reflecting our equilateral triangle with respect to *these*, fixed, axes of symmetry (Figure 11.67):



With a push pin punch a hole in the center O of the equilateral triangle, making sure that that triangle can rotate freely about its center in the cardboard plane (Figure 11.68):



Put this regular 3-gon thus produced aside for a moment. $\square$

In the next exercise we will go through the similar steps but for a square.

Exercise 11.49: using the Euclidean tools (a compass and a straightedge) alone:

- construct a regular 4-gon (a square)
- construct the center, O , of a regular 4-gon
- construct the bisectors of the interior angles of a regular 4-gon

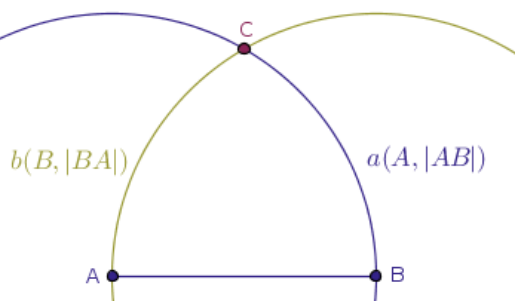
Using the construction algorithm thus generated, cut a physical copy (a model) of a regular 4-gon out of either a heavy or project paper or a sheet of thin but firm, preferably transparent, plastic.

Solution: of this exercise follows in the footsteps of the previous exercise.

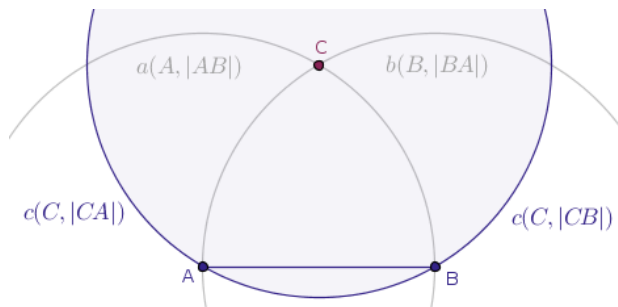
I. Mark and connect two distinct points $A \neq B$ with an arbitrary non zero distance between them. That distance $|AB|$ will be the length of the sides of the sought after regular 4-gon and the points A and B will right away represent the two vertices of the sought-after regular 4-gon (Figure 11.69):



II. Draw a (blue) circle $a(A, |AB|)$ centered at the point A with the radius $|AB|$ and a (green) circle $b(B, |BA|)$ centered at the point B with the radius $|BA|$. Mark just one point where the circles a and b intersect as C (Figure 11.70):

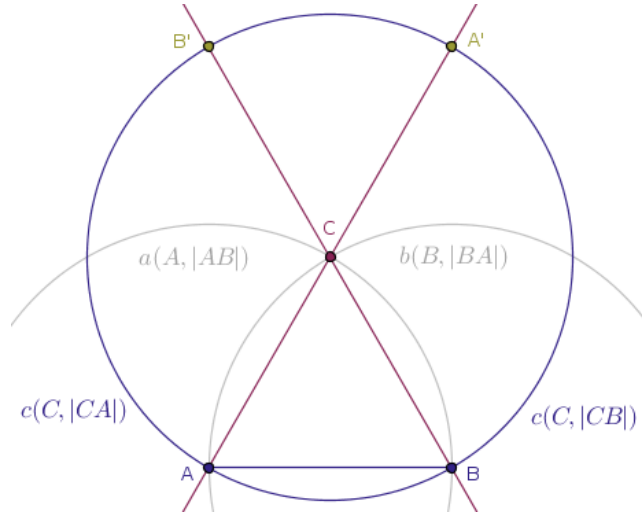


III. Draw a (blue) circle $c(C, |CB| = |CA|)$ centered at the vertex C with the radius $|CB| = |CA| = |AB|$ (Figure 11.71):



IV. Draw a straight line through the points A and C until that straight line intersects the circle c at the additional point A' .

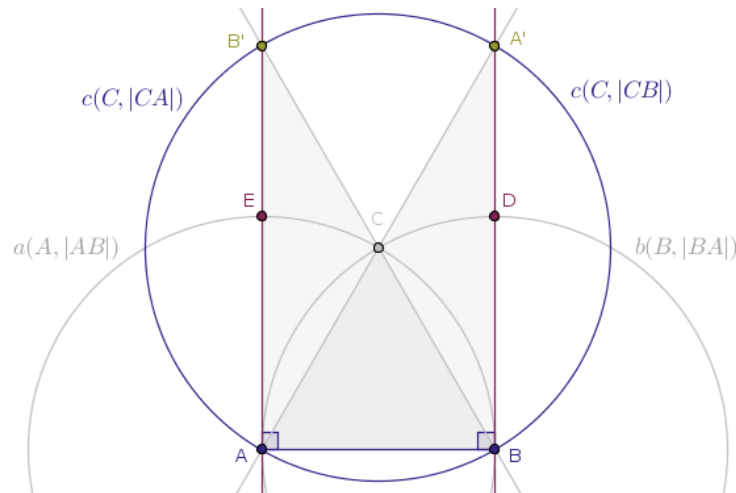
Draw a straight line through the points B and C until that straight line intersects the circle c at the additional point B' (Figure 11.72):



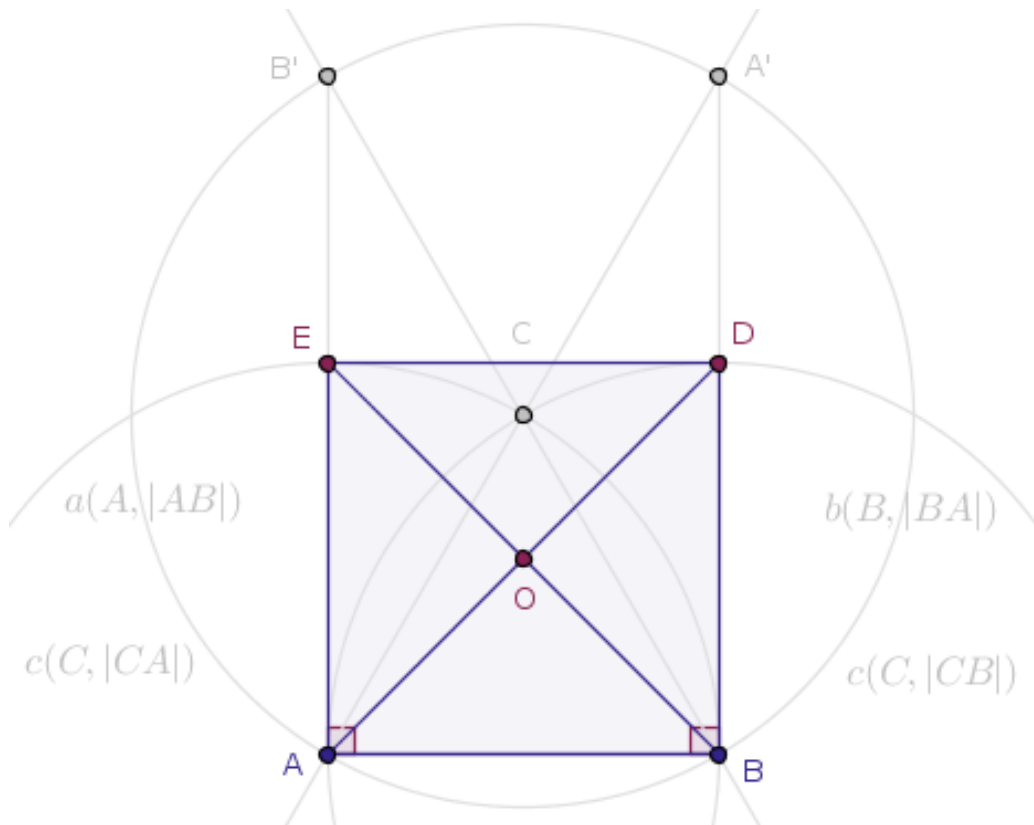
Note that by Euclid's Book 3 Proposition 31, the (not shown) angles ABA' and BAB' are right and that is exactly what we need for our square.

V. Draw a straight line through the points A and B' until that straight line intersects the circle a at the point E interior to the circle c .

Draw a straight line through the points B and A' until that straight line intersects the circle b at the point D interior to the circle c (Figure 11.73):



VI. Connect the points D and E thus constructed (Figure 11.74):



The, shown in solid blue in the diagram above, figure $ABDE$ thus obtained is the sought-after regular 4-gon or a square.

The diagonals AD and BE of the said square will intersect at the sought-after center of that square O and they will also be the sought-after bisectors of the interior angles of that square.

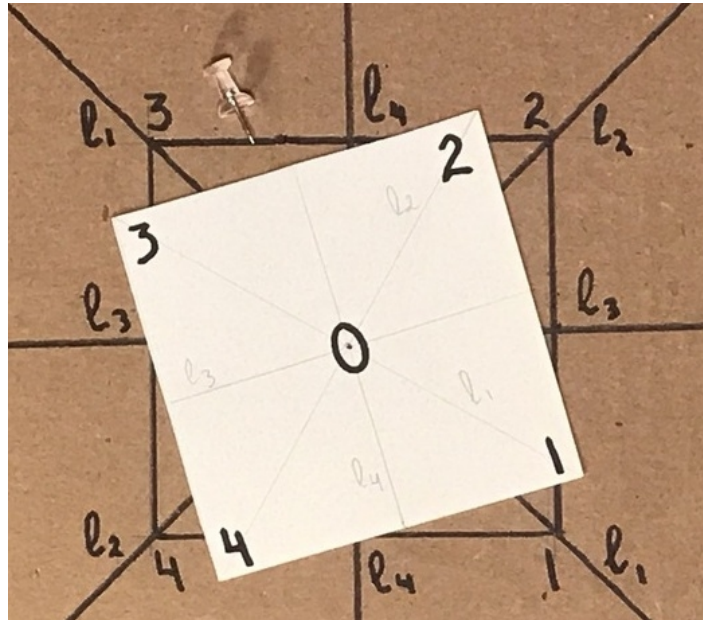
Cut a physical model of a regular 4-gon thus constructed from paper or, clear, plastic and mark its vertices with 1, 2, 3 and 4:

- on one side of the transparent piece of plastic or
- on both sides of paper, making sure that the names of these vertices on either surface agree - a vertex of your regular 4-gon that carries the name of 1 on one side of the paper should also be named as 1 on the opposite side of that paper and so on

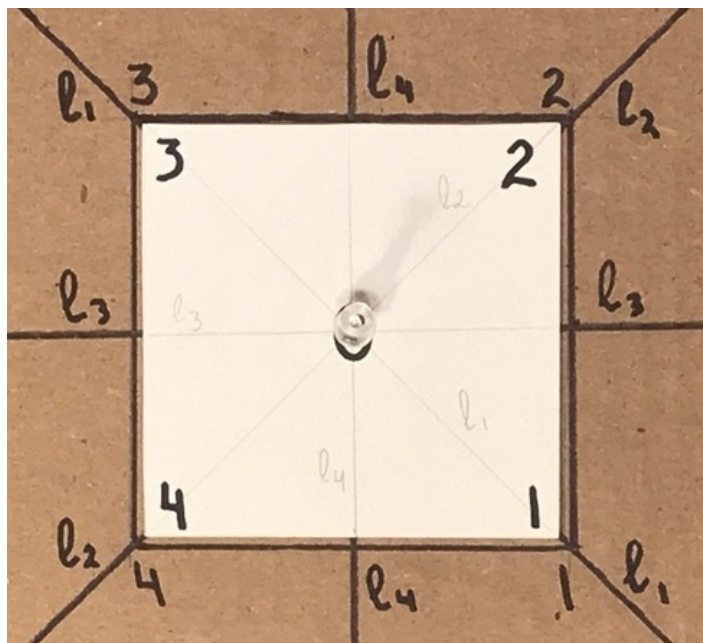
Put the newly obtained regular 4-gon on the cardboard workbench and trace the square's silhouette on it.

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Jot down the names of the vertices of the *fixed* or *reference* square on the cardboard in accordance with the names of the moving square and, anticipating the future work, draw all four *fixed* or *stationary* axes of symmetry $l_{1,2,3,4}$ of the square on the cardboard - later on we will be reflecting our square with respect to *these*, fixed, axes of symmetry (Figure 11.75):



With a push pin punch a hole in the center O of your square, making sure that that square can rotate freely about its center in the cardboard plane (Figure 11.76):



As the subtitle of this discussion goes, *omnia mea mecum porto*, meaning that *all that is mine I carry with me*, we produced everything that we need and we are now ready to get back to group-theoretic work.

Leading Up To The Study Of Cyclic Groups

Exercise 11.50: using the ideas presented and developed in the **Rectangle, Anyone?** chapter as a guide, investigate on your own whether the (rigid) rotations of an equilateral triangle under a mystery operation can form a group - if they can then, using the ideas presented at the end of the **Difficult, Giant, Leap Forward** chapter as a guide, construct a multiplication table of that group.

Try to answer the following questions on your own:

- Is that group finite or infinite?
- What is the order of that group?
- Is that group commutative or not?
- What is a characteristic feature of that particular group?

Solution: let us agree that the mystery operation in this case will be *composition of congruence rotations* of an equilateral triangle in the parent plane about its center, O , and that the following symbols will designate the elements of the candidate set of rotational congruence motions:

- the symbol e will stand for any rotation of an equilateral triangle in the parent plane about its center by any angle from the following infinite set E :

$$E = \{2\pi k, k = 0, \pm 1, \pm 2, \pm 3, \dots\} \quad (23)$$

- the symbol a will designate any rotation of an equilateral triangle in the parent plane about its center by any angle from the following infinite set A :

$$A = \{2\pi/3 + 2\pi k, k = 0, \pm 1, \pm 2, \pm 3, \dots\} \quad (24)$$

- the symbol b will designate any rotation of an equilateral triangle in the parent plane about its center by any angle from the following infinite set B :

$$B = \{4\pi/3 + 2\pi k, k = 0, \pm 1, \pm 2, \pm 3, \dots\} \quad (25)$$

Intuitively and informally, then, the above symbol e represents no rotation at all, effectively, and is known in geometry as *the identity transformation* because such a transformation returns the object being transformed into its original state precisely.

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The above symbol a represents *one third* of a complete revolution or rotation and the above symbol b represents *two thirds* of a complete revolution of an equilateral triangle in the parent plane about its center (counterclockwise).

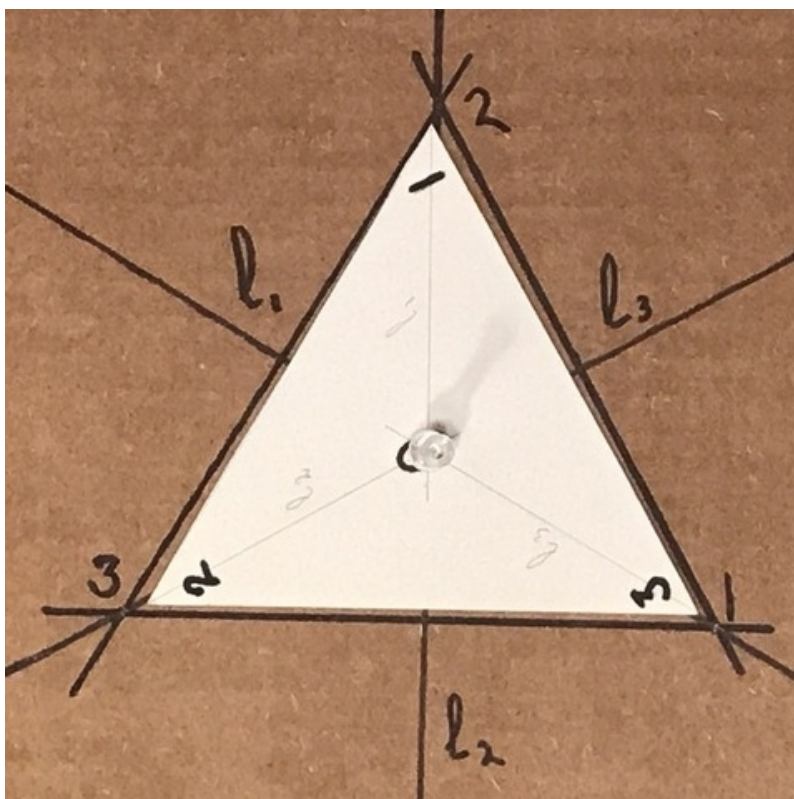
Now that we have a clear mental image of which symbol means what, we work more formally by rapidly jumping back and forth between the abstract symbols (e, a, b) and the concrete actions that these abstract symbols represent (the rotations of a triangle by certain angles).

I. Tentatively, the above element e represents *the identity* of this potential group because a rotation of an equilateral triangle in the parent plane about its center by any angular measure from the set E takes the vertices of that triangle back into themselves:

$$e : 1 \rightarrow 1, \quad 2 \rightarrow 2, \quad 3 \rightarrow 3$$

implying that the above preimage and image sets are one and the same set of vertices of an equilateral triangle $\{1, 2, 3\}$.

II. Tentatively, the above element a represents a non-identity of this potential group because a rotation of an equilateral triangle in the parent plane about its center by any angular measure from the set A (Figure 11.77):



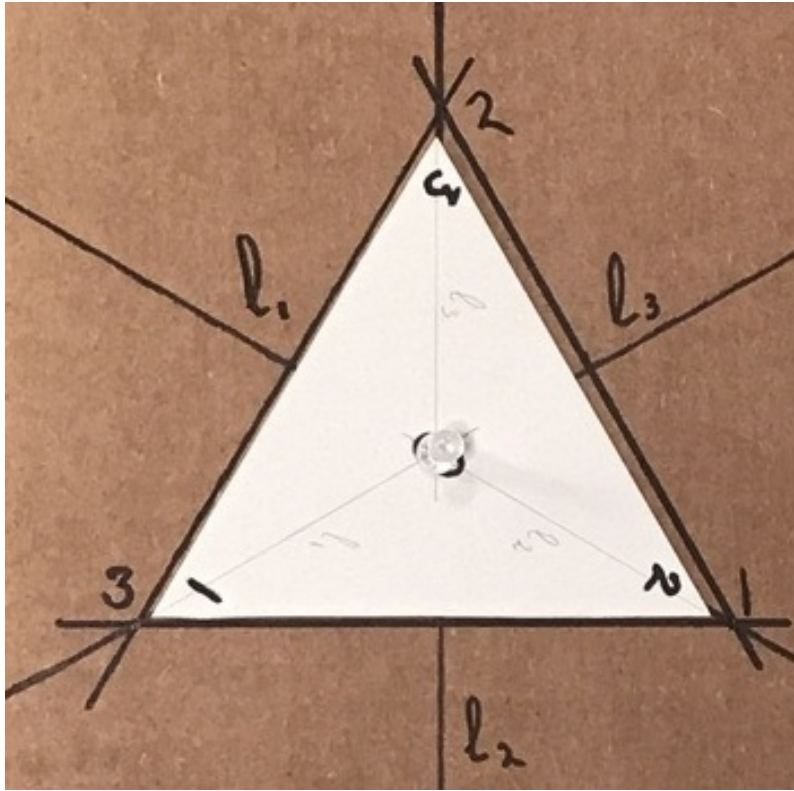
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takes the vertices of that triangle into the following set that is still comprised of the vertices of the same triangle:

$$a : 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1$$

We remind our readers that a rotation by a positive angular measure, such as $2\pi/3$ radians, is carried out *counterclockwise by default*, in the absence of other qualifying information.

III. Tentatively, the above element b represents another non identity of this potential group because a rotation of an equilateral triangle in the parent plane about its center by any angular measure from the set B (Figure 11.78):



takes the vertices of that triangle into the following set that is still comprised of the vertices of the same triangle:

$$b : 1 \rightarrow 3, 2 \rightarrow 1, 3 \rightarrow 2$$

and there are no other potential candidate elements for this potential group because the rotation of an equilateral triangle in the parent plane about its center by $6\pi/3 = 2\pi$ radians is a rotation of the said triangle by an angular measure taken from the set E and so on.

In other words, the rotations of an equilateral triangle in the parent plane about its center captured above represent *all* the planar *rotational* rigid congruence motions of a regular 3-gon - the planar rotations that bring the said regular 3-gon into coincidence with itself.

If, with our group-theoretic travel kit shown in Figure 11.67 on hand, we study the way the above rotational congruence motions of a regular 3-gon e , a and b *interact with each other*, that is if we study *the results* of the consecutive congruence rotations of the said equilateral triangle taken in all the possible ordered pairs $ee, ea, eb, ae, aa, ab, be, ba, bb$, of which, as we remember from **Exercise 5.1.1**, there should be exactly $3^2 = 9$, then we will discover that:

- the congruence rotation of an equilateral triangle e by 0 or any other integral multiple of 2π radians paired up with any other congruence rotation x of that triangle in any order will amount to the said rotation x (which may or may not be e itself):

$$ee = e, \quad ea = ae = a, \quad eb = be = b$$

- two consecutive rotations a of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians amount to a rotation of that triangle in the same plane about the same center by $2\pi/3 + 2\pi/3 = 4\pi/3$ radians, which is symbolized as b :

$$a \circ a = a \cdot a = aa = a^2 = b$$

- two consecutive rotations b of an equilateral triangle in the parent plane about its center by $4\pi/3$ radians amount to a rotation of that triangle in the same plane about the same center by $4\pi/3 + 4\pi/3 = 8\pi/3 \equiv 2\pi/3$ radians, which is symbolized as a :

$$b \circ b = b \cdot b = bb = b^2 = a$$

because, clearly, the angular measure of $8\pi/3$ radians can be represented in a way:

$$\frac{8\pi}{3} = \frac{6\pi + 2\pi}{3} = 2\pi + \frac{2\pi}{3}$$

that fits an angular measure from the above set A perfectly (if $k = 1$).

IV. We also see that the consecutive rotation of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians first and by $4\pi/3$ radians next or vice versa amounts to the identity transformation, symbolized as e , since:

$$\frac{2\pi + 4\pi}{3} = \frac{4\pi + 2\pi}{3} = \frac{6\pi}{3} = 2\pi$$

and, therefore, in symbols we have it that: $ab = ba = e$, which, basically, tells us that the elements a and b of our candidate set are *the inverses* of each other - in order to neutralize the rotational disturbance

introduced by the rotation of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians, all we have to do is rotate the same triangle, in the same plane, about the same center by $4\pi/3$ radians and conversely.

In order to neutralize the rotational disturbance introduced by the rotation of an equilateral triangle in the parent plane about its center by $4\pi/3$ radians, all we have to do is rotate the same triangle, in the same plane, about the same center by $2\pi/3$ radians (counterclockwise).

Armed with these very important miniature discoveries, we are now able to construct the following multiplication table of what we can now convincingly call *a group* in general or *the group of rotational symmetries of an equilateral triangle* in particular (Figure 11.79):

$\circ$	$e_O^{0^\circ}$	$a_O^{120^\circ}$	$b_O^{240^\circ}$
$e_O^{0^\circ}$	e	a	b
$a_O^{120^\circ}$	a	b	e
$b_O^{240^\circ}$	b	e	a

In **Exercise 11.43** we already proved that the operation of composition defined on the set of rigid planar motions, rotations including, is associative.

We know that that operation of composition acts on exactly two, no more and no less, rotational congruence motions of an equilateral triangle.

By manually examining or computing all the ordered pairs of consecutive rotational congruences of a regular 3-gon, we just proved that the operation of composition defined on the set $\{e, a, b\}$ is binary, since all the results of such compositions are again the elements of the same set $\{e, a, b\}$.

The element that we refer to as e is, clearly, the identity of this group.

The remaining two elements of the parent set, a and b , are the inverses of each other:

$$ab = ba = e$$

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This group is *finite*.

The order of that group is equal to 3.

Moreover, this group is commutative because, as is evident from our enumerate-all-ordered-group-products exercise, for any two elements x, y of the set $\{e, a, b\}$ it is the case that:

$$xy = yx$$

This particular group has one new interesting property that we haven't studied before and which we hope our readers can spot on their own.

Namely, as a honing-in question, ponder:

is it possible to render the contents of our multiplication table shown in Figure 11.79 above in terms of one element and one element only?

What element should that be? Correct - that element should be the element named a that corresponds to a rotation of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians because:

- that element a is represented in terms of itself as a
- the element b stands for *two* consecutive rotations of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians each and, thus, can be represented
 $asb = a \circ a = a \cdot a = aa = a^2$
- the identity element e stands for *three* consecutive rotations of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians each and, thus, can be represented as
 $e = a \circ a \circ a = a \cdot a \cdot a = aaa = a^3$

Thus, it becomes possible to render the multiplication table of this group as follows (Figure 11.80):

$\circ$	$e_O^{0^\circ}$	$a_O^{120^\circ}$	$b_O^{240^\circ}$
$e_O^{0^\circ}$	a^3	a	a^2
$a_O^{120^\circ}$	a	a^2	a^3
$b_O^{240^\circ}$	a^2	a^3	a

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If we write down as many terms of this *group-theoretic* geometric progression as we wish:

$$a, a^2, a^3, a^4, a^5, a^6, a^7, a^8, a^9, a^{10}, \dots$$

then we will quickly realize that not all the terms of this geometric progression are distinct.

In fact, if instead of the powers of a that are evenly divisible by 3 we write down the identity element to which such powers correspond:

$$a, a^2, e, a^4, a^5, e, a^7, a^8, e, a^{10}, \dots$$

then it will become apparent that all the powers of the resultant progression that leave the remainder of 1 upon division by 3 actually represent one and the same element a and that the powers that leave the remainder of 2 upon division by 3 actually represent one and the same element b , which, as we now understand it, also goes by the name of a^2 :

$$a, a^2, e, a, a^2, e, a, a^2, e, a, \dots$$

Intuitively speaking, then, such a progression *orbits* the same distinct elements with *the period* of 3.

But the same observation can be made about *the negative* integral powers of a just as well.

We already know that the element b is the inverse of the element a . Thus:

$$b = a^{-1}$$

What element of this group is hiding behind the second negative power of a :

$$a^{-2}$$

?

Correct, to the element that we refer to as a^{-2} there correspond two consecutive rotations of an equilateral triangle in the parent plane about its center by $4\pi/3$ radians each:

$$\frac{4\pi}{3} + \frac{4\pi}{3} = \frac{8\pi}{3} = \frac{2\pi + 6\pi}{3} = \frac{2\pi}{3} + 2\pi \equiv \frac{2\pi}{3}$$

which is equivalent to a single rotation of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians, which, in our notation, is the element a itself.

What about the sneaky animal hiding behind a^{-3} ?

Why, by the same reasoning, this is the identity element e :

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$$a^{-3} = e$$

and so on:

$$\dots, a^{-4}, a^{-3}, a^{-2}, a^{-1}, a, a^2, \dots$$

and it now seems only natural to *define* (!) the zero rotations of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians as the identity element e :

$$a^0 = e$$

We, thus, come across a feature of a group that we haven't seen before - in this particular group it is possible to:

express all the elements of a group only via the positive and negative integral powers of one element

Tuck this unit of technical knowledge in a corner of memory that is near by, as we will come back to it in the upcoming study of the so-called *cyclic groups*. $\square$

Exercise 11.51: using the ideas presented and developed in the **Rectangle, Anyone?** chapter as a guide, investigate on your own whether the rigid rotations of a square under some mystery operation can form a group.

If they can then, using the ideas presented at the end of the **Difficult, Giant, Leap Forward** chapter as a guide, construct a multiplication table of that group.

Is that group finite or infinite?

What is the order of that group?

Is that group commutative or not?

What characteristic feature of that particular group can you spot?

Solution: of this exercise is left to the reader.

Use the solution of **Exercise 5.4.3** and the ideas discussed and developed in the previous exercise to conclude that the set of *four* rotational congruence motions of a square forms a group under the operation of composition.

Naming the elements of that set in a predictable way:

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$$\{e, a, b, c\}$$

we show a multiplication table of that group for a reference with a correct response (Figure 11.81):

$\circ$	$e_O^{0^\circ}$	$a_O^{90^\circ}$	$b_O^{180^\circ}$	$c_O^{270^\circ}$
$e_O^{0^\circ}$	e	a	b	c
$a_O^{90^\circ}$	a	b	c	e
$b_O^{180^\circ}$	b	c	e	a
$c_O^{270^\circ}$	c	e	a	b

We, thus, see that that particular group is finite, of order 4 and is commutative.

We realize that all the elements of that group can also be represented only via some finite combinations of the same element, a , raised to positive or negative integral powers (Figure 11.82):

$\circ$	$e_O^{0^\circ}$	$a_O^{90^\circ}$	$b_O^{180^\circ}$	$c_O^{270^\circ}$
$e_O^{0^\circ}$	a^4	a	a^2	a^3
$a_O^{90^\circ}$	a	a^2	a^3	a^4
$b_O^{180^\circ}$	a^2	a^3	a^4	a
$c_O^{270^\circ}$	a^3	a^4	a	a^2

except that this time the smallest positive integer n for which the following relationship holds:

$$a^n = e \tag{26}$$

is equal to 4 and, thus, in the sequence of the consecutive integral powers of the element a the same distinct elements of that group repeat with the period of 4:

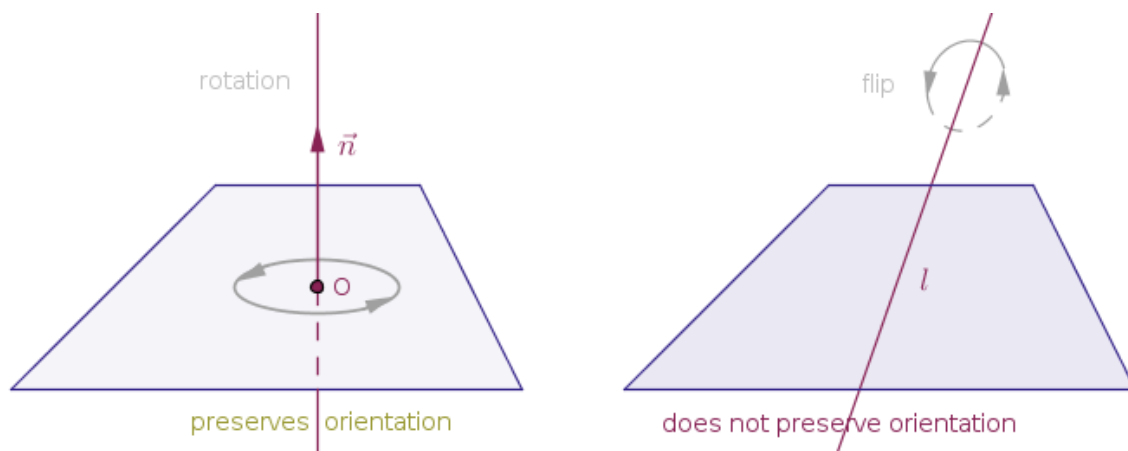
$$a, a^2, a^3, e, a^5, a^6, a^7, e, a^9, \dots$$

and so on. $\square$

Leading Up To The Study Of Subgroups

The rigid *rotational* congruence motion of regular polygons is not the only type of rigid motions that is capable of bringing such objects into coincidence with themselves.

Technically, when we discuss a *rotation* of a plane then by default we silently or implicitly assume that such a plane is rotated with respect to a *normal* $\vec{n}$ constructed to that plane through the point of rotation, O , as depicted on the left side of the diagram below (Figure 11.83):



Even though we, clearly, need the notion of the ambient 3-space in order to introduce the concept of a normal to a plane at a point, the actual rotation of the plane itself in this case has the following two properties:

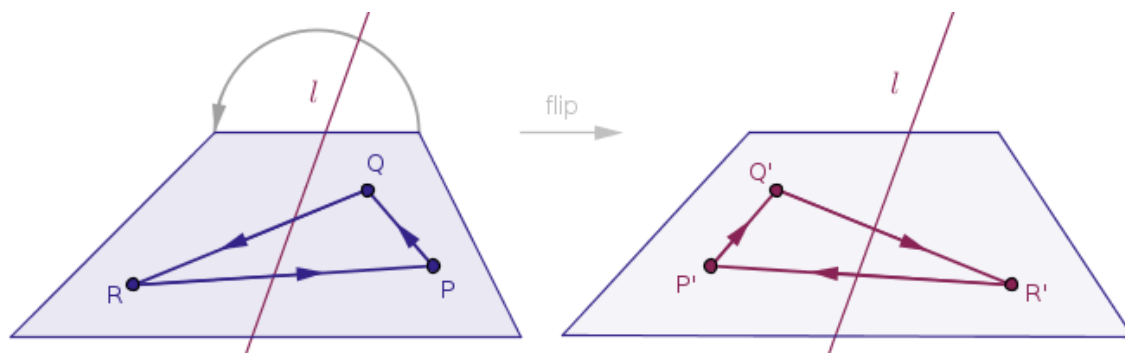
- it does *not step out* into the said ambient 3-space and
- it *does* preserve the orientation of the figures that are located in the plane that is being rotated

But it is also possible to rotate the same plane with respect to a straight line l that is *located in that plane*, as depicted on the right side of the diagram above.

In that case, however, such a rotation *will* use or *will* step out into the ambient 3-space, into which our plane is now embedded.

It is, of course, possible to rotate a plane about a straight line contained in it by any good old angular measure.

When such a plane is rotated about a straight line l , as shown above in the rightmost image, by $\pm\pi$ radians then such a rotation will bring that entire plane back into coincidence with itself (Figure 11.84):



but the resultant plane will, logically speaking, be *flipped up side down*.

That is why such a rotation of a plane when it steps out into the ambient 3-space and then returns back to its original position is referred to informally as *a flip*.

Such *flips* or such rotations, taking the straight line l into itself, do *not* preserve the orientation of the geometric figures contained in the rotated plane:

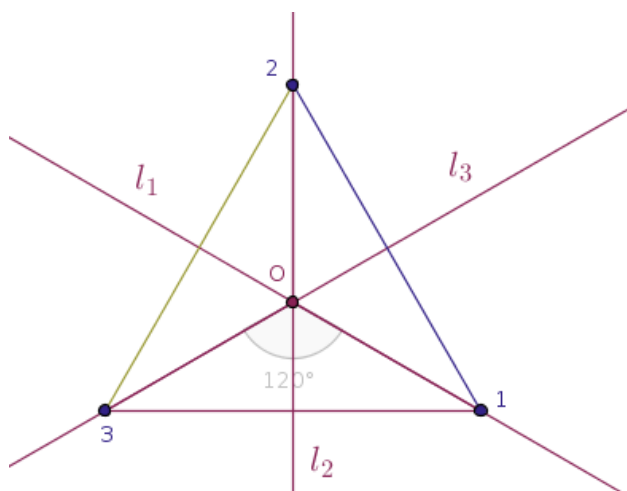
- if, with our both hands stretched out, we walk around the sample triangle PQR in the original plane from the vertex P to the vertex Q to the vertex R to the vertex P , then our *left* hand will always be *inside* of that triangle and our *right* hand will always be *outside* of that triangle, however
- if walk around the image triangle $P'Q'R'$ in the *flipped* plane *in the same fashion* from the vertex P' to the vertex Q' to the vertex R' to the vertex P' , then our *right* hand will always be inside of that triangle and our left hand will always be outside of the image triangle, which, clearly, violates the original *left-hand-inside* requirement

The above two types of rotations, the ones that do preserve and the ones that do not preserve the orientation of the underlying objects, can be captured analytically, via the corresponding, differing only by a sign in a certain place, equations, but we will not need this type of symbolic shuffling here.

In order to avoid having to deal with the ambient 3-space, we confine ourselves strictly to the parent plane and replace the above *flips* of a plane in 3-space with a strictly planar transformation known as *reflection of a plane in a straight line l* (in 2-space and 2-space only).

From the previous exercise we already know that reflections of a plane in a straight line do *not* preserve the orientation of the underlying figures, as explained above.

For example, let us agree to name *the axes of symmetry* of an equilateral triangle by the names of the vertices through which such axes pass (Figure 11.85):



All three such straight lines $l_{1,2,3}$ are contained in the parent plane of the given equilateral triangle.

To our group-theoretic menu of rigid congruence motions let us add one more such motion: the congruence reflections of a regular 3-gon in its three axis of symmetry.

If we name the elements of the set of rigid congruence motions of such a regular 3-gon as l_1 , l_2 and l_3 respectively then, as a formal exercise, let us ponder the following two questions:

Exercise 11.52: will such elements, taken in isolation, form a group under the operation of composition?

Will *all* congruence motions of an equilateral triangle form a group under composition?

Here by the word **all** we mean the congruence rotations and the congruence reflections of a regular 3-gon.

Solution: in order to answer these questions, let us construct the signatures of all the possible $3 \cdot 3 = 9$ ordered pairs of compositions of reflections of a regular 3-gon in its three axes of symmetry $l_{1,2,3}$.

First of all, we instantly see that:

- a composition of the identity transformation, e , of this potential group with the reflection of a regular 3-gon in any straight line $l_{1,2,3}$ amounts to the reflection of the said regular 3-gon in that straight line and that
- two consecutive reflections of a regular 3-gon in any one individual of its axes of symmetry results in the identity transformation, e

We, thus, begin developing a multiplication table of the new potential group (Figure 11.86):

$\circ$	e	$a_O^{120^\circ}$	$b_O^{240^\circ}$	l_1	l_2	l_3
e	e	a	b	l_1	l_2	l_3
$a_O^{120^\circ}$	a	b	e			
$b_O^{240^\circ}$	b	e	a			
l_1	l_1			e		
l_2	l_2				e	
l_3	l_3					e

Next, using our group-theoretic travel kit, we construct the signatures of the individual reflections of a regular 3-gon in its three axes of symmetry.

Namely, the reflection of a regular 3-gon in the axis of symmetry l_1 leaves the vertex named 1 fixed, while swapping the vertices 2 and 3, see Figure 11.85 above:

$$l_1 = \begin{cases} 1 \rightarrow 1 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{cases}$$

The reflection of a regular 3-gon in the axis of symmetry l_2 leaves the vertex named 2 fixed, while swapping the vertices 1 and 3:

$$l_2 = \begin{cases} 1 \rightarrow 3 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \end{cases}$$

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The reflection of a regular 3-gon in the axis of symmetry l_3 leaves the vertex named 3 fixed, while swapping the vertices 1 and 2:

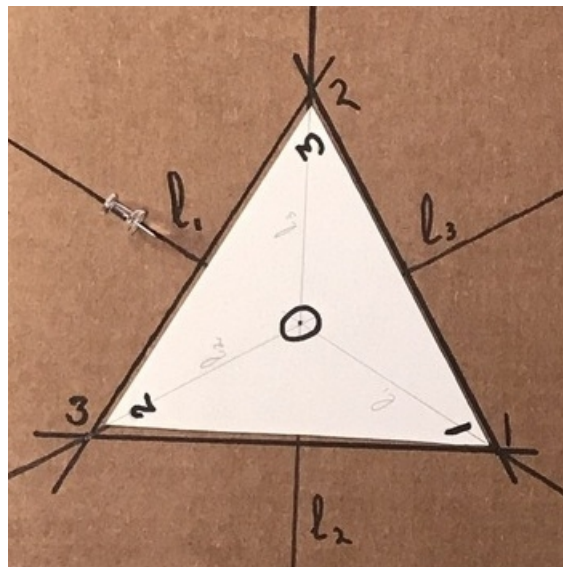
$$l_3 = \begin{cases} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{cases}$$

What about *the compositions* of these reflections?

Using our travel kit still, we discover that there is absolutely no hope for the this set of reflections alone to form a group under composition. That is. The signature of the composition $l_2 \circ l_1$ comes out to be (Figure 11.87):

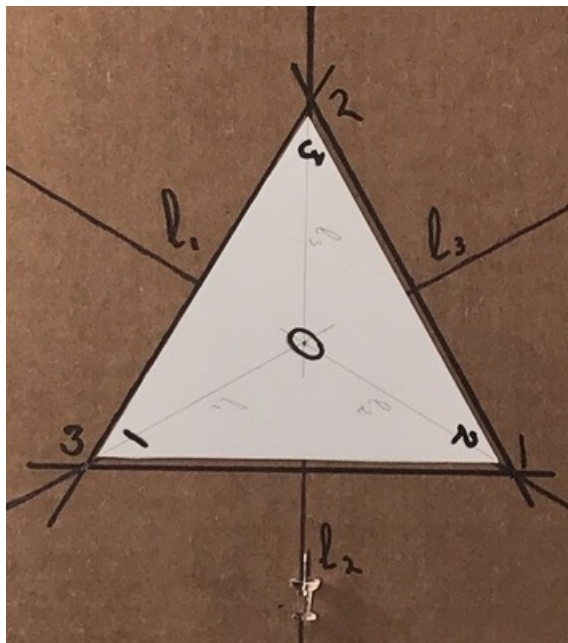
$$\begin{array}{c} 1 \rightarrow 3 \\ l_2 \circ l_1 = 2 \rightarrow 1 = b \\ 3 \rightarrow 2 \end{array}$$

or $(3, 1, 2)$ and that permutation of the names of the vertices of the object at hand corresponds to the rotation b because the reflection of an equilateral triangle in the stationary axis of symmetry l_1 takes its vertex 1 to the vertex 1, its vertex 2 - to the vertex 3 and its vertex 3 - to the vertex 2 (Figure 11.88):

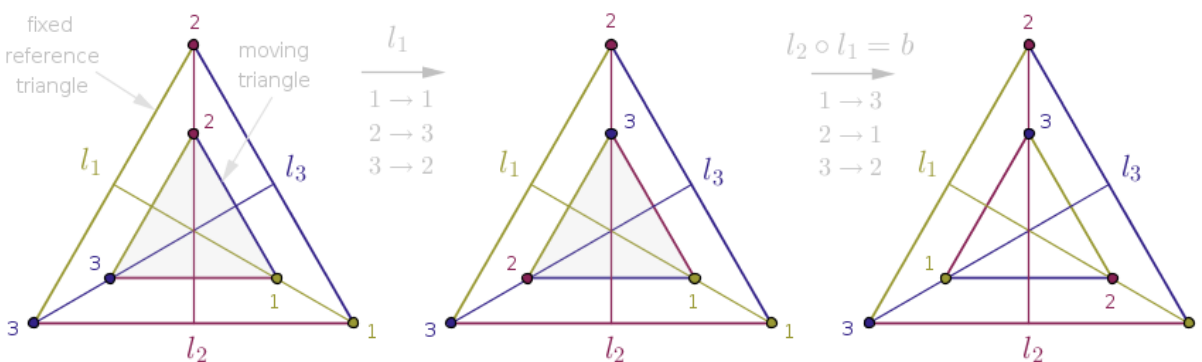


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while the consequent reflection of that triangle in the stationary axis of symmetry l_2 (that is drawn on the cardboard and that stays fixed at all times) takes the vertex 1 of that triangle to the vertex 3, the vertex 2 - to the vertex 1 and the vertex 3 - to the vertex 2 (Figure 11.89):



Merging the $l_2 \circ l_1$ composition into a single diagram, in which we have designated the three axes of symmetry of an equilateral triangle with different colors (Figure 11.90):



we see that such a composition results overall in the mapping of the triangle's vertices as shown in Figure 11.87 above.

Does that signature or the linear permutation $(3, 1, 2)$ of the vertices of a regular 3-gon $(1, 2, 3)$ under *any individual* reflection in its axes of symmetry $l_{1,2,3}$ exist?

In other words, does the reflection of an equilateral triangle in any of its axes of symmetry $l_{1,2,3}$ permute its vertices in the manner described above?

Nope, it does not - we can make that statement with confidence because we already computed the signatures of all the individual reflections $l_{1,2,3}$.

What gives then?

Ah. If we go back to the work that we have done in **Exercise 11.50**, then we will find that the signature $(3, 1, 2)$ is produced by *the rotation* that we named as b .

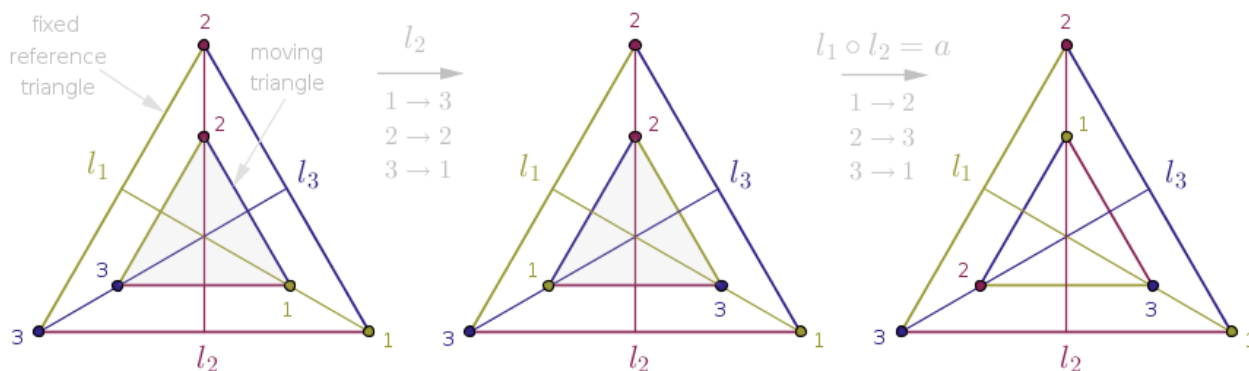
Bummer!

The composition of two reflections $l_2 \circ l_1$ takes us out of the set of such reflections and, necessarily, throws us on the rocks of *rotations*, thus, crushing any hopes of reflections of a regular 3-gon alone forming a group under composition.

Very well. Keep moving on. The signature of the composition $l_1 \circ l_2$ comes out to be (Figure 11.91):

$$\begin{aligned} 1 &\rightarrow 2 \\ l_1 \circ l_2 &= 2 \rightarrow 3 = a \\ 3 &\rightarrow 1 \end{aligned}$$

or $(2, 3, 1)$ and that is the rotation a because the reflection of an equilateral triangle in the stationary red axis of symmetry l_2 takes its vertex 1 to the vertex 3, its vertex 2 - to the vertex 2 and its vertex 3 - to the vertex 1 (Figure 11.92):



while the consequent reflection of that triangle in the stationary green axis of symmetry l_1 (that is drawn on the cardboard and that stays fixed at all times) takes the vertex 1 of that triangle to the vertex 2, the vertex 2 - to the vertex 3 and the vertex 3 - to the vertex 1, resulting overall in the mapping of the triangle's vertices as shown in Figure 11.91 above.

What are we to conclude from this experiment?

Well, since we see that it is the case that:

$$b = l_2 \circ l_1 \neq l_1 \circ l_2 = a$$

we conclude that if whatever we have on our hands is a group then that group is *not* commutative.

The signature of the composition $l_3 \circ l_1$ comes out to be (Figure 11.93):

$$\begin{array}{l} 1 \rightarrow 2 \\ l_3 \circ l_1 = 2 \rightarrow 3 = a \\ 3 \rightarrow 1 \end{array}$$

the rotation a , again - for the same reasons as explained above.

The signature of the composition $l_1 \circ l_3$ comes out to be (Figure 11.94):

$$\begin{array}{l} 1 \rightarrow 3 \\ l_1 \circ l_3 = 2 \rightarrow 1 = b \\ 3 \rightarrow 2 \end{array}$$

the rotation b .

Thus, we see that:

$$l_3 \circ l_1 \neq l_1 \circ l_3$$

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The signature of the composition $l_3 \circ l_2$ comes out to be (Figure 11.95):

$$\begin{array}{c}
 1 \rightarrow 3 \\
 l_3 \circ l_2 = 2 \rightarrow 1 = b \\
 3 \rightarrow 2
 \end{array}$$

the rotation b , again.

The signature of the composition $l_2 \circ l_3$ comes out to be (Figure 11.96):

$$\begin{array}{c}
 1 \rightarrow 2 \\
 l_2 \circ l_3 = 2 \rightarrow 3 = a \\
 3 \rightarrow 1
 \end{array}$$

the rotation a , implying that: $l_3 \circ l_2 \neq l_2 \circ l_3$.

When we, thus, develop the contents of the multiplication table of this potential group further (Figure 11.97):

$\circ$	e	$a_O^{120^\circ}$	$b_O^{240^\circ}$	l_1	l_2	l_3
e	e	a	b	l_1	l_2	l_3
$a_O^{120^\circ}$	a	b	e			
$b_O^{240^\circ}$	b	e	a			
l_1	l_1			e	a	b
l_2	l_2			b	e	a
l_3	l_3			a	b	e

then we discover that that table consists of several *blocks* - two such highlighted blocks, the upper-left or the North-Western block and the lower-right or the South-Eastern block, contain nothing more than the identity element and the rotations alone.

Our intuition now whispers to us that the remaining two blocks of such a table *may* consist of nothing but the reflections alone.

Is our gut right on the money?

Let us check.

The compositions of the rotation a of an equilateral triangle in the parent plane about its center by $2\pi/3$ radians with all of the reflections $l_{1,2,3}$ of the said triangle come out to be thoroughly not commutative:

$$a \circ l_1 = l_3 \neq l_2 = l_1 \circ a$$

$$a \circ l_2 = l_1 \neq l_3 = l_2 \circ a$$

$$a \circ l_3 = l_2 \neq l_1 = l_3 \circ a$$

Pictorially (Figure 11.98):

$\circ$	e	$a_O^{120^\circ}$	$b_O^{240^\circ}$	l_1	l_2	l_3
e	e	a	b	l_1	l_2	l_3
$a_O^{120^\circ}$	a	b	e	l_3	l_1	l_2
$b_O^{240^\circ}$	b	e	a			
l_1	l_1	l_2		e	a	b
l_2	l_2	l_3		b	e	a
l_3	l_3	l_1		a	b	e

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Likewise, the compositions of the rotation b of an equilateral triangle in the parent plane about its center by $4\pi/3$ radians with all of the reflections $l_{1,2,3}$ of the said triangle also turns out to be thoroughly not commutative:

$$b \circ l_1 = l_2 \neq l_3 = l_1 \circ b$$

$$b \circ l_2 = l_3 \neq l_1 = l_2 \circ b$$

$$b \circ l_3 = l_1 \neq l_2 = l_3 \circ b$$

Pictorially (Figure 11.99):

$\circ$	e	$a_O^{120^\circ}$	$b_O^{240^\circ}$	l_1	l_2	l_3
e	e	a	b	l_1	l_2	l_3
$a_O^{120^\circ}$	a	b	e	l_3	l_1	l_2
$b_O^{240^\circ}$	b	e	a	l_2	l_3	l_1
l_1	l_1	l_2	l_3	e	a	b
l_2	l_2	l_3	l_1	b	e	a
l_3	l_3	l_1	l_2	a	b	e

and our multiplication table of what we now can refer to, with confidence, as *a group* is complete.

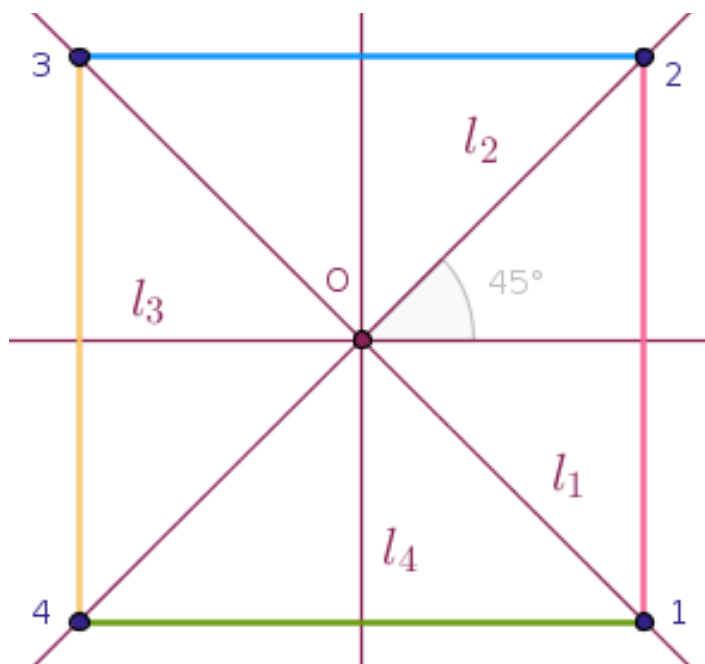
That group is finite, is of order 6 and is *not* commutative.

Observe that, since we proved earlier that the set $\{e, a, b\}$ of rigid rotational congruence motions of a regular 3-gon does form a group under composition on its own, that (mini me) group is now *contained within* the larger group of *all* the rigid congruence motions of a regular 3-gon, where by **all** congruence motions we mean the congruence *rotations* and the congruence *reflections* of an equilateral triangle.

That phenomenon of the containment of a smaller group *within* a larger group will pave the way toward the introduction of the concept of *a subgroup* of a group. $\square$

As a direct analog of the above equilateral triangle exercise, let us do the same exercise for a square.

Exercise 11.53: if we name the four axes of symmetry of a square as $l_{1,2,3,4}$ as follows (Figure 11.100):



then will the corresponding congruence reflections of a square form a group under the operation of composition?

Will **all** congruence motions of a square form a group under composition?

If the answer to the second question is *a yes*, then construct a multiplication table of that group.

Solution: in a way, we already solved this problem in **Exercise 5.4.3** from the **Early Examples** chapter.

However, in that exercise we were mostly guided by this very loose and intuitive *isomorphic argument*, which amounted to correlating the contra dance figures to the rigid symmetries of a square.

Even though such an argument was fairly straightforward, now we should approach *this* exercise, after all the material that we worked through, including the more rigorous discussion of the various types of group axioms, with a sense of a deeper understanding of what is required of us and exactly what goes into a correct solution of this problem.

First, using our group-theoretic travel kit, we construct the signatures of the individual congruence motions of a square whose vertices are named as 1, 2, 3 and 4.

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The rotation e of a square in the parent plane about its center by any angle from the infinite set E :

$$E = \{0 + 2\pi k, k \in \mathbb{Z}\}$$

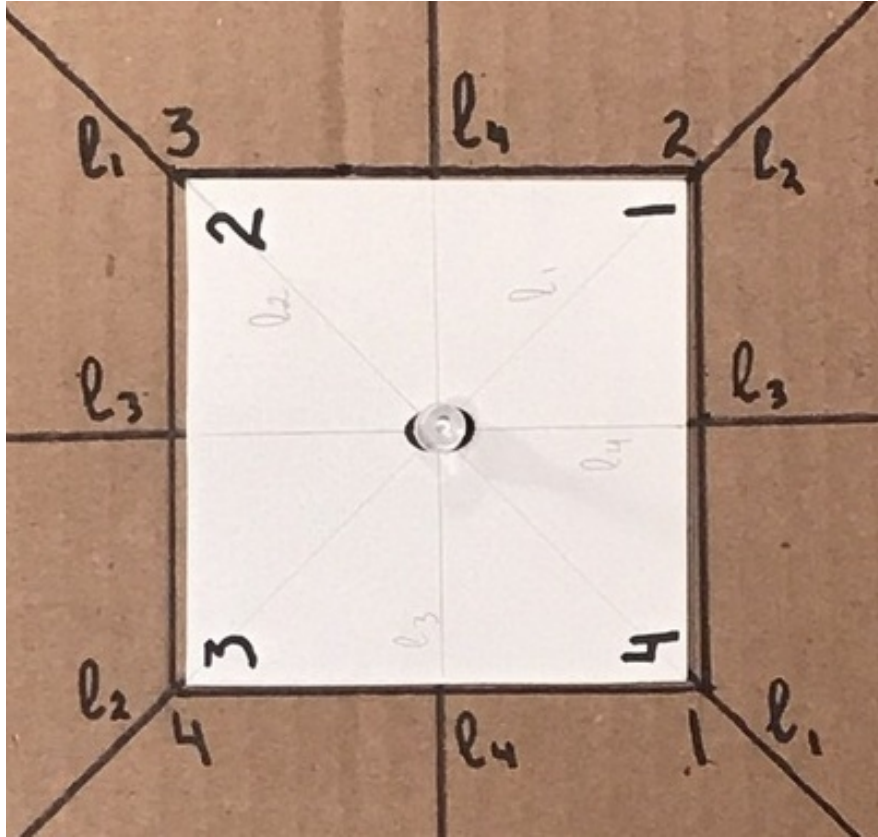
which, loosely, amounts to *no rotation at all*, takes the vertices of the said square into themselves:

$$e : 1 \rightarrow 1, 2 \rightarrow 2, 3 \rightarrow 3, 4 \rightarrow 4$$

The rotation a of a square in the parent plane about its center by any angle from the infinite set A :

$$A = \left\{ \frac{\pi}{2} + 2\pi k, k \in \mathbb{Z} \right\}$$

which, loosely, amounts to *one quarter* of the square's complete revolution (Figure 11.101):



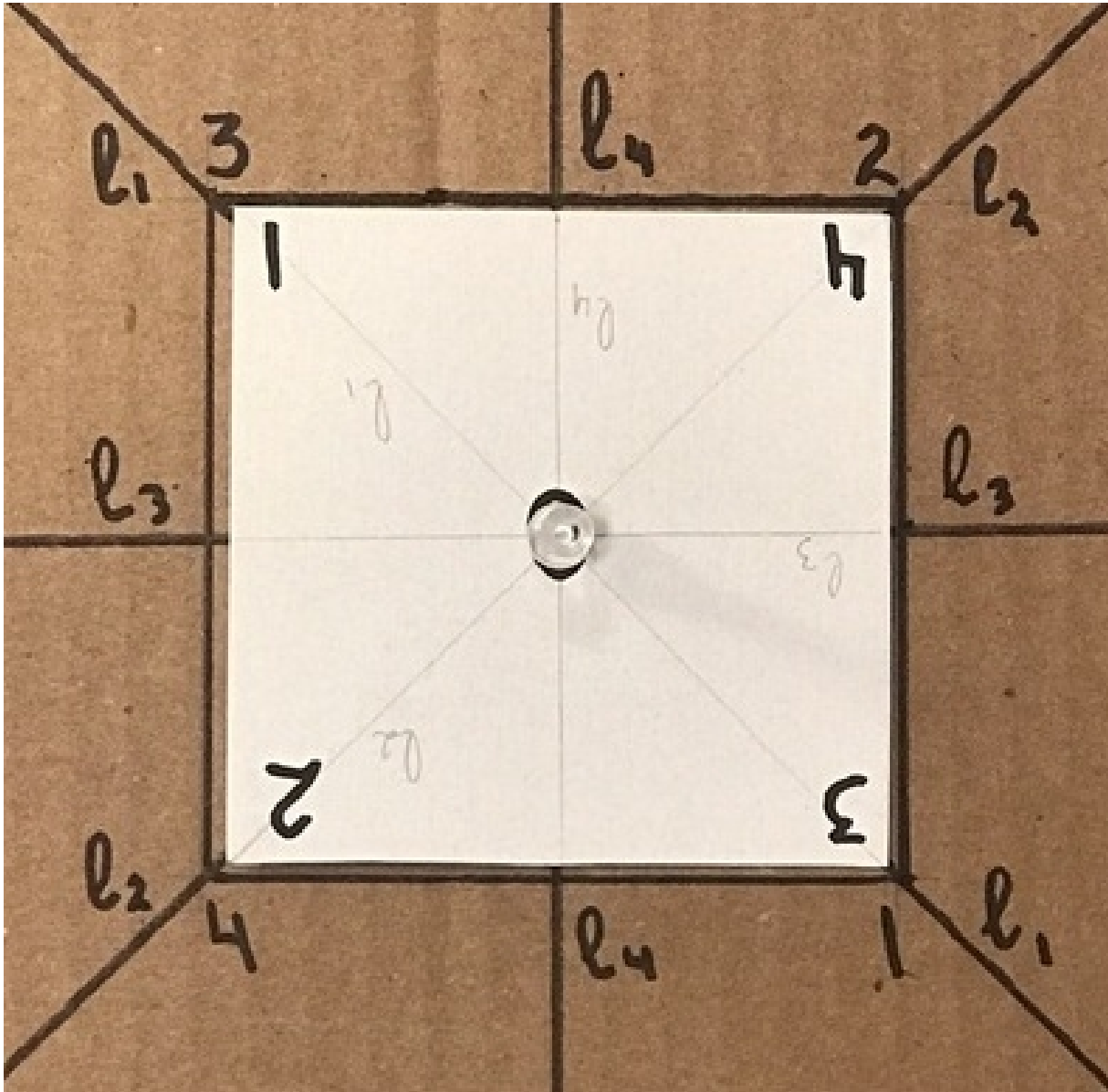
and shuffles the vertices of the said square as follows:

$$a : 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 1$$

The rotation b of a square in the parent plane about its center by any angle from the infinite set B :

$$B = \{\pi + 2\pi k, k \in \mathbb{Z}\}$$

which, loosely, amounts to *one half* of the square's complete revolution (Figure 11.102):



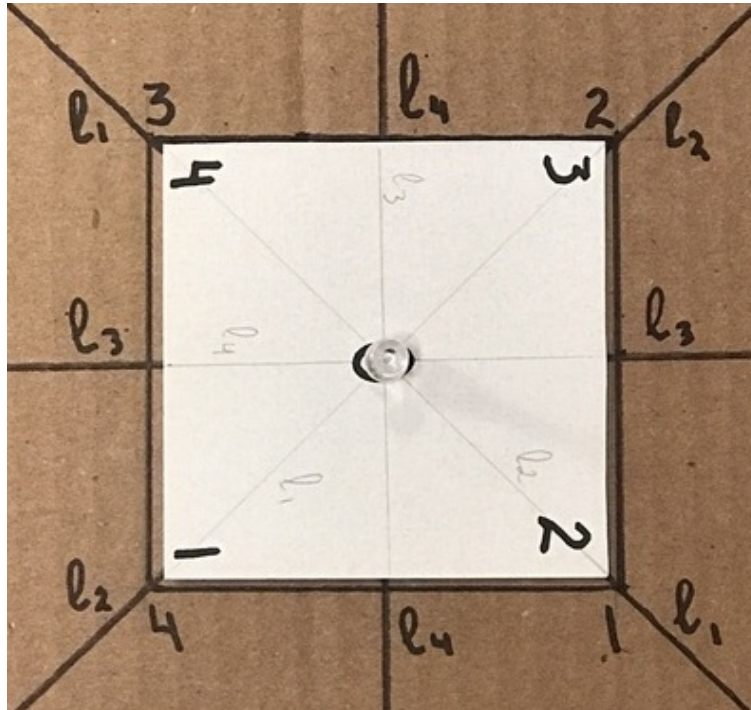
and shuffles the vertices of the said square as follows:

$$b : 1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 1, 4 \rightarrow 2$$

The rotation c of a square in the parent plane about its center by any angle from the infinite set C :

$$C = \left\{ \frac{3\pi}{2} + 2\pi k, k \in \mathbb{Z} \right\}$$

which, loosely, amounts to a *three quarters* of the square's complete revolution (Figure 11.103):



and shuffles the vertices of the said square as follows:

$$c : 1 \rightarrow 4, 2 \rightarrow 1, 3 \rightarrow 2, 4 \rightarrow 3$$

The reflection l_1 of a square does not move its vertices 1 and 3 but swaps the vertices 2 and 4 and, thus, generates the following signature:

$$l_1 : 1 \rightarrow 1, 2 \rightarrow 4, 3 \rightarrow 3, 4 \rightarrow 2$$

The reflection l_2 of a square does not move its vertices 2 and 4 but swaps the vertices 1 and 3 and, thus, generates the following signature:

$$l_2 : 1 \rightarrow 3, 2 \rightarrow 2, 3 \rightarrow 1, 4 \rightarrow 4$$

The reflection l_3 of a square moves all of its vertices and, thus, generates the following signature:

$$l_3 : 1 \rightarrow 2, 2 \rightarrow 1, 3 \rightarrow 4, 4 \rightarrow 3$$

The reflection l_4 of a square also moves all of its vertices and, thus, generates the following signature:

$$l_4 : 1 \rightarrow 4, 2 \rightarrow 3, 3 \rightarrow 2, 4 \rightarrow 1$$

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Using the above signatures of the individual rigid congruence motions of a square we, again, see that the reflections of a square alone have no hope of forming a group under composition, while all of the above congruence motions of a square do form a group under a composition with the following multiplication table (Figure 11.104):

$\circ$	e	$a_O^{90^\circ}$	$b_O^{180^\circ}$	$c_O^{270^\circ}$	l_1	l_2	l_3	l_4
e	e	a	b	c	l_1	l_2	l_3	l_4
$a_O^{90^\circ}$	a	b	c	e	l_3	l_4	l_2	l_1
$b_O^{180^\circ}$	b	c	e	a	l_2	l_1	l_4	l_3
$c_O^{270^\circ}$	c	e	a	b	l_4	l_3	l_1	l_2
l_1	l_1	l_4	l_2	l_3	e	b	c	a
l_2	l_2	l_3	l_1	l_4	b	e	a	c
l_3	l_3	l_1	l_4	l_2	a	c	e	b
l_4	l_4	l_2	l_3	l_1	c	a	b	e

Note again that the resultant multiplication table of the rigid congruence motions of a square consists of four blocks:

- the upper left or the North-Western block that contains only the rotations and the identity motion (no reflections)
- the lower right or the South-Eastern block that also contains only the rotations and the identity motion (no reflections)
- the upper right or the North-Eastern block that contains the reflections only (no rotations and no identity motion) and
- the lower left or the South-Western block that also contains the reflections only (no rotations and no identity motion)

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The contents of these blocks are consistent with the observation that we made in **Exercise 11.44** about the various compositions of rotations and reflections.

Observe that, since we proved earlier that the set $\{e, a, b, c\}$ of rigid rotational congruence motions of a regular 4-gon does form a group under composition on its own, that (mini me) group is now *contained within* the larger group of *all* the rigid congruence motions of a regular 4-gon: congruence rotations and congruence reflections.

Again, that phenomenon of the containment of a smaller group *within* a larger group will pave the way toward the introduction of the concept of a *subgroup* of a group. $\square$

Exercises 8: Modular Arithmetic, Part 1 Of 4

In the next three discussions we shall train our group-theoretic lens exclusively on a narrow type of numbers known as *integers* or *whole numbers*, as we consider two new types of operations known as *addition modulo n* and *multiplication modulo n* . In the process we will discover that these operations defined on certain sets of numbers form the very beautiful additive and multiplicative groups of integers modulo n .

The Division Algorithm Theorem

Let a blue line segment that is named b and that is b *integral* units of distance long play the role of a *yard stick*.

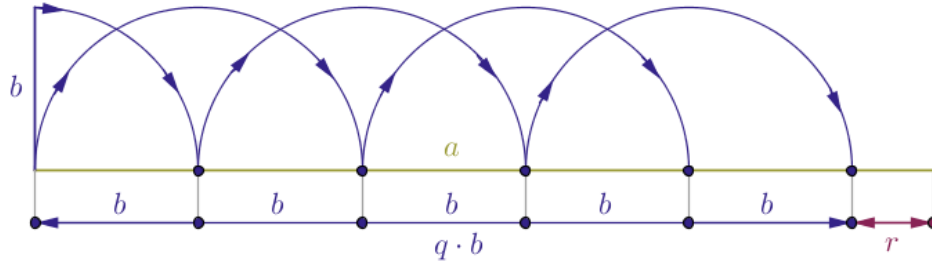
Let a green line segment that is named a and that is a integral units of distance long play the role of a mystery distance that we want to measure with the above yard stick, b (Figure 11.105):



Observe that since the actual numbers baked into a and b , such as 17 and 3, for example, are known to us, what we want to accomplish here is to see how many times the yard stick b folds into the distance a . In other words, in the upcoming experiment we want to investigate a specific relationship *between these two numbers*.

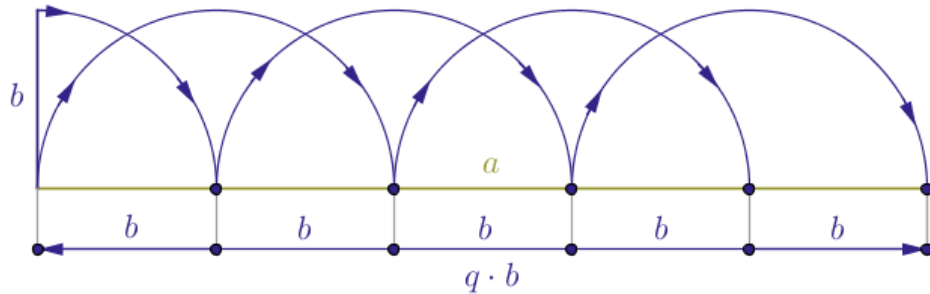
We will do so by simply walking the said yard stick b along the distance a , while making sure that that yard stick, b , ends its measuring journey strictly inside of the green distance being measured, a .

Align one end of the yard b stick with the beginning of the distance a , drop b onto a and count: *one*. Rotate b about its right, in this case, end clockwise until the traveling, left, end of b hits a and count: *two*. Keeping repeating this rotate-and-hit process until the next such maneuver will put the left extremity of b passed the end of a , at which point – stop and announce to the world the total number of flips, q , that b managed to make (Figure 11.106):



Consequently, after making the said q steps, each of which is, naturally, exactly b units of distance long, we will witness only one of only two possible outcomes:

- either we will have some, itsy bitsy, left over integral distance, $r < b$, not measured by b , as depicted in Figure 11.106 above, or
- we will not have any left over distance that is not measured by b , as depicted in the diagram below (Figure 11.107):



If we agree that:

- a , a *dividend*, is any whole number (negative, zero or positive)
- b , a *divisor*, is a strictly positive whole number
- q , a *quotient*, is any whole number and
- r , a *remainder* or a *residue*, is a non-negative whole number

then the distance-measuring process described above is a toy mechanical model of a theorem from elementary Number Theory that is known as *the Division Algorithm Theorem* or **DAT** for short: for any integers a , the dividend, and b , the divisor, with $b > 0$, there exist unique integers q , the quotient, and r , the remainder, such that:

$$a = q \cdot b + r \quad (27)$$

where the remainder r honors the:

$$0 \leq r < b \quad (28)$$

constraint.

Commentary: the word **algorithm** in the theorem's name Division Algorithm is a gross overpromise that delivers absolutely no algorithmic content because the traditional proof of that theorem is completely opaque and gives no algorithmic recipe for the actual recovery of the numbers q and r .

For example. If $a = 23$ and $b = 7$ then:

$$23 = 3 \cdot 7 + 2$$

implying that in this case we have for the quotient $q = 3$ and the remainder $r = 2 < 7$.

If $a = 54$ and $b = 3$ then:

$$54 = 18 \cdot 3 + 0 = 18 \cdot 3$$

implying that in this case we have for the quotient $q = 18$ and the remainder $r = 0 < 3$.

Thus, when an integer a is expanded with respect to the integer b as per the Division Algorithm formula (27), $a = q \cdot b + r$, in a such a way that the remainder r of that expansion is zero, $r = 0$:

$$a = q \cdot b$$

as it is the case in the last example above, then it is said that:

a is evenly divisible by b

or that:

a is divisible by b without the remainder

or that:

a is divisible by b

or that:

b divides a without the remainder

or that:

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b divides a evenly

or, simply, that:

b divides a

Our second outcome in the above toy mechanical model and the above numeric example, thus, translate into the following official definition:

Definition 20: the integer *a* is divisible by the integer $b \neq 0$ or the integer $b \neq 0$ divides the integer *a* if there exists an integer *q* such that $a = q \cdot b$.

Note that since for the duration of this and the next two discussions all the numbers under consideration are *whole*, let us agree to omit the qualifier *whole* or *integer* from all the subsequent formulations simply to avoid the cumbersome and often obscuring repetition and duplication of terms.

In other words, let it be written and let it be said that instead of the technically accurate but lengthy and awkward phrase:

the integer *b* integer-divides the integer *a* without the integer remainder

we shall use the short and succinct phrase:

b divides a

whose meaning is now unambiguously understood.

Consequently, the first outcome of our toy mechanical measuring process and the first numeric example above translate into the negation of the above *b-divides-a* definition as follows:

the integer *a* is not divisible by *b* or *b* does not divide *a* if there does not exist an integer *q* such that
$$a = q \cdot b$$

In other words, when the residue *r* in the workhorse formula (27) of DAT is distinct from zero, then *b* does *not* divide *a* and so on.

The Residue

It will very quickly become apparent that (his majesty) residue *r* in DAT is the first class group-theoretic citizen.

As such, we briefly highlight the important points about and the properties of the integer residues in the context of DAT.

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The above constraint (28), $0 \leq r < b$, imposed on the possible values of the remainder r is absolutely crucial and should not be overlooked - such a constraint, on the one hand, ensures that:

- in the Euclidean GCD algorithm every *next* problem becomes *smaller* than the previous one (due to the $r < b$ part) and that
- the said Euclidean GCD algorithm *terminates* in a finite number of steps with a correct result (due to the $0 \leq r$ part)

On the other hand, that constraint alone and single-handedly already generates a very important set of whole numbers that becomes the fodder for the theory of groups (and the theory of numbers).

Namely.

Note that in the Division Algorithm expansion (27), $a = q \cdot b + r$, the residue r is designed to *always be non-negative* - in the above example we, explicitly, do *not* multiplicatively expand 23 with respect to 7 with a *negative* residue:

$$23 = 4 \cdot 7 - 5$$

That is why in our toy model above we insisted on the yard stick to always end its measuring journey strictly inside of the distance being measured - if that requirement cannot be honored at all then that means that we make exactly zero steps with our yard stick:

$$3 = 0 \cdot 4 + 3$$

While it is also possible to generalize the above Division Algorithm Theorem for all types of whole numbers that the divisor b can take on - even the negative whole numbers, the residue r in that generalization is still kept non-negative:

for any integers a , the dividend, and b , the divisor, with $b \neq 0$, there exist unique integers q , the quotient, and r , the remainder, such that $a = q \cdot b + r$, where the remainder r honors the $0 \leq r < |b|$ constraint

For example:

$$61 = (-8) \cdot (-7) + 5, \quad 5 < |-7|$$

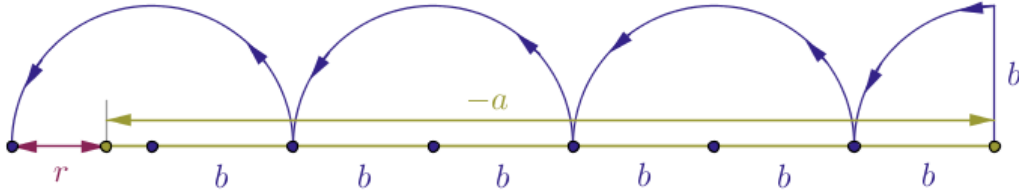
$$-59 = 9 \cdot (-7) + 4, \quad 4 < |-7|$$

$$-123 = 18 \cdot (-7) + 3, \quad 3 < |-7|$$

and so on.

Observe that when the dividend a is a *negative* whole number then our toy model breaks down because in order to honor the $0 \leq r$ constraint, we (now) have to make sure that one end of our yard stick always

lands *to the left* of the negative integer a or that we *overshoot* the $-a$ mark to the left a bit (Figure 11.108):



In other words, if, while seeking the remainder r of the division of a *positive* distance a by b , we want to make sure that our yard stick always ends its measuring journey strictly **inside** of the distance a , when we seek the remainder r of the division of a *negative* distance a by b , we want to make sure that our yard stick always ends its measuring journey strictly **outside** of the negative distance a .

As we noted earlier, since absolutely all the objects involved in the workhorse formula $a = q \cdot b + r$ of DAT are *integers*, the process of the recovery of the whole numbers q and r or the process of the decomposition or expansion of the integer a in terms of the respective integers q , b and r is in general referred to as *the integer division* and the branch of mathematics that studies such integer division and its properties is referred to as *the divisibility in the integers*.

From DAT it right away follows that while *the number* of possible residues of division of a varying integral quantity a by a fixed integer b is equal exactly to b , *the values* that these b remainders can possibly take on are the non-negative integers:

$$0, 1, 2, 3, \dots, b - 1$$

For example, if we fix b at, say, 3 then, upon the integer division by 3, across *all the integers* there will be exactly 3 distinct remainders, whose values will be 0, or 1, or 2:

$$6 = 2 \cdot 3 + 0, \quad r = 0$$

$$7 = 2 \cdot 3 + 1, \quad r = 1$$

$$8 = 2 \cdot 3 + 2, \quad r = 2$$

$$9 = 3 \cdot 3 + 0, \quad r = 0$$

By fixing b at 3 we just painted the entire set of integers, $\mathbb{Z}$, into just and only three *colors*:

- 1) the remainder-0-upon-division-by-3 color
- 2) the remainder-1-upon-division-by-3 color
- 3) the remainder-2-upon-division-by-3 color

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If we now pour all the integers of the same color into its own, dedicated, bag then we shall have just three bags - each such bag has its special name in mathematics and, overall, the set of residues $\{0, 1, 2\}$ also has its special name, see below*.

With each *color* we, thus, generated one and only *representative* element, a non-negative integer 0, or 1, or 2 (Figure 11.109):

$$\begin{aligned} 0 &\equiv \{ \dots, -18, -15, -12, -9, -6, -3, 0, 3, 6, 9, 12, 15, 18, \dots \} \\ 1 &\equiv \{ \dots, -17, -14, -11, -8, -5, -2, 1, 4, 7, 10, 13, 16, 19, \dots \} \\ 2 &\equiv \{ \dots, -16, -13, -10, -7, -4, -1, 2, 5, 8, 11, 14, 17, 20, \dots \} \end{aligned}$$

that plays the role of a *stand-in for an infinite number of integers when it comes to the remainders of the integer division by 3* (the DAT style).

That is, the number 0 **represents any** integer from the following infinite set of integers when it comes to the remainders upon the division by 3:

$$\{ \dots, -18, -15, -12, -9, -6, -3, 0, 3, 6, 9, 12, 15, 18, \dots \}$$

Namely, any integer from the above set, that our fancy decides to pick whatsoever, upon the integer division by 3 leaves the remainder of 0.

Officially such an infinite set of integers is designated as $0 + 3\mathbb{Z}$ or as just $3\mathbb{Z}$.

The number 1 also **represents any** integer from the following, different, infinite set of integers when it comes to the remainders upon the division by 3:

$$\{ \dots, -17, -14, -11, -8, -5, -2, 1, 4, 7, 10, 13, 16, 19, \dots \}$$

Namely, any integer from the above set, that our fancy decides to pick whatsoever, upon the integer division by 3 leaves the remainder of 1.

Officially such an infinite set of integers is designated as $1 + 3\mathbb{Z}$.

The number 2 also **represents any** integer from the following, different yet, infinite set of integers when it comes to the remainders upon the division by 3:

$$\{ \dots, -16, -13, -10, -7, -4, -1, 2, 5, 8, 11, 14, 17, 20, \dots \}$$

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Namely, any integer from the above set, that our fancy decides to pick whatsoever, upon the integer division by 3 leaves the remainder of 2. Officially such an infinite set of integers is designated as $2 + 3\mathbb{Z}$.

It goes without saying that it is absolutely possible to pick any good old finite positive integer n , fix it, paint the set of integers into n distinct colors as explained above, pour each integer of the same color into its own bag, generate n distinct **representatives** of the set of integers when it comes to the remainders upon the integer division by n , paint each such representative number on each corresponding bag and so on.

Before moving on, we want to make sure that the above ideas are settled firmly in our minds - in order to facilitate just that, we invite our readers to play with different types of whole numbers on their own, while computing a bunch of quotients and remainders of both negative and positive integers.

Gauss Does It, Again

So far we see that technically exactly *four* objects or *four* actors enter the stage when the divisibility in the integers show is put on:

- a dividend, a
- a divisor, b
- a quotient, q , and
- a residue or a remainder, r

However, on the suggestion of the 24-year-old German mathematician Carl Friedrich Gauss (1777–1855), in a certain context, the above quotient q is largely irrelevant - when that is the case, the modern discipline known as **The Theory of Congruences** is born.

In the context of the theory of congruences or in the context of the so-called *modular arithmetic* our divisor b is re-branded into what is known as a *modulus* or a fixed, strictly positive whole number, usually denoted as n or as m , for **m**odulus.

The main idea due to C. Gauss, then, is to compare *two* integers with respect to one another based solely on the remainder that these integers leave when divided by a fixed positive integer, a modulus, ignoring the respective quotients altogether.

According to Gauss himself:

“if a number n measures the difference between two numbers a and b , then a and b are said to be congruent with respect to n ; if not, incongruent”

The modern mathematics, surprisingly enough, has altered the above definition due to Gauss very little.

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According to one such **Definition 21**: two integers a and b are congruent modulo n , a fixed positive integer, if n divides the difference $a - b$, that is, if there exists an integer k such that $a - b = kn$.

Do notice how **Definition 21** depends on **Definition 20**.

According to yet another possible **Definition 22**: two integers a and b are congruent modulo n , a fixed positive integer, if a and b leave the same non-negative remainder upon division by n .

The above two definitions are actually equivalent.

Indeed.

If the integers a and b are congruent modulo n as per **Definition 21**, then there exists an integer k such that $a - b = kn$, implying that:

$$a = b + k \cdot n$$

But, being a perfectly good integer, b itself is the subject of the Division Algorithm Theorem, according to which, upon the division by n that integer b will leave a certain remainder, r :

$$b = q \cdot n + r$$

implying that:

$$a = q \cdot n + r + k \cdot n = (q + k) \cdot n + r$$

We remember that the operation of addition defined on the set of integers is binary: the sum of any two integers is, again, an integer, l : $q + k = l \in \mathbb{Z}$.

Thus:

$$a = l \cdot n + r$$

implying that with respect to the division by n , according to DAT, the integer a has exactly the same remainder, r , as the integer b .

From where our second definition of congruence of two integers, **Definition 22**, follows.

Conversely.

If the integers a and b are congruent modulo n as per **Definition 22**, then, as per DAT, the integer a is multiplicatively expandable with respect to n as follows:

$$a = q_a \cdot n + r, \quad 0 \leq r < n$$

and, likewise, the integer b is multiplicatively expandable with respect to n as follows:

$$b = q_b \cdot n + r, \quad 0 \leq r < n$$

Therefore, the said difference between a and b will be:

$$a - b = (q_a - q_b) \cdot n$$

But the operation of subtraction defined on the set of integers is binary: the difference of any two integers is, again, an integer, $l: q_a - q_b = l \in \mathbb{Z}$.

Thus:

$$a - b = l \cdot n$$

which, as per our toy mechanical model and **Definition 20**, means exactly that the fixed integer n , the modulus, divides the difference $a - b$.

From where the first definition of the congruence of two integers, **Definition 21**, follows.

As it is a tradition in mathematics, we use whichever definition is tactically more advantageous at the moment.

In literature, **Definition 21** is usually taken to be **the** definition of congruence of two integers and **Definition 22**, then, actually becomes an *if-and-only-if* theorem.

The Notation

In any case, there exist *two* related in form but different in meaning *modular arithmetic* notations that capture the above ideas symbolically and, importantly, both of these notations are completely void of the symbolic representation of the quotient, q .

That is, a *statement* of the fact that the remainder of division of a by b is r is rendered as follows:

$$a \bmod b = r \tag{29}$$

For example, the statement that 3 is the remainder of 7 integer-divided by 4 is captured symbolically thus:

$$7 \bmod 4 = 3$$

and is read as:

seven mod four is three

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meaning, as noted, that when 7 is integer-divided by 4 then the residue of 3 results.

In general, the symbolic expression $a \bmod b$ represents *an operation* or *a function*.

Separately, in order to convey the fact that the integers a and b are *congruent modulo n* the following, related by slightly different in form, notation is used:

$$a \equiv b \pmod{n} \quad (30)$$

For example, since 4 divides the difference $7-3 = 4$, we have:

$$7 \equiv 3 \pmod{4}$$

If a given fixed integer n does not divide the difference between the integers a and b then these integers are *incongruent modulo n* :

$$a \not\equiv b \pmod{n} \quad (31)$$

For example:

$$25 \not\equiv 13 \pmod{7}$$

because 7 does not integer-divide the difference $25-13 = 12$.

Since 0 is divisible by any positive integer n , the following symbolic expression:

$$a \equiv 0 \pmod{n}$$

simply means that n divides a or that a is divisible by n .

From the above notation, without any **ifs** and **buts**, it instantly follows, as a true identity, that:

$$n \equiv 0 \pmod{n}$$

because, as we remember, once we are given a modulus n , we are assured that that number n is strictly positive and, as such, it is always safe, legal and allowed to divide by a non-zero number.

The above identity, then, simply states that any non-zero integer divides itself.

In general, the symbolic expression $a \equiv b \pmod{n}$ represents *a relation*, which is not to be confused with *an operation* or *a function*.

Two Simple But Very Important Properties

The following two properties of the divisibility in the integers are, in general, quite helpful and our readers should already be able to prove them both on their own.

Exercise 11.54: if it is known that $a \equiv b \pmod{n}$ and that $c \equiv d \pmod{n}$, then what can be said about the sums $a + c$ and $b + d$ modulo n ?

Solution: of this exercise amounts to nothing more than the good old middle school algebra that is induced by **Definition 21**.

That is, if it is known that $a \equiv b \pmod{n}$, then by our first definition of congruence of two integers (**Definition 21**), there exists an integer k such that $a - b = kn$.

Likewise, if $c \equiv d \pmod{n}$, then there exists an integer l such that $c - d = ln$.

But then, via a straightforward rearrangement of integers, we have for the sums $a + c$ and $b + d$:

$$(a - b) + (c - d) = (a + c) - (b + d) = kn + ln = (k + l)n$$

which means, since $k + l$ is an integer, that n divides the difference between the numbers $(a + c)$ and $(b + d)$, which, by the first definition of congruence of two integers, implies that these integers are congruent modulo n :

$$a + c \equiv b + d \pmod{n} \tag{32}$$

We, thus, just proved the following miniature **theorem**: if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$. $\square$

For example: since $18 \equiv 2 \pmod{4}$ and $37 \equiv 1 \pmod{4}$, it follows that $18 + 37 \equiv 2 + 1 \pmod{4}$, which is another way of saying that $55 \equiv 3 \pmod{4}$.

Note that the above result can be verified by-hand by, first, computing the sum $18 + 37$ directly and, second, by *reducing the result modulo 4*.

What does the phrase *reduce the result modulo n* mean?

The phrase *reduce the result modulo n* simply means to compute the residue of division of an integer by n in the style of the operation captured symbolically in (29): $a \bmod b = r$.

In our example, since $55 = 13 \cdot 4 + 3$, the residue of 3 and the respective congruence relation $55 \equiv 3 \pmod{4}$ naturally follow.

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For the purposes of the upcoming discussion, we just:

added two integers, 18 and 37, modulo 4

and in general that fancy and scary phrase *the reduction of an integer a modulo n* is captured symbolically quite simply - almost like the expression shown in (29) but with the $= r$ part thrown out:

$$a \bmod n$$

but we understand that it means exactly that - the remainder r of division of a by n .

The last example, then, can be rendered as follows:

$$55 \bmod 4 = 3$$

and elsewhere in a proper context in place of the residue 3 we can simply write $55 \bmod 4$.

Why could we pull off such a maneuver?

In other words, why the residue of a sum modulo n is equal to the sum of the individual residues modulo n :

$$(a + b) \bmod n = [a \bmod n + b \bmod n] \bmod n$$

Well, using our toy mechanical model, the answer to that question is straightforward: if we measure the distance a with a yard stick n then the result of such a measurement will be recorded as follows:

$$a = q_a n + r_a$$

and if we measure the distance b with the same yard stick n then the result of such a measurement will be recorded as follows:

$$b = q_b n + r_b$$

But then for the compound *additive* distance $a + b$ we have:

$$a + b = (q_a + q_b)n + (r_a + r_b)$$

and since the object $q_a + q_b$ is an integer, it follows that that distance $(q_a + q_b)n$ will be measured by the yard stick n evenly or without the remainder, which a loose and an informal way of saying that n divides $(q_a + q_b)n$.

Therefore, reducing both sides of the last equation modulo n and remembering our designation of r in (29), we have exactly what we need:

$$(a + b) \bmod n = 0 + [a \bmod n + b \bmod n] \bmod n$$

It should now be clear that in the above relation only one integer, a or b , can be safely replaced with its residue modulo n :

$$(a + b) \bmod n = [a + b \bmod n] \bmod n$$

Note also that from DAT it instantly follows that if we add an exact multiple of n , say, kn , to any integer a , as in $a + kn$, then the result of the operation of division of the new integer $a + kn$ by n will not change:

$$(a + kn) \bmod n = [(q_a + k)n + r_a] \bmod n = r_a \bmod n$$

which is to say that:

$$(a + kn) \bmod n = a \bmod n$$

We will need these properties of modular arithmetic soon enough.

Exercise 11.55: if it is known that $a \equiv b \pmod{n}$ and that $c \equiv d \pmod{n}$, then what can be said about the products $a \cdot c$ and $b \cdot d$ modulo n ?

Solution: again, amounts to just a bit of a middle school algebra.

If it is known that $a \equiv b \pmod{n}$, then by our first definition of congruence of two integers (**Definition 21**), there exists an integer k such that $a - b = kn$, implying that $a = b + kn$.

Likewise, if $c \equiv d \pmod{n}$, then there exists an integer l such that $c - d = ln$, implying that $c = d + ln$.

But then for the products ac and bd we have:

$$ac = (b + kn)(d + ln) = bd + bln + dkn + klnn = bd + (bl + dk + kln)n$$

implying that:

$$ac - bd = (bl + dk + kln)n$$

which means, since $bl + dk + kln$ is an integer, that n divides the difference $(ac - bd)$, which, by definition, means that the products ac and bd are congruent modulo n :

$$ac \equiv bd \pmod{n} \tag{33}$$

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We, thus, just proved the following miniature **theorem**: if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$. $\square$

For example: since $18 \equiv 2 \pmod{4}$ and $37 \equiv 1 \pmod{4}$, it follows that $18 \cdot 37 \equiv 2 \cdot 1 \pmod{4}$, which is another way of saying that $666 \equiv 2 \pmod{4}$.

Note, again, that the above result can be verified by-hand by, first, computing the product $18 \cdot 37$ directly:

$$18 \cdot 37 = (20 - 2)(40 - 3) = 800 - 60 - 80 + 6 = 666$$

and, second, by reducing the result modulo 4:

$$666 = 166 \cdot 4 + 2$$

which is to say that:

$$666 \bmod 4 = 2$$

from where the claim follows.

For the purposes of the upcoming discussion, we just:

multiplied two integers, 18 and 37, modulo 4

Why could we pull off such a maneuver?

Again, using our toy mechanical model, we measure the distances a and b with the same yard stick n separately:

$$a = q_a n + r_a \quad \text{and} \quad b = q_b n + r_b$$

But then for the compound *multiplicative* distance $a \cdot b$ we have:

$$\begin{aligned} a \cdot b &= (q_a n + r_a)(q_b n + r_b) = \\ &= [q_a q_b n + q_a r_b + q_b r_a]n + r_a \cdot r_b \end{aligned}$$

and since the object $[q_a q_b n + q_a r_b + q_b r_a]$ is an integer (explain *why*), it follows that that distance $[q_a q_b n + q_a r_b + q_b r_a]n$ will be measured by the yard stick n evenly or without the remainder, which a loose and an informal way of saying that n divides $[q_a q_b n + q_a r_b + q_b r_a]n$.

Therefore, reducing both sides of the last equation modulo n and remembering our designation of r in (29), we have exactly what we need:

$$(a \cdot b) \bmod n = 0 + [a \bmod n \cdot b \bmod n] \bmod n$$

We will need this property of modular arithmetic soon enough also.

The Theory Of Groups, At Last

Armed even with this tiny bit of straightforward theory, we can already do some interesting group-theoretic work.

The congruence of any two integers modulo 1 is not particularly interesting because 1 divides all integers.

However, the moment we increase the modulus from 1 to 2, the things get very interesting fast.

As such, let us fix the modulus n at 2.

Then, by the Division Algorithm theorem, we know that upon division of any integer by 2 there can be only 2 remainders: 0 and 1.

Such a set of numbers, $\{0, 1\}$, is known as *the set of least non-negative residues modulo 2* and, consequently, a general set of residues G such that:

$$G = \{0, 1, 2, \dots, n - 1\} \tag{34}$$

is referred to as *the set of least non-negative residues modulo n* .

Moreover, by fixing the modulus n at 2, we actually painted all the integers in just two colors:

- 1) the 0-residue-upon-divison-by-2 color and
- 2) the 1-residue-upon-divison-by-2 color

If we now pour all the integers of the same color into their own bags then we shall have just two bags:

- 1) the **even** integers bag and
- 2) the **odd** integers bag

All the integers from the **even-integers** bag:

$$E = \{\dots, -14, -12, -10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10, 12, 14, \dots\}$$

leave the residue of 0 upon division by 2 the DAT style.

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All the integers from the **odd-integers** bag:

$$O = \{\dots, -13, -11, -9, -7, -5, -3, -1, 1, 3, 5, 7, 9, 11, 13 \dots\}$$

leave the residue of 1 upon division by 2 the DAT style.

In fact, if we mark the **even-integers** bag with 0 then any integers that we yank out of the **even-integers** bag will be *represented* by one and the same residue 0 when it comes to the division modulo 2.

Likewise, if we mark the **odd-integers** bag with 1 then any integers that we yank out of the **odd-integers** bag will be *represented* by one and the same residue 1 when it comes to the division modulo 2.

In other words, in the *division modulo 2* context, the numbers 0 and 1 become *the representatives* of the even and the odd integers respectively.

Officially, these bags, the 0-bag and the 1-bag, are known in mathematics as *the equivalence classes* modulo 2.

Such bags have a number of interesting properties: if an integer is even then it is not odd and if an integer is odd then it is not even - officially, *the intersection* of the sets E and O is the empty set.

Given any integer a and a modulus 2, it is possible to do two things uniquely:

- it is possible to determine which equivalence class that integer lives in via the $a \bmod 2$ operation and
- it is possible to *generate* that entire equivalence class by *adding* to a the integral multiples of 2 in order to move toward the positive infinity and by *subtracting* from a also the integral multiples of 2 in order to move toward the negative infinity. Both of these operations are captured succinctly with the following notation:

$$a + 2\mathbb{Z}$$

which is just as easy to decipher - to the integer a we add $2k$, where the integer k can be both negative and positive.

The above $a + 2\mathbb{Z}$ notation is a symbolic way to say that the linear distance between any two immediate neighbors in both equivalence classes E and O is exactly and always 2.

Moreover, as a straightforward exercise, it can be shown that our operation n *divides the difference* $(a - b)$ satisfies all three properties of *an equivalence relation*, because such an operation is Reflexive, Transitive and Symmetric.

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Such a formalism, however, will only obscure the current discussion and we will mostly shy away from it for now.

So how does the theory of groups dovetail into all of this?

Because we fired the actor that plays the role of the quotient, q , we now have only *three* objects to work with:

- 1) a dividend, a
- 2) a modulus, n
- 3) a residue, r

These three objects fall into place of a potentially *binary* operation quite neatly.

That is:

- since a modulus n is a *fixed* positive integer that does not vary, we leave it alone, thus reducing the number of varying objects to two
- we split the above single dividend a into *two* integers or two *operands*, a and b , of a mystery operation and
- we brand the above remainder r to be *the result* of that mystery operation

Exactly what this mysterious *operation* is?

This mysterious operation could be:

- either *the addition modulo n* or
- *the multiplication modulo n*

and we already understand exactly how to define such operations and exactly what these operations mean - we just have to find a suitable set on which we can unleash these operations.

In a bit more detail, the operation of *addition modulo n* simply means that we are to compute the sum of two or more integers the old-fashioned way and then we are to ... reduce that **sum** modulo n . That is all there is to it.

Likewise, the operation of *multiplication modulo n* simply means that we are to compute the product of two or more integers the old-fashioned way and then we are to ... reduce that **product** modulo n . That is all there is to it.

For example, officially, with $n = 2$ for any two integers a, b we define the operation $\circ$ that means *the addition modulo 2* as follows:

- $a \circ b = 0$ if $(a + b) \equiv 0 \pmod{2}$
- $a \circ b = 1$ if $(a + b) \equiv 1 \pmod{2}$

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If we agree to designate the operation of *addition modulo 2* symbolically as follows:

$$+_2$$

then alternatively and more succinctly we can define such an operation thus:

$$a +_2 b = (a + b) \bmod 2$$

Once such a definition is on the books and once the context of *yada yada modulo 2* has been established, it is a common practice in the literature to use the additive notation and simply write:

$$1 + 1 = 0 \quad \text{or} \quad 11 + 13 = 0$$

and:

$$1 + 0 = 1 \quad \text{or} \quad 21 + 8 = 1$$

understanding all along that the final result of an addition is always reduced modulo 2.

Likewise, for any two integers a, b we define the operation $\circ$ that means *the multiplication modulo 2* as follows:

- $a \circ b = 0$ if $(a \cdot b) \equiv 0 \pmod{2}$
- $a \circ b = 1$ if $(a \cdot b) \equiv 1 \pmod{2}$

If we agree to designate the operation of *addition modulo 2* symbolically as follows:

$$\cdot_2$$

then alternatively and more succinctly we can define such an operation thus:

$$a \cdot_2 b = (a \cdot b) \bmod 2$$

Again, once such a definition is on the books and once the context of *yada yada modulo 2* has been established, it is a common practice in the literature to use the multiplicative notation and simply write, in general:

$$2 \cdot 3 = 0 \quad \text{or} \quad 5 \cdot 7 = 1$$

understanding all along that the final result of a multiplication is always reduced modulo 2.

There we have it: a set, of residues modulo n in general, and the operation, the addition/multiplication modulo n in general.

Upon the happy marriage of the above operation(s) with the above set, is there any chance that such a dynamic duo will form a group?

We answer that question via a series of exercises.

Before we do that, however, please allow us to highlight one important aspect of the theorems that we proved in **Exercise 11.54** and **Exercise 11.55** one more time.

It may so happen that the numbers that we juggle will very well fall out of the set G of least non negative residues modulo n shown in (34).

For example: if $24 \equiv 8 \pmod{2}$ and $36 \equiv 14 \pmod{2}$ then, since $24 + 36 = 60$ and $8 + 14 = 22$, we have $60 \equiv 22 \pmod{2}$.

In the context of a certain problem the expression $60 \equiv 22 \pmod{2}$ may very well be left as is.

However, for the purposes of the upcoming exercises, due to the gift of the constraint (28) in DAT, $0 \leq r < n$, do not forget that it is always possible to reduce the integer 22 modulo 2 into 0 and, thus, whittle that number into the range of the respective set of least non-negative residues modulo 2.

Exercise 11.56: does the set $G = \{0, 1\}$ of least non-negative residues modulo 2 form a group under the operation of addition modulo 2?

Solution: as we explained the operation-formation process earlier, the operation of *addition modulo 2* acts on exactly two, no more and no less, integers a, b .

Since our residues can take on only two distinct values and exactly three elements participate in the associativity group axiom, it follows that for a brute-force manual verification of that group axiom we would have to generate and validate all the linear 3-permutations of 2 distinct items or:

$$2 \cdot 2 \cdot 2 = 2^3 = 8$$

additions modulo 2, just like we did that in **Exercise 11.40**.

In this particular case, it may seem like a reasonable approach:

$$0 + (0 + 0) = 0 = (0 + 0) + 0$$

$$1 + (1 + 1) = 1 + 0 = 1 = 0 + 1 = (1 + 1) + 1$$

but by now we quickly understand that as the number of residues grows with the modulus, we would have to generate and validate a combinatorially explosive number of additions modulo n , in general.

For example, for the additions modulo 3 we would have to generate and validate:

$$3^3 = 27$$

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strings of additions - each such string contains 4 operations of addition, 2 operations on either side of the equal sign.

Thus, in order to validate the associativity group axiom for the operation of addition modulo 3 we would have to crunch through:

$$3^3 \cdot 4 = 27 \cdot 4 = 108$$

additions modulo 3.

For the additions modulo 4 we would have to generate and validate:

$$4^3 = 64$$

strings of additions and crunch through $64 \cdot 4 = 256$ additions modulo 4, ... for the additions modulo 11 we would have to generate and validate:

$$11^3 = 1331$$

strings of additions and crunch through $1331 \cdot 4 = 5324$ additions modulo 11 and so on.

We, thus, see that the process of the brute-force verification of the associativity group axiom for the operation of *addition modulo n* is not scalable with n , is tedious and is error-prone.

It is tactically much more advantageous to use the tiny theorems that we proved earlier after **Exercise 11.54** and investigate the values of the following two *generic* expressions:

$$[(a + b) \bmod 2 + c] \bmod 2 \quad \text{and} \quad [a + (b + c) \bmod 2] \bmod 2$$

where a, b and c are any elements of the given set G .

If the two values above agree then the operation of *addition modulo 2* defined on the given set G is associative and if the above two values do not agree then the said operation is not associative.

Since we already showed that for the integers p, q and the modulus n it is the case that $(p + q) \bmod n = (p + q \bmod n) \bmod n$, it follows that in the above *first* composition or sequence of modulo 2 additions the role of the integer q is played by the traditional sum of integers $(a + b)$ and the role of the integer p is played by the integer c .

Therefore, reading the expression $(p + q) \bmod n = (p + q \bmod n) \bmod n$ right-to-left, we have:

$$[(a + b) \bmod 2 + c] \bmod 2 = [(a + b) + c] \bmod 2$$

But that traditional addition defined on the set of integers *is* associative, meaning that the following relation:

$$(a + b) + c = a + (b + c)$$

holds for all integers a, b, c .

Thus:

$$[(a + b) + c] \bmod 2 = [a + (b + c)] \bmod 2$$

Lastly, reading the expression $(p + q) \bmod n = (p + q \bmod n) \bmod n$ left-to-right, we see that the role of the integer q is now played by the integer $(b + c)$ and that the role of the integer p is played by a :

$$[a + (b + c)] \bmod 2 = [a + (b + c) \bmod 2] \bmod 2$$

which tells us that the values of the above two expressions are actually equal:

$$[(a + b) \bmod 2 + c] \bmod 2 = [a + (b + c) \bmod 2] \bmod 2$$

and that the operation of *addition modulo 2* defined on the set of integers G is associative.

But since in our reasoning above we did not rely on any specific properties of the modulus 2, it follows that that argument will work for any legal value of $n > 1$ (or 2, 3, 4, 5 and so on):

$$[(a + b) \bmod n + c] \bmod n = [a + (b + c) \bmod n] \bmod n$$

Commentary: in general, we have to get a bit more formal here because each integer from the set of least non-negative residues modulo n represents an infinite set of respective integers or an equivalence class modulo n .

Thus, the operation of addition modulo 2, or the operation of addition modulo n in general, is associative also.

So far so good.

Now we want to investigate whether the operation of addition modulo 2 over the set G is binary and whether each element from the set G possesses a unique inverse.

We remember that, as per **Definition 10**, the operation $\circ$ is binary if for any two elements a, b of the given set the uniquely determined element $c = a \circ b$ also belongs to the same set.

Because we already anticipate the investigation of the similar sets coupled with the operation of addition modulo 3 and modulo 4, we decide to work in a consistent fashion by constructing the maybe multiplication table of this maybe group.

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In such a potential table we quickly fill out the traditional topmost row and the traditional leftmost column (Figure 11.110):

$+$ mod 2	0	1
0	0	1
1	1	

because it is evident that:

- the sum of two even numbers is even and that
- the sum of an even and an odd number, taken in any order, is an odd number

Commentary: an alternative way to fill out not just the above three entries but *all* the entries in such a multiplication table is to use the above notion of *equivalence classes* in the following highly manual but straightforward process.

In order to find the result of the operation $0 + 0$ modulo 2, pick any integer from the set E as an integer that is *represented* by the number 0:

$$E = \{\dots, -18, -16, -14, -12, -10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10, 12, 14, 16, 18, \dots\}$$

Say, we pick the integer -12 for that purpose.

Then, since the second operand in the operation $0 + 0$ modulo 2 is also 0, pick any integer from the same set E .

Say, we pick the integer 16 for that purpose. Next, add these two integers together using the traditional operation of addition of integers:

$$-12 + 16 = 4$$

Scan all the equivalence classes on hand and find in which such class the integer 4 lives.

In our case, the integer 4 lives in the equivalence class E and that class is represented by the single integer, 0.

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Thus, we just unmasked the result of the operation of $0 + 0$ modulo 2.

Write the integer 0 at the intersection of the 0-row and the 0-column of the above multiplication table.

Floss. Rinse. Repeat. In order to find the result of the operation $0 + 1$ modulo 2, pick any integer, say, 8, from the set E . Pick any integer, say, -5 , from the set O . Add these two integers together the usual way: $8 + (-5) = 3$. Scan all the equivalence classes on hand and find in which such class the integer 3 resides.

Aha!

In this case the integer 3 dwells in the class O and that class is represented by the single residue, 1. Write the number 1 at the intersection of the 1-column and the 0-row of the multiplication table at hand.

Move on.

In order to fill out the last entry in our potential multiplication table we observe that under a traditional addition the result of adding two ones together is equal to 2:

$$1 + 1 = 2$$

But in the context of the current operation of *addition modulo 2*, the above result, when *reduced modulo 2*:

$$(1 + 1) \equiv 0 \pmod{2}$$

yields 0.

Thus, in simple terms, the sum of any two odd integers is an integer that is even, as was proved by Euclid in his *“Elements”*, Book 9 Proposition 22 (Figure 11.111):

$\begin{matrix} + \\ \text{mod } 2 \end{matrix}$	0	1
0	0	1
1	1	0

Huh.

It looks like we do have a group after all.

The identity element of this group is 0 and, since we remember that the identity of any group is its own inverse, we took care of the inverse of the first element of that group:

$$e \cdot e = e$$

The remaining and the only non identity element of this group is 1 and it is its own inverse:

$$1 \cdot 1 = 1 + 1 = 0 = e$$

Since all the popular group axioms of this unordered pair $\langle G; \circ \rangle$ are satisfied, the answer to this exercises is:

yes, the set G of least non negative residues modulo 2 is a group under the addition modulo 2

Such a group is finite, is of order 2 and is commutative, because the operation of addition defined over the set of integers is commutative:

$$(a + b) \bmod 2 = (b + a) \bmod 2$$

for any two elements a, b of G .

This group is also called *the additive group of integers modulo 2* and is symbolized in various ways as $(\mathbb{Z}/2\mathbb{Z}, +)$ or as $\mathbb{Z}/2\mathbb{Z}$ or as $\mathbb{Z}_2$ and so on - we are not pushing the symbolization of groups too much in this text precisely because of our earlier remark: here we are mostly after the understanding of *the essence* of a phenomenon; once such an understanding is developed, we can pick up and agree on any reasonable symbolization of it effortlessly.

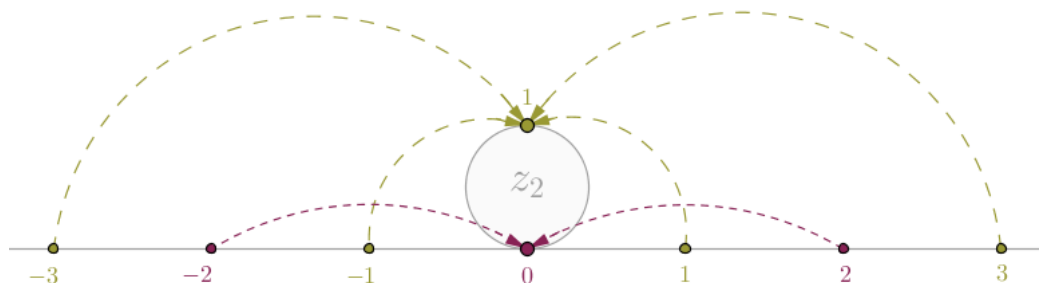
Also, remember that many abstract concepts in mathematics can be visualized with a mechanical-geometrical model to some degree.

Inventing such models is never a waste of time and is always beneficial because during the process of the invention of such visual models we kick the technical ideas of the phenomenon at hand around and look at them from various angles and, thus, *learn* the new material and reconcile it in our minds.

So how can we visualize these equivalence classes modulo 2 that play the vital role in our additive group of integer modulo 2?

Well, let us begin with a circle z_2 that is tangent to the number line at the point 0 and whose circumference is of length 2.

If the circumference of such a circle is of length 2 then it is possible to divide such a circumference in exactly 2 equal parts which will generate only one more integral point on that circle - a point marked as 1 (Figure 11.112):



Now, if we begin winding the right portion of the number line onto the circle z_2 like a rope in, say, counterclockwise direction, then all the positive *odd* integers will be sitting on the North Pole of the circle z_2 or on the mark 1, while all the positive *even* numbers will be sitting on the South Pole of the circle z_2 on the mark 0.

Likewise, if we begin winding the left portion of the number line onto the circle z_2 like a rope in, say, clockwise direction, then all the negative *odd* integers will be also sitting on the North Pole of the circle z_2 or on the mark 1, while all the negative *even* numbers will be also sitting on the South Pole of the circle z_2 on the mark 0.

The above fact that all the even numbers will be taken into the point marked 0 on the circle z_2 visualizes (to some degree):

- the equivalence class E and
- the fact that the residue of all such numbers modulo 2 is 0

The above fact that all the odd numbers will be taken into the point marked 1 on the circle z_2 visualizes (to some degree):

- the equivalence class O and
- the fact that the residue of all such numbers modulo 2 is 1

Your imagination is the limit with such models: we can, for example, replace the above *winding-the-number-line* process with its opposite.

That is. Imagine that the circle z_2 begins *rolling* without sliding and slipping along the number line in, say, rightward direction.

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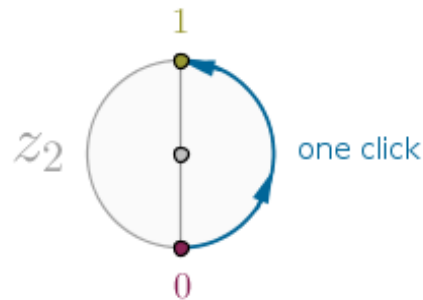
Then, the circle's red mark 0, loaded with the red paint, will color all the positive even numbers in red, while the circle's green mark 1, loaded with the green paint, will color all the positive odd numbers in green.

Replace the paint with the electric light bulbs of the appropriate color and so on.

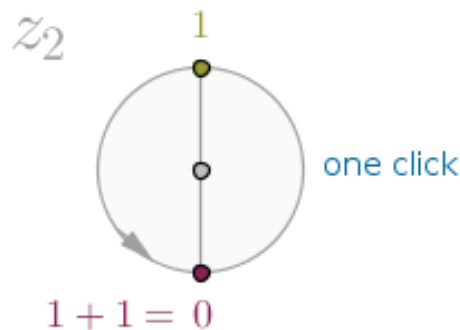
In any case, our circle z_2 is a convenient visual and mechanical model which can be used to generate all the entries in that group's multiplication table and which can be used to find the inverse of any given element of that group on demand.

Namely.

Let us agree that *one (positive) click* of the circle z_2 is a turn of the circle z_2 about its center counterclockwise in the parent plane by the angular distance between any two immediate neighbors on that circle - just like a click that we hear when we turn the mechanical dial on an old safe (Figure 11.113):



Then, in order to compute the result of, say, $1 + 1$ operation using the circle z_2 , we simply start with the point on that circle marked as 1 and then we see where that point will be taken after we do as many *clicks* as the second operand in that operation demands. In this particular case, since the second operand is 1, we do just 1 click - that click will take the point marked 1 into the point on the circle z_2 marked as 0 (Figure 11.114):



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Thus, 0 is the result of the operation $1 + 1$ (modulo 2).

In order to recover the inverse of any element on the circle z_2 , we inquire: given the element, say, 1, how many clicks of the circle z_2 will take *that* mark into the mark 0? Well, in this case, exactly *one* click will transform the mark 1 into the mark 0 on z_2 . Therefore, the inverse of the element 1 in this particular group is 1 and so on. $\square$

In one of our previous problem-solving sessions we, essentially, already came across such a group (Figure 11.35):

+	<i>E</i>	<i>O</i>		◦	<i>e</i>	<i>a</i>		XOR	0	1
<i>E</i>	<i>E</i>	<i>O</i>		<i>e</i>	<i>e</i>	<i>a</i>		0	0	1
<i>O</i>	<i>O</i>	<i>E</i>		<i>a</i>	<i>a</i>	<i>e</i>		1	1	0

except that then we worked in a loose even-odd fashion and now we have the means and a basic apparatus to work more formally and more rigorously.

Note, again, that the structure of our newly-found group is still the same as the structure that was depicted in the cherry-banana group (Figure 11.36):

◦	<i>e</i>	<i>x</i>		◦	<i>e</i>	<i>x</i>
<i>e</i>	<i>e</i>	<i>x</i>		<i>e</i>	<i>e</i>	<i>x</i>
<i>x</i>	<i>x</i>	?		<i>x</i>	<i>x</i>	<i>e</i>
◦				◦		
						
						

We, thus, came across yet another manifestation of the concept of *group isomorphism* - we looked at many groups of order 2, the underlying nature of the elements of all such groups and of the underlying nature of the operations defined on these elements of all such groups are wildly different.

However.

The structure of this entity that we refer to as *a group*, remains the same across all of the concrete examples of such a group.

In the next two discussions we will look at more examples of sets of residues modulo n under the operations of addition and multiplication modulo n .

Exercises 9: Modular Arithmetic, Part 2 Of 4

In this section we continue looking at the sets of least non-negative residues modulo n under the operations of addition and multiplication modulo n .

Even though we successfully solved the previous exercise, **Exercise 11.56**, we are still not really sure about what is going on with the additive inverses in the additive groups of integers modulo n . To that end, we solve a couple of more related problems.

Exercise 11.57: does the set $G = \{0, 1, 2\}$ of least non-negative residues modulo 3 form a group under the operation of addition modulo 3?

Solution: we define the given operation of *addition modulo 3* over the given set G , $\circ$, as follows:

- $a \circ b = 0$ if $(a + b) \equiv 0 \pmod{3}$
- $a \circ b = 1$ if $(a + b) \equiv 1 \pmod{3}$
- $a \circ b = 2$ if $(a + b) \equiv 2 \pmod{3}$

We already know that by so doing, we partition the entire set of integers into three infinite sets or three equivalence classes modulo 3, each of which is represented by only one number or only one remainder upon division by 3 the DAT style:

$$0 : \{ \dots, -15, -12, -9, -6, -3, 0, 3, 6, 9, 12, 15, \dots \}$$

$$1 : \{ \dots, -14, -11, -8, -5, -2, 1, 4, 7, 10, 13, 16, \dots \}$$

$$2 : \{ \dots, -13, -10, -7, -4, -1, 2, 5, 8, 11, 14, 17, \dots \}$$

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From the solution of **Exercise 11.56** we already know that the current operation, the addition modulo 3, defined on the above set G is associative.

We also feel that that operation is binary - we verify that hunch by constructing the potential multiplication table of this potential group and filling out its first few *usual suspects* entries (Figure 11.115):

$+$ $\text{mod } 3$	0	1	2
0	0	1	2
1	1		
2	2		

For the remaining entries we work using the above definition of addition modulo 3. We, thus, compute:

$$(1 + 1) \equiv 2 \pmod{3}, \quad (1 + 2) \equiv 0 \pmod{3}, \quad (2 + 1) \equiv 0 \pmod{3}, \quad (2 + 2) \equiv 1 \pmod{3}$$

and fill out the rest of our multiplication table (Figure 11.116):

$+$ $\text{mod } 3$	0	1	2
0	0	1	2
1	1	1	2
2	2	2	0

Alternatively, it is possible to fill out the above multiplication table via the custom-tailored process described in the previous discussion.

That is. In order to find the result of the operation $a + b$ modulo 3:

- pick any integer k from the equivalence class modulo 3 to which the integer a belongs
- pick any integer l from the equivalence class modulo 3 to which the integer b belongs
- add the integers k and l the old-fashioned way, $k + l = x$
- find the equivalence class to which the integer x belongs
- claim *the representative* of that equivalence class modulo 3 to be the sought-after result of the operation $a + b$ modulo 3

For example, in order to find the result of the operation $1 + 2$, we pick the integer -11 from the equivalence class 1 and we pick the integer -7 from the equivalence class 2.

Then, the traditional sum $-11 + (-7)$ comes out to be equal to -18 and we see that the integer -18 lives in the equivalence class 0.

Thus, the result of the operation $1 + 2$ is 0, in the modulo 3 context.

Very good.

Let us analyze what we just discovered.

Not only the operation of addition modulo 3 defined on the set G of least non-negative residues modulo 3 is binary, every element of the set G also possesses a *unique inverse* under that operation:

- the inverse of the element 0 is that element itself, $0 + 0 = 0$
- the inverse of the element 1 is the element 2, $1 + 2 = 0$ and
- the inverse of the element 2 is the element 1, $2 + 1 = 0$

Since all the popular group axioms in this problem are satisfied, we answer this question in the positive:

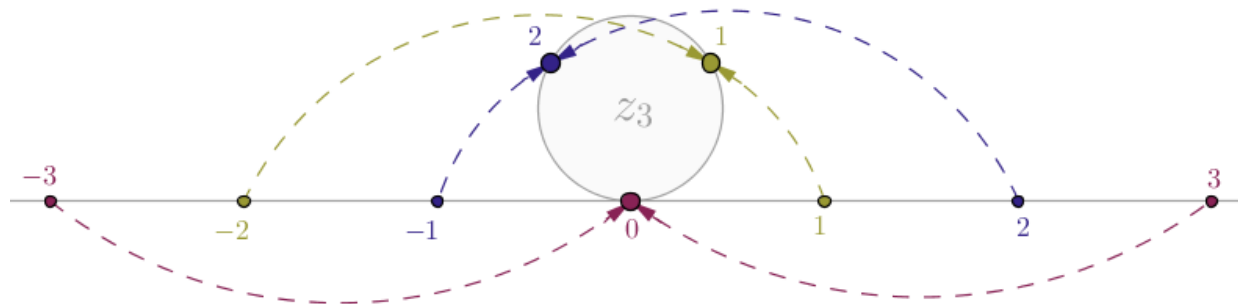
yes, the set G of least non-negative residues modulo 3 is a group under the addition modulo 3

That group is finite, is of order 3 and is commutative.

This group is also called *the additive group of integers modulo 3* and is symbolized in various ways as $(\mathbb{Z}/3\mathbb{Z}, +)$, or as $\mathbb{Z}/3\mathbb{Z}$, or as $\mathbb{Z}_3$ and so on.

Using the mechanical-geometrical model described in the previous discussion, we extend that model into the modulo 3 context by constructing a circle z_3 that:

- is tangent to the number line at the point 0 and
- whose circumference is equal to 3 units of length and
- that is divided into 3 equal parts of length 1 each (Figure 11.117):



Note that *the circular* distance between any two immediate neighbors of z_3 , or the distance measured over the circumference of z_3 , is exactly 1 unit of distance long.

Therefore.

If we wind the *positive* portion of the number line over the circle z_3 as a rope *counterclockwise* then:

- all the red numbers 0, 3, 6 and so on will be sitting on that circle z_3 at the point 0 - these are the numbers that are divisible by 3 and represent the positive portion of the above equivalence class 0 modulo 3
- all the green numbers 1, 4, 7 and so on will be sitting on that circle z_3 at the point 1 - these are the numbers that leave the residue of 1 upon division by 3 and such numbers represent the positive portion of the above equivalence class 1 modulo 3
- all the blue numbers 2, 5, 8 and so on will be sitting on that circle z_3 at the point 2 - these are the numbers that leave the residue of 2 upon division by 3 and such numbers represent the positive portion of the above equivalence class 2 modulo 3

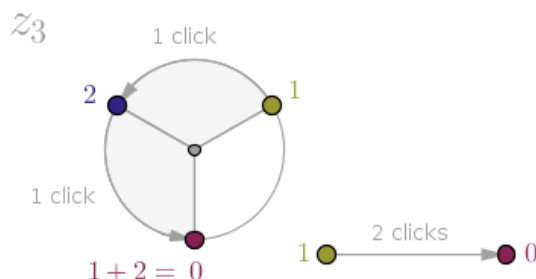
Consequently, if we wind the *negative* portion of the number line over the circle z_3 as a rope *clockwise* then:

- all the red numbers 0, -3 , -6 and so on will be sitting on that circle z_3 at the point 0 - these are the numbers that are divisible by 3 and represent the negative portion of the above equivalence class 0 modulo 3
- all the green numbers -1 , -4 , -7 and so on will be sitting on that circle z_3 at the point 2 - these are the numbers that leave the residue of 2 upon division by 3 and such numbers represent the negative portion of the above equivalence class 2 modulo 3
- all the blue numbers -2 , -5 , -8 and so on will be sitting on that circle z_3 at the point 1 - these are the numbers that leave the residue of 1 upon division by 3 and such numbers represent the negative portion of the above equivalence class 1 modulo 3

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Which such a device in place it becomes possible to rapidly fill in the values of the above multiplication table by following the mechanical procedure already described if we agree that one z_3 -click constitutes one turn of the circle z_3 about its center counterclockwise in the parent plane by the angular distance between any two immediate neighbors on that circle.

That is. Say, we wish to find the result of the $1 + 2$ operation on z_3 . First, we find the point 1 on the circle z_3 . Then, we execute 2 clicks of that circle because 2 is the second operand in the sum requested. Such 2 clicks will take the point 1 on z_3 into the point 0 on z_3 (Figure 11.118):



Thus, 0 is the result of the operation $1 + 2$ on z_3 . Likewise, in order to find the inverse of a given element of G on z_3 we simply ask the following question: how many clicks will take the corresponding point on z_3 into the point 0?

For example, from the previous example we already see that 2 clicks will take the point 1 into the point 0 on z_3 . Thus, the element 2 is the inverse of the element 1 in this particular group.

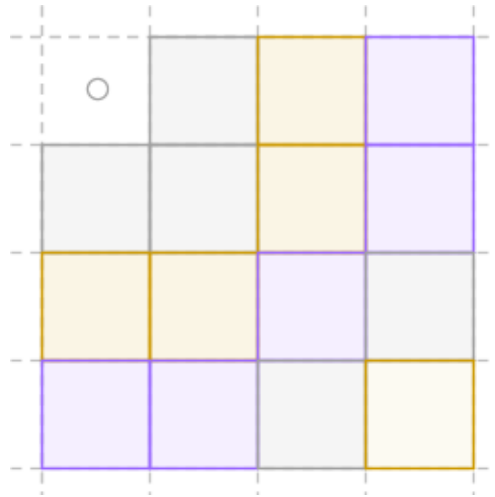
How many clicks of z_3 will take its point 2 into the point 0? Why, 1 click: the element 1 is the inverse of the element 2 in this group and so on. $\square$

It should come as no surprise that the group of order 3 that we just discovered has the same *structure* as another group of order 3 that we already studied in **Exercise 11.50**, the group of rotational congruence motions of a regular 3-gon (Figure 11.79):

$\circ$	$e_O^{0^\circ}$	$a_O^{120^\circ}$	$b_O^{240^\circ}$
$e_O^{0^\circ}$	e	a	b
$a_O^{120^\circ}$	a	b	e
$b_O^{240^\circ}$	b	e	a

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Observe yet again that if we remove the named elements from both multiplication tables shown above then both such tables will leave the same colorful trail (Figure 11.119):



That *sameness* of the above colorful trail tells us that *the structure* of these two groups is the same - we remember that that is always the case when the order of a group is a prime number and 3 is certainly prime.

These two groups, thus, are *isomorphic* with respect to each other or are *essentially the same*.

In any case, observe one more time how the theory of groups discards and eats away what it bravely qualifies as *the external minutiae* - the underlying nature of the elements and the underlying nature of the operation that acts on these elements - and leaves only the quintessential core, the group's *structure*, on the table for us to study.

The group elements in the first case were *the planar congruence rotations* of a regular 3-gon and the group elements in the second case were *the residues* of integer division by 3.

Can the underlying nature of such elements get any more different?

The group operation in the first case was *the composition* of the said planar congruence rotations of a regular 3-gon and the group operation in the second case was *the addition modulo 3*.

Can the underlying nature of such operations get any more different?

"Yaaaawn", says the theory of groups, "I don't really care about all of this. I only care about *how the said elements interact with each other under the said operation*."

Note, however, that our mechanical-geometrical model of circles and ropes of the group Z_3 in some vague way brings the group of rotational symmetries of an equilateral triangle and the group Z_3 closer together.

Exercise 11.58: does the set $G = \{0, 1, 2, 3\}$ of least non negative residues modulo 4 form a group under the operation of addition modulo 4?

Solution: we define the given operation of *addition modulo 4* over the given set $G, \circ$, as follows:

- $a \circ b = 0$ if $(a + b) \equiv 0 \pmod{4}$
- $a \circ b = 1$ if $(a + b) \equiv 1 \pmod{4}$
- $a \circ b = 2$ if $(a + b) \equiv 2 \pmod{4}$
- $a \circ b = 3$ if $(a + b) \equiv 3 \pmod{4}$

By so doing we partition the entire set of integers into four infinite sets or four equivalence classes modulo 4, each of which is represented by a single number or by the remainder of the division of any integer from the respective set by 4 the DAT style:

$$0 : \{ \dots, -16, -12, -8, -4, 0, 4, 8, 12, 16, \dots \}$$

$$1 : \{ \dots, -15, -11, -7, -3, 1, 5, 9, 13, 17, \dots \}$$

$$2 : \{ \dots, -14, -10, -6, -2, 2, 6, 10, 14, 18, \dots \}$$

$$3 : \{ \dots, -13, -9, -5, -1, 3, 7, 11, 15, 19, \dots \}$$

and so on. From the solution of **Exercise 11.56** we already know that the current operation, the addition modulo 4, defined on the above set G is associative. We also feel that that operation is binary - we verify that hunch by constructing the potential multiplication table of this potential group and filling out its first few usual suspects entries (Figure 11.120):

$\begin{smallmatrix} + \\ \text{mod } 4 \end{smallmatrix}$	0	1	2	3
0	0	1	2	3
1	1			
2	2			
3	3			

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Again, we compute the remaining entries in the above table by relying on the definition of the current operation:

$$(1 + 1) \equiv 2 \pmod{4}, \quad (1 + 2) \equiv 3 \pmod{4}, \quad (1 + 3) \equiv 0 \pmod{4}$$

So far so good (Figure 11.121):

$\begin{smallmatrix} + \\ \text{mod } 4 \end{smallmatrix}$	0	1	2	3	
0	0	1	2	3	
1	1	1	2	3	0
2	2	2			
3	3	3			

Next:

$$(2 + 1) \equiv 3 \pmod{4}, \quad (2 + 2) \equiv 0 \pmod{4}, \quad (2 + 3) \equiv 1 \pmod{4}$$

Excellent (Figure 11.122):

$\begin{smallmatrix} + \\ \text{mod } 4 \end{smallmatrix}$	0	1	2	3	
0	0	1	2	3	
1	1	1	2	3	0
2	2	2	3	0	1
3	3	3			

Lastly:

$$(3 + 1) \equiv 0 \pmod{4}, \quad (3 + 2) \equiv 1 \pmod{4}, \quad (3 + 3) \equiv 2 \pmod{4}$$

Can't be better (Figure 11.123):

$+ \pmod{4}$	0	1	2	3
0	0	1	2	3
1	1	1	2	3
2	2	2	3	0
3	3	3	0	1

Commentary: alternatively, it is possible to fill out the above multiplication table via the custom-tailored process described in the previous discussion - in order to find the result of the operation $a + b$ modulo 4 :

- pick any integer k from the equivalence class modulo 4 to which the integer a belongs
- pick any integer l from the equivalence class modulo 4 to which the integer b belongs
- add the integers k and l the old-fashioned way, $k + l = x$
- find the equivalence class modulo 4 to which the integer x belongs
- claim *the representative* of that equivalence class to be the sought-after result of the operation $a + b$ modulo 4

Again, we see that the identity e of this group is the element 0 and, in any group, the identity is its own inverse.

We also see that all the remaining, non identity, elements of this group do possess a unique inverse. The element 2 is its own inverse and the elements 1 and 3 are the inverses of each other.

Since all the popular group axioms in this problem are satisfied, we answer this question in the positive:

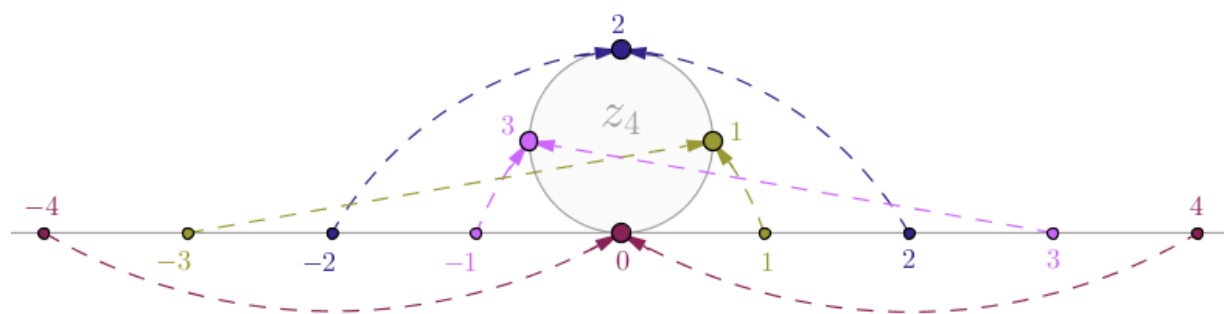
yes, the set G of least non-negative residues modulo 4 is a group under the addition modulo 4

That group is finite, is of order 4 and is commutative.

This group is also called *the additive group of integers modulo 4* and is symbolized in various ways as $(\mathbb{Z}/4\mathbb{Z}, +)$, or as $\mathbb{Z}/4\mathbb{Z}$, or as $\mathbb{Z}_4$ and so on.

We feel confident about adjusting the mechanical-geometrical model described in the previous discussion for the modulo 4 context by constructing a circle z_4 that:

- is tangent to the number line at the point 0 and
- whose circumference is equal to 4 units of length and
- that is divided into 4 equal parts, 1 unit of length each (Figure 11.124):



We also understand how the circle z_4 models all the equivalence classes modulo 4.

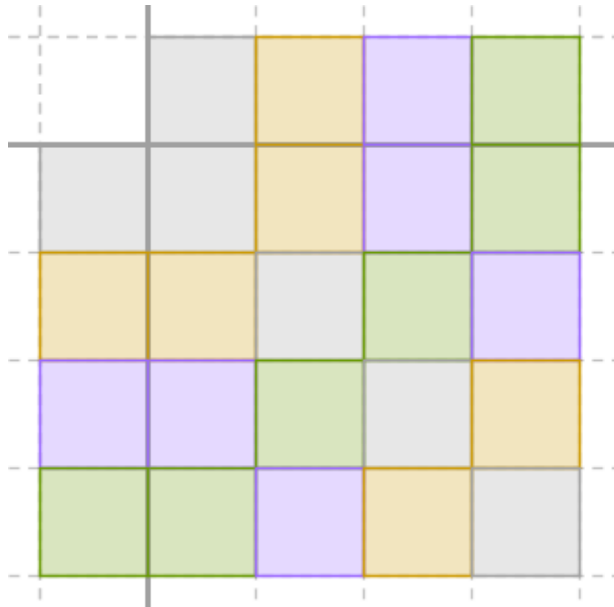
We understand how to perform the additions modulo 4 with z_4 and we understand how to find the inverses of the elements of the given set G under the operation of addition modulo 4 with the circle z_4 - the smallest number of clicks that takes a given element of G on z_4 into the point marked as 0 on z_4 and so on. $\square$

Have we already seen *an essentially the same* group on four symbols before? Well. In our very first, **We See The Light** discussion of the **Early Examples** chapter, we came across a duplex light switch group, a particular instance of the Klein four-group, that has the following multiplication table (Figure 6.9):

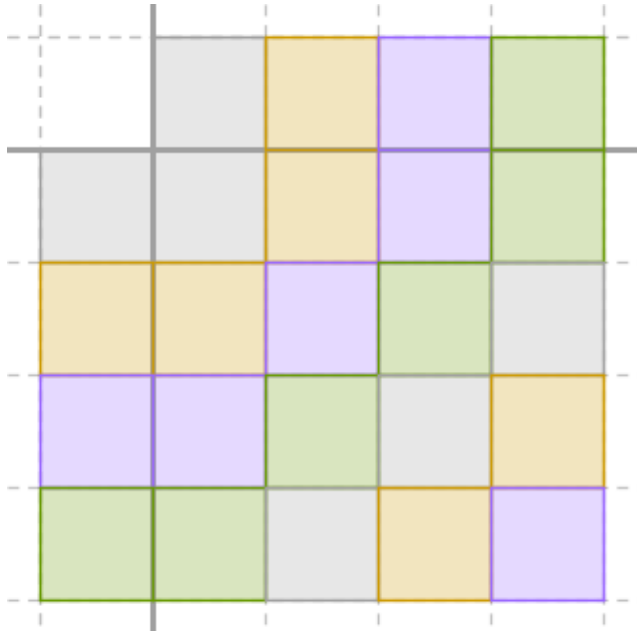
	e	l	r	b
e	e	l	r	b
l	l	e	b	r
r	r	b	e	l
b	b	r	l	e

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whose colorful trail or portrait, with all the symbols removed from it (Figure 6.13):



is quite different from the colorful trail generated by the above group of least non-negative residues modulo 4 under the addition modulo 4 (Figure 6.15):



Have we seen *that* multiplication table before?

Sure.

In one of our previous discussions we have constructed the multiplication table of the group of planar rotational congruence motions of a square (Figure 11.81):

$\circ$	$e_O^{0^\circ}$	$a_O^{90^\circ}$	$b_O^{180^\circ}$	$c_O^{270^\circ}$
$e_O^{0^\circ}$	e	a	b	c
$a_O^{90^\circ}$	a	b	c	e
$b_O^{180^\circ}$	b	c	e	a
$c_O^{270^\circ}$	c	e	a	b

This state of group-theoretic affairs should come as no surprise because the order of all of these groups, 4, is a number that is *composite*, $4 = 2 \cdot 2$, and is *not* prime.

In any case, we now see what is going on with the inverses of the groups of least non-negative residues modulo n under the operation of addition modulo n and our mechanical-geometrical model of such groups is especially helpful.

Under the *traditional* addition defined on the set of integers, the additive inverse of a fixed integer a is such an integer x that with a sums to zero:

$$a + x = x + a = 0$$

From such a definition of a traditional additive inverse we conclude that the inverse x is the integer a taken with the minus sign: $x = (-a) \in \mathbb{Z}$.

By analogy, as we extend the traditional addition defined on the set of integers into the addition modulo n , it seems only natural to define the additive inverse modulo n of a fixed integer a to be such an integer x that when summed with a is congruent to zero modulo n :

$$a + x \equiv x + a \equiv 0 \pmod{n}$$

In other words, given an integer a , which, as per DAT, is expandable multiplicatively with respect to n as $a = q_a n + r_a$, $0 \leq r_a < n$, its inverse x under the operation of addition modulo n , when added to a , must make that sum $a + x$ be divisible by n :

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$$a + x \equiv 0 \pmod{n}$$

But the integer x itself is expandable as per DAT with respect to n as follows:

$$x = q_x n + r_x, \quad 0 \leq r_x < n$$

Thus, in the sum $a + b$:

$$a + b = q_a n + r_a + q_x n + r_x = (q_a + q_x)n + r_a + r_x$$

the first summand, $(q_a + q_x)n$, being an integral multiple of n , is already divisible by n .

Hence, for that sum $a + b$ to be divisible by n , the sum of the remainders $(r_a + r_x)$ must be divisible by n , implying that that sum must be an integral multiple of n also:

$$r_a + r_x = qn$$

for an integer q .

However, the constraint (28) of DAT imposed on the possible values of the residues r_a and r_x tells us that that sum is bounded from above by $2n$:

$$0 \leq r_a + r_x < 2n$$

In other words, the mystery integer q cannot even touch the value of 2. Hence, the sum of residues $(r_a + r_x)$ must be equal to n and this line of number-theoretic reasoning motivates the arithmetic of addition modulo n that we carried out with our mechanical model of a circle touching a number line.

Officially, then, we see that it is possible to define the said operation of addition modulo n over a set $G = \{0, 1, 2, \dots, n-1\}$ of least non negative residues modulo n as follows:

$$a \circ b = k \quad \text{if} \quad (a + b) \equiv k \pmod{n}$$

for all $k = 0, 1, 2, \dots, n-1$.

Then, given a fixed element k of the set $G = \{0, 1, 2, \dots, n-1\}$ of least non negative residues modulo n , its inverse x must satisfy the following equations:

$$k + x \equiv x + k \equiv 0 \pmod{n}$$

But in our previous discussion we already observed that a positive integer n always divides itself:

$$n \equiv 0 \pmod{n}$$

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Juxtaposing the above two relations against each other, we conclude that given a fixed integer k from G , its inverse x must satisfy the following equation:

$$k + x = n$$

from where:

$$x = (n - k) \bmod n = (n \bmod n + (-k)) \bmod n = (-k) \bmod n$$

and the reason why the above last transition is valid is because n divides n and, thus, the result of the operation n modulo n is just 0.

In other words, using the mechanical-geometric model of the group Z_4 shown in Figure 11.124, for example, if $k = 1$ then the inverse element x of 1 is simply $(-1) \bmod 4$, which is 3. Indeed, $1 + 3 = 0$ modulo 4.

If $k = 0$ then the inverse element x of 0 is $(4-0) = 4 \bmod 4 = 0$ and so on.

In general, we already proved that given the set G of least non-negative residues modulo n :

$$G = 0, 1, 2, 3, \dots, n-1, \quad n > 1$$

and the operation of *addition modulo n* defined on the elements of G :

$$a \circ b = a +_n b = (a + b) \bmod n$$

such an operation is associative.

Moreover, such an operation is clearly *binary* because the constraint (28) from the Division Algorithm Theorem insures that the residue $k \bmod n$ of division of any integer k by n always honors the $0 \leq k \bmod n < n$ constraint and, thus, belongs back to the set G .

The above set G under the operation of *addition modulo n* possess a unique identity element, 0, and each element k of G under addition modulo n possess a unique inverse $x = (n - k) \bmod n = (-k) \bmod n$.

Therefore, the set of least non-negative residues modulo n under addition modulo n forms a finite group known as **an additive group of integers modulo n** .

The order of that group is n and, because essentially the operation of addition defined on the set of integers is commutative, that group is also commutative or Abelian.

Such a group is symbolized as Z/nZ , or as $(Z/nZ, +)$, or as Z_n and an inverse x of an element a of that group is called **the additive inverse of a modulo n** or **the additive inverse of a in Z/nZ** .

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Next, we want to investigate whether specifically the set of least non-negative residues modulo n under the operation of *multiplication modulo n* is a group.

To that end, we define that operation of multiplication modulo n for any two integers a and b in a natural and a predictable way:

$$a \circ b = a \cdot_n b = (a \cdot b) \bmod n$$

Such an operation is clearly binary because the constraint (28) from the Division Algorithm Theorem insures that the residue $k \bmod n$ of division of any integer k by n always honors the $0 \leq k \bmod n < n$ constraint. A product of any two integers $a \cdot b = k$ is, again, an integer and its residue modulo n , thus, belongs back to the set G .

Is the composition of two consecutive multiplications modulo n associative?

In other words, is the order in which the individual multiplications modulo n in such a composition are evaluated relevant?

Well, we know that the integer d can be expanded multiplicatively with respect to the modulus n as per DAT as follows:

$$d = q_d n + r_d, \quad 0 \leq r_d < n$$

Therefore, for the product of two integers $c \cdot d$ we have:

$$c \cdot d = c \cdot (q_d n + r_d) = c q_d n + c r_d$$

Applying the $\bmod n$ operation to the last equation, we see:

$$(c \cdot d) \bmod n = (c \cdot r_d) \bmod n = (c \cdot d \bmod n) \bmod n$$

that indeed, under multiplication modulo n it is safe and legal to replace any one factor in the product $c \cdot d$ with its individual residue modulo n , $d \bmod n$, for example:

$$(c \cdot d) \bmod n = (c \cdot d \bmod n) \bmod n$$

But then, for the said composition of two consecutive multiplications modulo n , we have, reading the last equality right-to-left and agreeing that the role of the integer d in that equality is played by the integer $a \cdot b$ below:

$$([(a \cdot b) \bmod n] \cdot c) \bmod n = [(a \cdot b) \cdot c] \bmod n$$

But the operation of multiplication defined on the set of integers **is** commutative and for any three integers a, b, c it is the case that:

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$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

Therefore:

$$[(a \cdot b) \cdot c] \bmod n = [a \cdot (b \cdot c)] \bmod n$$

and we now read the above equality left-to-right, agreeing that the role of the integer d in that equality is played by the integer $b \cdot c$ below:

$$[a \cdot (b \cdot c)] \bmod n = (a \cdot [(b \cdot c) \bmod n]) \bmod n$$

We, thus, arrive at a relation:

$$([(a \cdot b) \bmod n] \cdot c) \bmod n = (a \cdot [(b \cdot c) \bmod n]) \bmod n$$

that shows that the operation of multiplication modulo n for any integers is associative.

At this point, it pays to carry out a bunch of sample computations in order to test the associativity of the operation of multiplication modulo n in practice.

For example: for $a = 7, b = 6$ and $n = 4$ we have: $7 \bmod 4 = 3$ and $6 \bmod 4 = 2$.

Then: $(7 \cdot 6) \bmod 4 = 42 \bmod 4 = 2$. Check.

On the other hand: $(7 \cdot 2) \bmod 4 = 14 \bmod 4 = 2$. Check.

And: $(3 \cdot 6) \bmod 4 = 18 \bmod 4 = 2$. Check.

Finally: $(3 \cdot 2) \bmod 4 = 6 \bmod 4 = 2$ and so on.

Very good.

What about the identity element of such a potential group?

In order to answer that question, let us do the following

Exercise 11.59: does the set $G = \{0, 1\}$ of least non negative residues modulo 2 form a group under multiplication modulo 2?

Solution: well, we already understand that the operation of multiplication modulo 2 defined on the set G is both binary and associative.

Does that set G under the given operation possess a unique identity element?

Does every element of that set under that operation possess a unique inverse?

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In order to answer these questions, we construct the multiplication table of the given operation defined over the given set of numbers (Figure 11.125):

	*		
	mod 2	0	1
0		0	0
1		1	1

We now see that the element 1 of G is shaping up to be the unique identity of this maybe a group, because:

$$1 \cdot_2 1 = (1 \cdot 1) \bmod 2 = 1$$

and:

$$1 \cdot_2 0 = (1 \cdot 0) \bmod 2 = 0, \quad 0 \cdot_2 1 = (0 \cdot 1) \bmod 2 = 0$$

However, if 1 is the unique identity of this maybe-a-group, then it necessarily must be the case that:

$$0 \cdot_2 0 = (0 \cdot 0) \bmod 2 = e = 1$$

Is it the case, though?

Nope, it is not because $0 \cdot 0 = 0$, always. Therefore, a unique inverse of the element 0 of G under multiplication modulo 2 does not exist and it is also sometimes said that in that case the element 0 is not invertible.

Since the third group axiom is not satisfied, it follows that we answer the question of this exercise in the negative:

no, the set of least non-negative residues $G = \{0, 1\}$ modulo 2 is not a group under multiplication modulo 2

and we understand what was the root cause of such a failure - the absence of a unique inverse of the element 0 of G . $\square$

What are we to conclude from this exercise?

Well, we quickly realize that no matter the actual value of the modulus n , for as long as 0 is present in the candidate set G , such a set will never form a group under multiplication modulo n because such an element is not invertible under the given operation.

We do not even have to construct the multiplication tables of the operation multiplication modulo n over the set of least non negative residues modulo n for larger values of n in order to see it in our mind's eye that the 0-row and the 0-column in such a table will always be filled with nothing but zeros.

Even though we have not officially studied the structure of a multiplication table of a group yet, we constructed enough such tables in order to spot one interesting feature that is present in all such tables:

every row in a multiplication table of a group is some linear rearrangement of the table's column headers
and every column in a multiplication table of a group is some linear rearrangement of the table's row headers

With 0 as the entry in a table's row header and a table's column header the above feature of a multiplication table of a group is no longer there!

Conclusion: if there is any hope for the set of least residues modulo n to form a group under multiplication modulo n , then we must necessarily remove 0 from that set, turning it into the set of least **positive** residues modulo n :

$$G = \{1, 2, 3, \dots, n - 1\}$$

Is the removal of 0 from the set of least non negative residues modulo n enough for the consequent formation of a group under multiplication modulo n ?

We discover the answer to that question via the following three exercises.

Exercise 11.60: does the set $G = \{1\}$ of least positive residues modulo 2 form a group under multiplication modulo 2?

Justify your response.

Solution: cutting to the chase, we know that the given operation defined on the elements of the given set is both binary and associative.

What about the unique identity and inverses for all the elements of G ?

By adjusting the above multiplication table ever so slightly, we realize that yes, in this case not only we have a multiplication table of a given operation defined on the elements of a given set but we do have a multiplication table of a group.

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That is, even though we have only one row and only one column in our table, that table does represent a group (Figure 11.126):

	$*$ mod 2	1
1		1

indeed. The operation of multiplication modulo 2 defined on the set $G = \{1\}$ is binary and associative.

The unique identity element of that group is 1 and its unique inverse is 1. Therefore, since all the group axioms are satisfied, we answer the question in this exercise in the positive:

yes, the set $G = \{1\}$ of least positive residues modulo 2 forms a group under multiplication modulo 2

That group is finite, is of order 1 and is commutative.

In order to not spoil the fun of the upcoming miniature discovery, we will not give the complete official name of this particular group yet. An incomplete name of that group is: the multiplicative group of integers modulo 2, symbolized as $(\mathbb{Z}/2\mathbb{Z}, *)$, or as $(\mathbb{Z}/2\mathbb{Z}, \cdot)$, or as $(\mathbb{Z}/2\mathbb{Z}, \times)$, or as $(\mathbb{Z}/2\mathbb{Z})^*$ and so on.

In any case, did we just get lucky? Or is there a bigger something at play here?

Well, let us do some more investigations. $\square$

Exercise 11.61: does the set $G = \{1, 2\}$ of least positive residues modulo 3 form a group under multiplication modulo 3?

Solution: again, cutting to the chase, we know that the given operation defined on the elements of the given set is both binary and associative.

What about the unique identity and inverses for all the elements of G ?

Using the definition of the current operation, $a \circ b = a \cdot_3 b = (a \cdot b) \bmod 3$, we crunch the numbers at hand in a straightforward manner:

$$(1 \cdot 1) \bmod 3 = 1 \bmod 3 = 1$$

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$$(1 \cdot 2) \bmod 3 = 2 \bmod 3 = 2$$

$$(2 \cdot 1) \bmod 3 = 2 \bmod 3 = 2$$

$$(2 \cdot 2) \bmod 3 = 4 \bmod 3 = 1$$

and fold the results into the following multiplication table (Figure 11.127):

$\begin{smallmatrix} * \\ \hline \text{mod } 3 \end{smallmatrix}$	1	2
1	1	2
2	2	1

Hm.

Would you look at that?

It seems that we do have a group in this case after all.

The operation of multiplication modulo 3 defined on the set $G = \{1, 2\}$ is binary and associative.

The unique identity element of that group is 1 and its unique inverse is 1.

The unique inverse of the element 2 is that element itself, 2.

Therefore, since all the group axioms are satisfied, we answer the question in this exercise in the positive:

yes, the set $G = \{1, 2\}$ of least positive residues modulo 3 forms a group under multiplication modulo 3

That group is finite, is of order 2 and is commutative.

Again, an incomplete name of that group is: the multiplicative group of integers modulo 3, symbolized as $(\mathbb{Z}/3\mathbb{Z}, *)$, or as $(\mathbb{Z}/3\mathbb{Z}, \cdot)$, or as $(\mathbb{Z}/3\mathbb{Z}, \times)$, or as $(\mathbb{Z}/3\mathbb{Z})^*$ and so on.

Did we just get lucky once more? Or is there a bigger something at play here?

Well, let us do some more investigations. $\square$

Exercise 11.62: does the set $G = \{1, 2, 3\}$ of least positive residues modulo 4 form a group under multiplication modulo 4?

Solution: again, cutting to the chase, we know that the given operation defined on the elements of the given set is both binary and associative. Or is it?

What about the unique identity and inverses for all the elements of G ?

Using the definition of the current operation, $a \circ b = a \cdot_4 b = (a \cdot b) \bmod 4$, we crunch some numbers and construct the following multiplication table of that operation defined over the elements of the given set (Figure 11.128):

$\begin{smallmatrix} * \\ \text{mod } 4 \end{smallmatrix}$	1	2	3
1	1	2	3
2	2	0	2
3	3	2	1

The computations in this case are also straightforward.

The usual suspects that live in the topmost row and the leftmost column of the table cause no trouble:

$$(1 \cdot 1) \bmod 4 = 1 \bmod 4 = 1$$

$$(1 \cdot 2) \bmod 4 = 2 \bmod 4 = 2$$

$$(1 \cdot 3) \bmod 4 = 3 \bmod 4 = 3$$

and so on.

However, as we wade deeper into the above multiplication table, we discover a problem in the works:

$$2 \circ 2 = 2 \cdot_4 2 = (2 \cdot 2) \bmod 4 = 4 \bmod 4 = 0$$

which means that the product $2 \circ 2$ generates a value, 0, that takes us out of the parent set $G = \{1, 2\}$, since, clearly, 0 is not in that set - remember that in our very first experiment we already decided to expel zero from G .

And even though all the remaining products *compute*:

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$$(2 \cdot 3) \bmod 4 = 6 \bmod 4 = 2$$

$$(3 \cdot 2) \bmod 4 = 6 \bmod 4 = 2$$

$$(3 \cdot 3) \bmod 4 = 9 \bmod 4 = 1$$

that does not change the fact that the universal requirement of the definition of a binary operation is no longer satisfied: for it is not the case that for any two integers a, b from the set $G = \{1, 2, 3\}$ the result of the operation $(a \cdot b) \bmod 4$ belongs back to G .

Namely, when $a = b = 2$ then $(2 \cdot 2) \bmod 4 = 0$ and 0 does not belong to G .

Therefore, one of the very first requirements for a group operation to be binary is violated. Hence, we answer the question in this exercise in the negative:

no, the set $G = \{1, 2, 3\}$ of least positive residues modulo 4 does not form a group under multiplication modulo 4

and we understand what was the root cause of such a failure - the absence of a unique multiplicative inverse modulo 4 of the element 2 of G .

Hang on a second, though.

As a cancellation prize for our contestant, we see that if we throw this bad boy, the element 2, out of G , then we actually do obtain a group of order 2 (Figure 11.129):

$\begin{smallmatrix} * \\ \hline \text{mod } 4 \end{smallmatrix}$	1	3
1	1	3
3	3	1

that is isomorphic to the group that we constructed in the previous exercise.

Such a maneuver should make us go ... hm. $\square$

Food for thought. Do realize that in these exercises we are not looking for all the ways in which a set of integers or, to be more official and generic, a set of equivalence classes modulo n can be beaten into a submission of forming a group under multiplication modulo n .

In this series of exercises we are only and solely after a criterion or a bunch of criterions according to which this specific and particular set G :

$$G = \{1, 2, 3, \dots, n - 1\}$$

will definitely form a group under multiplication modulo n .

Once we hone in on such a criterion(s) and once we generate several beautiful proofs where such proofs are needed, we will be able to cast our group-theoretic net wider.

So far, our experiments with such sets produced the results that are mixed - sometimes such a set G does form a group under multiplication modulo n and sometimes - it does not.

What is going on here?

Clearly, the answer to that question pivots on the property(ies) of the modulus n .

At this point we will let our readers think on their own for the time being - maybe it is a good idea to keep on experimenting and maybe it is a good idea to construct a few more multiplication tables for the moduli $n = 5$ and $n = 6$, say.

We invite our readers to gather their own thoughts and bring them to our next section of the book.

Exercises 10: Modular Arithmetic, Part 3 Of 4

In this block of exercises, which showcase how seemingly different areas of mathematics are interconnected, we continue looking at the very beautiful multiplicative groups of integers modulo n .

In the exercises **11.59** though **11.62** we discovered that this particular set of positive integers G_n :

$$G_n = \{1, 2, 3, \dots, n - 1\}$$

for the values of $n = 2, 3$ and 4 sometimes does form a group under multiplication modulo n and sometimes it does not.

After constructing the respective multiplication tables for the operation of multiplication modulo 5 defined over the set $G_5 = \{1, 2, 3, 4\}$, which does form a finite Abelian group of order 4 (Figure 11.130):

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$\begin{smallmatrix} * \\ \text{mod } 5 \end{smallmatrix}$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

and the operation of multiplication modulo 6 defined over the set $G_6 = \{1, 2, 3, 4, 5\}$, which does not form a group (Figure 11.131):

$\begin{smallmatrix} * \\ \text{mod } 6 \end{smallmatrix}$	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

we develop the following hunch:

- every time when the modulus n is a number that is prime, $n = p$, the respective set $G_p = \{1, 2, 3, \dots, p - 1\}$ does form a group under multiplication modulo p and
- every time when the modulus n is a number that is not prime or is composite then the respective set $G_n = \{1, 2, \dots, n - 1\}$ does not form a group under multiplication modulo n but a diminished set $G_{n'}$, a set G_n from which all the elements not coprime with n are thrown out, does form a group under multiplication modulo n

Indeed.

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If we analyze the multiplication table of the set G_6 under multiplication modulo 6 shown above, then we will quickly see that not only the elements 2, 3 and 4 of that set do not possess the multiplicative inverses modulo 6, but the rows and the columns in which these elements live are not some linear rearrangements of the table's column and row headers, as is the case with the rest of the elements of G_6 , the elements 1 and 5.

Commentary: again, we see that the elements 1 and 5 alone do form a finite Abelian group of order 2 under multiplication modulo 6 if we throw the bad boys 2, 3 and 4 out of G_6 .

Instead of probing all the elements 1, 2, 3, 4 and 5 in some order, the contents of the row and the contents of the column of the element 3 lazily flip-flop through only 0 and 3.

This is bad news for our hero: first of all, 0 takes us out of the parent set G_6 and, second of all, where the hell did the elements 1, 2, 4 and 5 disappear?

Let us think.

What is one property that all these elements, 2, 3 and 4, have in common?

Correct, all these numbers are not coprime with their modulus, 6.

We remind our readers that a positive **prime** number is a positive integer $p > 1$ that has exactly and only two positive divisors, 1 and p itself.

The first few prime numbers are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53

A positive **composite** number is a positive integer $n > 1$ that has at least one positive divisor other than 1 and n itself.

The first few composite numbers are:

4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25

The positive integer 1 is neither prime nor composite and according to the Fundamental Theorem of Arithmetic, or the **FTA** for short, any whole number $n > 1$ is either prime or composite, in which case such a number is representable as a unique product of primes if the order of the said primes in the said product is ignored.

Two integers a and b are **coprime** or **relatively prime** if the largest positive integer that divides both of them, or the greatest common divisor $\gcd(a, b)$ of these integers, is 1 and not coprime or not relatively prime otherwise.

For example, the integers $16 = 2 \cdot 2 \cdot 2 \cdot 2$ and $21 = 3 \cdot 7$ are coprime because their largest common factor is 1, while the integers 6 and 30 are not coprime, since the largest integer that divides both 6 and 30 is $6 \neq 1$.

In any case, from the definition of a number that is composite it instantly follows that when the modulus n of a potential multiplicative group of integers modulo n is a number that is composite then there always exist (by definition) two non trivial positive integers x and y such that neither of them is the (positive) unity:

$$x \neq 1 \text{ and } y \neq 1$$

and each of which is strictly less than n :

$$x < n \text{ and } y < n$$

which multiply out exactly to (a composite) number n :

$$x \cdot y = n$$

This is bad group-theoretic news because under the current operation of multiplication modulo n that fact, just as instantly, implies that when the modulus n is composite then the above positive integers x and y will always be in the parent set $G_n = \{1, 2, 3, \dots, n-1\}$ and that the remainder of division of n by n , generated by the product $x \cdot y$, will be zero:

$$x \circ y = x \cdot_n y = (x \cdot y) \bmod n = n \bmod n = 0$$

which takes us out of the parent set $G_n = \{1, 2, 3, \dots, n-1\}$, which does not contain 0, and that fact alone crushes any hopes for the set G_n to form a group under multiplication modulo n .

At this point we see that we have a decision to make: we either keep the set $G_p = \{1, 2, \dots, p-1\}$ as is and demand that the modulus n must be a prime number p or we alter our set G_n by applying a certain subtractive action to it and removing certain elements from it.

In the latter case - exactly which elements are we to remove from G_n and why?

Ah.

We already proved that the operation of multiplication modulo n defined on the set of integers is certainly associative, regardless of whether n is prime or composite.

We also see that the element 1 of the set G_n will most certainly be the unique identity element of a potential multiplicative group modulo n .

The only two tactical questions that remain unresolved so far are:

- when is the candidate operation of multiplication modulo n is binary and
- when do the multiplicative inverses modulo n exist?

We, first, address the question of whether the candidate operation of multiplication modulo n defined on the elements of the set $G_n = \{1, 2, \dots, n - 1\}$ is binary or not.

Is It Binary and When Is It Binary?

Let us assume for a moment that the modulus n of the operation multiplication modulo n is a prime number p .

Then, the candidate set G_p will have the following $(p - 1)$ positive integers in it:

$$G_p = \{1, 2, 3, \dots, p - 1\}$$

From the above definition of a prime number it follows that in that case each and every element of that set will be coprime with p .

In other words, the largest positive integer that divides any element a of the set G_p and the prime p is 1.

Moreover, because no element of the set G_p has the prime p as its factor, **the product** of any two such elements, $a \cdot b$, will also never have p as **its** factor.

Why is that?

Because prime numbers in general and the above Fundamental Theorem of Arithmetic are defined and framed in terms of a product and, specifically, not a sum of integers.

Recall that in the above multiplicative group of integers modulo 5 its two elements 2 and 3 very well **sum** to $5 = 2 + 3$ and such a sum is certainly divisible by 5, since it has 5 as its factor.

However, **the product** of these elements, $2 \cdot 3 = 6$, each of which happens to be a prime number, does not have the prime factor of 5 in it and, thus, such a product will never be divisible by 5.

As we climb to larger and larger primes, it may very well happen that both elements in the product $a \cdot b$ are composite, as is the case in, say, the multiplicative group of integers modulo 7, where the product $4 \cdot 6 = 24$ is made up of composite numbers, but that does not change the fact that the result of any

such product will still not have p , or 7 in this particular case, as one of its factors because the Fundamental Theorem of Arithmetic guarantees us that any composite number is made up of prime factors in a unique way, if the order of these factors is ignored.

Therefore, if the integer a does not have p as its factor and if the integer b does not have p as its factor then the product $a \cdot b$ will not have p as its factor and, therefore, such a product $a \cdot b$ will never be divisible by p .

For a more formal treatment of this idea the so-called Euclid's Lemma, which states that if a prime p divides the product $a \cdot b$ of two integers a and b then p divides at least one of those integers a and b , can be used as follows.

We remember that a direct statement, IF P THEN Q, and its contrapositive, IF NOT Q THEN NOT P, are logically equivalent.

A contrapositive of a given statement is constructed by swapping and negating the premise and the conclusion of that statement.

Hence, in this case, the contrapositive of the statement of Euclid's Lemma becomes: if p does not divide the integer a and p does not divide the integer b then p does not divide the product of those integers $a \cdot b$.

In any case, the remainder r of the division of such a product by a prime p will never be zero.

But by the constraint (28) of DAT, such a remainder is still guaranteed to be strictly less than p .

The last two observations lead us to conclude that the operation of multiplication modulo a prime p is always well-defined:

$$0 < (ab) \bmod p < p$$

and generates a result that always belongs exactly to the set $G_p = \{1, 2, 3, \dots, p-1\}$.

Officially, for any two integers a, b from the set $G_p = \{1, 2, 3, \dots, p-1\}$ the result c of the operation $\circ$ such that $c = a \circ b = a \cdot_p b = (ab) \bmod p$ belongs back to the set G_p .

Hence, by definition, for a prime number p the operation of multiplication modulo p defined over the set $G_p = \{1, 2, 3, \dots, p-1\}$ is binary.

Multiplicative Inverses Modulo n

We now have only one question to resolve: exactly what is this animal who wants to be called the multiplicative inverse modulo n of some fixed integer a and exactly when such an integer does possess the multiplicative inverse modulo n ?

It turns out that it is possible to answer the above question by using the machinery of what we refer to as the elementary theory of numbers or by using the machinery of what we refer to as the theory of groups.

Because this book is dedicated primarily to the theory of groups only, we will discuss the number-theoretic criterion for the existence of the multiplicative inverses modulo n without delving deep into the proofs of the required theorems, but we will generate two beautiful group-theoretic proofs of the fact that the multiplicative inverses modulo p of the set $G_p = \{1, 2, 3, \dots, p-1\}$ exist whenever p is prime.

Moreover, the upcoming justifications of the existence of multiplicative inverses modulo n by the means of both the theory of numbers and the theory of groups highlight an interesting and perhaps an unusual for a side observer feature of the grown-up mathematics - both such proofs only demonstrate the existence of such inverses and provide no concrete algorithmic recipe for their recovery given the specific instances of the integer a and the modulus n .

In order to facilitate our understanding of multiplicative inverses modulo n at a deeper level, we shall also look at a purely computational means of their recovery - the means that hail from the vaguely defined mathematical discipline of the theory of numbers.

Definition

Recall that under the operation of addition modulo n we defined the additive inverse modulo n , the integer x , as follows:

$$a + x = x + a \equiv 0 \pmod{n}$$

By analogy with a traditional addition, under which such a sum must be simply equal to zero, under addition modulo n such a sum must be equal to zero modulo n and translated into the language of congruences, such a definition tells us that the sum of a fixed integer and its additive inverse modulo n must be congruent to zero modulo n .

By a similar line of reasoning, if under a traditional multiplication defined on the set of, say, rational numbers, the product of a fixed rational number a with its multiplicative inverse x must be equal to 1, as in $a \cdot x = x \cdot a = 1$, then, by definition, the multiplicative inverse modulo n of a fixed integer a must be an integer x that, when multiplied with a on its either side, is congruent to 1 modulo n :

$$x \cdot a \equiv a \cdot x \equiv 1 \pmod{n}$$

Such a definition, basically, claims that the integers ax and 1 must be congruent modulo n .

Existence, A Number-Theoretic Proof

From the definition of congruence of two integers modulo n due to Carl Gauss, we remember that the integers ax and 1 are congruent modulo n if that integer n divides the difference of these integers, $ax - 1$.

Further, from the definition of integer division, we remember that symbolically the phrase “the integer n divides the integer $(ax - 1)$ ” is rendered as follows:

$$ax - 1 = kn, \quad k \in \mathbb{Z}$$

Adding 1 to both sides of the above equation, we obtain:

$$ax = kn + 1$$

and adding the integer $(-kn)$ to both sides of the last equation, we arrive at a result:

$$ax - kn = 1 \tag{35}$$

that can be interpreted as follows:

a fixed integer a has a multiplicative inverse modulo n , designated as x , if and only if 1 can be represented as a linear combination of the said fixed integer a and the said modulus n

Commentary: in general, the equation of the form $ax + by = c$ is called “a linear Diophantine equation in two unknowns”, x and y , where the integers a , b and c are taken to be known and fixed ahead of time.

The bridge that connects what we refer to as “the theory of groups” to what we refer to as “the theory of numbers” is the following theorem from the elementary theory of numbers.

Theorem: two integers a and b , not both zero, are coprime if and only if there exist integers x and y such that $1 = ax + by$.

In general, if the integers a and b are coprime then the very existence of the above integers x and y is guaranteed by the so-called **Bezout’s Lemma**, which states that: for any two integers a and b , not both zero, there exist integers x and y such that $\gcd(a, b) = ax + by$.

It looks like in our case we will be demanding that the integers a and b are coprime.

Hence, their greatest common divisor is the positive unity, $\gcd(a, b) = 1$.

Therefore, there exist integers x and y such that $1 = ax + by$.

Speaking of mechanical-geometrical models of mathematical concepts: the above Bezout's Lemma can be used in order to create the following mechanized toy model/puzzle.

A grasshopper, sitting at the origin, 0, of the whole numbers line, will jump a integral units of distance if (s)he pushes with her/his, say, left legs and (s)he will jump b integral units of distance if (s)he pushes with her/his right legs.

Question: which points on the whole numbers line can this grasshopper possibly visit if (s)he jumps in the fashion described above?

Answer: by jumping in the fashion described above, this grasshopper can clearly visit the points on the whole numbers line whose coordinates are the integral multiples of $ax + by$, where the integers x and y , which simply tell us how many a -jumps and how many b -jumps our grasshopper has respectively made, can be negative or positive.

If, for example, the integer x is negative then our grasshopper made x jumps a units of distance long to the left, in the negative direction that is, and if the integer x is positive then our grasshopper made x jumps a units of distance long to the right, in the positive direction that is.

>Likewise, if the integer y is negative then our grasshopper made y jumps b units of distance long to the left, in the negative direction that is, and if the integer y is positive then our grasshopper made y jumps b units of distance long to the right, in the positive direction that is.

Since the total distance $ax + by$ covered by our grasshopper is none other than the greatest common divisor of the integers a and b , it follows that that grasshopper can only visit the points on the whole numbers line that are the integral multiples of $\gcd(a, b)$.

Consequently, if the integers a and b are coprime then our grasshopper can visit any (integral) point on the whole numbers line: $0, \pm 1, \pm 2, \pm 3$ and so on.

Because we have to keep the size of this discussion manageable, we will omit the complete proof of the above theorem but that does not change the fact that that theorem provides a concrete tactical criterion for the existence of the multiplicative inverses modulo n .

Namely:

the multiplicative inverse modulo n of a fixed integer a exists precisely when the integers a and n are coprime or when $\gcd(a, n) = 1$ and it does not exist if the integers a and n are not coprime or when $\gcd(a, n) \neq 1$

In general, the equation of the form $a \cdot x \equiv b \pmod{n}$, for some fixed integer b , is called a **linear congruence**.

In the elementary theory of numbers there exists the following theorem that spells out the condition when such a congruence has a solution.

Theorem: the linear congruence $a \cdot x \equiv b \pmod{n}$ has a solution if and only if the greatest common divisor $\gcd(a, n)$ of the integers a and n divides the integer b . If $\gcd(a, n)$ divides b then such a congruence has exactly $\gcd(a, n)$ mutually congruent solutions.

Again, in order to keep the size of this discussion manageable, we will omit the proof of this theorem.

Still, because in our case we demand that the integers a and n are coprime, their greatest common divisor is the positive unity: $\gcd(a, n) = 1$.

Therefore, by the above theorem, **our** linear congruence $a \cdot x \equiv 1 \pmod{n}$ has exactly one solution!

In other words: if the integers a and n are coprime then the linear congruence $a \cdot x \equiv 1 \pmod{n}$ has a unique solution.

This is exactly what we need in the theory of groups, **the unique** multiplicative inverses modulo n .

But we must be frank and honest with our readers. In the above line of reasoning we did not show the complete proofs of all the intermediate theorems from the theory of numbers that lead us up to the above conclusion.

Since this is not a space about the theory of numbers but is a space about the theory of groups, in the next discussion we will generate the complete and our own group-theoretic proofs of the existence of the modular inverses modulo p for prime numbers p .

The Computational Mechanics

As we noted earlier, neither the number-theoretic nor the group-theoretic proofs of the existence of multiplicative inverses modulo n provide a concrete algorithmic recipe for the recovery of the multiplicative inverses modulo n given the specific instances of the integers a and n .

We conclude this discussion with a detailed look at exactly the procedural recipe that allows us to find the multiplicative inverse modulo n of a fixed integer a coprime with n , regardless of whether the integer n itself is prime or composite.

Such a computational recipe amounts to rolling back a finite number of equalities generated by a well-understood and a well-studied algorithm known as the “Euclidean GCD Algorithm”, where the abbreviation GCD stands for the “Greatest Common Divisor”.

Even though there exists historical evidence that the algorithm for the recovery of the greatest common divisor of two integers predates Euclid of Alexandria, these days that algorithm carries the name of the ancient Greek mathematician and is known as the Euclidean Algorithm.

However, in order to avoid any confusion with the Division Algorithm, let us agree, for the time being, to refer to the above Euclidean Algorithm as “the Euclidean GCD Algorithm”.

It so happens that unlike the impostor Division Algorithm Theorem, the Euclidean GCD Algorithm is exactly what its name advertises, an absolutely wonderful, elegant and beautiful description of a finite number of steps that halt or terminate with the correct result, the greatest common divisor of two integers (not both zero).

The Euclidean GCD Algorithm is based on or is motivated by the following lemma.

Lemma: if the integers a, b are such that $a = qb + r$ with $0 \leq r < |b|$, then $\gcd(a, b) = \gcd(b, r)$.

For the sake of brevity we shall omit the proof of that lemma that captures or explains the reductive and the iterative essence of the Euclidean GCD Algorithm.

What matters for our group-theoretic purposes here is that tactically, if we roll the iterative equalities of the Euclidean GCD Algorithm in reverse order then in a very straightforward and easy-to-comprehend but tedious-to-execute fashion we recover exactly the integers x and k in the Diophantine equation shown in (35) and by so doing we compute the sought-after multiplicative inverse modulo n of the integer a : we do not really care about the value of the integer k but the value of the integer x is exactly the sought-after multiplicative inverse modulo n of a (possibly shifted over by enough copies of n if x comes out to be negative).

Let us see how that may work in practice for some toy numbers hand-picked for demonstration purposes.

The Inverse Modulo p , A Sample Computation

The reason why earlier we claimed that the Euclidean GCD Algorithm is reductive is because its very first step in the recovery of the greatest common divisor of two integers a and b , motivated by the above Lemma, amounts to the multiplicative decomposition of the integer a with respect to the integer b the DAT style:

$$a = q \cdot b + r, \quad 0 \leq r < |b|$$

In such a decomposition, we remember, the remainder r is guaranteed to be strictly smaller than $|b|$.

But because $\gcd(a, b)$ is equal to $\gcd(b, r)$, we now managed to make a new but an equivalent problem that is smaller in size. In other words, we reduced the size of the current problem.

If the above remainder r happens to be equal to zero right away, if $a = qb$ that is, then the integer b clearly divides the integer a , that integer b is the sought-after greatest common divisor of a and b and the algorithm halts with the correct result in a finite number of steps, one.

If the remainder r of the above expansion is not zero then, and this is the ingenious part of this algorithm, we simply apply the same Lemma to the new smaller numbers b and r by:

- promoting the integer b to become the new integer a and by
- promoting the integer r to become the new integer b

In other words, we iterate our original decomposition $a = qb + r$ by substituting:

$$a \leftarrow b \text{ and } b \leftarrow r$$

and by multiplicatively decomposing the new integer a with respect to the new and smaller integer b the DAT-style:

$$a = qb + r, 0 \leq r < b$$

We, thus, keep iterating the above decompose-and-promote steps until at a certain stage the remainder r vanishes.

The last non-zero remainder in the above decompose-and-promote chain is the sought-after greatest common divisor of the original integers a and b .

Why is the Euclidean GCD Algorithm guaranteed to halt in a finite number of steps with the correct result?

The Euclidean GCD Algorithm is guaranteed to halt in a finite number of steps with the correct result because the integer b is finite and because the remainder r in the $a = qb + r$ decomposition cannot step into the negative integers territory, in which case the Euclidean GCD Algorithm would simply diverge and never halt.

Since the value of r diminishes with each iteration of the Euclidean GCD Algorithm, it follows that that remainder r will turn into 0 in a finite number of iterations and the algorithm will halt, in at most b iterations but usually that happens much faster.

The French mathematician Gabriel Lamé (1795–1870) showed that the actual number of steps required for the Euclidean GCD Algorithm to halt is at most five times the number of digits in the smaller integer.

Let us see how this reductive and iterative process works in practice when we want to compute the greatest common divisor of the integers $a = 53$ and $b = 8$.

Because $\gcd(|a|, |b|) = \gcd(a, b)$, there is no harm in assuming that $a \geq b > 0$.

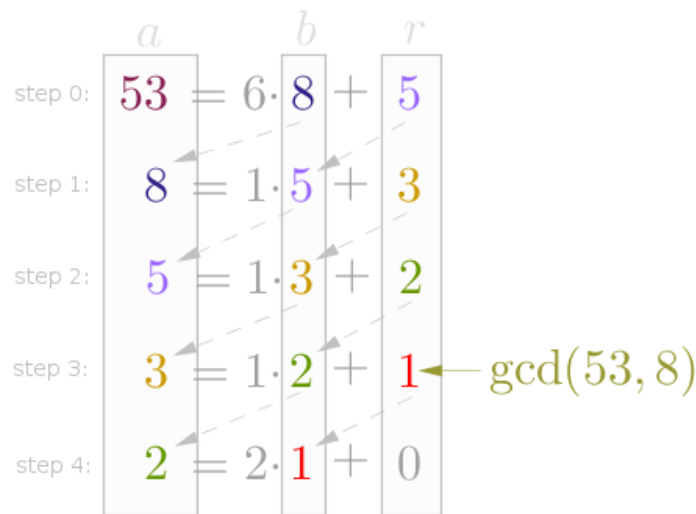
Thus, in order to find the greatest common divisor of the integers 53 and 8, we, first, multiplicatively decompose the integer 53 with respect to the integer 8 as follows:

$$53 = 6 \cdot 8 + 5$$

Because the remainder $r = 5$ in this decomposition is not zero, we promote the integer $8 = b$ to play the role of a and we promote the integer $5 = r$ to play the role of b , as we multiplicatively decompose 8 with respect to 5 as follows:

$$8 = 1 \cdot 5 + 3$$

and so on until we arrive at the decomposition in which the remainder of such decompositions is zero (Figure 11.132):



at which point the Euclidean GCD Algorithm halts with the correct result of 1 in no more than five steps, as predicted by the above theorem due to Lamer:

$$\gcd(53, 8) = 1$$

and which is the very last non-zero remainder of the multiplicative decomposition of the integer $a = 3$ with respect to the integer $b = 2$.

A Little Extra For Experts

With just one line of `printf()` effort, see the line 17 below, the C code below can be used in order to compute of the greatest common divisor of two integers.

Feel free to modify that code as you see fit:

```
#include <stdio.h>
#include <stdlib.h>

extern int
main( int argc, char *argv[] )
{
    int  r;
    int  a = atoi( argv[ 1 ] );
    int  A = a;
    int  b = atoi( argv[ 2 ] );
    int  B = b;

    while ( b != 0 )
    {
        r = a % b; /* a = qb + r */
        printf( "%d = %d*d + %d\n", a, (a-r)/b, b, r );
        a = b; /* Turn old 'b' into new 'a' */
        b = r; /* Turn old 'r' into new 'b' */
    }

    printf( "gcd(%d, %d) = %d\n", A, B, a );

    return 0;
}
```

Save the above code in the `gcd.c` file and compile it as follows:

```
gcc -g -o gcd gcd.c
```

Have fun witnessing how the GCD of two integers travels from one decomposition to another by running the above program as follows:

```
./gcd 53 8
53 = 6*8 + 5
8 = 1*5 + 3
5 = 1*3 + 2
3 = 1*2 + 1
2 = 2*1 + 0
gcd(53, 8) = 1
```

and so on.

Now that we have generated a set of equalities shown in Figure 11.132 above, we can roll them in reverse order and compute the multiplicative inverse modulo 53 of the fixed integer 8 as follows.

Starting with the last equality that generated the correct GCD of two integers:

$$3 = 1 \cdot 2 + 1$$

express that GCD in terms of the remaining members of that equality:

$$1 = 3 - 1 \cdot 2$$

Note that the last equality above already contains the blueprint of the Diophantine equation shown in (35):

$$1 = ax - kn$$

Then, working in reverse order with respect to the order of the equalities generated during the computation of $\gcd(53, 8)$, simply keep substituting the remainders from these equalities back into the equality $1 = 3 - 1 \cdot 2$ until the original equality $53 = 6 \cdot 8 + 5$ is reached.

Namely, from the above “step 2” equality $5 = 1 \cdot 3 + 2$, express the remainder 2 as follows:

$$2 = 5 - 1 \cdot 3$$

and substitute that expression for 2 back into $1 = 3 - 1 \cdot 2$:

$$1 = 3 - (5 - 1 \cdot 3) = -5 + 2 \cdot 3$$

Doing so, introduces “the previous” remainder, 3, into the works: floss, rinse, repeat.

From the above “step 1” equality $8 = 1 \cdot 5 + 3$, express that remainder 3 as follows:

$$3 = 8 - 1 \cdot 5$$

and substitute that expression for 3 back into our equation modeled after (35):

$$1 = -5 + 2 \cdot 3 = -5 + 2 \cdot (8 - 1 \cdot 5) = 2 \cdot 8 - 3 \cdot 5$$

and so on. Keep repeating these steps until you arrive at an equality that contains the original integers 53 and 8. Such an equality:

$$1 = 8 \cdot 20 - 3 \cdot 53 = ax - kn$$

reveals exactly the multiplicative inverse modulo 53 of the fixed integer 8 to be 20:

$$8 \cdot 20 \equiv 1 \pmod{53}$$

It pays to capture all these steps graphically, in a table of sort (Figure 11.133):

$\gcd(53, 8)$

from step 3: $1 = 3 - 1 \cdot 2$ from step 2: $2 = 5 - 1 \cdot 3$

$= 3 - 1 \cdot (5 - 1 \cdot 3)$

$= -5 + 2 \cdot 3$ from step 1: $3 = 8 - 1 \cdot 5$

$= -5 + 2 \cdot (8 - 1 \cdot 5)$

$= 2 \cdot 8 - 3 \cdot 5$ from step 0: $5 = 53 - 6 \cdot 8$

$= 2 \cdot 8 - 3 \cdot (53 - 6 \cdot 8)$

$= 20 \cdot 8 - 3 \cdot 53$

Thus, the next time you will be working with the multiplicative group of integers modulo 53, you will already know the multiplicative inverse modulo 53 of the element 8 of that group, which is 20.

It should be noted that the above chain of equalities generated by the Euclidean GCD Algorithm can grow to a considerable length.

On the one hand, **the length** of such a chain has nothing to do with the perceived conceptual complexity of the overall computation because the production of each such equality is very simple and is very easy to comprehend.

On the other hand, the manual computations involved in the above process are tedious and error-prone.

Once we understand **why** and **when** such a computation produces a correct multiplicative inverse modulo n , we can put together a little bit of C code in order to automate the process:

```
#include <stdio.h>
#include <stdlib.h>

/*
  A recursive implementation of the
  extended Euclidean GCD algorithm
*/
extern int
gcdX( int a, int b, int* x, int* y )
{
    /* Halt when the remainder vanishes */
    if ( a == 0 )
    {
        *x = 0;
        *y = 1;
        return b;
    }

    int    x1;
    int    y1;
    int    gcd = gcdX( b % a, a, &x1, &y1 );

    *x = y1 - (b/a) * x1;
    *y = x1;

    return gcd;
}

extern int
mInvModN( int a, int n )
{
    int    x;
    int    y;
    int    minmodn;
    int    gcd = gcdX( a, n, &x, &y );

    if ( gcd != 1 )
    {
        printf( "The integers %d ad %d are not coprime:\n", n, a );
        printf( "\tno multiplicative inverse modulo %d for %d exists\
n", n, a );
    }
}
```

```

    }
    else
    {
        printf( "1 = %d * (%d) + (%d) * %d\n", a, x, y, n );
        minmodn = ( x % n + n ) % n;
        if ( x < 0 )
        {
            printf( "Note that (%d) mod %d = ((%d) %% %d + %d) mod %d\n",
                x, n, x, n, n, minmodn );
            printf( "(%d) %% %d = %d in C. Thus\n", x, n, x % n );
        }
        printf( "The multiplicative inverse modulo %d of %d is %d\n",
            n, a, minmodn );
    }
}

extern int
main( int argc, char *argv[] )
{
    int    a = atoi( argv[ 1 ] );
    int    n = atoi( argv[ 2 ] );

    mInvModN( a, n );

    return 0;
}

```

Save the above code in, say, the `mimodn.c` file and compile it like so:

```
gcc -g -o mimodn mimodn.c
```

and execute it like so:

```

./mimodn 8 53
1 = 8 * (20) + (-3) * 53
The multiplicative inverse modulo 53 of 8 is 20

```

Note that the order in which the arguments to the `mimodn` program are given is very important:

- the first argument of that program is the fixed integer, a , whose multiplicative inverse modulo n we seek
- the second argument of that program is the modulus, n

If the integers a and n are not coprime then the program will fail with an informative message:

```
./mimodn 8 52
The integers 52 and 8 are not coprime:
no multiplicative inverse modulo 52 for 8 exists
```

It is perfectly fine to swap the order of the program's arguments:

```
./mimodn 53 8
1 = 53 * (-3) + (20) * 8
Note that (-3) mod 8 = ((-3) % 8 + 8) mod 8 = 5
(-3) % 8 = -3 in C. Thus
The multiplicative inverse modulo 8 of 53 is 5
```

but then it should be understood exactly what is being calculated by the that program.

Observe that when the inverse x comes out to be negative then it is never a problem because by adding to a negative integer enough moduli n , such an inverse can always be brought into the non-negative integers territory.

In the next discussion we will present two group-theoretic proofs of the existence of multiplicative inverses modulo p for prime p of the integers from the set $G_p = \{1, 2, 3, \dots, p-1\}$.

Exercises 11: Modular Arithmetic, Part 4 Of 4

With this discussion we conclude our introductory investigation of the multiplicative groups of integers modulo n by inviting our readers to generate at least two different, elementary, proofs of the existence of multiplicative inverses modulo p of all the positive integers $1, 2, 3, \dots, p-1$ under multiplication modulo p when p is prime.

In the previous discussion we took a cursory look at the proof of the existence of such inverses that is based purely on the machinery of the theory of numbers and we also constructed a mechanized tool, `mimodn`, that can compute such inverses programmatically, on demand.

Now, however, is the time to operate solely or mostly within the (vague) boundaries of the theory of groups.

A hint for the diligent students of mathematics who heed the advice of the “What Makes It Perfect” chapter: a contradiction (for the first proof) and the Pigeonhole Principle (for the second proof).

Exercise 11.63: show in at least two different group-theoretic ways that every element of the set $G_p = \{1, 2, 3, \dots, p-1\}$ possesses a unique multiplicative inverse modulo p under the operation of multiplication modulo p for p prime.

Proof 1: uses the technique of “a contradiction” as follows.

Consider the $(p-1)$ positive integers of the given set G_p :

$$G_p = \{1, 2, 3, \dots, p-1\} \quad (36)$$

Let a be an integer from (36) chosen or fixed at will.

What is the exact value of a ?

We do not know. That value can be **any** integer from the set (36).

In order to prove that that integer a possess a unique left-sided multiplicative inverse modulo p under multiplication modulo p for p prime we construct exactly $(p-1)$ left-sided products of the above integers from (36) with a :

$$1 \circ a, 2 \circ a, 3 \circ a, \dots, (p-1) \circ a \quad (37)$$

where by the product $\circ$ we, of course, mean the current operation of multiplication modulo p :

$$(1 \cdot a) \bmod p, (2 \cdot a) \bmod p, (3 \cdot a) \bmod p, \dots, ((p-1) \cdot a) \bmod p \quad (38)$$

that simply computes and returns the remainder upon division by p of the resultant integer $k \cdot a$ for all $k = 1, 2, 3, \dots, p-1$.

Since all the integers in (36), including the fixed integer a , are coprime with p , from Euclid’s Lemma it follows that none of the above abstract products will be equal to 0, since p does not divide any of the said integers $k \cdot a$ for $k = 1, 2, 3, \dots, p-1$.

Commentary: recall the tactical maneuver from our previous discussion. A direct statement and its contrapositive are logically equivalent. Hence, the traditional or the direct statement of Euclid’s Lemma, if a prime p divides the product of two integers $a \cdot b$ then p divides a or b or both, has the following logical equivalent: if a prime p does not divide the integer a and p does not divide the integer b then p does not divide the product of these integers $a \cdot b$.

But by the constraint (28) of DAT, all the remainders shown in (38) are strictly less than p .

Hence, for all the integers $k = 1, 2, 3, \dots, p-1$ we have:

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$$0 < (k \cdot a) \bmod p < p$$

Moreover, we claim that all of the abstract products shown in (37) or the concrete products shown in (38) are distinct and we use the technique of a proof by contradiction* to show that.

To that end, assume that some of the products shown in (37) or (38) are not distinct.

That means that there exist at least two distinct integers j and l :

$$0 < j < p, \text{ and } 0 < l < p$$

for which the above remainders $(j \cdot a) \bmod p$ and $(l \cdot a) \bmod p$ from (36) are the same.

Without loss of generality, it is safe to assume that, being distinct, the integers j and l are sorted as follows: $j > l$ (if it is not the case that $j > l$ then it is always possible to swap the values of these variables so that $j > l$).

Then, for the difference $j - l$ we have:

$$0 < j - l < p$$

If the integers $(j \cdot a) \bmod p$ and $(l \cdot a) \bmod p$ are not distinct:

$$(j \cdot a) \bmod p = (l \cdot a) \bmod p$$

then it means that the prime modulus p divides the difference $j \cdot a - l \cdot a$, implying that:

$$j \cdot a - l \cdot a = mp$$

for some integer m .

But since the fixed integer a in the above difference is one and the same, it can be factored out:

$$(j - l) \cdot a = mp$$

and in that case we managed to reduce “a difference” of two integers to “a product” of two integers.

But to a product of two integers we can apply our favorite, by now, Euclid’s Lemma:

if a prime p divides the product $(j - l) \cdot a$ of two integers $(j - l)$ and a then p divides $(j - l)$ or a or both

In our case that means that the prime p must divide the strictly smaller than p positive integer a , which is absurd, or the prime p must divide the strictly smaller than p positive integer $(j - l)$, which is also absurd.

Thus, by assuming that the integers in (37) (or in (38)) are not all distinct, we arrive at a contradiction. Therefore, such an assumption is false and indeed, all the products in (37) (or in (38)) are, in fact, distinct.

Hence, since all such products belong back to the set G_p , it follows that exactly one such product must be equal to 1, which officially means that among the integers of the set G_p there exists a unique integer x such that:

$$(x \cdot a) \bmod p = 1$$

which is another way of saying that:

$$xa \equiv 1 \pmod{p}$$

the integer x is the unique left-sided multiplicative inverse modulo p of the chosen integer a and that the notation of a^{-1} for x makes sense:

$$a^{-1} \circ a = 1 = e$$

Since we did not make any special assumptions in the above line of reasoning as it pertains to the **left**-sided inverse x of a and since the traditional multiplication of integers is commutative, by the same argument that integer a possesses the same unique **right**-sided inverse x :

$$(a \cdot x) \bmod p = 1$$

since if p divides the product $a \cdot (j - l)$ then it must divide either a or $(j - l)$ or both and so on.

Altogether, because the sample integer a was chosen at will, this proves that every element of the set $G_p = \{1, 2, 3, \dots, p - 1\}$ possesses a unique multiplicative inverse modulo p under the operation of multiplication modulo p for p prime. $\square$

In the above proof, what exactly is this mysterious inverse x of a ?

We do not know.

But such is the magic of the grown-up mathematics that is filled with these so-called **existence theorems** or theorems that do prove the existence of certain mathematical objects that have certain properties but do not provide any concrete recipe for the recovery or the construction of such objects.

Proof 2: pivots on the so -called Pigeonhole Principle, which in its simplest formulation states that: if $(n + 1)$ balls are to be distributed across n boxes then at least one box will have at least two balls in it, as follows.

In the given set of $(p - 1)$ positive integers G_p :

$$G_p = \{1, 2, 3, \dots, p - 1\}$$

put the integer 1 aside, seeing that $1 \cdot 1 \equiv 1 \pmod{p}$, and from the remaining $(p - 2)$ integers choose an integer a at will.

Constructing exactly p traditional self-products or powers of a as follows:

$$a^1, a^2, a^3, \dots, a^{p-1}, a^p \tag{39}$$

we see that since the integer a is coprime with p , none of the above products is divisible by p by our favorite Euclid's Lemma.

Again, since p does not divide a , p does not divide the product $a \cdot a = a^2$. Since p does not divide a and p does not divide a^2 , p does not divide their product $a \cdot a^2 = a^3$ and, likewise, p does not divide the product $a \cdot a^3 = a^4$ so on.

Model time: note how the above argument can be modeled with a train that carries its own rails and first, puts these rails in front of it, then, rolls over the just installed track and then, installs the next set of rails and so on.

Officially, for no integer $k = 1, 2, 3, \dots, p$ the prime p divides a^k .

Therefore, upon the integer division by p , as per DAT, the above p powers of a shown in (39) will generate exactly $(p - 1)$ remainders:

$$1, 2, 3, \dots, p - 1$$

which gives us the ability to treat the above p powers of a in (39) as the proverbial “balls” and the above $(p - 1)$ remainders modulo p of these powers as the proverbial “boxes” from the Pigeonhole Principle.

Since we have to distribute p balls, the powers of a , across $(p - 1)$ boxes, the remainders modulo p of these powers, by the said Pigeonhole Principle, at least one box, a remainder modulo p , will contain at least two balls, the powers of a .

Which is to say that at least two distinct powers k and $m \neq k$ of the chosen integer a will generate the same remainder upon division by p when both integers k and m belong to the set $1, 2, 3, \dots, p$:

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$$a^k \bmod p = a^m \bmod p$$

Without loss of generality, assume that the integers k and m are sorted as follows:

$$0 < k < m \leq p$$

But then, by an argument already familiar to us, it follows that the prime modulus p must divide “the difference” of the integers a^m and a^k :

$$a^m - a^k = a^m(a^{m-k} - 1) \equiv 0 \pmod{p}$$

and we understand that:

$$a^m \not\equiv 0 \pmod{p}, a^k \not\equiv 0 \pmod{p}, a^m - a^k > 0$$

But since the prime p does not divide the integer a^m , by the same Euclid’s Lemma, it follows that the prime p must divide the above difference of a^{m-k} and 1:

$$a^{m-k} - 1 \equiv 0 \pmod{p}$$

which means that the integers a^{m-k} and 1 are congruent modulo p :

$$a^{m-k} \equiv 1 \pmod{p} \tag{40}$$

Let the integer x be the remainder of division of the integer a^{m-k-1} by the prime p :

$$a^{m-k-1} \equiv x \pmod{p} \tag{41}$$

Commentary: because of the $0 < k < m$ constraint, it follows that the difference $(m - k - 1)$ can be 0 at its lowest if the integer m is just one greater than the integer k . Hence, in that, worst case, scenario $a^0 = 1$ is still an integer from the set G_p .

In the **Exercise 11.55** we, cleverly, have shown that:

$$\text{if } a \equiv b \pmod{n} \text{ and } c \equiv d \pmod{n}, \text{ then } ac \equiv bd \pmod{n}$$

In our case, it is clear that the integer a is congruent to itself modulo p :

$$a \equiv a \pmod{p} \tag{42}$$

Therefore, multiplying the congruences in (41) and (42), we find:

$$a^{m-k} \equiv ax \pmod{p} \tag{43}$$

and by comparing the congruences in (40) and (43) we conclude that:

$$ax \equiv 1 \pmod{p}$$

which tells us that x is the unique **right**-sided multiplicative inverse modulo p of the integer a of G_p under multiplication modulo p for p prime.

Consequently, x is the unique left-sided multiplicative inverse modulo p of the integer a of G_p :

$$xa \equiv 1 \pmod{p}$$

and, thus, the element a of G_p chosen at will does possess a unique multiplicative inverse modulo p for p prime, implying that indeed, every element of the set $G_p = \{1, 2, 3, \dots, p-1\}$ possesses a unique multiplicative inverse modulo p under the operation of multiplication modulo p for p prime. $\square$

Again, exactly what this mystery inverse x is?

We do not know, but we proved that it must exist nonetheless.

Reflection On The Experience

Adhering to the advice given in the **What Makes It Perfect** chapter, we inquire: what was the tactical recipe that worked for each of the above proofs?

The essence of the first proof was a contradiction plus the Euclid's Lemma.

The essence of the second proof was the Pigeonhole Principle plus the Euclid's Lemma.

What was the commonality between the above two proofs: the Euclid's Lemma.

No wonder - the Euclid's Lemma captures one of the fundamental properties of prime numbers.

Separately, note well that across the above two proofs we have uncovered enough tactical material to generate a group-theoretic proof of the Fermat's Little Theorem. We will examine such a proof, essentially due to Lagrange, in the near future.

Gathering all the work that we have done so far, we see that the operation of multiplication modulo p defined on the set $G_p = \{1, 2, 3, \dots, p-1\}$ for p prime is both binary and associative.

Under the said operation the element 1 of G_p is the unique identity e and every element of G_p possess a unique inverse.

Since all the popular group axioms in the case are satisfied, it follows that under the operation of multiplication modulo p the said set G_p forms a finite commutative group of order $(p - 1)$, where p is a prime number.

When p is understood to be a prime number then such a group is known as **the multiplicative group of integers modulo p** symbolized as $(\mathbb{Z}/p\mathbb{Z}, \times)$, or as $(\mathbb{Z}/p\mathbb{Z})^*$, or as $(\mathbb{Z}/p\mathbb{Z})^\times$ and so on.

If the modulus n at hand is not prime but is composite then the official name of the respective group is **the multiplicative group of integers modulo n that are coprime with n** .

In that case we now understand exactly why and how such a group is formed: if n is composite then from the set of positive integers $G_n = \{1, 2, 3, \dots, n - 1\}$ we are to throw out all the integers that are not coprime with n because all such integers are not invertible under multiplication modulo n due to the fact that the needed integers x, y , of which the integer x is the inverse, that make the equality $1 = ax + ny$ true do not exist.

Such groups are also Abelian and finite but their order is not $(n - 1)$ but exactly the number of positive integers smaller than and coprime with n , a number returned by the so-called Euler's totient function $\phi(n)$, which is $(p - 1)$ when n is a prime number p .

For example, in order to form a multiplicative group of integers modulo 10 for the composite integer $n = 10 = 2 \cdot 5$, from the corresponding set $G_{10} = \{1, 2, 3, \dots, 9\}$ we are to throw out the integers 2, 4, 5, 6 and 8, all of which are not coprime with 10.

Consequently, the diminished set $G'_{10} = \{1, 3, 7, 9\}$ will form a multiplicative group of integers modulo 10 under the operation of multiplication modulo 10 and the order of that group will be equal to $\phi(10) = 4$.

Clearly, when the modulus n is a prime number p then the said totient function $\phi(p)$ will return $(p - 1)$ and we, thus, see a path from here to there that will present a coherent and a uniform view of the multiplicative groups of integers modulo n in general.

Also, remember that officially the contents of the above sets G_p and G'_n are not simple integers but, rather, are infinite sets of integers known as the equivalence classes modulo p or modulo n .

The multiplicative groups of integers modulo p for p prime have a number of interesting properties.

Here is one.

First, construct the multiplication table of the group $(\mathbb{Z}/7\mathbb{Z}, \times)$ by placing its elements in the table's row and column headers in their natural order.

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Once that is done (Figure 11.134):

$\ast$ mod 7	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

ponder, what patterns do we see?

If we dim the lights in the entire table except for its top and bottom **rows** then we will see that the contents of these two rows will be sorted in exactly the opposite order (Figure 11.135):

$\ast$ mod 7	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Hm.

What is going on here?

Did we just get lucky with the prime 7 or is this **the** property of the appropriately arranged multiplication tables of such groups for all primes?

Tell us what you think.

Exercise 11.64: will it always be the case that the contents of the top and the bottom row of a multiplication table of the multiplicative group of integers modulo p for p prime are in opposite order when the contents of the table's column and row headers are arranged in their natural order?

Solution: well, let us think together.

Even though we are yet to officially study the generic properties of a multiplication table of a group, we already understand that the contents of such a table are not uniquely defined in the sense of the immediate arrangement of its elements in the table's column and row headers.

In other words, the elements in the column and row headers of such a table can be shuffled in a linear fashion in any which legal way, one element per cell and so on.

However, if we agree to place the positive integers of the set $G_p = \{1, 2, 3, \dots, p-1\}$ in the said column and row headers in their natural ordering, as is shown in the Figure 11.135, then what can be said about the contents of the top row of that table?

Well, in that case the entries of the top row of the said table are generated by the group products $1 \circ k$ for k varying from 1 to $p-1$.

Since the integer 1 is the identity of the group at hand, it follows that all of the above products will amount to k , since all such integers are strictly less than p and by DAT for all such integers we will have:

$$k = 0 \cdot p + k$$

Hence, $k \bmod p = k$ and the said entries mimic the first consecutive natural numbers $1, 2, 3, \dots, p-1$.

What about the contents of the bottom row of our table?

Such contents are generated by the group product $(p-1) \circ k$ for all k varying from 1 to $p-1$.

For the first entry in the bottom row of our table we will have:

$$[(p-1) \cdot 1] \bmod p = [p-1] \bmod p = -1$$

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Huh?

Didn't we agree earlier that the remainders upon integer division must always be non-negative?

Yes we did and the above result should cause no trouble. Recall the idea of the equivalence classes modulo n .

In which equivalence class modulo p does the integer -1 live?

In order to answer that question all we have to do is add our modulus p to -1 enough times, which is just once:

$$-1 + p = p - 1 > 0$$

Commentary: in the sample group above we have: $p - 1 = 7 - 1 = 6$.

Thus, the first, leftmost, entry in the bottom row of our multiplication table will or, in fact, **must** always be $p - 1$, which is the last, rightmost, entry in the table's top row.

For the second entry in the bottom row of our table we will have:

$$[(p - 1) \cdot 2] \bmod p = [2p - 2] \bmod p = -2$$

all because p divides its own multiple, $2p$.

The equivalence class modulo p in which the integer -2 lives, as we now understand, is $p - 2$.

Hence, the next entry in the bottom row of our multiplication table will always be $p - 2$, which is the second to last entry in the table's top row.

For the third entry in the bottom row of our table we will have:

$$[(p - 1) \cdot 3] \bmod p = [3p - 3] \bmod p = -3$$

all because p divides its own multiple, $3p$, and so on.

The last, rightmost, entry in the bottom row of our multiplication table will be:

$$[(p - 1) \cdot (p - 1)] \bmod p = [(p - 1)^2] \bmod p = [p \cdot p - 2 \cdot p + 1] \bmod p = 1$$

all because p divides both p^2 and $2p$.

We, thus, answer the given question in the positive:

yes, if the contents of the column and row headers of a multiplication table of the multiplicative group of integers modulo p for p prime are arranged in their natural order, then the contents of that table's top and the bottom rows will always be sorted in the opposite order

which completes this exercise. $\square$

In fact, if, by analogy, we dim the lights in the entire table at hand yet one more time then we will see that the contents of the table's leftmost and rightmost **columns** will also be sorted in exactly the opposite order (Figure 11.136):

$\ast_{\text{mod } 7}$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

We leave the proof of this property of the said multiplication table to the reader as another useful exercise.

Thus, we see that in an appropriately arranged multiplication table of the multiplicative group of integers modulo p for p prime it is always possible to fill out the contents of its, so to speak, outer ribbon mechanically without the computations of any kind.

Now that the outer ribbon is in place, what else do we see?

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Now that the outer ribbon is in place, we see a smaller interior square whose smaller outer ribbon also has the same property (Figure 11.137):

$\begin{smallmatrix} * \\ \text{mod } 7 \end{smallmatrix}$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

and so on, recursively.

Exercises 12: Fermat's Little Theorem, A Group-Theoretic Proof Of

One of the main reasons why many generic and abstract statements and proofs in higher mathematics sometimes seem so impenetrable, unpredictable and almost magical, as in *where did that come from*, is because such lines of reasoning simply erase an enormous, centuries-in-the-making, amount of the experimental evidence that was generated by-hand by our predecessors and present only the tail end of that evidence at best.

As Sherlock Holmes so eloquently put it, when we remove the entire train of thought that lead to a certain conclusion and present only the problem and its solution, then such a solution comes across as nothing short of a miracle.

It should be understood well, however, that the Fermats, the Eulers and the Gausses of the past did not have access to the modern electronic computers and they did all their computations by-hand, with pen and paper.

Yes, these people covered pages and pages with numerous mathematical experiments, spotted certain patterns in the resultant numbers and then, as an aftermath, tried to slide a powerful mathematical apparatus underneath such results in order to explain them theoretically, in general and rigorously.

The sleek, polished and smooth proofs shown in the modern textbooks did not pop into the mind of a certain mathematician in their final instance all at once. A huge amount of **connecting the hand bone to the brain bone** work preceded many, if not all, of these theoretic discoveries.

It is no wonder that in the absence of such computational work the modern students of mathematics, force-fed large amounts of abstract and complicated proofs, find mathematics difficult and, at times, completely opaque.

In this discussion, however, we shall take exactly that, the mathematical experiments first, approach of building a bridge from pure experiment to pure theory.

It will be very exciting.

To that end, below we present the C source code of the `flts` program, for **F**ermat's **L**ittle **T**heorem **S**tudy:

```
#include <stdio.h>
#include <stdlib.h>

extern int
residueModP( int base, int exp, int p)
{
    int res = 1;

    //
    // Reduce the base mod p
    //
    base = base % p;

    while ( exp > 0 )
    {
        //
        // If the exponent is odd, multiply
        // the base by the residue modulo p
        //
        if ( exp & 1 )
            res = (res * base) % p;

        // The exponent is now even
    }
}
```

```

        exp = exp >> 1;
        base = (base * base) % p;
    }

    return res;
}

extern int
main( int argc, char* argv[] )
{
    int      i;
    int      res;
    int      exp;
    int      p = atoi( argv[ 1 ] );

    for ( i = 1; i < p; i++ )
    {
        res = 0;
        exp = 1;
        printf( "%d: ", i );
        do
        {
            res = residueModP( i, exp, p );
            printf( "%d%s", res, res != 1 ? ", " : "" );
            exp++;
        }
        while ( res != 1 );
        printf( "\n" );
    }

    return 0;
}

```

Save the shown code in the flts.c disk file and build it like so now:

```
gcc -g -o flts flts.c
```

Anyone who does not have access to a UNIX/Linux machine can use any on-line C compiler to build and run the flts program like so:

```

./flts 5
1: 1
2: 2, 4, 3, 1
3: 3, 4, 2, 1

```

4: 4, 1

We shall see in a moment what the above output means and how it was created.

Our `flts` program accepts only one command-line argument, a prime number p that becomes the seed of the corresponding multiplicative group of integers modulo p or $(\mathbb{Z}/p\mathbb{Z})^\times$, known more formally as the multiplicative group of equivalence classes modulo p .

In our previous discussion we already proved that the set of positive integers $G_p = \{1, 2, 3, \dots, p-1\}$ under the operation of multiplication modulo p forms a finite Abelian group of order $p-1$.

The Fermat's Little Theorem can be worded in a number of different but equivalent ways. Here is one: if a is any integer not divisible by a prime p then the number $a^{p-1} - 1$ is divisible by p .

In symbols:

$$a^{p-1} \equiv 1 \pmod{p}$$

which is simply another way of saying that for some integer k it is that case that:

$$a^{p-1} - 1 = k \cdot p$$

The Fermat's Little Theorem, or the **FLT** for short, can be proved via the machinery of the theory of groups in a number of different ways. One such proof, for example, relies heavily on the concept of a subgroup of a group, which we have not studied yet.

In this discussion, however, we will present a group-theoretic proof of FLT that must surely be historically interesting for the students of mathematics who are geared toward the idea of discovering things on their own. A proof that, going all the way back to the very roots of the birth of the theory of groups, is based on the ideas put forward by the French mathematician Lagrange (1736–1813).

The upcoming proof may come across as a bit loose by the modern standards and that is perfectly fine because we want to expose exactly how the grown-up mathematics is made, loosely and informally first, rigorously and formally later.

Now, since the Fermat's Little Theorem deals with whole positive powers of an arbitrary integer a , we first make an intuitive and loose connection with the powers of the elements of a multiplicative group of integers modulo p , where for the rest of this discussion we agree that:

- the symbol p means **a positive prime** and that
- the symbol a means **an integer not divisible by p**

As a fair warning, the value of the above integer a during the course of this discussion will be jumping back and forth between the set G_p and **any** integer not divisible by p required by the **FLT**.

This is a very important first step because we remember the crucial fact that all the integers in the above set G_p are relatively prime with the modulus p .

Since in the above **FLT** formulation we are demanding that an arbitrary integer a is coprime with p , from the Euclid's Lemma, it right away follows that no integral power of a will be divisible by p either. In other words, for no positive integer n the integer a^n is divisible by p and we already saw the justification of that fact in our previous discussion.

That is: if p does not divide a then, by the said Euclid's Lemma, p does not divide the product $a \cdot a = a^2$. Likewise, p does not divide the product $a^2 \cdot a = a^3$ and so on.

That, in turn, means that no positive whole power of a , that is no number of the form a^n for a whole positive integer n , will ever leave the remainder of 0 upon division by p .

But such powers are still perfect integers that are subject to the Division Algorithm Theorem and according to the constraint of that theorem, the remainder of division of such integers by p must be strictly less than p , which is always a finite positive whole number:

$$0 < a^n \bmod p < p$$

We, thus, realize that the above powers a^n modulo p of an arbitrary integer a not divisible by p are in some sense equivalent to the elements of the set G_p .

Therefore, we would like to see if it is possible to drag an arbitrary integer a coprime with p through the group $(\mathbb{Z}/p\mathbb{Z})^\times$ in order to prove the **FLT**.

In other words, we would like to see if it is possible to put our integer a onto the $(\mathbb{Z}/p\mathbb{Z})^\times$ train and take it all the way home.

Thinking in that direction, we develop a hunch that the residues $a^n \bmod p$ will have no choice but repeat themselves eventually, as the exponent n grows.

Why is that?

Because even though the prime p is completely arbitrary, it is still a number that is finite. And if that number is finite then the number of elements in the set G_p or the number of possible residues $a^n \bmod p$ will also be finite.

Thus, as we compute the residues upon division by p of the higher and higher consecutive powers a^n , these residues cannot vary forever. They can vary only within a finite set of values, the elements of the set G_p .

We reason, then, that there must exist positive integers n and $m > n$ such that:

$$a^n \bmod p = a^m \bmod p$$

But we remember that if the integers a and p are coprime then the multiplicative inverse modulo p of a exists and conversely, if such an inverse exists then the integers at hand are coprime.

Therefore, we can multiply the last equality through by the inverse of a from the left once:

$$a^{-1}a^n \bmod p = a^{n-1} \bmod p = a^{m-1} \bmod p$$

and twice:

$$a^{n-2} \bmod p = a^{m-2} \bmod p$$

and thrice:

$$a^{n-3} \bmod p = a^{m-3} \bmod p$$

and so on n times:

$$a^0 = 1 \bmod p = a^{m-n} \bmod p$$

and realize that not only our residues $a^n \bmod p$ must somehow orbit the integers in G_p , but they must do so in a very peculiar way, a way in which these residues eventually converge onto 1.

We now switch from an arbitrary integer a to the elements of the set G_p and test our hunch numerically by generating a large body of mathematical evidence.

This is where our `flts` program comes in. In the old days we would crunch these numbers by hand, which wouldn't be the worst thing by the way, but these days we can simply give a prime p to an instance of a Turing machine and that machine will:

- step through all the positive integers $1, 2, 3, \dots, p-1$ of the group $(\mathbb{Z}/p\mathbb{Z})^\times$
- for each such integer k compute the residues k^n modulo p for the consecutive positive integers n starting from 1
- print the above residues and continue until
- the said residue comes out to be 1, at which point the program will move onto the next integer k until all the integers in the said group are thus processed

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Let us see how this works in practice. Run the `flts` program for $p = 2$:

```
./flts 2
1: 1
```

For $p = 3$:

```
./flts 3
1: 1
2: 2, 1
```

For $p = 5$:

```
./flts 5
1: 1
2: 2, 4, 3, 1
3: 3, 4, 2, 1
4: 4, 1
```

For $p = 7$:

```
./flts 7
1: 1
2: 2, 4, 1
3: 3, 2, 6, 4, 5, 1
4: 4, 2, 1
5: 5, 4, 6, 2, 3, 1
6: 6, 1
```

For $p = 11$:

```
./flts 11
1: 1
2: 2, 4, 8, 5, 10, 9, 7, 3, 6, 1
3: 3, 9, 5, 4, 1
4: 4, 5, 9, 3, 1
5: 5, 3, 4, 9, 1
6: 6, 3, 7, 9, 10, 5, 8, 4, 2, 1
7: 7, 5, 2, 3, 10, 4, 6, 9, 8, 1
8: 8, 9, 6, 4, 10, 3, 2, 5, 7, 1
9: 9, 4, 3, 5, 1
10: 10, 1
```

And for $p = 13$:

```
./flts 13
1: 1
2: 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, 1
3: 3, 9, 1
4: 4, 3, 12, 9, 10, 1
5: 5, 12, 8, 1
6: 6, 10, 8, 9, 2, 12, 7, 3, 5, 4, 11, 1
7: 7, 10, 5, 9, 11, 12, 6, 3, 8, 4, 2, 1
8: 8, 12, 5, 1
9: 9, 3, 1
10: 10, 9, 12, 3, 4, 1
11: 11, 4, 5, 3, 7, 12, 2, 9, 8, 10, 6, 1
12: 12, 1
```

Let us take a short break and wrap our brains around what we see.

Concentrate on, say, the last group $(\mathbb{Z}/13\mathbb{Z})^\times$.

We see that the consecutive whole powers of the element 2 of that group $2, 2^2, 2^3, 2^4$ modulo 13 and so on orbit the entire set G_{13} or all of its 12 residues:

$$2 \bmod 13 = 2, 2^2 \bmod 13 = 4, \dots, 2^7 \bmod 13 = 11, \dots, 2^{12} \bmod 13 = 1$$

However, the consecutive whole powers of the element 3 of that group 3 and 3^2 modulo 13 orbit only two elements of the set G_{13} , the residues 3 and 9, before converging onto 1:

$$3 \bmod 13 = 3, 3^2 \bmod 13 = 9, 3^3 \bmod 13 = 1$$

The consecutive whole powers of the element 4 of that group $4, 4^2, 4^3, 4^4, 4^5$ modulo 13 orbit only five elements of the set G_{13} , the residues 4, 3, 12, 9 and 10, before converging onto 1:

$$4 \bmod 13 = 4, 4^2 \bmod 13 = 3, \dots, 4^5 \bmod 13 = 10, 4^6 \bmod 13 = 1$$

The consecutive whole powers of the element 5 of that group $5, 5^2, 5^3$ modulo 13 orbit only three elements of the set G_{13} , the residues 5, 12 and 8, before converging onto 1:

$$5 \bmod 13 = 5, 5^2 \bmod 13 = 12, 5^3 \bmod 13 = 8, 5^4 \bmod 13 = 1$$

The consecutive whole powers modulo 13 of the elements 6, 7 and 11 of that group probe all the residues in G_{13} , while the consecutive whole powers modulo 13 of the elements 8, 9, 10 and 12 of that group probe only 3, 2, 5 and 1 residues in G_{13} respectively, before converging onto 1.

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Hm. This is interesting. Let us play with the `flts` program some more.

For $p = 17$ we have:

```
./flts 17
1: 1
2: 2, 4, 8, 16, 15, 13, 9, 1
3: 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1
4: 4, 16, 13, 1
5: 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7, 1
6: 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3, 1
7: 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5, 1
8: 8, 13, 2, 16, 9, 4, 15, 1
9: 9, 13, 15, 16, 8, 4, 2, 1
10: 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12, 1
11: 11, 2, 5, 4, 10, 8, 3, 16, 6, 15, 12, 13, 7, 9, 14, 1
12: 12, 8, 11, 13, 3, 2, 7, 16, 5, 9, 6, 4, 14, 15, 10, 1
13: 13, 16, 4, 1
14: 14, 9, 7, 13, 12, 15, 6, 16, 3, 8, 10, 4, 5, 2, 11, 1
15: 15, 4, 9, 16, 2, 13, 8, 1
```

and for $p = 19$ we find:

```
./flts 19
1: 1
2: 2, 4, 8, 16, 13, 7, 14, 9, 18, 17, 15, 11, 3, 6, 12, 5, 10, 1
3: 3, 9, 8, 5, 15, 7, 2, 6, 18, 16, 10, 11, 14, 4, 12, 17, 13, 1
4: 4, 16, 7, 9, 17, 11, 6, 5, 1
5: 5, 6, 11, 17, 9, 7, 16, 4, 1
6: 6, 17, 7, 4, 5, 11, 9, 16, 1
7: 7, 11, 1
8: 8, 7, 18, 11, 12, 1
9: 9, 5, 7, 6, 16, 11, 4, 17, 1
10: 10, 5, 12, 6, 3, 11, 15, 17, 18, 9, 14, 7, 13, 16, 8, 4, 2, 1
11: 11, 7, 1
12: 12, 11, 18, 7, 8, 1
13: 13, 17, 12, 4, 14, 11, 10, 16, 18, 6, 2, 7, 15, 5, 8, 9, 3, 1
14: 14, 6, 8, 17, 10, 7, 3, 4, 18, 5, 13, 11, 2, 9, 12, 16, 15, 1
15: 15, 16, 12, 9, 2, 11, 13, 5, 18, 4, 3, 7, 10, 17, 8, 6, 14, 1
16: 16, 9, 11, 5, 4, 7, 17, 6, 1
17: 17, 4, 11, 16, 6, 7, 5, 9, 1
18: 18, 1
```

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Our readers are encouraged to not stop here but keep generating more and more numerical evidence with the `flts` program for themselves.

We, however, shall go back to the previous group, the printout that shows the structure of the $(\mathbb{Z}/17\mathbb{Z})^\times$ group as it pertains to the consecutive whole powers of its elements modulo 17.

This time such powers of, say, the element 2 modulo 17 converge onto 1 in 8 steps and 8, clearly, divides the order of that group, 16:

$$2^8 = 1 \bmod 17$$

What will happen to the next consecutive powers of 2 modulo 17?

Let us see by crunching some numbers ourselves, with a calculator perhaps, omitting the obvious modulo 17 operation below:

$$2^9 = 2, 2^{10} = 4, 2^{11} = 8, 2^{12} = 16$$

$$2^{13} = 15, 2^{14} = 13, 2^{15} = 9, 2^{16} = 1$$

Aha!

Just like we thought. The consecutive whole powers of 2 modulo 17 not only converge onto 1, but they orbit the respective residues **in the same trajectory** or **in a cycle of length 8** that still divides the order of that group, 16.

Let us work more closely with the set of the above residues, which, for the sake of the argument, we shall designate as the set A :

$$A = \{1, 2, 4, 8, 9, 13, 15, 16\}, |A| = 8$$

The cardinality of that set is less than the cardinality of the parent set: $|A| = 8 < 16 = |G_{17}|$.

Some elements that are present in G_{17} are missing from the set A . Let us take any such missing element - say, 3.

Here is the key observation that was made by the French mathematician Lagrange.

Essentially, using these concrete numbers as the playground, Lagrange suggested that if we multiply all the elements in A through by the missing element 3:

$$3A = \{3, 6, 12, 24, 27, 39, 45, 48\}$$

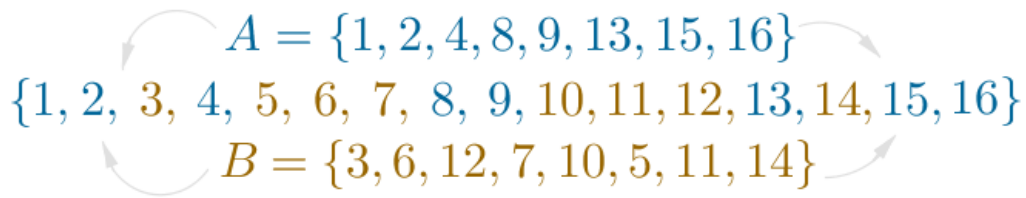
and then compute the residues of these elements modulo 17 then all such residues will be distinct from their counterparts in A and that they **may** complement that set A to the complete set G_{17} .

Will it be the case?

Let us check by putting the residues of the elements of the set $3A$ modulo 17 into its own dedicated set B :

$$B = \{3, 6, 12, 7, 10, 5, 11, 14\}, |B| = 8$$

If we unionize the sets A and B then, indeed, we shall get (Figure 11.138):



the entire set G_{17} .

Amazing!

More evidence.

From the same group $(\mathbb{Z}/17\mathbb{Z})^\times$ we pick the set of its powers of 4 modulo 17:

$$A = \{1, 4, 13, 16\}, |A| = 4$$

What elements that are present in the set G_{17} are absent from the above set A ? Quite a few. Say, we pick the missing element ... 5.

Then:

$$5A = \{5, 20, 65, 80\}$$

and modulo 17 these numbers yield:

$$B = \{5, 3, 14, 12\}, |B| = 4$$

Unionized, the above two sets produce:

$$A \cup B = \{1, 3, 4, 5, 12, 13, 14, 16\}$$

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OK. The element 2 of G_{17} is also missing from the set A . Thus:

$$2A = \{2, 8, 26, 32\}$$

and modulo 17 these integers give:

$$C = \{2, 8, 9, 15\}$$

Unionized, all three sets A , B and C produce:

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 8, 9, 12, 13, 14, 15, 16\}$$

Very well. The element 7 is still missing from A :

$$7A = \{7, 28, 91, 112\}$$

and modulo 17 these numbers give:

$$D = \{7, 11, 6, 10\}$$

Unionized, all four sets A , B , C and D produce:

$$A \cup B \cup C \cup D = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\} = G_{17}$$

the parent set G_{17} in its entirety.

Playing a little Lagrange of some two hundred years ago, we are onto something here!

So if we were Joseph-Louis Lagrange, then what should we do next?

Correct. Just about now we should switch from pure experiment to pure theory and make an attempt to prove the Fermat's Little Theorem by the means of the theory of groups.

How can we do that?

Well, first we go back to an arbitrary integer a coprime with p . Such an integer now may or may not belong to the set G_p .

Mimicking the set of residues $2^k \bmod p$ from the group $(\mathbb{Z}/17\mathbb{Z})^\times$, we agree that the positive integer l is **the very fist** or **the smallest** power that turns our integer a into 1 modulo p :

$$a^l \bmod p = 1 \tag{44}$$

which means that now we have a set of residues A :

$$A = \{a, a^2, a^3, \dots, a^{l-1}, a^l\} \quad (45)$$

where we understand the shortcut notation above to actually mean yada-yada modulo p .

In our numerical experiments we **saw** that all these residues are distinct.

We now **think** or **hypothesize** that all these residues **must be** distinct in general and always.

Can we prove that hypothesis rigorously?

Yes.

How?

Using the same tactical vehicle that we already used in our previous discussion, a contradiction.

Namely. Assume that the residues in A are not all distinct.

Then, among the residues of A there will exist at least two, say, a^n and a^m with $0 < n < m \leq l$, that are the same:

$$a^n \bmod p = a^m \bmod p$$

But, since a and p are coprime, the multiplicative inverse modulo p of the integer a exists and we can always multiply the above equality from the left by that inverse n times:

$$1 = a^{m-n}$$

The integer $(m - n)$, however, is strictly smaller than the integer l :

$$m - n < l$$

and that means that the residue 1 will be generated earlier than we agreed - not by the l -th power of a but by the smaller $(m - n)$ the power of a .

Contradiction!

Therefore, our original assumption that the elements of the set A shown in (45) are not distinct was false and, consequently:

all the residues in the set A are distinct

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Commentary: it is very important to understand exactly why we were able to pull off this contradiction in the first place. Because of the existence of the multiplicative inverses.

Since we agreed that $a^l = 1$, we can re-write our set A as follows:

$$A = \{1, a, a^2, a^3, \dots, a^{l-1}\}$$

How many elements are in that set?

Well, there are exactly l elements in A :

$$|A| = l$$

Now, in our numerical experiments we saw that in the same group $(\mathbb{Z}/17\mathbb{Z})^\times$, for example, the residues modulo 17 generated by the consecutive whole powers of the elements 3, 5, 6, 7, 10, 11, 12 and 14 probed the entire set G_{17} :

3 : 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1

5 : 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7, 1

6 : 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3, 1

7 : 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5, 1

10 : 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12, 1

11 : 11, 2, 5, 4, 10, 8, 3, 16, 6, 15, 12, 13, 7, 9, 14, 1

12 : 12, 8, 11, 13, 3, 2, 7, 16, 5, 9, 6, 4, 14, 15, 10, 1

14 : 14, 9, 7, 13, 12, 15, 6, 16, 3, 8, 10, 4, 5, 2, 11, 1

Such residues were linearly permuted in some funny ways but that does not matter.

Walking the bridge from pure experiment to pure theory, we, thus, argue that we now have two and only two cases: the case when $l = p - 1$ and the case when $l < p - 1$ (we understand the it cannot be the case that $l > p - 1$).

In other words, if, in the first case, it so happens that the set A already has exactly $l = p - 1$ elements in it then the FLT is already proved:

$$a^l = a^{p-1} \equiv 1 \pmod{p}$$

What are we to do in the second case when the set A is genuinely shorter than the parent set G_p ?

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Use the idea proposed by Lagrange!

If the set A is genuinely shorter than the parent set G_p then, clearly, some element b that is in G_p is not in the set A .

Very well.

Take that missing element b , multiply all the elements in A by b and compute the residues $b \cdot a^k \bmod p$, constructing the set B :

$$bA = B = \{b, ba^2, ba^3, \dots, ba^{l-1}\}$$

Hypothesis, driven by the numerical experiments: 1) all the elements in the set B are distinct among themselves and 2) they are all different from the elements of the set A (officially, the intersection of the sets A and B is the empty set: $A \cap B = \emptyset$).

Proof?

By contradiction: first, assume that it is not the case that all the elements in the set B are distinct. Then, for some two elements of that set we shall have, say:

$$ba^2 = ba^3$$

Again, the integer b being coprime with p , has a multiplicative inverse b^{-1} . Multiply the last equality by that inverse from the left to get:

$$a^2 = a^3$$

which means that some two elements in the set A are the same.

But we already proved that all the elements in the set A are distinct.

Contradiction!

Next, assume that some element in the set B , such as ba^k , is equal to some element in A , such as a^m :

$$ba^k = a^m$$

Inverses to the rescue, yet again: by applying the inverse a^{-1} to the above equality from the right, we can always whittle it down to:

$$b = a^{m-k}$$

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implying that b is an element of the set A because even if the integer $(m - k)$ comes out to be negative, we can always add enough copies of l to it in order to make it non-negative.

But we agreed that that element b is exactly what is **not** in A .

Contradiction!

If it so happens that the sets A and B now make up the parent set G_p in its entirety then how many residues will these two sets have?

Correct, these two sets will have $l + l = 2l$ residues between them.

And what does that mean?

That means that, since these residues cover the entire set G_p , the number of elements in the parent set G_p will be the exact multiple of l :

$$p - 1 = l \cdot 2$$

just like we saw that scenario in our experiments with the powers of 2 in the group $(\mathbb{Z}/17\mathbb{Z})^\times$.

Very well. But what if the sets A and B do not cover the parent set G_p in its entirety? Just like we saw with the powers of 4 in the same group?

Floss. Rinse. Repeat: in that case there will be an element c in the set G_p that is not in the set A and not in the set B , multiply all the elements in A by c modulo p :

$$cA = C = \{c, ca^2, ca^3, \dots, ca^{l-1}\}$$

and prove in the same fashion that all the elements in C are distinct among themselves and that the intersection of all three sets is the empty set, no element in C is in A or in B . For example, in order to show that no element in C is in B , assume the opposite and let:

$$ca^3 = ba^5$$

say. But by applying the right inverse of a to the above equality, we arrive at a contradiction:

$$c = ba^{5-3} = ba^2 \in B$$

because we agreed that the element c is not in the set B to begin with.

If these three sets cover the parent set G_p in its entirety then:

$$p - 1 = l \cdot 3$$

and if not then pick the next missing element, d , that is not in A , not in B , not in C and repeat the above arguments, pretty much, verbatim.

But such a game of hide and seek cannot last forever because the set G_p is finite - sooner or later, in exactly q such games we will exhaust all the residues across the q sets each of which is made in the image of the sets B, C and so on.

Thus:

$$p - 1 = l \cdot q, \quad q = 1, 2, 3, 4, \dots \quad (46)$$

But then, what is this power of a :

$$a^{p-1} = ?$$

This power of a , according to the rules of traditional exponentiation, is, via (46):

$$a^{p-1} = a^{l \cdot q} = [a^l]^q$$

However, earlier, in (44), we agreed that $a^l = 1$. Hence:

$$a^{p-1} = a^{l \cdot q} = [a^l]^q = 1^q = 1 \pmod{p}$$

which completes the group-theoretic proof of the Fermat's Little Theorem. $\square$

Who was the central character of the above line of reasoning, aside from Lagrange, of course?

Its majesty **inverse**.

Playing with numbers or conducting experiments with the actual numbers is often a rich vein, if not to say a gold mine of mathematical ideas.

Let us agree, for the sake of brevity, to refer to the smaller sets of numbers that we see in the output of our `flts` program, technically **the residues** $a^k \pmod{p}$, as just **strings**.

For example, in the group $(\mathbb{Z}/17\mathbb{Z})^\times$ we have a string generated by the powers of 2:

$$2 : 2, 4, 8, 16, 15, 13, 9, 1$$

and a string generated by the powers of 3:

$$3 : 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1$$

and a string generated by the powers of 4:

$$4 : 4, 16, 13, 1$$

The length of the first string is 8, which divides 16; the length of the second string is 16, which divides 16 and the length of the third string is 4, which divides 16.

We see that sometimes such strings are genuinely shorter than the parent set, as in the first and the third examples, and sometimes such strings are equal in length to the size of the parent set, as in the second example.

Let us now do some more experiments.

Exercise 11.65: do the residues in the above 4-string, the integers 1, 4, 13 and 16 of the group $(\mathbb{Z}/17\mathbb{Z})^\times$ that is, taken in an isolation, form a group under multiplication modulo 17?

Solution: well, constructing the multiplication table of this set of integers under the given operation (Figure 11.139):

$\begin{smallmatrix} * \\ \text{mod } 17 \end{smallmatrix}$	1	4	13	16
1	1	4	13	16
4	4	16	1	13
13	13	1	16	4
16	16	13	4	1

we quickly see that **yes**, the shorter string of integers $\{1, 4, 13, 16\}$ taken in an isolation does form a group under multiplication modulo 17. $\square$

But now let us bring that set of integers back into the context of the parent set G_{17} .

What can be said about the set $\{1, 4, 13, 16\}$?

Every element of the set $\{1, 4, 13, 16\}$ is also an element of the set G_{17} , but not necessarily conversely, since some integers that are in the set G_{17} are missing from the set $\{1, 4, 13, 16\}$.

Under the operation of the parent set or the parent group, the set $\{1, 4, 13, 16\}$ does form a smaller group or **a group contained within a group** in general or **a genuinely smaller group contained in a genuinely larger group**.

Such a **group contained within a group** is now called a **subgroup of a group** and it plays an important role in the overall study of groups because some groups possess a structure that can be described in terms of subgroups and the purely formal and the theoretic notions now begin to pour out of the above numerical experiments as if it were a horn of plenty:

- we arrived at the notion of a **subgroup of a group**
- the mini-me sets $bA = \{ba, ba^2, \dots, ba^{l-1}\}$, $cA = \{ca, ca^2, \dots, ca^{l-1}\}$ and such that we used in the above proofs are what is now formally known as **the cosets**, technically **the left cosets of the group G with respect to the subgroup A**
- we see that **every group is a subgroup of itself** (recall the powers of, say, 3 in the group $(\mathbb{Z}/17\mathbb{Z})^\times$)
- we realize that every group has a **trivial subgroup** consisting of its identity e
- we, thus, can introduce the notion of a **proper subgroup of a group** or a subgroup of a group that does not consist of all the elements of the parent group
- we see that the identity of a subgroup is the identity of the parent group
- we can now introduce the concept of a **closure** of a subset **with respect to** or **under** an operation
- we, next, realize that a finite group has no proper subgroups if and only if its order is a prime number, p , and
- we, pretty much, proved, loosely, the so-called **Lagrange's Theorem**, which states that the order of a subgroup of a finite group always divides the order of its parent group

Wow. Look at how much material we managed to walk across the bridge that connects pure experiment with pure theory.

If witnessing the birth of a mathematical discipline is not exciting then we do not know what is.

As a parting informal assignment, we encourage our readers to crunch some numbers on their own and prove to themselves that the proper substrings of residues that we generated with our `flts` program all, indeed, form the respective (sub)groups when governed by the operation of their parent set.

For example, the string of residues:

$$8 : 8, 7, 18, 11, 12, 1$$

generated by the element 8 of the group $(\mathbb{Z}/19\mathbb{Z})^\times$ does form a group under the operation of multiplication modulo 19.

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That means that whenever any two numbers from the above string are multiplied together and the residue of that product modulo 19 is obtained then that residue will, again, be one of the elements in that string (Figure 11.140):

$\ast$ mod 19	1	7	8	11	12	18
1	1	7	8	11	12	18
7	7	11	18	1	8	12
8	8	18	7	12	1	11
11	11	1	12	7	18	8
12	12	8	1	18	11	7
18	18	12	11	8	7	1

On a separate note, observe how such an informal and loose study of the technical essence of certain concepts makes the consequent consumption of the purely theoretical material a breeze.

Exercises 13: General Theory

So far our group-theoretic vocabulary is rather shallow and that is perfectly fine because even though the number of the group-theoretic notions that we can juggle freely is small, we understand all such notions well or well enough.

We understand what a mathematical group is, we know that there exist groups that are finite and that are infinite, we know what the order of a finite group is and we know that there exist groups that are commutative and that are not commutative.

In all the previous exercises we looked at specific and concrete instances of groups with the purpose of developing a better feel for what groups are and how to work with them.

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Now, however, we will flex the group-theoretic muscle just a bit by looking at not specific instances of groups but, rather, by making sweeping conclusions about all the relevant groups.

Doing so should bring to light one of the fundamental ideas of the theory of groups: using the group-theoretic methods, it is possible to draw the generic conclusions about groups **without knowing anything about the underlying nature of the elements and of the operation of such groups**.

Exercise 11.66: say, we have, for definitiveness, a finite group G in which the product of any of its element a with itself, $a \circ a$, is always the group's identity e :

$$a \circ a = e, \forall a \in G$$

Have we seen such groups already? Sure. For example, we have seen such a group in our very first, **We See The Light**, discussion of what a group is (Figure 6.9):

	e	l	r	b
e	e	l	r	b
l	l	e	b	r
r	r	b	e	l
b	b	r	l	e

and in the **Group Theory Takes A Ride** discussion (Figure 6.10):

	e	f	s	c
e	e	f	s	c
f	f	e	c	s
s	s	c	e	f
c	c	s	f	e

and in the **Rectangle, Anyone** discussion (Figure 6.11):

	e	r	v	h
e	e	r	v	h
r	r	e	h	v
v	v	h	e	r
h	h	v	r	e

and so on.

Now let us abstract ourselves from the specific underlying nature of the elements and of the operations of these groups and wonder:

what can be said about the commutativity of such groups?

Solution: well, in order to make any headway while answering the above question, we must, first, unpack the definition of what a commutative group is.

We remember, that as per the corresponding definition, a group G is commutative if for any two of its elements a and b it is the case that $a \circ b = b \circ a$.

From the above definition it instantly follows that if in a group G there exists at least one pair of elements a, b such that $a \circ b \neq b \circ a$ then such a group is **not** commutative, one bad apple spoils the entire barrel, so to speak.

We, next, juxtapose what we have against what we need by formalizing what we have: we have a group G in which its **any** element a multiplies with itself to the group's identity e :

$$a \circ a = a^2 = e, \forall a \in G \quad (47)$$

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In a formal proof we would not do that, but as an informal and loose introduction into such a proof we, in order to make the things painfully clear, recall that in the above dual light switch group, for example, we had:

$$l \circ l = l^2 = e, \quad r \circ r = r^2 = e, \quad b \circ b = b^2 = e \quad (48)$$

Thus, the current question becomes: if we have a group G with the property captured in (47) (and specifically in (48)), then can we show that for any two elements a and b of G it must necessarily be the case that:

$$a \circ b = b \circ a$$

or that in such a group there must exist at least one pair of elements a, b such that:

$$a \circ b \neq b \circ a$$

Commentary: as our readers can see, we are keeping true to our original agenda of exposing our students to the ways in which the grown-up mathematics is done. That is, in grown-up mathematics the open-endedness of questions is a given.

Very well.

Again, even though we are about to formally study the properties of a multiplication table of a group, even a cursory look at such tables shown above tells us that yes, in at least these three specific instances all the groups **are** indeed commutative.

Why is that?

Because all the respective group products are symmetric or **the same** with respect to the main, North-West to South-East, diagonal of the corresponding multiplication table.

We, thus, form a hunch: yes, so long that a group G has the above property, it must always be the case, such a group **must be** commutative!

A hunch is a good thing. But can we show that that hunch is on the money by constructing a correct formal and rigorous proof?

Hm. We now move closer to formal and rigorous symbolic shuffling by forming the following statement:

if in a group G for all of its elements x it is the case that $x \circ x = x^2 = e$ then such a group is necessarily commutative

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In our **Take 2** discussion of various types of group axioms we noted that the following tactical maneuver may be helpful for generating a group-theoretic proof: instead of working with a monolithic object, such as a or x , try to find an advantageous compound object instead.

Namely, if we are to show that if for all the elements x in a group G it is the case that $x \circ x = e$ then that group is commutative then we can stare at the following line:

$$a \circ b = b \circ a$$

all day long and not make any headway whatsoever.

But apparently, by the standing assumption, the property of G captured in (47) has enough juice to flip the elements a and b around.

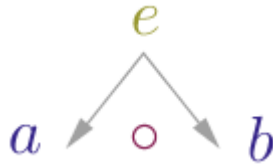
So what gives?

Ah. We always have the group's identity at our disposal, that's one.

Two, the group's identity figures prominently in the group's property. Clearly, if we are to make any progress here, then we have to splice that identity e into the works somehow.

But how?

Well, we can wedge the identity e into the works as follows (Figure 11.141):



because every element of a group possesses both the right-sided and the left-sided identity; recall our **Take 3** and **Take 4** discussions of the group axioms.

In particular, we can claim that, since a is an element of a group G , then it is the case that $a = a \circ e$ or we can claim that, since b is an element of a group G , then it is the case that $b = e \circ b$.

It is not entirely clear **why** we wedged e in there in this particular way. Why, for example, we did not tack it on either side of the product $a \circ b$? Our readers should try that maneuver themselves.

In either case, the following transition is perfectly valid and legal:

$$a \circ b = a \circ e \circ b \tag{49}$$

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Excellent. But now, we sweep the given property of G into the net: **any** (!) element x of G is known to produce e upon the group product with itself:

$$x \circ x = x^2 = e \quad (50)$$

Thus, putting (50) into (49), we have:

$$a \circ b = a \circ e \circ b = a \circ x^2 \circ b \quad (51)$$

Commentary: did you catch what just transpired here? Yes, we switched from a monolithic object, e , to its compound representation, $x \circ x$.

Next, we rely on one of the fundamental properties of a group: the group product of any two of its elements is, again, some element of that group.

Since a and b are both elements of a group G , it follows that their product $a \circ b$ is also a certain element, x , of the same group G .

But we are told that G has this peculiar property that *any* of its elements, including x , multiplies out to e :

$$x \circ x = x^2 = (a \circ b)^2 = (a \circ b) \circ (a \circ b) \quad (52)$$

Hence, putting (52) into (51), we find:

$$\begin{aligned} a \circ b &= a \circ e \circ b = a \circ x^2 \circ b = \\ &= a \circ (a \circ b)^2 \circ b = a \circ (a \circ b) \circ (a \circ b) \circ b \end{aligned} \quad (53)$$

Now take a moment to amuse yourself with the fact that on the LHS of (53) we started with just two objects, a and b , and on the RHS of (53) we wound up with a small zoo of them.

Next, we use the theorem **GTT1**, which we proved in the **Associativity** chapter:

$$(a_1 \circ \dots \circ a_m) \circ (a_{m+1} \circ \dots \circ a_n) = a_1 \circ \dots \circ a_n$$

as we make the next transition:

$$a \circ (a \circ b) \circ (a \circ b) \circ b = a \circ a \circ b \circ a \circ b \circ b$$

By relying on the fact that the group operation $\circ$ is associative, we can introduce the following parenthesis (see the proof of the above theorem for more details):

$$a \circ a \circ b \circ a \circ b \circ b = (a \circ a) \circ (b \circ a \circ b \circ b)$$

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Since we know that $a \circ a = e$, we arrive at:

$$(a \circ a) \circ (b \circ a \circ b \circ b) = e \circ (b \circ a \circ b \circ b)$$

But we know that the animal in the above rightmost parentheses is still an element of the same group G . Hence, multiplied by that group's identity from the left it is not changed:

$$e \circ (b \circ a \circ b \circ b) = b \circ a \circ b \circ b$$

Likewise, we now grab the two rightmost elements in the above string of products:

$$b \circ a \circ b \circ b = b \circ a \circ (b \circ b)$$

But we are told that $b \circ b = e$. Hence:

$$b \circ a \circ (b \circ b) = b \circ a \circ e$$

Since the group operation is associative, it follows that:

$$b \circ a \circ e = (b \circ a) \circ e = b \circ a$$

which shows that:

if in a group G for all of its elements x it is the case that $x \circ x = x^2 = e$ then such a group is necessarily commutative

and we are done. $\square$

The awesome penetrating power of the theory of groups should be dawning on our readers just about now: say, one fine day we are walking outside and then it happens. A group is falling onto us from the clear blue sky. Bam! Maybe that group fell out of theoretical physics, maybe that group fell out of crystallography or from chemistry. Who knows? We do not know and we do not care.

We look at the elements of that group, we see that every element of that group multiplies with itself to the group's identity and that is it - we claim that that group must be commutative and we did not even know anything about the underlying nature of its elements and its operation!

Amazing.

Exercise 11.67: say, we grab any strictly positive integer $n = 1, 2, 3, 4$ and so on and we only postulate that whatever such number our heart picks out, it is **finite**.

Affirm or refute the following statement: a group of order n , where n is any finite strictly positive integer, exists.

Hint 1: analyze the solution of **Exercise 11.40**.

Hint 2: if binary strings of length $n > 0$ form a group under the bit-wise XOR operation, then what can be said about the existence of the groups of order 2^n ?

Solution: again, in the daily life of grown-up mathematics the open-endedness of questions that these mathematicians work on is a given.

In this exercise we are asked to investigate a pretty sweeping, deep and exciting question.

As it is customary in grown-up mathematics, it is not even clear how can we attack such a question? From what end should we even begin chewing on it? What is the battle plan here?

Well, taking a cue from the first hint, we realize that in **Exercise 11.40** we proved that the finite binary strings of length $n > 0$ form a finite, Abelian, group under the XOR operation for any positive integer n .

Taking a cue from the second hint, we recall that while working on that exercise, we deduced that the order of the above group is 2^n .

In other words, by exhibiting a concrete example of a group of order 2^n , we actually showed that for any positive integer n a group of order 2^n exists.

By analogy, in this particular exercise, can we use some concrete and specific group as a vehicle that will allow us to conclude that yes, indeed, for any positive integer n a group of order n exists?

What could be such a group?

Well, the n -roots of unity come to mind, the complex numbers $z_{n,k}$ that are the roots of the following equation:

$$z^n - 1 = 0$$

For any positive integer n there are exactly n **distinct** such roots and the second index, k , in the notation above is used to enumerate all of those, as in $k = 0, 1, 2, 3, \dots, n-1$:

$$z_{n,0} = 1, z_{n,1} = \exp\left(\frac{2i\pi}{n}\right), z_{n,2} = \exp\left(\frac{4i\pi}{n}\right), \dots, z_{n,n-1} = \exp\left(\frac{2i(n-1)\pi}{n}\right)$$

Under what operation can such numbers form a group to begin with?

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We quickly rule out the addition of complex numbers as a candidate operation, because even though the addition defined on the set of complex numbers is an operation that is certainly associative, that operation is not binary in this particular case: the sum of, say, the second roots of unity, $1 + (-1) = 0$, takes us out of the parent set, a set that does not contain the element 0.

What about the operation of multiplication of complex numbers?

Under the operation of multiplication the things look promising: clearly, 1 will be the candidate identity of this particular group, since 1 is the multiplicative identity for all complex numbers, not just the n -th roots of unity.

Is the operation of multiplication of the n -th roots of unity associative?

It certainly is because the multiplication defined on the set of all complex numbers is associative.

Is the operation of multiplication of the n -th roots of unity binary?

Well, if a and b are two n -th roots of unity then it is the case that:

$$a^n = 1 \text{ and } b^n = 1$$

Therefore, for the product of two arbitrary n -th roots of unity we have:

$$(a \cdot b)^n = a^n \cdot b^n = 1 \cdot 1 = 1$$

Aha, yes, indeed, the operation of multiplication of the n -th roots of unity is binary, since the product of any two such roots is again an n -th root of unity.

What about the multiplicative inverses?

If a is an arbitrary n -th root of unity, then, being a non zero complex number. $a \neq 0$, the reciprocal complex number a^{-1} is always well-defined and is unique. But then:

$$[a^{-1}]^n = a^{-n} = \frac{1}{a^n} = \frac{1}{1} = 1$$

Thus, the multiplicative inverse a^{-1} of an n -th root of unity a is also an n -th root of unity that is unique for a .

Very nice. We have now shown that under the operation of multiplication of complex numbers the set of n -th roots of unity forms a finite, Abelian, group of order n .

But that fact implies that for every positive integer n a group of order n exist. $\square$

Take a good note of this tactical maneuver that is quite popular in grown-up mathematics. For such a maneuver to succeed we, relying on our creative juices, must fish out a successful candidate from the web of our personal knowledge, where the word “web” describes the corpus of our technical knowledge rather accurately: a collection of fuzzy and seemingly disparate nodes with various ways of traveling across them.

Exercise 11.68: what can be said about the solutions of the following, left-sided, **group equation**:

$$x \circ a = b \tag{54}$$

where x, a and b are the elements of a group G ?

Solution: from our **Take 5** discussion of various types of group axioms, we remember that the algebraic structure that we are studying, a group, can very well be defined with just a pair of axioms: the axiom of associativity of the group operation and the axiom of the existence of the solutions of the $a \circ x = b$ and $y \circ a = b$ group equations.

In the actual wording of the above second axiom we did not use that exact language since we demanded that for any two elements a, b of a set G it is always possible to find the elements x and y from the same set G such that $a \circ x = b$ and $y \circ a = b$.

This is perfectly fine because while constructing an axiom, we want to reduce the number of terms that must be understood a priori.

But that does not change the fact that what we are dealing with here is called a **group equation** or a construct in which the elements a and b of some group G are known and in which the element x of the same group G is sought-after.

We refer to such an equation as **left**-sided because the unknown element x in it is located to the left of the known element a .

If we take the above two group axioms as **the** group-formation axioms then the existence of the element x that satisfies (54) is already baked right into the heart and soul of a group and we only have to investigate the uniqueness of such an element.

If we take the more traditional three group axioms as the group-formation axioms then for the existence of a solution x we would, instead, rely on the existence of the unique inverses of the elements that participate in the formation of the given group.

Namely, since a is an element of a group G then a possesses a unique two-sided inverse a^{-1} .

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Let us multiply the equation (54) through by that inverse from the right:

$$x \circ a \circ a^{-1} = b \circ a^{-1}$$

Since the group operation $\circ$ is associative, it follows that we can evaluate the above LHS group product in any order we wish and we wish to multiply out its two rightmost elements first:

$$x \circ a \circ a^{-1} = x \circ (a \circ a^{-1}) = x \circ e$$

But by assumption, x must be some element of the group G . As such, that element, like all of its brothers and sisters, when acted upon by the identity, e , of its group, G , will remain the same:

$$x \circ e = e \circ x = x$$

That fact, in turn, yields:

$$x = b \circ a^{-1} = ba^{-1} \tag{55}$$

Since b is an element of a group G and a^{-1} is an element of the same group G , it follows, via the fact that the group operation is binary, that their product is also some element of the same group G and, hence, it exists:

$$ba^{-1} \in G$$

Now that the existence of the sought-after element x in (55) is established, we wonder if such an element is unique or if there are any other contenders that will happily solve the group equation in (54)?

In order to settle the matter one way of the other, we argue by contradiction: let us assume that there exists some other element $y \neq x$ in G such that $y \circ a = b$.

But, 1) being an element of G , the two-sided identity e of G does not move y anywhere, 2) the element a of G possesses a right-sided inverse a^{-1} and 3) the group operation is associative:

$$y = y \circ e = y \circ (a \circ a^{-1}) = (y \circ a) \circ a^{-1} = b \circ a^{-1}$$

Huh. Indeed, the element y , then, must be equal to the element x - contradiction! Therefore, the solution x of the group equation shown in (54) exists and is unique. $\square$

Note that all three popular group axioms were needed in the above proof.

Exercise 11.69: affirm or refute the following idea: if for the elements a, b, c of a group G it is the case that $b \circ a = c \circ a$ then it necessarily must be the case that $b = c$.

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Solution: well, by solving the previous exercise, we see that the above statement is certainly true because since a is an element of a group, G , then a possesses a unique two-sided inverse a^{-1} such that:

$$a \circ a^{-1} = a^{-1} \circ a = e$$

where e is the group's identity.

Consequently, we have the right to multiply the given group equation $ba = ca$ by the inverse of a from the right:

$$b \circ a \circ a^{-1} = c \circ a \circ a^{-1}$$

and use the fact that the group operation, $\circ$, is associative, meaning that we have the right to evaluate the above group products in any order that we wish and we wish to evaluate the above rightmost products first:

$$b \circ (a \circ a^{-1}) = c \circ (a \circ a^{-1})$$

which means that after carrying out such a multiplication, we shall arrive at the following equivalent equation:

$$b \circ e = c \circ e$$

But b and c are the elements of the same group, G . Hence, when acted upon by the identity, e , of that group, G , from either side, these elements remain unchanged:

$$b \circ e = b = c = c \circ e$$

Therefore, the idea given in this exercise is correct:

if a, b, c are the elements of a group G then from the fact that $b \circ a = c \circ a$ it follows that $b = c$

and that idea is otherwise known as the Group Right Cancellation Law or the Right Cancellation Law in a Group. $\square$

Note, again, that all three popular group axioms were needed in the above proof.

Exercise 11.70: what can be said about the solutions of the following, right-sided, group equation:

$$a \circ x = b \tag{56}$$

where x, a and b are the elements of a group G ?

Solution: of this exercise is left to the reader - generate your own line of reasoning that shows that the solution sought-after, x , always exist, is unique and is given by:

$$x = a^{-1} \circ b \quad (57)$$

Make sure the your reasoning matches the chosen set of group axioms. $\square$

Exercise 11.71: affirm or refute the following idea:

if for the elements a, b, c of a group G it is the case that $a \circ b = a \circ c$ then it necessarily must be the case that $b = c$

Solution: of this exercise if left to the reader - generate your own line of reasoning that affirms the idea that:

if for the elements a, b, c of a group G it is the case that $a \circ b = a \circ c$ then it necessarily must be the case that $b = c$

that is otherwise known as the Group Left Cancellation Law or the Left Cancellation Law in a Group. $\square$

Make sure that all three popular group axioms are needed in your proof.

Exercise 11.72: solve each of the following group equations by writing down the explicit shape of the unknown x in terms of the known elements.

1) $xab = c$; 2) $axb = c$; 3) $abx = c$; 4) $x^{-1} = abc$; 5) $a = bx^2$ if $x^3 = e$; 6) $x^3 = d$ if $x^4 = e$

Solution: of this exercise is left mostly to the reader.

Observe that if in the group equations shown in (54) and (56) we have the simplest, in some sense, constructs that can be modeled with Shrek's favorite analogy - a single-layer onion, in these group equations we have an onion that has more layers but, still, we know all the weaponry that we need in order to solve these equations - we simply peel all the layers in a given onion one-by-one by multiplying the equation at hand by a certain inverse from a certain side.

For example, in order to solve the given group equation 1), we multiply its both sides from the right by the inverse of b first:

$$xabb^{-1} = cb^{-1}$$

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justifying such a maneuver (why is it legal to do that?), stating the existence of the unique identity in a group, stating the associativity of the group operation and so on, we conclude that:

$$xa = cb^{-1}$$

and then, saying all the correct words that need to be said, we multiply the resultant intermediate group equation by the inverse of a from the right:

$$xaa^{-1} = cb^{-1}a^{-1}$$

to unmask the unique solution sought-after:

$$x = cb^{-1}a^{-1}$$

and so on.

Below, for a reference with a correct response, we list the answers for the remaining group equations: 2) $x = a^{-1}cb^{-1}$; 3) $x = b^{-1}a^{-1}c$; 4) $x = c^{-1}b^{-1}a^{-1}$; 5) $x = a^{-1}b$; 6) $x = d^{-1}$;

For the group equation 4) conduct two experiments:

- by multiplying both sides of the said equation by x from the right and
- by multiplying both sides of the said equation by x from the left

and then answer the following question: does the side from which the given group equation is multiplied by x affect the structure of the solution?

For the group equations 5) and 6) answer the following additional questions: do we have the same freedom to multiply these equations by x from either side as we had with the equation 4? Why? *\square*

Exercise 11.73: by answering the following questions on your own, develop and formulate a general hypothesis and then refute or affirm it.

Given a compound group element $(a_1 \circ a_2)$, find the structure of its inverse in terms of the known elements a_1 and a_2 .

Given a compound group element $(a_1 \circ a_2 \circ a_3)$, find the structure of its inverse in terms of the known elements a_1 , a_2 and a_3 .

Given a compound group element $(a_1 \circ a_2 \circ a_3 \circ a_4)$, find the structure of its inverse in terms of the known elements a_1 , a_2 , a_3 and a_4 .

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Given a compound group element $(a_1 \circ a_2 \circ a_3 \circ a_4 \circ a_5)$, find the structure of its inverse in terms of the known elements a_1, a_2, a_3, a_4 and a_5 .

And so on.

Solution: of this exercise should be more than straightforward - after we solved all the previous ones.

Since a_1 and a_2 are the elements of some group, G , it follows that their group product $a_1 \circ a_2$ not only belongs to the same group, G , but is an element of that group that is unique.

By the third group axiom, taken from the popular set of group axioms, that element, $(a_1 \circ a_2)$, also has a unique inverse x such that:

$$(a_1 \circ a_2) \circ x = x \circ (a_1 \circ a_2) = e \quad (58)$$

where e is the group's unique identity element.

How can we recover the explicit structure of the element x expressed in terms of the elements a_1 and a_2 ?

Well, we already know - all we have to do here is peel a finite number of onion layers, right?

For example, let us work with the right-sided group equation shown in (58):

$$(a_1 \circ a_2) \circ x = e \quad (59)$$

We know that we have the legal right to multiply the equation (59) through from the left by the inverse of the element a_1 because such a unique inverse exists:

$$a_1^{-1} \circ (a_1 \circ a_2) \circ x = a_1^{-1} \circ e$$

But in the theorem **GTT1** we already proved that precisely because the group operation is associative, we can move the single pair of parenthesis through a long group product in any which legal way.

In this case, we shift the above parenthesis leftward once on the left and we invoke the unique identity of a group property on the left of the last equation:

$$(a_1^{-1} \circ a_1) \circ a_2 \circ x = a_1^{-1} \circ e = a_1^{-1}$$

Next, we invoke the property of an inverse element:

$$a_1^{-1} \circ a_1 = e$$

on the LHS of the above equation:

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$$(e) \circ a_2 \circ x = a_1^{-1} \circ e = a_1^{-1}$$

and then we move the above parenthesis rightward twice:

$$e \circ (a_2 \circ x) = a_1^{-1} \circ e = a_1^{-1}$$

By assumption, x is an element of the same group, G . Hence, the product $a_2 \circ x$ is also a unique element of the same group. Since e is the, both-sided, identity of that group, it will leave the said product in-place:

$$a_2 \circ x = a_1^{-1}$$

Lastly, we repeat the same experiment with the element a_2 , pretty much, verbatim but we still encourage our readers to say all the correct words that need to be said in order to justify every step of the symbolic shuffling:

$$x = a_2^{-1} \cdot a_1^{-1} \tag{60}$$

It may seem like we are spending way too much energy to justify the obvious, but “obvious” is the biggest hurdle and source of grief in grown-up mathematics for nothing is obvious and everything needs to be proved and justified, especially in the theory of groups in which even the tiny-teeny symbolic shuffling has an enormous amount of power.

In any case, after we think that we have found the structure of the inverse sought-after, we verify it:

$$\begin{aligned} (a_1 \circ a_2) \circ x &= (a_1 \circ a_2) \circ (a_2^{-1} \cdot a_1^{-1}) = \\ &= a_1 \circ (a_2 \circ a_2^{-1}) \cdot a_1^{-1} = \\ &= a_1 \circ e \cdot a_1^{-1} = \\ &= a_1 \circ a_1^{-1} = e \end{aligned}$$

observing that x in this case matches the x in the equation (59) perfectly.

But we could have started working with the left-sided version of the equation shown in (58):

$$x \circ (a_1 \circ a_2) = e \tag{61}$$

Will this approach affect the structure of the inverse x found in (60)?

No, it will not.

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Because in this case we would first multiply (61) through from the right by the inverse of a_2 :

$$x \circ a_1 = a_2^{-1}$$

and then we would multiply the above intermediate group equation through from the right by the inverse of a_1 :

$$x = a_2^{-1} \circ a_1^{-1} \quad (62)$$

only to observe that the structure of the sought-after inverse x shown in (62) matches the structure of the inverse x shown in (60) perfectly.

We, thus, conclude that the structure of the inverse of the compound group element $(a_1 \circ a_2)$ depends solely on the structure of the said element, $(a_1 \circ a_2)$, and nothing else.

Now that we know the structure of the inverse of a two-element group product, we investigate the structure of the inverse of a three-element group product:

$$(a_1 \circ a_2 \circ a_3) \circ x = e = x \circ (a_1 \circ a_2 \circ a_3)$$

Here, however, we can reduce the amount of work that we have to do by a maneuver that naturally sets up a consequent proof by induction on n , since we already proved the validity of the $n = 2$ case and we want to prove validity of the $n + 1 = 2 + 1 = 3$ case.

>Commentary: pedantically, the $n = 1$ case is vacuously true because $[a_1]^{-1} = a_1^{-1}$.

Namely, we already claim that the structure of the sought-after inverse x in this case will be:

$$x = a_3^{-1} \circ a_2^{-1} \circ a_1^{-1} \quad (63)$$

and in order to show that that is indeed the case, we simply partition our 3-products into the previous case 2-products like so:

$$(A_2 \circ a_3) \circ x = e \quad \text{and} \quad x = a_3^{-1} \cdot A_2^{-1} \quad (64)$$

where we agree that:

$$A_2 = a_1 \cdot a_2 \quad \text{and} \quad A_2^{-1} = a_2^{-1} \circ a_1^{-1}$$

But that way what we have in (64) is the previous, 2-product, scenario, whose solution has been found and verified and we spell out that solution in the given parameters:

$$x = a_3^{-1} \circ A_2^{-1} = a_3^{-1} \circ (a_2^{-1} \circ a_1^{-1}) = a_3^{-1} \circ a_2^{-1} \circ a_1^{-1}$$

as claimed.

What just transpired here is following: using the validity of a result for n , we showed that the sought-after result holds for $n + 1$.

It should now be clear that the inverse x_4 of a four-product element will be given by:

$$x_4 = a_4^{-1} \circ a_3^{-1} \circ a_2^{-1} \circ a_1^{-1}$$

because such a product and its candidate inverse can be partitioned as follows:

$$A_3 \circ a_4 \quad \text{and} \quad x_4 = a_4^{-1} \circ A_3^{-1}$$

where the structure of A_3 is given by $a_1 \circ a_2 \circ a_3$ and the structure of its inverse A_3^{-1} has been proved to be $a_3^{-1} \circ a_2^{-1} \circ a_1^{-1}$.

The inverse x_5 of a five-product element will be given by:

$$x_5 = a_5^{-1} \circ a_4^{-1} \circ a_3^{-1} \circ a_2^{-1} \circ a_1^{-1}$$

and so on.

In general, then, given an n -product of n group elements a_1 through a_n :

$$a_1 \circ a_2 \circ \dots \circ a_n \tag{65}$$

its inverse x_n will be:

$$x_n = a_n^{-1} \circ a_{n-1}^{-1} \circ \dots \circ a_1^{-1} \tag{66}$$

for all positive integers n .

Moreover, if c is some element of the same group, then we arrive at the second law or rule of removing the parenthesis in a lengthy group product:

$$c(a_1 a_2 \dots a_{n-1} a_n)^{-1} = c a_n^{-1} a_{n-1}^{-1} \dots a_2^{-1} a_1^{-1} \tag{67}$$

which, basically, means that:

the inverse of a group product is the group product of the inverses taken in reverse order

□

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We would like to stress it one more time that all the generic symbolic shuffling in the theory of groups has to be done with utter care precisely because the theory of groups has such a humongous penetrating power precisely because it is such an abstract discipline that completely ignores the underlying nature of the group elements and of the group operation.

As such, all the results obtained through the said generic symbolic shuffling are universal.